Foundations of Algebraic Geometry

Alexandre Grothendieck

English Edition

Eight exposés, their errata, and the 1962 Comments

Contents

1 Exposé 149 — Duality Theorems for Coherent Algebraic Sheaves **1**
1. The Ext's of sheaves of modules ([2], Chapters 3 and 4) . . . 1
2. Composition law in Ext . . . 3
3. Results in local cohomology . . . 4
4. Cohomology class associated with a subvariety . . . 7
5. The duality theorem . . . 10
6. The duality theorem for singular varieties . . . 12
7. Poincaré duality . . . 14
8. Generalization of the duality theorem . . . 15

Erratum to Exposé no. 149 **20**

2 Exposé 182 — Formal Geometry and Algebraic Geometry **21**
1. Schemes . . . 21
2. Formal schemes . . . 22
3. The three fundamental theorems . . . 23
4. Application to the connectedness theorem and Zariski's Main Theorem . . . 25
5. Application to the cohomological study of proper flat morphisms . . . 26
6. Application to existence and uniqueness theorems for sheaves and schemes over a complete J-adic ring . . . 28
7. Application to the theory of moduli . . . 31
8. Application to the fundamental group . . . 32

Erratum to Exposé no. 182 **40**

3 Exposé 190 – Descent Technique I **41**
A. Preliminaries on categories . . . 42
1. Fibered categories and descent data . . . 42
2. Exact diagrams and strict epimorphisms . . . 44
3. Application to étale morphisms . . . 47
4. Relations with 1-cohomology . . . 48
B. Descent by faithfully flat morphisms . . . 52
1. Statement of the descent theorems . . . 52
2. Descent of properties of morphisms . . . 54
3. Descent by finite faithfully flat morphisms . . . 55
4. Application to rationality criteria . . . 57
5. Restricting the base in an abelian scheme . . . 58
6. Criteria for local triviality and isotriviality . . . 59

Erratum to Exposé no. 190 **61**

4 Exposé 195 – Descent Technique II **62**
A. Representable and pro-representable functors . . . 62
1. Representable functors . . . 62
2. Pro-representable functors and pro-objects . . . 63
3. Characterization of pro-representable functors . . . 65
4. Groups of Galois type and pro-algebraic groups . . . 66
5. Example: formal varieties . . . 66
B. The two existence theorems . . . 68
C. Applications to some special cases . . . 69
1. General remarks on functors represented by preschemes . . . 69
2. Hom, Pi, and Aut schemes . . . 70
3. Picard schemes . . . 72
4. Formal moduli of a variety . . . 73
5. Extension of coverings . . . 75

Erratum to Exposé no. 195 **78**

5 Exposé 212 – Quotient preschemes **79**
Introduction . . . 79
1. Equivalence relations and effective equivalence relations . . . 79
2. Example: preschemes finite over S . . . 81
3. The case of an operator group . . . 82
4. Pre-equivalence relations . . . 83
5. Quotient by a finite flat equivalence relation . . . 86
6. Quotient by a proper flat equivalence relation . . . 87
7. Applications . . . 89
8. A conjecture . . . 90
Bibliography . . . 91

Erratum to Exposé no. 212 **92**

6 Exposé 221 – Hilbert schemes **93**
Introduction . . . 93
1. Bounded families of sheaves: permanence properties . . . 93
2. Bounded families and the Hilbert polynomial . . . 96
3. Hilbert schemes: definition and existence theorem . . . 99
4. Variants . . . 104
5. Infinitesimal study of Hilbert schemes . . . 105
6. Relations with the norm and symmetric products . . . 108
7. Complements and questions . . . 109
Bibliography . . . 110
Addendum . . . 110

Erratum to Exposé no. 221 **111**

7 Exposé 232 – Picard schemes: existence theorems **112**
1. Relative Picard groups and functors . . . 112
2. Relations among the various relative and absolute Picard groups . . . 114
3. The main existence theorem: statement . . . 116
4. Relative Cartier divisors and projective bundles . . . 117
5. Proof of the main existence theorem . . . 119

6. Relative existence theorems . . . 121
Bibliography . . . 124

Erratum to Exposé no. 232 **125**

8 Exposé 236 – Picard schemes: general properties **126**
0. Complements to the preceding exposé ([3], V)$^{(1)}$ . . . 126
1. Topological properties of commutative group preschemes . . . 127
2. Application to local properties of Picard schemes . . . 132
3. The canonical abelian subscheme of $\mathrm{Pic}_{X/S}$, and the Albanese scheme . . . 135
4. The finiteness theorem for the Picard scheme . . . 139
Bibliography . . . 140

Erratum and supplement to Exposé no. 236 **141**

9 1962 Comments **144**
Notice . . . 144
Abbreviations used . . . 144

Chapter 1

Exposé 149 — Duality Theorems for Coherent Algebraic Sheaves

BOURBAKI SEMINAR
(May 1957)

DUALITY THEOREMS FOR COHERENT ALGEBRAIC SHEAVES

by Alexandre GROTHENDIECK

The results that follow, inspired by Serre's "algebraic duality theorem", were found in the winter of 1955 and the winter of 1956. They are established very simply with the aid of fairly elementary results on the cohomology of projective spaces [3], and through intensive use of the homological algebra of Cartan–Eilenberg, in the form given in [2].

1. The Ext's of sheaves of modules ([2], Chapters 3 and 4)

Let X be a topological space equipped with a sheaf $\underline{O}$ of rings with identity (not necessarily commutative). We consider the abelian category $\mathcal{C}^{\underline{O}}$ of sheaves of $\underline{O}$-modules, also called $\underline{O}$-Modules. Every object of this category is known to admit an injective resolution, which makes it possible to define in it the Ext functors having the familiar formal properties. More precisely, to avoid confusion, we denote by $\mathrm{Hom}_{\underline{O}}(X; A, B)$, or simply $\mathrm{Hom}(X; A, B)$, the abelian group of $\underline{O}$-homomorphisms from A to B, whereas $\underline{\mathrm{Hom}}_{\underline{O}}(A, B)$ will denote the sheaf of germs of homomorphisms from A to B ($A, B \in \mathcal{C}^{\underline{O}}$). For fixed $A \in \mathcal{C}^{\underline{O}}$, we define functors h_A and $\underline{h}_A$, taking values respectively in the category $\mathcal{C}$ of abelian groups and the category $\mathcal{C}^{\underline{\mathbf{Z}}}$ of sheaves of abelian groups on X, by the formulas

$$h_A(B) = \mathrm{Hom}_{\underline{O}}(X; A, B), \qquad \underline{h}_A(B) = \underline{\mathrm{Hom}}_{\underline{O}}(A, B). \tag{1.1}$$

The functors h_A and $\underline{h}_A$ are covariant left-exact functors. We consider their right derived functors, denoted respectively by $\mathrm{Ext}^p_{\underline{O}}(X; A, B)$ and $\underline{\mathrm{Ext}}^p_{\underline{O}}(A, B)$. Thus, by definition,

$$\begin{cases} \mathrm{Ext}^p_{\underline{O}}(X; A, B) = (R^p h_A)(B) = H^p(\mathrm{Hom}_{\underline{O}}(X; A, C(B))), \\ \underline{\mathrm{Ext}}^p_{\underline{O}}(A, B) = (R^p \underline{h}_A)(B) = H^p(\underline{\mathrm{Hom}}_{\underline{O}}(A, C(B))), \end{cases} \tag{1.2}$$

where the symbol R^p denotes passage to right derived functors, and where $C(B)$ denotes an arbitrary injective resolution of B in $\mathcal{C}^{\underline{O}}$. Let

$$\Gamma : \mathcal{C}^{\underline{\mathbf{Z}}} \longrightarrow \mathcal{C}$$

denote the "sections" functor. Recall that its right derived functors are denoted by

$$B \longmapsto H^p(X, B).$$

170

$$H^p(X, B) = (R^p\Gamma)(B) = H^p(\Gamma(C(B))). \tag{1.3}$$

Clearly $h_A = \Gamma \underline{h}_A$; on the other hand, one verifies that $\underline{h}_A$ takes injective objects to Γ-acyclic objects. The familiar conclusion is the following.

PROPOSITION 1. — For every $\underline{O}$-Module A, there exists a cohomological spectral functor on $\mathcal{C}^{\underline{O}}$ converging to the graded functor $(\mathrm{Ext}^*_{\underline{O}}(X; A, B))$, whose initial term is

$$E_2^{p,q}(A, B) = H^p(X, \underline{\mathrm{Ext}}^q_{\underline{O}}(A, B)). \tag{1.4}$$

From it one obtains edge homomorphisms and a five-term exact sequence, which we shall not write down.

COROLLARY 1. — If A is locally isomorphic to $\underline{O}^n$, then there are canonical isomorphisms

$$\mathrm{Ext}^p_{\underline{O}}(X; A, B) \xleftarrow{\sim} H^p(X, \underline{\mathrm{Hom}}_{\underline{O}}(A, B)) \tag{1.5}$$

(given by edge homomorphisms of the spectral sequence). In particular, there is a canonical isomorphism

$$\mathrm{Ext}^p_{\underline{O}}(X; \underline{O}, B) = H^p(X, B). \tag{1.6}$$

To use these results, one must be able to describe the $\underline{\mathrm{Ext}}^p_{\underline{O}}(A, B)$. These functors are computed locally: if U is an open subset of X, then

$$\underline{\mathrm{Ext}}^p_{\underline{O}}(A, B)|U = \underline{\mathrm{Ext}}^p_{\underline{O}|U}(A|U, B|U),$$

as follows from the fact that the restriction to U of an injective $\underline{O}$-Module is an injective $(\underline{O}|U)$-Module. Furthermore, for fixed $x \in X$, there are functorial homomorphisms

$$\underline{\mathrm{Hom}}_{\underline{O}}(A, B)_x \longrightarrow \mathrm{Hom}_{\underline{O}_x}(A_x, B_x), \tag{1.7}$$

which extend uniquely to a homomorphism of cohomological functors in B:

$$\underline{\mathrm{Ext}}^p_{\underline{O}}(A, B)_x \longrightarrow \mathrm{Ext}^p_{\underline{O}_x}(A_x, B_x). \tag{1.8}$$

PROPOSITION 2. — If, in a neighborhood of x, A is isomorphic to the cokernel of a homomorphism $\underline{O}^m \longrightarrow \underline{O}^n$, then (1.8) is an isomorphism for every p.[1] This holds in particular if A is a
171 coherent $\underline{O}$-Module [3].

PROPOSITION 3. — Let $L_* = (L_i)$ be a left resolution of the $\underline{O}$-Module A by $\underline{O}$-Modules, each locally isomorphic to some $\underline{O}^n$. Then $\underline{\mathrm{Ext}}^*_{\underline{O}}(A, B)$ identifies with $H^*(\underline{\mathrm{Hom}}_{\underline{O}}(L_*, B))$, and $\mathrm{Ext}^*_{\underline{O}}(X; A, B)$ identifies with the hypercohomology of X with respect to the complex $\underline{\mathrm{Hom}}_{\underline{O}}(L_*, B)$.

The proof is standard: consider the bicomplex $\underline{\mathrm{Hom}}_{\underline{O}}(L_*, C(B))$, where $C(B)$ is an injective resolution of B, and the natural homomorphisms from $\underline{\mathrm{Hom}}_{\underline{O}}(L_*, B)$ and $\underline{\mathrm{Hom}}_{\underline{O}}(A, C(B))$ into this bicomplex.

Finally, let us point out that the two Ext functors introduced in (1.2) are not only cohomological functors in B, but cohomological bifunctors, covariant in B and contravariant in A.

[1] The printed text reads "(1.7)"; the presence of p and the preceding sentence require "(1.8)."

2. Composition law in Ext

The results of this section are due independently to CARTIER and YONEDA; see an exposé by CARTIER [1] for details. Let $\mathcal{C}$ be an abelian category, and let K and L be two graded objects in $\mathcal{C}$. We denote by $\mathrm{Hom}(K, L)$ the graded abelian group whose component of degree n consists of the homogeneous homomorphisms of degree n from K to L (that is, systems (u_i) of homomorphisms $K^i \longrightarrow L^{i+n}$).

If K and L are complexes (with differential operators of degree $+1$, to fix ideas), we equip $\mathrm{Hom}(K, L)$ with the differential operator given by

$$\delta(u) = du + (-1)^{n+1}ud, \qquad \text{where } n = \deg(u), \tag{2.1}$$

making it a complex with differential of degree $+1$. The cycles of degree n are the operators of degree n that anticommute with d (as homogeneous operators).[2] We may therefore consider $H^*(\mathrm{Hom}(K, L))$; it is an invariant of the homotopy types of K and L, and may be denoted by $H^*(K, L)$.

Given a third complex M, composition of homomorphisms defines a pairing

$$\mathrm{Hom}(K, L) \times \mathrm{Hom}(L, M) \longrightarrow \mathrm{Hom}(K, M)$$

compatible with the differentials, hence, after passing to cohomology, pairings

$$H^*(K, L) \times H^*(L, M) \longrightarrow H^*(K, M), \tag{2.2}$$

denoted $(u, v) \longmapsto vu$. These pairings satisfy the evident associativity property. In particular, $H^*(K, K)$ is an associative graded ring with identity, and $H^*(K, L)$ (respectively $H^*(L, K)$) is a graded right (respectively left) module over this ring, and so on. In degree 0, (2.2) reduces to composition of morphisms of complexes. Finally, an exact sequence of complexes

$$0 \longrightarrow K' \longrightarrow K \longrightarrow K'' \longrightarrow 0,$$

such that K'^i is a direct summand of K^i for every i, gives rise to an exact sequence of complexes 172
of groups $\mathrm{Hom}(K'', L)$, and so on, and hence to a connecting operator

$$H^i(K', L) \longrightarrow H^{i+1}(K'', L).$$

The connecting operators associated with an exact sequence in L are defined similarly. The pairings (2.2) are compatible, in the usual sense, with these connecting operators.

Now suppose that $\mathcal{C}$ is a category in which every object A admits an injective resolution $C(A)$. One then finds, using one of the many variants of the bicomplex theorem, that

$$H^*(C(A), C(B)) = H^*(\mathrm{Hom}(C(A), C(B)))$$

is canonically isomorphic to

$$H^*(\mathrm{Hom}(A, C(B))) = \mathrm{Ext}^*(A, B).$$

The connecting operators considered above give the connecting homomorphisms for Ext. Moreover, the pairings (2.2) here give associative pairings compatible with the connecting homomorphisms:

$$\mathrm{Ext}^*(A, B) \times \mathrm{Ext}^*(B, C) \longrightarrow \mathrm{Ext}^*(A, C). \tag{2.3}$$

[2] The printed text reads "with u"; equation (2.1) requires "with d."

In particular, $\mathrm{Ext}^*(A, A)$ is an associative graded ring with identity, and so on. (One proves similarly that the Ext functors operate on the derived functors of an arbitrary functor; we shall not use this fact here.)

When the category in question is the category $\mathcal{C}^{\underline{O}}$ of $\underline{O}$-Modules on X, one therefore obtains pairings

$$\mathrm{Ext}^p_{\underline{O}}(X; A, B) \times \mathrm{Ext}^q_{\underline{O}}(X; B, C) \longrightarrow \mathrm{Ext}^{p+q}_{\underline{O}}(X; A, C), \tag{2.4}$$

which may be made explicit as indicated above. Moreover, the same developments, with the category of abelian groups replaced by the category of sheaves of abelian groups on X and the functors Hom replaced by the functors $\underline{\mathrm{Hom}}$, also define pairings having the same formal properties and, this time, of a "local nature":

$$\underline{\mathrm{Ext}}^p_{\underline{O}}(A, B) \times \underline{\mathrm{Ext}}^q_{\underline{O}}(B, C) \longrightarrow \underline{\mathrm{Ext}}^{p+q}_{\underline{O}}(A, C). \tag{2.5}$$

These are sharpened by observing that the homomorphisms (1.8) are compatible with the pairings
173 between the Ext groups.

Finally, recall that there is also a multiplicative structure among the functors $H^p(X, A)$ (the cup product). One then finds that the spectral sequences of Proposition 1 are compatible with the multiplicative structures. More precisely, there is a pairing of the spectral sequences $E(A, B)$ and $E(B, C)$ into the spectral sequence $E(A, C)$. At the abutment it reduces to the pairing between global Ext groups, while on the initial term it is the pairing deduced, in the right-hand side of (1.4), from the cup product and the pairings of local Ext groups. In particular, it follows that the edge homomorphisms

$$\mathrm{Ext}^n_{\underline{O}}(X; A, B) \longrightarrow H^0(X, \underline{\mathrm{Ext}}^n_{\underline{O}}(A, B)) \tag{2.6}$$

and

$$H^n(X, \underline{\mathrm{Hom}}_{\underline{O}}(A, B)) \longrightarrow \mathrm{Ext}^n_{\underline{O}}(X; A, B) \tag{2.7}$$

are compatible with the multiplicative structures. Thus, if one restricts to sheaves locally isomorphic to some $\underline{O}^m$, this completely describes composition between global Ext groups in terms of the cup product, in view of the isomorphism (1.5).

3. Results in local cohomology

Let A be a commutative ring with identity, equipped with an ideal J. For a varying A-module M, we shall define functorial homomorphisms

$$\begin{cases} \mathrm{Ext}^p_A(A/J, M) \longrightarrow \mathrm{Hom}_A\left(\bigwedge^p J/J^2, M \otimes A/J\right), \\ \mathrm{Tor}^A_p(A/J, M) \longleftarrow \left(\bigwedge^p J/J^2\right) \otimes \mathrm{Hom}_A(A/J, M), \end{cases} \tag{3.1}$$

where the tensor and exterior products are taken over the ring A. Notice also that J/J^2 is in fact an A/J-module, and that its exterior powers as an A-module or as an A/J-module are the same. Defining the homomorphisms (3.1) amounts to defining, for every system $\underline{x} = (x_1, \ldots, x_p)$ of elements of J, homomorphisms $\varphi_{\underline{x}}$:

$$\begin{cases} \varphi_{\underline{x}} : \mathrm{Ext}^p_A(A/J, M) \longrightarrow M \otimes A/J, \\ \varphi_{\underline{x}} : \mathrm{Hom}_A(A/J, M) \longrightarrow \mathrm{Tor}^A_p(A/J, M), \end{cases} \tag{3.2}$$

174 in such a way that the following conditions are satisfied:

i. $\varphi_{x_1,\dots,x_p}$ depends in an alternating A-multilinear fashion on the system of $x_i \in J$.

ii. $\varphi_{x_1,\dots,x_p}$ is zero whenever one of the x_i lies in J^2.

In fact, ii follows from i, since $a\varphi_{\underline{x}} = 0$ for $a \in J$, as is seen by observing that all the modules in (3.2) are annihilated by J.

To define $\varphi_{\underline{x}}$, consider the complex $K_{\underline{x}}$ whose underlying A-module is $\bigwedge A^p$ and whose differential is contraction $i_{\underline{x}}$ by $\underline{x}$, regarded as the linear form on A^p with components $x_1, \dots, x_p$. The differential has degree -1, the degrees are nonnegative, and the homology of this complex in dimension 0 is $A/(x_1A + \cdots + x_pA)$. Since the x_i belong to J, this gives an augmentation $K_{\underline{x},0} \longrightarrow A/J$. Thus $K_{\underline{x}}$ appears as an augmented free complex with augmentation module A/J. In the familiar way, one obtains homomorphisms

$$\mathrm{Ext}^*_A(H_0(K_{\underline{x}}), M) \longrightarrow H^*(\mathrm{Hom}_A(K_{\underline{x}}, M))$$

and

$$\mathrm{Tor}^A_*(H_0(K_{\underline{x}}), M) \longleftarrow H_*(K_{\underline{x}} \otimes_A M).$$

Composing these with the homomorphisms on Ext and Tor induced by the augmentation homomorphism $H_0(K_{\underline{x}}) \longrightarrow A/J$ gives homomorphisms

$$\begin{cases} \psi_{\underline{x}} : \mathrm{Ext}^*_A(A/J, M) \longrightarrow H^*(\mathrm{Hom}_A(K_{\underline{x}}, M)), \\ \psi_{\underline{x}} : \mathrm{Tor}^A_*(A/J, M) \longleftarrow H_*(K_{\underline{x}} \otimes_A M). \end{cases} \tag{3.3}$$

One sees immediately that in the top dimension p, the homology of the right-hand sides is $M/(x_1M + \cdots + x_pM)$ (respectively, the set of elements of M annihilated by every x_i). Since the x_i lie in J, one obtains homomorphisms

$$\begin{cases} H^p(\mathrm{Hom}_A(K_{\underline{x}}, M)) \longrightarrow M \otimes A/J, \\ \qquad H_p(K_{\underline{x}} \otimes_A M) \longleftarrow \mathrm{Hom}_A(A/J, M). \end{cases} \tag{3.4}$$

Composing the homomorphisms (3.3) and (3.4) gives the homomorphisms (3.2) that we wished to define. Verifying i is tedious but presents no difficulty. 175

PROPOSITION 4. — Let A be a commutative ring with identity, and let $(x_1, \dots, x_p)$ be a sequence of elements of A such that, for $1 \leq i \leq p$, the image of x_i in the quotient of A by the ideal generated by $(x_1, \dots, x_{i-1})$ is not a zero divisor. Let J be the ideal generated by the x_i. Then J/J^2 is a free (A/J)-module with basis the canonical images of the x_i, the complex $K_{\underline{x}}$ is a free resolution of A/J, and for every A-module M, the homomorphisms (3.1) in degree p are bijective. The same is true of the analogous homomorphisms defined in arbitrary degrees i, provided that $JM = 0$.

(The essential point, from which all the others follow, is the acyclicity of $K_{\underline{x}}$, a well-known fact under the stated hypotheses.)

COROLLARY 1. — With A and J as above, suppose in addition that A is a regular affine algebra of dimension n over a perfect field k, and that A/J is a regular affine algebra. Denote by $\Omega^1(A)$ and $\Omega^1(A/J)$ the modules of Kähler differentials. Then there is a canonical isomorphism compatible with localization,

$$\mathrm{Ext}^p_A(\Omega^{n-p}(A/J), \Omega^n(A)) = A/J. \tag{3.5}$$

Indeed, $\Omega^{n-p}(A/J)$ is a free (A/J)-module of rank 1, and likewise $\Omega^n(A)$ is a free A-module of rank 1.[3] Thus the left-hand side equals

$$\mathrm{Ext}^p_A(A/J, A) \otimes \Omega^{n-p}(A/J)' \otimes \Omega^n(A)$$

[3]The printed text says "rank n." Both modules of top-degree forms have rank 1, as is also required by the identification with $\bigwedge^p(J/J^2)$ below.

(where the prime denotes the dual (A/J)-module). The tensor product of the last two factors identifies with $\bigwedge^p(J/J^2)$, so the whole expression identifies with

$$\operatorname{Ext}_A^p\left(A/J, \bigwedge^p(J/J^2)\right),$$

and therefore, by the proposition, with

$$\operatorname{Hom}_A\left(\bigwedge^p J/J^2, \bigwedge^p J/J^2\right),$$

that is, with A/J. In particular, $\operatorname{Ext}_A^p(\Omega^{n-p}(A/J), \Omega^n(A))$ contains a distinguished element corresponding to the identity element of A/J, called the *fundamental class of the ideal J in A*. (It can in fact be defined under considerably broader hypotheses.) Corollary 1 can be put in a more geometric and more global form:

COROLLARY 2. — Let X be a nonsingular variety over an algebraically closed field k, let Y be a closed nonsingular subvariety of X, let $\underline{O}_X$ be the sheaf of local rings of X, and let $\underline{O}_Y$ be the sheaf of local rings of Y, regarded as a quotient sheaf of $\underline{O}_X$. Let n be the dimension of X and $n-p$ that
176 of Y.

Let Ω_X (respectively Ω_Y) be the sheaf of germs of regular differential forms on X (respectively Y). Then there are canonical isomorphisms

$$\underline{\operatorname{Ext}}^p_{\underline{O}_X}(\Omega_Y^{n-p}, \Omega_X^n) = \underline{O}_Y, \tag{3.6}$$

or equivalently

$$\underline{\operatorname{Ext}}^p_{\underline{O}_X}(\underline{O}_Y, \Omega_X^n) = \Omega_Y^{n-p}. \tag{3.6 bis}$$

Formula (3.6 bis) can serve as the definition of Ω_Y^{n-p} when Y is a singular variety. More precisely:

PROPOSITION 5. — Let Y be an algebraic subset of dimension $q = n - p$ in a nonsingular algebraic variety X of dimension n. Let F be a coherent algebraic sheaf on X whose support is contained in Y, and let L be a locally free algebraic sheaf on X. Then the sheaves $\underline{\operatorname{Ext}}^i_{\underline{O}_X}(F, L)$ vanish for $i < p$, while for $i = p$ there is a canonical isomorphism

$$\underline{\operatorname{Ext}}^p_{\underline{O}_X}(F, L) = \underline{\operatorname{Hom}}_{\underline{O}_X}\left(F, \underline{\operatorname{Ext}}^p_{\underline{O}_X}(\underline{O}_X/J, L)\right), \tag{3.7}$$

where J denotes an arbitrary sheaf of ideals on X that annihilates F and whose zero set is Y. In particular, if F is a coherent algebraic sheaf on Y, then

$$\underline{\operatorname{Ext}}^p_{\underline{O}_X}(F, L) = \underline{\operatorname{Hom}}_{\underline{O}_Y}\left(F, \underline{\operatorname{Ext}}^p_{\underline{O}_X}(\underline{O}_Y, L)\right). \tag{3.7 bis}$$

Finally, still taking F to be a coherent algebraic sheaf on Y, the sheaves

$$E^i(F) = \underline{\operatorname{Ext}}^{p+i}_{\underline{O}_X}(F, \Omega_X^n)$$

do not depend on the choice of embedding of the algebraic space Y into a nonsingular variety X.

Since the question is local, we may suppose that X is affine and $L = \underline{O}_X$. This reduces matters to a question in commutative algebra, indeed in local algebra: if A is a regular local ring and M an A-module whose support has dimension $\leq q = n - p$, one must prove that $\operatorname{Ext}_A^i(M, A) = 0$ for $i < p$, and that

$$\operatorname{Ext}_A^p(M, A) = \operatorname{Hom}_A(M, \operatorname{Ext}_A^p(A/J, A)),$$

where J is any ideal of "dimension" $\leq q$ that annihilates M. For the first assertion, argue by induction on q: an immediate dévissage reduces to the case $M = A/J$; replacing J by a smaller ideal, using the induction hypothesis and the exact sequence of Ext groups, then reduces to the case where J is generated by a "system of parameters," as in Proposition 4, where the result is
177 immediate. The preceding result implies that if J is an ideal

fixed of "dimension" $\leq q$, then the contravariant functor $E(M) = \mathrm{Ext}^p_A(M, A)$ on the category of A/J-modules is left exact; moreover, it takes direct sums to direct products, from which it follows readily that

$$E(M) = \mathrm{Hom}_A(M, E(A/J)).$$

[4] Finally, the last assertion of Proposition 5 is subtler and follows from an intrinsic characterization of the $E^i(F)$ by means of a local duality theorem that cannot be given here.

COROLLARY. — Denote by ω_Y^q the sheaf $\underline{\mathrm{Ext}}^p_{\underline{O}_X}(\underline{O}_Y, \Omega^n_X)$. Then, for coherent algebraic sheaves F on Y, there is a functorial isomorphism

$$\underline{\mathrm{Ext}}^p_{\underline{O}_X}(F, \Omega^n_X) = \underline{\mathrm{Hom}}_{\underline{O}_X}(F, \omega^q_Y). \tag{3.8}$$

4. Cohomology class associated with a subvariety

Throughout what follows, X denotes an algebraic set of dimension n, defined over a field k, which we suppose algebraically closed for simplicity. Except in Section 6, X is assumed nonsingular. We denote by $\underline{O}_X$ the sheaf of local rings on X, and by

$$\Omega^*_X = \bigoplus_p \Omega^p_X$$

the sheaf of germs of differential forms on X.[5] If Y is a closed subset of X, coherent algebraic sheaves on Y are identified with coherent sheaves on X that vanish off Y; this applies in particular to $\underline{O}_Y$ and Ω_Y.

LEMMA 1. — Let F be a coherent algebraic sheaf on X whose support has dimension $\leq n - p$, and let L be a locally free coherent algebraic sheaf on X. Then $\mathrm{Ext}^i_{\underline{O}_X}(X; F, L)$ vanishes for $i < p$, and there is a canonical isomorphism

$$\mathrm{Ext}^p_{\underline{O}_X}(X; F, L) = H^0(X, \underline{\mathrm{Ext}}^p_{\underline{O}_X}(F, L)). \tag{4.1}$$

If F is a coherent algebraic sheaf on a closed subset Y of X of dimension $\leq n - p$, there is a canonical isomorphism

$$\underline{\mathrm{Ext}}^p_{\underline{O}_X}(F, L) = \underline{\mathrm{Hom}}_{\underline{O}_X}(F \otimes L' \otimes \Omega^n_X, \omega^{n-p}_Y)\,, \tag{4.1 bis}$$

where ω^{n-p}_Y is the sheaf on Y defined in the corollary to Proposition 5 (and identifies with Ω^{n-p}_Y if Y is nonsingular).

Formula (4.1) is an immediate consequence of the spectral sequence of Proposition 1 and Proposition 5; by formula (3.8), one may write 178

$$\begin{aligned}
\underline{\mathrm{Ext}}^p_{\underline{O}_X}(F, L) &= L \otimes (\Omega^n_X)' \otimes \underline{\mathrm{Ext}}^p_{\underline{O}_X}(F, \Omega^n_X) \\
&= L \otimes (\Omega^n_X)' \otimes \underline{\mathrm{Hom}}_{\underline{O}_X}(F, \omega^q_Y) \\
&= \underline{\mathrm{Hom}}_{\underline{O}_X}(F \otimes L' \otimes \Omega^n_X, \omega^q_Y)\,,
\end{aligned}$$

[4] The printed text has $E(A)$. In the category under consideration the representing object is A/J, and the equality needed for (3.7) is $E(A/J)$.

[5] The printed text uses a union sign; the graded structure is the direct sum of the sheaves Ω^p_X.

where $q = n - p$, which gives formula (4.1 bis).

In particular, take $F = \underline{O}_Y$ and $L = \Omega^p_X$. Since $\Omega^n_X \otimes (\Omega^p_X)' = \Omega^{n-p}_X$, one obtains a canonical isomorphism

$$\mathrm{Ext}^p_{\underline{O}_X}(X; \underline{O}_Y, \Omega^p_X) = \mathrm{Hom}_{\underline{O}_X}(X; \Omega^{n-p}_X, \omega^{n-p}_Y). \tag{4.2}$$

Now suppose for simplicity that Y is nonsingular, so that $\omega^{n-p}_Y = \Omega^{n-p}_Y$. There is a natural homomorphism from Ω^{n-p}_X to Ω^{n-p}_Y, hence a canonical section s_Y of the sheaf $\underline{\mathrm{Ext}}^p_{\underline{O}_X}(\underline{O}_Y, \Omega^p_X)$. If all components of Y have dimension $n-p$, we call it the *fundamental section* of the sheaf $\underline{\mathrm{Ext}}^p_{\underline{O}_X}(\underline{O}_Y, \Omega^p_X)$. By (4.1), this section defines an element of $\mathrm{Ext}^p_{\underline{O}_X}(X; \underline{O}_Y, \Omega^p_X)$. The natural homomorphism $\underline{O}_X \longrightarrow \underline{O}_Y$ in turn defines a homomorphism

$$\mathrm{Ext}^p_{\underline{O}_X}(X; \underline{O}_Y, \Omega^p_X) \longrightarrow \mathrm{Ext}^p_{\underline{O}_X}(X; \underline{O}_X, \Omega^p_X) = H^p(X, \Omega^p_X).$$

Thus one obtains an element of $H^p(X, \Omega^p_X)$, denoted $P_X(Y)$ and called the *cohomology class of* Y *in* X. It is therefore obtained from the section s_Y of $\underline{\mathrm{Ext}}^p_{\underline{O}_X}(\underline{O}_Y, \Omega^p_X)$ through the following diagram of homomorphisms:

$$\begin{aligned} H^p(X, \Omega^p_X) &= \mathrm{Ext}^p_{\underline{O}_X}(X; \underline{O}_X, \Omega^p_X) \\ &\longleftarrow \mathrm{Ext}^p_{\underline{O}_X}(X; \underline{O}_Y, \Omega^p_X) \xrightarrow{\sim} H^0\Big(X, \underline{\mathrm{Ext}}^p_{\underline{O}_X}(\underline{O}_Y, \Omega^p_X)\Big). \end{aligned} \tag{4.3}$$

Call a *nonsingular cycle of dimension* $n - p$ an element of the free abelian group generated by the irreducible nonsingular subvarieties of dimension $n - p$ in X. Then the function $Y \longmapsto P(Y)$ extends to a homomorphism from the group of nonsingular cycles of dimension $n-p$ on X to the group $H^p(X, \Omega^p_X)$.

Let Z^{n-p} and $Z'^{n-p'}$ be nonsingular cycles of dimensions $n - p$ and $n - p'$. They are said to *intersect transversely* if every component of Z intersects every component of Z' transversely. The cycle $Z.Z'$ is then defined; it is a nonsingular cycle of dimension $n-p-p'$. With this understood,
179 we have the

THEOREM 1. — If Z^{n-p} and $Z'^{n-p'}$ are nonsingular cycles that intersect transversely, then

$$P_X(Z.Z') = P_X(Z).P_X(Z'), \tag{4.4}$$

where the product on the right is the cup product

$$H^p(X, \Omega^p_X) \times H^{p'}(X, \Omega^{p'}_X) \longrightarrow H^{p+p'}(X, \Omega^{p+p'}_X).$$

(We assume that X is isomorphic to a locally closed subset of a projective space.)

This last hypothesis is used only to conclude that every coherent algebraic sheaf on X is a quotient of a locally free coherent algebraic sheaf (SERRE), and hence admits a left resolution by locally free sheaves. To prove Theorem 1, we may suppose that Z and Z' are irreducible nonsingular subvarieties Y and Y' meeting transversely. Let L_* be a left resolution of $\underline{O}_Y$ by locally free sheaves. By Proposition 3, the diagram of homomorphisms (4.3) then identifies with

$$R^p\Gamma(\Omega^p_X) \xleftarrow{\alpha} \mathbb{R}^p\Gamma\Big(\underline{\mathrm{Hom}}_{\underline{O}_X}(L_*, \Omega^p_X)\Big) \xrightarrow[\sim]{\beta} \Gamma\Big(H^p(\underline{\mathrm{Hom}}_{\underline{O}_X}(L_*, \Omega^p_X))\Big).$$

Here Γ is the "group of sections" functor on the category of sheaves of abelian groups on X, $\mathbb{R}^p\Gamma$ denotes its hypercohomology in dimension p, and $R^p\Gamma$ its p-th derived functor. For simplicity, we may suppose that $L_0 = \underline{O}_X$ and that the augmentation $L_0 \longrightarrow \underline{O}_Y$ is the natural homomorphism (which is permissible). Then α is induced by the homomorphism of complexes $\underline{O}_X \longrightarrow L_*$ (with $\underline{O}_X$ regarded as a complex concentrated in degree 0), taking account of the equality $\mathbb{R}^p\Gamma(K) =$

$R^p\Gamma(K_0)$ when K is a complex of sheaves concentrated in degree 0. The homomorphism β is the familiar edge homomorphism. Consider the analogous diagram for a locally free resolution L'_* of $\underline{O}_{Y'}$, and then the following commutative diagram of pairings:

$$\begin{array}{ccccc} R^p\Gamma(\Omega_X^p) & \longleftarrow \mathbb{R}^p\Gamma\Big(\underline{\mathrm{Hom}}_{\underline{O}_X}(L_*, \Omega_X^p)\Big) & \xrightarrow{\sim} \Gamma H^p\Big(\underline{\mathrm{Hom}}_{\underline{O}_X}(L_*, \Omega_X^p)\Big) \\ \times & \times & \times \\ R^{p'}\Gamma(\Omega_X^{p'}) & \longleftarrow \mathbb{R}^{p'}\Gamma\Big(\underline{\mathrm{Hom}}_{\underline{O}_X}(L'_*, \Omega_X^{p'})\Big) & \xrightarrow{\sim} \Gamma H^{p'}\Big(\underline{\mathrm{Hom}}_{\underline{O}_X}(L'_*, \Omega_X^{p'})\Big) \\ \downarrow & \downarrow & \downarrow \\ R^{p+p'}\Gamma(\Omega_X^{p+p'}) & \longleftarrow \mathbb{R}^{p+p'}\Gamma\Big(\underline{\mathrm{Hom}}_{\underline{O}_X}(L_* \otimes L'_*, \Omega_X^{p+p'})\Big) & \xrightarrow{\sim} \Gamma H^{p+p'}\Big(\underline{\mathrm{Hom}}_{\underline{O}_X}(L_* \otimes L'_*, \Omega_X^{p+p'})\Big). \end{array} \tag{4.5}$$

180

The pairings in the two right-hand columns are induced by the pairing of complexes of sheaves

$$\underline{\mathrm{Hom}}_{\underline{O}_X}(L_*, \Omega_X^p) \times \underline{\mathrm{Hom}}_{\underline{O}_X}(L'_*, \Omega_X^{p'}) \longrightarrow \underline{\mathrm{Hom}}_{\underline{O}_X}(L_* \otimes L'_*, \Omega_X^{p+p'}),$$

defined using the exterior product $\Omega_X^p \times \Omega_X^{p'} \longrightarrow \Omega_X^{p+p'}$. The pairing in the first column is the cup product (relative to the exterior product). I claim that the last row of (4.5) identifies with the diagram of isomorphisms analogous to (4.3), with Y replaced by $Y \cap Y'$ and p by $p + p'$. For this it suffices to show that $L \otimes L'$ is an (evidently locally free) resolution of $\underline{O}_{Y \cap Y'}$. Indeed,

$$H_0(L \otimes L') = \underline{O}_Y \otimes \underline{O}_{Y'} = \underline{O}_{Y \cap Y'}$$

and

$$H_i(L \otimes L') = \mathrm{Tor}_i^{\underline{O}_X}(\underline{O}_Y, \underline{O}_{Y'}) = 0 \qquad (i > 0),$$

because Y and Y' meet transversely. Theorem 1 now follows from the formula

$$s_Y . s_{Y'} = s_{Y.Y'} \tag{4.6}$$

(where the product on the left is that of the last column of (4.5)). Formula (4.6), being purely local, is verified without difficulty by taking for L_* and L'_* the resolutions considered in Proposition 4. One proves similarly (and more easily) that $Z \longmapsto P_X(Z)$ is compatible with Cartesian products:

$$P_{X \times X'}(Z \times Z') = P_X(Z) \otimes P_{X'}(Z') \tag{4.7}$$

(this formula is valid when Z, respectively Z', is a nonsingular cycle on the nonsingular variety X, respectively X', with $Z \times Z'$ regarded as a nonsingular cycle on $X \times X'$). From (4.4) and (4.7), it follows that $P_X(Z)$ is also compatible with inverse image under a morphism of nonsingular varieties $f : X \longrightarrow X'$:

$$P_X(f^{-1}(Z')) = f^*(P_{X'}(Z')), \tag{4.8}$$

provided Z' is a nonsingular cycle on X' to which f is "transverse," that is, the graph of f is transverse to the cycle $X \times Z'$ in $X \times X'$.

181

COROLLARY 1. — Let X and X' be two nonsingular locally closed varieties in a projective space, with X' complete. Let U be a nonsingular cycle on $X \times X'$, and let a and b be two points of X' such that U meets the cycles $X \times (a)$ and $X \times (b)$ transversely. Let Z and Z' be the nonsingular cycles on X such that

$$Z \times (a) = (X \times (a)).U, \qquad Z' \times (b) = (X \times (b)).U.$$

Then

$$P_X(Z) = P_X(Z').$$

Indeed, let $f_a : X \longrightarrow X \times X'$ be defined by $f_a(x) = (x, a)$. Formula (4.8) gives $P(Z) = f_a^*(P(U))$, and similarly $P(Z') = f_b^*(P(U))$. Using the Künneth formula

$$H^*(X \times X', \Omega^*_{X\times X'}) = H^*(X, \Omega^*_X) \otimes H^*(X', \Omega^*_{X'})$$

and the fact that $H^0(X', \underline{O}_{X'})$ consists only of scalars, one readily obtains $f_a^* = f_b^*$, proving the assertion.

For every $x \in X$, the point (x) is a nonsingular subvariety of X of codimension n, and therefore defines an element ε_x of $H^n(X, \Omega^n_X)$. If X is a nonsingular projective variety, Corollary 1 shows that ε_x does not depend on the chosen point x; it is denoted ε_X and called the *fundamental class* of $H^n(X, \Omega^n_X)$.

REMARK. — To obtain a satisfactory theory, one would have to define $P_X(Z)$ for arbitrary cycles Z and prove Theorem 1 for a proper intersection of cycles. (At the time this exposé was written, this had not yet been done in full generality.) Assuming this has been done, Corollary 1 becomes: if Z and Z' are two algebraically equivalent cycles, then $P_X(Z) = P_X(Z')$ (a statement that does not seem to follow from the preceding results, even when Z and Z' are nonsingular).

5. The duality theorem

In this section, X denotes a nonsingular projective variety of dimension n.

THEOREM 2. — The fundamental class ε_X of $H^n(X, \Omega^n_X)$ is a basis of this vector space.

(The proof will be given below.) By the preceding theorem, we may therefore identify $H^n(X, \Omega^n_X)$ with the field k. Now consider the pairings introduced in Section 2, which in particu-
182 lar give a pairing

$$\mathrm{Ext}^p_{\underline{O}_X}(X; \underline{O}_X, F) \times \mathrm{Ext}^{n-p}_{\underline{O}_X}(X; F, \Omega^n_X) \longrightarrow \mathrm{Ext}^n_{\underline{O}_X}(X; \underline{O}_X, \Omega^n_X),$$

that is,

$$H^p(X, F) \times \mathrm{Ext}^{n-p}_{\underline{O}_X}(X; F, \Omega^n_X) \longrightarrow H^n(X, \Omega^n_X). \tag{5.1}$$

In view of Theorem 2, this pairing defines a homomorphism

$$\mathrm{Ext}^{n-p}_{\underline{O}_X}(X; F, \Omega^n_X) \longrightarrow \big(H^p(X, F)\big)'. \tag{5.2}$$

This homomorphism is functorial in F and commutes with the connecting homomorphisms associated with exact sequences $0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$.

THEOREM 3. — The homomorphism (5.2) is an isomorphism.

In particular, this recovers Serre's result:

COROLLARY. — Let E be an algebraic vector bundle on X, and let $\underline{O}_X(E)$ be the sheaf of germs of regular sections of E. Then there are canonical isomorphisms

$$\big(H^p(X, \underline{O}_X(E))\big)' = H^{n-p}(X, \Omega^n_X \otimes \underline{O}_X(E')). \tag{5.3}$$

It suffices to apply Theorem 3 and Corollary 1 to Proposition 1.

Theorems 2 and 3 will follow from the following assertion:

(D) The following homomorphism

$$\mathrm{Ext}^{n-p}_{\underline{O}_X}(X; F, \Omega^n_X) \longrightarrow \big(H^p(X, F)\big)' \otimes L, \qquad L = H^n(X, \Omega^n_X), \tag{5.2 bis}$$

induced by the pairing (5.1), is an isomorphism.

Let us first show that (D) implies Theorem 2. Let $k_x = \underline{O}_{(x)}$ be the sheaf of local rings of the variety reduced to the point $x \in X$. Consider the canonical homomorphism $\underline{O}_X \longrightarrow k_x$ and the associated homomorphism

$$H^0(X, \underline{O}_X) \longrightarrow H^0(X, k_x). \tag{5.3 bis}$$

Its transpose identifies with the homomorphism

$$\mathrm{Ext}^n_{\underline{O}_X}(X; k_x, \Omega^n_X) \otimes L' \longrightarrow \mathrm{Ext}^n_{\underline{O}_X}(X; \underline{O}_X, \Omega^n_X) \otimes L' \tag{5.4}$$

induced by the homomorphism on Ext^n associated with $\underline{O}_X \longrightarrow k_x$:

$$\mathrm{Ext}^n_{\underline{O}_X}(X; k_x, \Omega^n_X) \longrightarrow \mathrm{Ext}^n_{\underline{O}_X}(X; \underline{O}_X, \Omega^n_X) = H^n(X, \Omega^n_X). \tag{5.5}$$

183

Since (5.3 bis) is an isomorphism,[6] so is (5.4), and hence so is (5.5). Since $s_{(x)}$ is a basis of $\mathrm{Ext}^n_{\underline{O}_X}(X; k_x, \Omega^n_X)$ by (4.2), it follows that its image ε_X is a basis of $H^n(X, \Omega^n_X)$.

It remains to prove assertion (D), which will follow in a purely formal way from the elementary facts summarized in the following lemmas. Here X is assumed to be a closed subset (singular or not) of projective space P of dimension r. We use $\underline{O}_P(m)$ for the sheaf on P denoted $\underline{O}(m)$ in [3], and $\underline{O}_X(m)$ for the analogous sheaf on X.

LEMMA 2. — Assertion (D) is true if $X = P$ and $F = \underline{O}_P(m)$.

This lemma is verified by a direct calculation. The explicit calculation of $H^i(P, \underline{O}_P(m))$ is given in [3], but it can be carried out more elementarily. The computation of the cup product

$$H^i(P, \underline{O}_P(m)) \times H^j(P, \underline{O}_P(m')) \longrightarrow H^{i+j}(P, \underline{O}_P(m + m'))$$

needed to make the pairing (5.1) explicit presents no difficulty.[7]

LEMMA 3. — Every coherent algebraic sheaf F on X is isomorphic to a quotient sheaf of a sheaf $\underline{O}_X(-m)^k$, where m may be taken as large as desired.

This follows from the fact that $F \otimes \underline{O}_X(m)$ is "generated by its sections" for m large; cf. [3].

LEMMA 4. — Let $i > 0$. Then $H^{r-i}(P, \underline{O}_P(-m)) = 0$ for m large; and for every coherent algebraic sheaf B on X, one has

$$\mathrm{Ext}^i_{\underline{O}_X}(X; \underline{O}_X(-m), B) = 0$$

for m large.

The first assertion follows from the explicit calculations mentioned above. For the second, note that there is an isomorphism

$$\mathrm{Ext}^i_{\underline{O}_X}(X; \underline{O}_X(-m), B) = H^i(X, B \otimes \underline{O}_X(m))$$

(Proposition 1, Corollary 1), whence the conclusion by a well-known result of [3]. Combining the two preceding lemmas, we obtain the

COROLLARY. — Let $i > 0$. Then the functor $F \longmapsto H^{r-i}(P, F)$ on the category of coherent algebraic sheaves on P is coeffaceable, and so is the functor $\mathrm{Ext}^i_{\underline{O}_X}(X; F, B)$ on the category of coherent algebraic sheaves on X.

184

LEMMA 5. — Let A and B be two coherent algebraic sheaves on X, and set $A(m) = A \otimes \underline{O}_X(m)$. Then, for m sufficiently large, the canonical homomorphism

$$\mathrm{Ext}^1_{\underline{O}_X}(X; A(-m), B) \longrightarrow H^0(X, \underline{\mathrm{Ext}}^1_{\underline{O}_X}(A(-m), B)) = H^0(X, \underline{\mathrm{Ext}}^1_{\underline{O}_X}(A, B)(m))$$

[6] The printed text gives a second formula "(5.3)" on page 182 and refers to it here; the identifier "(5.3 bis)" distinguishes this formula from the first (5.3).

[7] The printed text has $\underline{O}_P(r + r')$; the two factors $\underline{O}_P(m)$ and $\underline{O}_P(m')$ require $\underline{O}_P(m + m')$.

is an isomorphism.

This follows at once from the spectral sequence of Proposition 1 applied to $A(-m)$ and B, since one then has

$$E_2^{p,q}(A(-m),B) = H^p(X, \underline{\mathrm{Ext}}^q_{\underline{O}_X}(A(-m),B)) = H^p(X, \underline{\mathrm{Ext}}^q_{\underline{O}_X}(A,B)(m)),$$

which vanishes for $p > 0$ and m large.

We now prove (D) in the case $X = P$. First we prove that (5.2 bis) is an isomorphism for $p = n$. Both sides are then left-exact functors (since $H^{r+1}(P,F) = 0$), so Lemma 3 shows that it suffices to prove the assertion for $F = \underline{O}_P(-m)$; in that case it is contained in Lemma 2. The homomorphisms (5.2 bis) are functorial and compatible with the connecting homomorphisms, and for $p < n$ its two sides are coeffaceable functors of F (Corollary to Lemma 4). A standard argument therefore shows that (5.2 bis) is an isomorphism for every p. This proves the duality theorem for projective space. Now let X be arbitrary but nonsingular. By the duality theorem for P, there is an isomorphism

$$H^n(X,F)' = H^n(P,F)' = \mathrm{Ext}^{r-n}_{\underline{O}_P}(P; F, \Omega^r_P).$$

By Lemma 1 (no. 4), the last term identifies with

$$\mathrm{Hom}_{\underline{O}_P}(P; F, \omega^n_X) = \mathrm{Hom}_{\underline{O}_X}(X; F, \Omega^n_X) = \mathrm{Ext}^0_{\underline{O}_X}(X; F, \Omega^n_X),$$

and hence there is an isomorphism

$$H^n(X,F)' = \mathrm{Hom}_{\underline{O}_X}(X; F, \Omega^n_X) = \mathrm{Ext}^0_{\underline{O}_X}(X; F, \Omega^n_X). \tag{5.6}$$

Taking $F = \Omega^n_X$ gives an isomorphism

$$\gamma : H^n(X, \Omega^n_X) \xrightarrow{\sim} k. \tag{5.7}$$

185

One checks that the isomorphism (5.6) is precisely (5.2 bis) for $p = n$ after setting $L = k$ by means of (5.7). Consequently, (5.2 bis) is an isomorphism for $p = n$. To prove that it is an isomorphism for every p, it again suffices to prove that, for $p < n$, the two sides of (5.2 bis) are coeffaceable functors of F, and a fortiori (in view of Lemma 3) that both sides vanish when $F = \underline{O}_X(-m)$ with m large. For the first side this follows from Lemma 4; for the second, using the duality theorem on P, we write

$$H^p(X, \underline{O}_X(-m))' = \mathrm{Ext}^{r-p}_{\underline{O}_P}(P; \underline{O}_X(-m), \Omega^r_P).$$

The second side vanishes for $p < n$ and m large, as follows from Lemma 5 (with $X = P$ there) and from the fact that, as a coherent algebraic sheaf on P, $\underline{O}_X$ has cohomological dimension $\leq r - n$ (since X is nonsingular), whence

$$\underline{\mathrm{Ext}}^{r-p}_{\underline{O}_P}(\underline{O}_X, \Omega^r_P) = 0 \qquad \text{for } p < n.$$

6. The duality theorem for singular varieties

Let X be a closed subset of dimension n of projective space P of dimension r. Formula (5.6) must here be written

$$H^n(X,F)' \simeq \mathrm{Hom}_{\underline{O}_X}(X; F, \omega^n_X) = \mathrm{Ext}^0_{\underline{O}_X}(X; F, \omega^n_X), \tag{6.1}$$

where we have set

$$\omega^n_X = E^0(\underline{O}_X) = \underline{\mathrm{Ext}}^{r-n}_{\underline{O}_P}(\underline{O}_X, \Omega^r_P). \tag{6.2}$$

As stated in Proposition 5, the sheaf so introduced does not in fact depend on the chosen embedding of X into a nonsingular variety P. Taking $F = \omega_X^n$ in (6.1), we obtain

$$H^n(X, \omega_X^n)' \simeq \mathrm{Hom}_{\underline{O}_X}(X; \omega_X^n, \omega_X^n), \tag{6.2 bis}$$

[8] and hence a distinguished element of $H^n(X, \omega_X^n)'$ corresponding to the identity homomorphism of ω_X^n:

$$\gamma : H^n(X, \omega_X^n) \longrightarrow k. \tag{6.3}$$

186

Now consider the pairings defined by composition of Ext:

$$H^p(X, F) \times \mathrm{Ext}^{n-p}_{\underline{O}_X}(X; F, \omega_X^n) \longrightarrow H^n(X, \omega_X^n), \tag{6.4}$$

and compose them with the homomorphism γ of (6.3). This gives functorial homomorphisms, compatible with the boundary operators,

$$\mathrm{Ext}^{n-p}_{\underline{O}_X}(X; F, \omega_X^n) \longrightarrow H^p(X, F)' \tag{6.5}$$

(generalizing (5.2)). One checks that for $p = n$ this gives the isomorphism (6.1). With this understood, we have the following result.

THEOREM 3 bis. — The following four conditions on X are equivalent for a given integer $k \geq 0$:

i. The functorial homomorphism (6.5) is an isomorphism for $n - k \leq p \leq n$.

ii. One has $H^p(X, \underline{O}_X(-m)) = 0$ for m large and $n - k \leq p < n$.

iii. The functor $H^p(X, F)$ on the category of coherent algebraic sheaves on X is coeffaceable for $n - k \leq p < n$.

iv. One has $E^i(\underline{O}_X) = \underline{\mathrm{Ext}}^{r-n+i}_{\underline{O}_P}(\underline{O}_X, \Omega_P^r) = 0$ for $0 < i \leq k$.

PROOF. — *i.*⇒*ii.* follows from Lemma 4, *ii.*⇒*iii.* from Lemma 3, and *iii.*⇒*i.* by a well-known standard argument, since the two sides of (6.5) are then coeffaceable functors for $n - k \leq p < n$ (the first by Lemma 4). Finally, let us prove *ii.*⇔*iv.* This follows from the corollary to the next proposition.

PROPOSITION 6. — Let F be a coherent algebraic sheaf on X, and let i be an integer. Then, for m sufficiently large, there is an isomorphism

$$H^i(X, F(-m))' \simeq H^0(X, E^{n-i}(F)(m)), \tag{6.6}$$

where we set (compare no. 3, Proposition 5)

$$E^j(F) = \underline{\mathrm{Ext}}^{r-n+j}_{\underline{O}_P}(F, \Omega_P^r). \tag{6.7}$$

Indeed, by the duality theorem on P, the first side of (6.6) is isomorphic to $\mathrm{Ext}^{r-i}_{\underline{O}_P}(P; F(-m), \Omega_P^r)$, so (6.6) follows from Lemma 5. 187

COROLLARY. — One has $H^i(X, F(-m)) = 0$ for m large if and only if $E^{n-i}(F) = 0$.

Recall that the $E^j(F)$ do not in fact depend on the projective embedding under consideration. The condition in the corollary is purely local. Consequently, if it holds for F, it holds for every sheaf locally isomorphic to a sheaf F^n. In particular, if this condition holds for $\underline{O}_X$, it holds for every locally free coherent algebraic sheaf. This is the case, for example, for every $i < n$ if X is

[8] The printed text also numbers this formula "(6.2)"; the suffix "bis" distinguishes the two occurrences.

nonsingular; for $i = 0$ if no component of X is reduced to a single point; and for $i = 0, 1$ if X is normal and all its components have dimension > 1 (cf. [3]).[9] For this to hold for every $i < k$, it is necessary and sufficient, by definition, that the local rings $\underline{O}_x$ $(x \in X)$ have "homological codimension" $\geq k$ (see [4] for this notion). If $k = n$, Theorem 3 bis says that this means the duality theorem is true for X, i.e. that (6.5) is an isomorphism for every p and every F. Many equivalent conditions on the local rings $\underline{O}_x$ can be given for this to hold (Nagata), for example that they satisfy the Cohen–Macaulay equidimensionality theorem. It holds, for example, if X is locally a "complete intersection" in a nonsingular ambient variety.

7. Poincaré duality

Let X be a nonsingular projective variety of dimension n. Then

$$H^*(X) = H^*(X, \Omega_X^*)$$

is a finite-dimensional anticommutative bigraded algebra, which we grade by total degree, so that $H^{p,q}(X) = H^p(X, \Omega_X^q)$ has degree $p+q$. The degrees of $H^*(X)$ range from 0 to $2n$. By Theorem 2 and the corollary to Theorem 3, $H^*(X)$ is a "Poincaré algebra" of dimension $2n$: that is, $H^{2n}(X)$ is equipped with an isomorphism to the base field k, and the product

$$H^m(X) \times H^{2n-m}(X) \longrightarrow H^{2n}(X) = k$$

is a duality between $H^m(X)$ and $H^{2n-m}(X)$. Moreover, if Y is another nonsingular projective variety, the Künneth formula for coherent algebraic sheaves gives

$$H^*(X \times Y) = H^*(X) \otimes H^*(Y), \tag{7.1}$$

an isomorphism compatible with the Poincaré-algebra structures. Furthermore, as a commutative algebra, $H^*(X)$ is a contravariant functor of X: a morphism $f : Y \longrightarrow X$ defines in the evident
188 way a homomorphism of graded algebras

$$f^* : H^*(X) \longrightarrow H^*(Y). \tag{7.2}$$

Since these are Poincaré algebras, transposition gives a homomorphism of vector spaces

$$f_* : H^*(Y) \longrightarrow H^*(X). \tag{7.3}$$

We saw in Section 4 that the effect of f^* on the cohomology classes corresponding to nonsingular cycles has the geometric interpretation of taking the cohomology classes corresponding to their inverse images. In the present setting it is important to show that (7.3) similarly corresponds to the "direct image" operation on cycles. Under suitable nonsingularity conditions, at least, this follows from the following special case.

THEOREM 4. — If f is the inclusion of a nonsingular subvariety Y^m of X^n into X^n, then, writing 1_Y for the unit element of $H(Y)$, one has

$$f_*(1_Y) = P_X(Y), \tag{7.4}$$

where the right-hand side is the cohomology class in X associated with Y.

This formula is equivalent to

$$\langle \xi^{m,m}, P_X(Y) \rangle \varepsilon_Y = f^*(\xi^{m,m}) \qquad (\xi^{m,m} \in H^m(X, \Omega_X^m)), \tag{7.4 bis}$$

[9] The printed text says here "S is normal"; the paragraph involves only the variety X.

[10] where ε_Y is the fundamental element in $H^m(Y, \Omega_Y^m)$. For nonsingular projective varieties this gives a new definition of the cohomology class associated with Y. To prove Theorem 4, Theorem 3 lets us interpret the transpose of the homomorphism

$$H^m(X, \Omega_X^m) \longrightarrow H^m(Y, \Omega_Y^m) = H^m(X, \Omega_Y^m)$$

as the homomorphism

$$\begin{aligned} H^{n-m}(X, \Omega_X^{n-m}) &\simeq \operatorname{Ext}_{\underline{O}_X}^{n-m}(X; \Omega_X^m, \Omega_X^n) \\ &\longleftarrow \operatorname{Ext}_{\underline{O}_X}^{n-m}(X; \Omega_Y^m, \Omega_X^n) \simeq \operatorname{Hom}_{\underline{O}_X}(X; \Omega_Y^m, \Omega_Y^m). \end{aligned} \tag{7.5}$$

One checks that the element 1_Y of the dual of $H^m(Y, \Omega_Y^m)$ identifies with the element of the right-hand side corresponding to the identity endomorphism of Ω_Y^m, and also that the image of this element in $H^{n-m}(X, \Omega_X^{n-m})$ is indeed $P_X(Y)$. These compatibilities, which could have been stated already in Section 4, can be formulated—and in fact hold—for arbitrary nonsingular varieties, the second 189

following, for example, from the commutativity of the following diagram of canonical endomorphisms:

$$\left\{ \begin{array}{ccccc} \operatorname{Ext}_{\underline{O}_X}^{n-m}(X; \Omega_X^m, \Omega_X^n) & \longleftarrow & \operatorname{Ext}_{\underline{O}_X}^{n-m}(X; \Omega_Y^m, \Omega_X^n) & \simeq & \operatorname{Hom}_{\underline{O}_X}(X; \Omega_Y^m, \Omega_Y^m) \\ \downarrow & & & & \downarrow \\ \operatorname{Ext}_{\underline{O}_X}^{n-m}(X; \underline{O}_X, \Omega_X^{n-m}) & \longleftarrow & \operatorname{Ext}_{\underline{O}_X}^{n-m}(X; \underline{O}_Y, \Omega_X^{n-m}) & \simeq & \operatorname{Hom}_{\underline{O}_X}(X; \Omega_Y^m, \Omega_Y^m) \end{array} \right. \tag{7.6}$$

This gives an exact analogue of the formalism of Poincaré duality on compact oriented varieties. In particular, Theorem 4 determines the cohomology class associated with the diagonal of $X \times X$. A well-known argument then gives, for example, a Lefschetz formula.

THEOREM 5. — Let f be an endomorphism of a nonsingular projective variety X whose fixed points all have multiplicity 1. Then the number of these fixed points is congruent, modulo the characteristic of k, to the alternating sum of the traces of the endomorphisms of the $H^i(X)$ induced by f.

The restriction imposed on f is due to the difficulties mentioned in the remark in Section 4. Note, however, that the Lefschetz formula remains true if f is "homotopic" to an endomorphism all of whose fixed points have multiplicity 1.

8. Generalization of the duality theorem

Let X be a nonsingular algebraic variety such that every coherent algebraic sheaf F on X is isomorphic to a quotient of a locally free coherent algebraic sheaf (as is the case if X is locally closed in projective space). Then every coherent algebraic sheaf F on X admits a finite resolution L by locally free sheaves; for any two such resolutions one can always find a third and a homomorphism from it to each of the first two compatible with the augmentations. Likewise, if L is a finite locally free resolution of F and a homomorphism $F' \longrightarrow F$ is given, there exists a finite locally free resolution L' of F' and a homomorphism $L' \longrightarrow L$ compatible with $F' \longrightarrow F$, which may be assumed surjective if $F' \longrightarrow F$ is surjective. Given two integers $r, s \geq 0$, this makes it possible to define two cohomological multifunctors in the arguments $A_1, \ldots, A_r; B_1, \ldots, B_s$, 190

[10] The printed text has $f_*(\xi^{m,m})$. Since $\xi^{m,m} \in H^m(X, \Omega_X^m)$ and both sides belong to $H^m(Y, \Omega_Y^m)$, the required map is f^*.

which range over the category of coherent algebraic sheaves on X and take values, respectively, in the category of coherent algebraic sheaves on X and in the category of modules over $H^0(X, \underline{O}_X)$, by the formulas

$$\begin{cases} \underline{T}_r^{s*}(A_1, \dots, A_r; B_1, \dots, B_s) = H^*\Big(\underline{\mathrm{Hom}}_{\underline{O}_X}(L(A_1) \otimes \cdots \otimes L(A_r), L(B_1) \otimes \cdots \otimes L(B_s))\Big), \\ T_r^{s*}(A_1, \dots, A_r; B_1, \dots, B_s) = \mathbf{R}^*\Gamma\Big(\underline{\mathrm{Hom}}_{\underline{O}_X}(L(A_1) \otimes \cdots \otimes L(A_r), L(B_1) \otimes \cdots \otimes L(B_s))\Big). \end{cases} \tag{8.1}$$

Here $L(F)$ denotes a finite locally free resolution of the coherent algebraic sheaf F, and $\mathbf{R}^*\Gamma(K)$ denotes the hypercohomology of X with respect to the complex of sheaves K. If r or s is zero, the tensor product of the $L(A_i)$, respectively of the $L(B_j)$, is replaced by $\underline{O}_X$. In particular, $\underline{T}_0^0$ and T_0^0 are graded functors with no arguments: $\underline{T}_0^0$ is concentrated in degree 0, where it is the sheaf $\underline{O}_{X'}$ while T_0^0 is identified with $H^*(X, \underline{O}_X)$. The independence of the right-hand sides of (8.1) from the chosen resolutions is in any event evident for the first line (the question then being local), and for the second follows from the preceding general remarks, taking into account the hypercohomology spectral sequence of the complex of sheaves

$$K = \underline{\mathrm{Hom}}_{\underline{O}_X}(L(A_1) \otimes \cdots, L(B_1) \otimes \cdots),$$

which abuts to the hypercohomology of X with respect to K. Its initial term is $H^p(X, H^q(K))$, i.e.

$$E_2^{p,q} = H^p\big(X, (\underline{T}_r^s)^{(q)}(A_1, \dots, A_r; B_1, \dots, B_s)\big). \tag{8.2}$$

It follows that this spectral sequence itself is independent of the chosen resolutions, and its abutment is the first member of (8.1). The connecting operators in the various arguments A_i, B_j are readily defined, since every exact sequence $0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$ can be resolved by an exact sequence of finite locally free complexes.

On the system of functors $\underline{T}_r^{s*}$, respectively T_r^{s*}, one defines operations analogous to those of tensor calculus, whose definitions follow immediately from (8.1). Thus there is a composition (generalizing the one considered in Section 2)

$$\begin{aligned} &T_r^{s*}(A_1, \dots, A_r; B_1, \dots, B_s) \times T_{r'}^{s'*}(A'_1, \dots, A'_{r'}; B'_1, \dots, B'_{s'}) \\ &\quad \longrightarrow T_{r+r'}^{s+s'*}(A_1, \dots, A_r, A'_1, \dots, A'_{r'}; B_1, \dots, B_s, B'_1, \dots, B'_{s'}), \end{aligned} \tag{8.3}$$

satisfying the evident associativity and compatibility properties with respect to functorial homomorphisms, connecting homomorphisms, and spectral sequences. There are likewise symmetry operations, whose explicit description is left to the reader. In addition, there is a contraction oper-
191 ation whenever one of the arguments A_i

is equal to one of the arguments B_j:

$$\begin{aligned} &T_r^{s*}(A_1, \dots, A_{i-1}, C, A_{i+1}, \dots, A_r; B_1, \dots, B_{j-1}, C, B_{j+1}, \dots, B_s) \\ &\quad \longrightarrow T_{r-1}^{s-1*}(A_1, \dots, \widehat{A_i}, \dots, A_r; B_1, \dots, \widehat{B_j}, \dots, B_s). \end{aligned} \tag{8.4}$$

Furthermore, if an argument A_i is a locally free sheaf, it may be removed provided that one of the A_j $(j \neq i)$ is replaced by $A_j \otimes A_i$, or one of the B_k by $B_k \otimes A'_i$ (where $A'_i = \underline{\mathrm{Hom}}_{\underline{O}_X}(A_i, \underline{O}_X)$); there is an analogous calculation rule when one of the arguments B_i is locally free. In particular, an argument equal to $\underline{O}_X$ may always be removed. If all the arguments are locally free except for

at most one of the B_i, the rule just stated gives a functorial isomorphism

$$T_r^{s*}(A_1, \dots, A_r; B_1, \dots, B_s) = H^*(X, A_1' \otimes \cdots \otimes A_r' \otimes B_1 \otimes \cdots \otimes B_s) \tag{8.5}$$

(since this reduces to the case $r = 0$, $s = 1$, where it is immediate; one may also use directly the spectral sequence with initial term (8.2)). Operations corresponding to the preceding ones are defined for the $\underline{T}_r^s$. The relations among the various operations so introduced are the same as for the analogous operations in tensor calculus.

Let n be the dimension of X. Successively applying a tensor composition (8.3) and contractions (8.4) to the repeated arguments gives a pairing

$$(T_r^s)^p(A_1, \dots; B_1, \dots) \times (T_s^r)^{n-p}(B_1, \dots; A_1, \dots, A_r \otimes \Omega_X^n) \longrightarrow H^n(X, \Omega_X^n). \tag{8.6}$$

THEOREM 6. — If X is a nonsingular projective variety, then the pairings (8.6) are dualities.

This follows in a purely formal way from the corollary to Theorem 3. Indeed, it follows readily from that corollary that if K is a complex of locally free coherent algebraic sheaves, then the hypercohomology of X with respect to K is in duality with the hypercohomology of X with respect to $K' \otimes \Omega_X^n$ through the natural pairings

$$\mathbf{R}^p\Gamma(K) \times \mathbf{R}^{n-p}\Gamma(K' \otimes \Omega_X^n) \longrightarrow \mathbf{R}^n\Gamma(\Omega_X^n) = H^n(X, \Omega_X^n). \tag{8.7}$$

This is seen by using the spectral sequence with initial term $H^p(X, H^q(K))$[11] and the analogous spectral sequence for $K' \otimes \Omega_X^n$. Theorem 6 follows from the preceding result by using definition (8.1). 192

REMARKS

1. For the definitions preceding Theorem 6, it was not necessary for X to be nonsingular, since it was not essential to work only with finite resolutions. But if X is singular, one can no longer assert a priori that the $(\underline{T}_r^s)^p(A_1, \dots; B_1, \dots)$ are coherent sheaves, because the complex of sheaves

$$\underline{\mathrm{Hom}}_{\underline{O}_X}(L(A_1) \otimes \cdots, L(B_1) \otimes \cdots)$$

will have infinitely many components of any given total degree.

2. It is readily checked that in the formulas (8.1), one of the $L(B_i)$ may be replaced by B_i. In view of Proposition 3, this shows that

$$\begin{cases} \underline{T}_1^{1*}(A; B) = \underline{\mathrm{Ext}}^*_{\underline{O}_X}(A, B), \\ T_1^{1*}(A; B) = \mathrm{Ext}^*_{\underline{O}_X}(X; A, B). \end{cases} \tag{8.8}$$

In particular, taking $r = s = 1$ and $A_1 = \underline{O}_X$ in (8.6) recovers Theorem 3. Formula (8.8) also implies

$$T_0^{1*}(B) = H^*(X, B), \qquad T_1^{0*}(A) = \mathrm{Ext}^*_{\underline{O}_X}(X; A, \underline{O}_X).$$

3. Formula (8.1) shows that in general the functors $\underline{T}_r^{s*}$ and T_r^{s*} have components in both positive and negative degrees. Using Remark 2, one sees that if X has dimension n, then the nonzero components of $\underline{T}_r^{s*}$ lie between $-(s-1)n$ and rn if $s > 0$, and between 0 and rn if $s = 0$; the nonzero components of T_r^{s*} lie between $-(s-1)n$ and $(r+1)n$ if $s > 0$, and between 0 and $(r+1)n$ if $s = 0$ (and, unless I am mistaken, if $r > 0$, even between $-(s-1)n$ and rn, respectively between 0 and rn).

[11] The printed text has $H^p(H^q(X, K))$; the initial term is that of the spectral sequence (8.2), namely $H^p(X, H^q(K))$.

BIBLIOGRAPHY

[1] Cartier (Pierre). — Les groupes $\mathrm{Ext}^s(A, B)$, Séminaire A. Grothendieck : Algèbre homologique, t. 1, 1957, no. 3.
[2] Grothendieck (Alexandre). — Sur quelques points d'algèbre homologique, *Tohoku Math. J.*, vol. 9, 1957, pp. 119–183.
[3] Serre (Jean-Pierre). — Faisceaux algébriques cohérents, *Annals of Math.*, vol. 61, 1955, pp. 197–278.
[4] Serre (Jean-Pierre). — Sur la dimension homologique des anneaux et des modules noethériens, *Proc. Intern. Symp. on Alg. Number Theory* [1955, Tokyo and Nikko]. — Tokyo, Science Council of Japan, 1956, pp. 175–189.

ADDENDUM

The difficulties noted in the remark on page 13 have now been completely resolved, through an extension of the duality theorems to arbitrary varieties (with arbitrary singularities). For this theory, which can be formulated in the general framework of the "theory of schemes," we refer to the "Éléments de Géométrie algébrique" of J. Dieudonné and A. Grothendieck, currently in preparation. Some indications may be found in:

Grothendieck (A.). — Cohomology theory of algebraic varieties, *International Congress of Mathematicians*, 1958, Edinburgh (to appear).

[April 1959]

Erratum to Exposé no. 149

298 **1. Duality theorem for coherent algebraic sheaves.**

[Bourbaki Seminar, vol. 9, 1956/57, no. 149, 25 pp.]

Remark on page 14. — As I pointed out in my lecture at the International Congress of Mathematicians in 1958,[1] the questions raised here are now completely resolved.

The reader will find further information on the duality of coherent sheaves in *loco citato*[1], pp. 112–115, in EGA III, Part 2 (in preparation), and, in the meantime, in SGA 1962. A more systematic treatment will be found in a later chapter of EGA (Chapter IX in the planned outline).

[1]Grothendieck (Alexander). — The cohomology theory of abstract algebraic varieties, *Proceedings of the International Congress of Mathematicians* [1958, Edinburgh], pp. 103–118. — Cambridge, at the University Press, 1960.

Chapter 2

Exposé 182 — Formal Geometry and Algebraic Geometry

193

BOURBAKI SEMINAR
(May 1959)

FORMAL GEOMETRY AND ALGEBRAIC GEOMETRY

by Alexander GROTHENDIECK

1. Schemes

An affine algebraic space defined over a field k is known to be essentially determined by its affine algebra A (the ring of regular functions defined over k), with morphisms of algebraic spaces $X \longrightarrow Y$ corresponding bijectively to homomorphisms of k-algebras $A(Y) \longrightarrow A(X)$. The affine algebra corresponding to an algebraic space is a finitely generated k-algebra and, from the "classical" point of view, has no nilpotent elements; conversely, every algebra of this type is obtained as the affine algebra of an algebraic space defined over k. There is then a familiar dictionary for interpreting situations concerning affine algebraic spaces in terms of commutative algebra. It has long been observed that this yields more general statements, since it is usually no longer necessary to suppose that the rings in question are of the type just considered, the Noetherian hypothesis most often being sufficient. In particular, whether or not a base field is specified, there is no reason to exclude the case in which these rings contain nilpotent elements. Until now, geometers have refused to take these indications into account and have persisted in restricting themselves to affine algebras without nilpotent elements, i.e. to algebraic spaces whose structure sheaves contain no nilpotent elements (and, most often, even to "absolutely irreducible" algebraic spaces). The speaker believes that this state of mind has been a serious obstacle to the development of genuinely natural methods in algebraic geometry.

Let A be a commutative ring. It is well known that the set $X = \mathrm{Spec}(A)$ of prime ideals of A carries a natural topology, the "Zariski topology" or spectral topology. Moreover, there is a sheaf of commutative rings $\underline{O}_X$ on X whose stalk at $\mathfrak{p} \in X$ is the localized ring $A_{\mathfrak{p}}$ and whose ring of sections identifies with A. Thus X becomes a *ringed space*, called the *prime spectrum* of A. A ring homomorphism $f : A \longrightarrow B$ defines a morphism of ringed spaces $f' : \mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$ whose underlying map of sets is simply $\mathfrak{p} \longmapsto f^{-1}(\mathfrak{p})$. The homomorphisms 194

of ringed spaces from $\mathrm{Spec}(B)$ to $\mathrm{Spec}(A)$ obtained in this way are exactly those for which the homomorphisms $\underline{O}_x \longrightarrow \underline{O}_y$ ($x = f'(y)$) are local (i.e. the inverse image of the maximal ideal is the maximal ideal).

An *affine scheme* is a ringed space isomorphic to some $\mathrm{Spec}(A)$, and a *prescheme* is a locally affine ringed space, i.e. one in which every point has an open neighborhood that is an affine scheme for the induced structure. Morphisms of preschemes are defined in the evident way; locally they correspond to ring homomorphisms.

When a prescheme S is fixed and morphisms of preschemes $X \longrightarrow S$ are considered, S plays the role of a base field or base ring (or better, of a base space in a fibration). One then says that X is an S-prescheme; if $S = \mathrm{Spec}(A)$, this also means that $\underline{O}_X$ is a sheaf of A-algebras. Thus every prescheme may be regarded uniquely as a $\mathbf{Z}$-prescheme. The S-preschemes naturally form a category; moreover, one proves that the product of any two objects X, Y in this category always exists, and it is denoted $X \times_S Y$. This notion of product makes it possible to define base change in an S-prescheme corresponding to a morphism $S' \longrightarrow S$: indeed, $X \times_S S'$ may be regarded as an S'-prescheme.

One says that X is *separated* over S if the diagonal of $X \times_S X$ is closed. A *scheme* is a prescheme separated over $\mathbf{Z}$; it is then separated over anything. For simplicity, from now on we shall speak only of schemes, which we shall moreover assume *Noetherian*, i.e. finite unions of affine open subsets that are spectra of Noetherian rings. One says that X is *of finite type over* S if, for every affine open subset U of S, its inverse image in X is a finite union of affine open subsets whose rings are finite-type algebras over the ring of U. It is such S-schemes that lend themselves to a properly geometric study. In particular, for every $s \in S$, the fiber $f^{-1}(s)$ of X over s is an algebraic scheme over the residue field $\kappa(s)$ of the local ring $\underline{O}_s$ of s in S. Thus X may, to a certain extent, be regarded as a family of "algebraic spaces" $f^{-1}(s)$, with the parameter s ranging over S (i.e., from the local point of view, over the set of prime ideals of a given ring). The $\kappa(s)$ may of course have different characteristics. If $S = \mathrm{Spec}(k)$, where k is a field, one essentially recovers the usual notion of an "algebraic space,"
195 with the sole difference that the structure sheaf may now have nilpotent elements.

Drawing on familiar notions, one defines a *projective morphism*, and more generally a *proper morphism*. Such a morphism is of finite type; moreover, it maps closed subsets to closed subsets and retains this property under arbitrary base change.

If X is a scheme (Noetherian, as always), the sheaf $\underline{O}_X$ is a *coherent sheaf of rings* in the sense of [2]. Thus the coherent sheaves of modules on X are also precisely those sheaves locally isomorphic to the cokernel of a homomorphism $\underline{O}_X^m \longrightarrow \underline{O}_X^n$.

2. Formal schemes

Let X be a scheme and X' a closed subset of X. There is then a coherent subsheaf J of $\underline{O}_X$ such that $X' = \mathrm{Supp}(\underline{O}_X/J)$ (and there is even a largest such subsheaf). Equipped with $\underline{O}_X/J$, X' becomes a scheme, denoted X_0; such a scheme is called a "closed subscheme of X." For every n, one may also consider X' equipped with $\underline{O}_X/J^{n+1}$, denoted X_n; this is a closed subprescheme of X whose underlying set is still X' but which has a different structure sheaf, namely $\underline{O}_{X_n} = \underline{O}_X/J^{n+1}$. The $\underline{O}_{X_n}$ plainly form an inverse system of sheaves of rings on X', whose inverse limit $\overline{\underline{O}}_X$ is called the *formal completion of* $\underline{O}_X$ *along* X'. Equipped with this sheaf of rings, X' is called the *formal completion of* X *along* X'; it is therefore a ringed space but not, in general, a scheme. For every coherent sheaf F on X, one may similarly consider the formal completion $\overline{F} = \varprojlim_n F_n$ of F along X' (where $F_n = F \otimes_{\underline{O}_X} \underline{O}_X/J^{n+1}$); it is a sheaf of modules on $\overline{X}$. Its sections are called the "formal sections of F along X'" and identify with the elements of $\varprojlim_n \Gamma(X', F_n)$. For $F = \underline{O}_{X'}$, one obtains the "holomorphic functions" of X along X' in Zariski's sense (whose terminology we

shall not follow because it conflicts with classical terminology).

A *formal scheme* (understood to mean Noetherian) is a topological space $\mathfrak{X}$ equipped with a sheaf of topological rings $\underline{O}_{\mathfrak{X}}$ satisfying the following condition: there is an isomorphism of sheaves of topological rings $\underline{O}_{\mathfrak{X}} = \varprojlim_n \underline{O}_{n'}$ where the $\underline{O}_n$ form an inverse system of sheaves of rings on $\mathfrak{X}$, each making $\mathfrak{X}$ into a scheme $\mathfrak{X}_n$, and such that for $m \geq n$ the homomorphism $\underline{O}_m \longrightarrow \underline{O}_n$ is surjective and has kernel J_m^{n+1}, where J_m is 196

the kernel of $\underline{O}_m \longrightarrow \underline{O}_0$. One then proves that $\underline{O}_{\mathfrak{X}}$ is a *coherent sheaf of Noetherian local rings*.

By the definitions, a formal completion $\overline{X}$ as above is a formal scheme, and conversely every formal scheme is *locally* of this type. In fact, specifying an *affine* formal scheme (i.e. one for which $\mathfrak{X}_0$ is affine, which implies that all the $\mathfrak{X}_n$ are affine) is equivalent to specifying a Noetherian topological ring that is separated and complete for the J-adic topology.

The usual definitions for ordinary schemes—morphism, morphism of finite type, proper morphism, and so forth—extend without difficulty to formal schemes.

3. The three fundamental theorems

Let $f : X \longrightarrow Y$ be a proper morphism of schemes (Noetherian, as always), let Y' be a closed subset of Y, and let X' be its inverse image in X; consider the corresponding formal completions $\overline{Y}$ and $\overline{X}$. Then f induces a morphism $\overline{f} : \overline{X} \longrightarrow \overline{Y}$ of formal schemes, which is itself proper. If F is a coherent sheaf on X, then $\overline{F}$ is a coherent sheaf on $\overline{X}$. In Theorem 1, the original X, Y, F are forgotten and only the proper morphism $\overline{f}$ of formal schemes and the coherent sheaf $\overline{F}$ on $\overline{X}$ are retained. (The speaker has, however, written out a complete proof only in the case that begins with X, Y, f, F.)

THEOREM 1 (finiteness theorem). —

i. The $R^q\overline{f}_*(\overline{F})$ are coherent sheaves on $\overline{Y}$.

ii. The natural homomorphisms

$$R^q\overline{f}_*(\overline{F}) \longrightarrow \varprojlim_n R^q f_{n*}(F_n)$$

are isomorphisms.

In this statement, a coherent subsheaf J of $\underline{O}_Y$ defining Y' is supposed chosen; its inverse image gives a coherent subsheaf of $\underline{O}_X$ defining X', and hence the definition of F_n, X_n, Y_n and $f_n : X_n \longrightarrow Y_n$ as in Section 2. The minor changes needed to make these notations explicit are evident if one starts with an arbitrary proper morphism of two formal schemes.

Theorem 1 concerned only "formal cohomology." The following theorem 197

relates it to "algebraic cohomology" and is akin to a well-known theorem of Serre [4] comparing algebraic and analytic cohomology.

THEOREM 2 (First comparison theorem). — The $R^q f_*(F)$ are coherent sheaves on Y (which is a special case of Theorem 1), and the natural homomorphisms

$$\overline{R^q f_*(F)} \longrightarrow \varprojlim_n R^q f_{n*}(F_n)$$

are isomorphisms.

COROLLARY 1. — There are canonical isomorphisms

$$\overline{R^q f_*(F)} = R^q\overline{f}_*(\overline{F}).$$

For $q = 0$, this corollary is a generalization of Zariski's "fundamental theorem of holomorphic functions," from which we shall deduce a generalization of Zariski's "connectedness theorem." Note, moreover, that although Theorem 1 (*ii*) is trivial for $q = 0$, the same is by no means true of Theorem 2 or of its equivalent formulation (Corollary 1). In fact, the proof proceeds by descending induction on q (being trivial for large q, since both sides then vanish), and the case $q = 0$ thus appears as the last step of the induction and therefore, so to speak, as the "most difficult" case.

COROLLARY 2. — Suppose $Y = \mathrm{Spec}(A)$, with Y' defined by the ideal J of A. Then, for every coherent sheaf F on X, the $H^q(X, F)$ are finite-type A-modules whose J-adic completions are the $H^q(\overline{X}, \overline{F})$.

Finally, applying this corollary to $H = \underline{\mathrm{Hom}}_{\underline{O}_X}(F, G)$ gives:

COROLLARY 3. — Suppose $Y = \mathrm{Spec}(A)$, with Y' defined by the ideal J of A. If F, G are two coherent sheaves on X, then $\mathrm{Hom}(F, G)$ is a finite-type A-module whose J-adic completion identifies with $\mathrm{Hom}(\overline{F}, \overline{G})$.

The natural map $\mathrm{Hom}(F, G) \longrightarrow \mathrm{Hom}(\overline{F}, \overline{G})$ is of course the one that assigns to a homomorphism $u : F \longrightarrow G$ its extension "by continuity" $\overline{u} : \overline{F} \longrightarrow \overline{G}$ (with which convention $\overline{F}$ becomes a functor of F).

Now suppose that A is separated and complete for its J-adic topology. The preceding Corollaries 2 and 3 then give

$$H^q(X, F) = H^q(\overline{X}, \overline{F}), \qquad \mathrm{Hom}(F, G) = \mathrm{Hom}(\overline{F}, \overline{G}).$$

198

This last identity shows that the category of coherent sheaves on X identifies with a *subcategory* (whose morphisms are the induced morphisms) of the category of coherent sheaves on $\overline{X}$. In fact, one has the stronger statement:

THEOREM 3. — A sheaf of modules on $\overline{X}$ is coherent if and only if it is isomorphic to a sheaf of the form $\overline{F}$, where F is a coherent sheaf on X (determined up to canonical isomorphism by Theorem 2, Corollary 3). [Recall that now $Y = \mathrm{Spec}(A)$, where A is a Noetherian topological ring that is separated and complete for the J-adic topology.]

COROLLARY 1. — The closed subschemes of X correspond bijectively to the closed formal subschemes of $\overline{X}$.

Indeed, they correspond to the coherent subsheaves of $\underline{O}_X$, respectively of $\overline{\underline{O}}_X$. Viewing the graphs of morphisms as closed subschemes, one deduces from Corollary 1:

COROLLARY 2. — Let X, Z be two schemes proper over A (a Noetherian J-adic ring that is separated and complete). Then the map $g \longmapsto \overline{g}$ defines a bijection between the Y-morphisms from X to Z and the $\overline{Y}$-morphisms from $\overline{X}$ to $\overline{Z}$.

In other words, algebraic schemes proper over A appear as a *subcategory* (whose morphisms are the induced morphisms) of the category of formal schemes proper over $\overline{Y}$. It should be noted, however, that there are formal schemes proper over $\overline{Y}$ that are not "algebraizable," i.e. are not isomorphic to any $\overline{X}$ with X proper over A (just as there are compact complex analytic varieties that do not come from algebraic varieties defined over the complex numbers). Such formal schemes arise naturally in the "theory of moduli." Let us nevertheless note an interesting special case in which a formal scheme is algebraizable:

THEOREM 4. — Let A be a complete Noetherian local ring with residue field k, and let $\mathfrak{X}$ be a formal scheme proper over A (equipped with its $\mathfrak{m}(A)$-adic topology). Suppose:

i. that the local rings of $\underline{O}_{\mathfrak{X}}$ are flat A-modules, or equivalently, if $\underline{O}_{\mathfrak{X}}$ and A are equipped with the filtrations defined by the powers of the maximal ideal of A, that the associated graded objects satisfy

$$\mathrm{gr}(\underline{O}_{\mathfrak{X}}) \simeq \mathrm{gr}^0(\underline{O}_{\mathfrak{X}}) \otimes_k \mathrm{gr}(A).$$

ii. regarding $\mathfrak{X}_0 = \mathfrak{X} \otimes_A k$ as an algebraic scheme over k, one has

$$H^2(\mathfrak{X}_0, \underline{O}_{\mathfrak{X}_0}) = 0.$$

iii. $\mathfrak{X}_0$ is projective.

Under these conditions, $\mathfrak{X}$ is algebraizable; more precisely, it is isomorphic to $\overline{X}$, where X is a projective A-scheme.

Conditions (*ii*) and (*iii*) hold in particular when $\mathfrak{X}_0$ is a nonsingular curve over k, and Theorem 4 can therefore be applied, in particular, in the "theory of moduli" for curves of given genus... Let us indicate how Theorem 4 is proved. One shows (cf. Proposition 3 below) that (*i*) and (*ii*) imply that every coherent sheaf on $\mathfrak{X}_0$ which is locally isomorphic to the structure sheaf is obtained by reduction from a sheaf of the same kind on $\mathfrak{X}$. Starting with an "ample" sheaf on $\mathfrak{X}_0$ (one exists by (*iii*)), one then lifts it to an invertible sheaf on $\mathfrak{X}$; using Theorem 1, one proves that a multiple of this sheaf defines an immersion of $\mathfrak{X}$ into the formal completion of a scheme $\mathbf{P}^r_A$ (of "projective type" and dimension r over A).

For the proofs of Theorems 1 through 3, see [1]. 199

4. Application to the connectedness theorem and Zariski's Main Theorem

Let $f : X \longrightarrow Y$ be a proper morphism of schemes. By the finiteness theorem, $f_*(\underline{O}_X) = \mathcal{A}$ is a coherent sheaf on Y; it is moreover a sheaf of commutative algebras, and therefore corresponds to a Y-scheme $g : Y' \longrightarrow Y$ finite over Y (defined by the conditions that it be affine over Y, i.e. that the inverse image of an affine open subset be affine, and that $g_*(\underline{O}_{Y'}) = \mathcal{A}$). It is immediate that f then factors canonically as $f = gf'$, where $f' : X \longrightarrow Y'$ is a morphism from X to Y' satisfying $f'_*(\underline{O}_X) = \underline{O}_{Y'}$. This factorization of f is called the *Stein factorization* of f. Applying the first comparison theorem, Corollary 1, to f' and to the part of Y' reduced to a point y', one finds that $f'^{-1}(y') = X'$ is connected (otherwise the formal sections of X along X' would not form a local ring, whereas the completion of $f'_*(\underline{O}_X)_{y'} = \underline{O}_{Y',y'}$ is local!). We have proved:

THEOREM 5 (Zariski's "connectedness theorem"). — Let $f : X \longrightarrow Y$ be a proper morphism. Then f factors uniquely (up to isomorphism) as $f = gf'$, where $g : Y' \longrightarrow Y$ is finite and $f' : X \longrightarrow Y'$ satisfies $f'_*(\underline{O}_X) = \underline{O}_{Y'}$ (whence $g_*(\underline{O}_{Y'}) = f_*(\underline{O}_X)$). The fibers of f' are connected; that is, the set of connected components of a fiber $f^{-1}(y)$ of f is in bijection with the set of points of Y' above y, i.e. with the set of maximal ideals of $f_*(\underline{O}_X)_y$.

The usual variants of the connectedness theorem follow immediately. We note here only the following. 200

COROLLARY 1. — A point x of X is isolated in its fiber $f^{-1}(y)$ if and only if the fiber $f'^{-1}(y')$ (where $y' = f'(x)$) is reduced to x, or equivalently if f' induces an isomorphism from a neighborhood of x onto a neighborhood of y'. The set of these points is an open subset U, and f' induces an isomorphism from U onto an open subset of Y'.

To show that f' is a local isomorphism at x, observe that f' induces an isomorphism $\underline{O}_{y'} \longrightarrow \underline{O}_x$, as follows from $f'_*(\underline{O}_X) = \underline{O}_{Y'}$ and from the fact that (since f' is a closed map whose fiber over y' is reduced to x) the sets $f'^{-1}(V)$ run through a fundamental system of neighborhoods of x as V runs through a fundamental system of neighborhoods of y'. The following result, due to Chevalley in the "geometric" case, follows at once.

COROLLARY 2. — A morphism f is finite if and only if it is proper and its fibers are finite.

Indeed, in that case f' is an isomorphism by the preceding result.

Let $f : X \longrightarrow Y$ be a morphism not necessarily proper, but suppose that X is contained as an open subset in a proper Y-scheme $\tilde{f} : \widetilde{X} \longrightarrow Y$ (as will be the case, in particular, if f is quasi-projective). Applying Corollary 1, one sees that $\tilde{f}'$ induces an isomorphism from the set U of points of X isolated in their fiber onto an open subset of Y' (and that U is indeed open). This gives the following global form of Zariski's Main Theorem.

THEOREM 6 (Zariski's Main Theorem). — Let $f : X \longrightarrow Y$ be a morphism of finite type. Then the set U of points of X that are isolated in their fiber is open; and if f is separated,[1] U is Y-isomorphic to an open part of a scheme Y' finite over Y.

Since a morphism of finite type is locally affine and hence, *a fortiori*, locally quasi-projective, the usual local forms of the Main Theorem follow at once from Theorem 6.

5. Application to the cohomological study of proper flat morphisms

Let $f : X \longrightarrow Y$ be a proper morphism, and let F be a coherent sheaf on X, assumed flat over Y, i.e. the F_x are flat modules over the rings $\underline{O}_y$ $(y = f(x))$. Equivalently, for every $y \in Y$, if F is filtered along the fiber $f^{-1}(y)$ by the $\mathfrak{m}_y^n F$ (where $\mathfrak{m}_y$ is the maximal ideal of $\underline{O}_y$), the associated graded object is isomorphic to $(F/\mathfrak{m}_y F) \otimes_{\kappa(y)} \mathrm{gr}(\underline{O}_y)$; in other words,

$$\mathfrak{m}_y^n F/\mathfrak{m}_y^{n+1} F = F_y \otimes_{\kappa(y)} (\mathfrak{m}_y^n/\mathfrak{m}_y^{n+1})$$

for every integer n. Here X_y denotes the fiber $f^{-1}(y)$ regarded as a proper scheme over the residue field $\kappa(y)$ of y, and F_y is the sheaf $F/\mathfrak{m}_y F$ induced by F on X_y. Taking this isomorphism and Theorem 2 into account, one obtains bounds, and sometimes computations, for the $R^q f_*(F)$ near y from the cohomology of X_y with coefficients in F_y. Here Theorem 2 takes the form

$$\widehat{R^q f_*(F)}_y = \varprojlim_n H^q(X_y, F/\mathfrak{m}_y^n F).$$

201 We shall mention only the following consequence.

PROPOSITION 1. — Let $f : X \longrightarrow Y$ be a proper morphism and let F be a coherent sheaf on X, flat over Y. Let $y \in Y$ and let q be an integer, and suppose that $H^q(X_y, F_y) = 0$. Then $R^q f_*(F)$ vanishes in a neighborhood of y, and for every n the natural homomorphism

$$R^{q-1} f_*(F)_y \longrightarrow H^{q-1}(X_y, F/\mathfrak{m}_y^n F)$$

is surjective.

In particular, if f is a flat morphism (i.e. $\underline{O}_X$ is flat over Y), then every locally free coherent sheaf F on X is flat over Y. Let F, G be two such sheaves. Applying Proposition 1 to $\underline{\mathrm{Hom}}_{\underline{O}_X}(F, G)$ with $q = 1$, one obtains:

THEOREM 7. — Let f be a proper flat morphism, let F, G be two locally free coherent sheaves on X, let $y \in Y$, and suppose that

$$H^1(X_y, \underline{\mathrm{Hom}}_{\underline{O}_{X_y}}(F_y, G_y)) = 0.$$

Then every homomorphism $u_0 : F_y \longrightarrow G_y$ is induced by a homomorphism

$$u : F|V \longrightarrow G|V, \qquad V = f^{-1}(U),$$

[1] The 1962 erratum replaces the printed hypothesis "f is quasi-projective" by the weaker hypothesis "f is separated," by SGA VIII, 6.2: every separated quasi-finite morphism is quasi-projective.

where U is a neighborhood of y.

COROLLARY. — If u_0 is an isomorphism (respectively a monomorphism, an epimorphism), then so is u for sufficiently small U.

In particular: 202

COROLLARY 2. — Let E_0 be a locally free coherent sheaf on X_y such that

$$H^1(X_y; \underline{\mathrm{Hom}}_{\underline{O}_{X_y}}(E_0, E_0)) = 0.$$

Then two locally free sheaves whose restrictions to X_y are isomorphic to E_0 are isomorphic in a neighborhood of X_y.

Thus:

COROLLARY 3. — Suppose that $H^1(X_y, \underline{O}_{X_y}) = 0$. Then two invertible sheaves on X (i.e. sheaves locally isomorphic to $\underline{O}_X$) whose restrictions to X_y are isomorphic are themselves isomorphic.

It follows that:

PROPOSITION 2. — Let Y be a connected scheme and let E be a locally free coherent sheaf on Y. Consider the associated projective-space bundle $X = \mathbf{P}(E)$, equipped with its familiar invertible sheaf $\underline{O}_X(1)$. Then every invertible sheaf L on X is isomorphic to a sheaf of the form

$$f^*(L') \otimes \underline{O}_X(n),$$

where L' is an invertible sheaf on Y and n is an integer. The integer n is uniquely determined, and L' is determined up to isomorphism.

The preceding Corollary 3 proves that L is isomorphic to some $\underline{O}_X(n)$ in a neighborhood of every fiber. The rest is more or less formal.

Proposition 2 makes it possible to determine the Y-morphisms from $X = \mathbf{P}(E)$ to another projective bundle. In particular, one finds:

COROLLARY. — Let u be an automorphism of $X = \mathbf{P}(E)$. Then there exist an invertible sheaf L' on Y and an isomorphism $v : E \longrightarrow E \otimes L'$ such that u is the corresponding isomorphism

$$\mathbf{P}(E) \xrightarrow{\sim} \mathbf{P}(E \otimes L') = \mathbf{P}(E);$$

the pair (v, L') is determined up to isomorphism.

Let Γ be the set of classes of invertible bundles L' on Y such that $E \otimes L'$ is isomorphic to E. Its elements are torsion: indeed, if n is the rank of E, taking nth exterior powers shows that one must have $L'^{\otimes n} \simeq \underline{O}_Y$. The corollary may therefore be expressed by saying that there is an exact sequence of groups

$$e \longrightarrow \mathrm{Aut}(E)/\Gamma(Y, \underline{O}_Y^*) \longrightarrow \mathrm{Aut}_Y(X) \longrightarrow \Gamma \longrightarrow e$$

(which may also be deduced from the cohomology exact sequence arising from the following exact sequence of sheaves of groups

$$e \longrightarrow \underline{O}_Y^* \longrightarrow \underline{\mathrm{Aut}}(E) \longrightarrow \underline{\mathrm{Aut}}_Y(X) \longrightarrow e,$$

where $\underline{O}_Y^*$ is the sheaf of "units" of $\underline{O}_Y$, identified with the center of $\underline{\mathrm{Aut}}(E)$). 203

6. Application to existence and uniqueness theorems for sheaves and schemes over a complete J-adic ring

Theorem 7 gave a uniqueness result for locally free coherent sheaves, using Theorems 1 and 2. Using Theorem 3, we shall now deduce existence theorems for sheaves, morphisms of schemes, or schemes. In what follows, A denotes a Noetherian local ring that is separated and complete. The general method always consists in carrying out formal constructions, which amounts essentially to doing algebraic geometry over an Artinian ring, and then using the three fundamental theorems to draw conclusions of an "algebraic" nature.

PROPOSITION 3. — Let $\mathfrak{X}$ be a formal scheme proper and flat over A, and let F_0 be a locally free sheaf on X_0 such that

$$H^2(X_0, \underline{\mathrm{Hom}}_{\underline{O}_{X_0}}(F_0, F_0)) = 0.$$

Then there exists a locally free sheaf F on $\mathfrak{X}$ whose restriction to X_0 is isomorphic to F_0. (Moreover, this F is unique up to isomorphism if $H^1(X_0, \underline{\mathrm{Hom}}_{\underline{O}_{X_0}}(F_0, F_0)) = 0$.)

One constructs, successively, locally free sheaves F_n on the X_n, each inducing the preceding one. The construction of F_n encounters an obstruction in

$$H^2(X_0, \underline{\mathrm{Hom}}_{\underline{O}_{X_0}}(F_0, F_0)) \otimes_{A/J} (J^n/J^{n+1}),$$

which therefore vanishes by hypothesis. Applying Theorem 3 now gives:

COROLLARY 1. — Let X be a scheme proper and flat over A, and let F_0 be as above. Then there exists a locally free sheaf F on X whose restriction to X_0 is isomorphic to F_0. Moreover, F is unique up to isomorphism if $H^1(X_0, \underline{\mathrm{Hom}}_{\underline{O}_{X_0}}(F_0, F_0)) = 0$.

Let X_0 be a scheme of finite type over the field k. Suppose that X_0 is nonsingular (by which we mean absolutely nonsingular) over k, but not necessarily proper over k. Let A be a local Artinian ring with residue field k. We are interested in finding schemes X flat over A such that $X \otimes_A k = X_0$ (this is the starting point of the "theory of moduli," or of "variations of structure" of X_0). Giving such an X is equivalent to giving, on the topological space X_0, a sheaf $\underline{O}_X$ endowed with the following structures:

i. It is a sheaf of A-algebras.

ii. It is endowed with an augmentation homomorphism $\underline{O}_X \longrightarrow \underline{O}_{X_0}$ (compatible with the A-algebra structures), subject to the following conditions: the augmentation induces an isomorphism $\underline{O}_X \otimes_A k \xrightarrow{\sim} \underline{O}_{X_0}$; and $\underline{O}_X$ is flat over A, i.e. the graded object associated to $\underline{O}_X$, filtered by the powers of the maximal ideal $\mathfrak{m}$ of A, is isomorphic to $\mathrm{gr}^0(\underline{O}_X) \otimes_k \mathrm{gr}(A)$; equivalently, there are isomorphisms

$$\mathfrak{m}^n \underline{O}_X / \mathfrak{m}^{n+1} \underline{O}_X = \underline{O}_{X_0} \otimes_k (\mathfrak{m}^n/\mathfrak{m}^{n+1}).$$

204 The fundamental fact is the following.

THEOREM 8. — Let X_0 be a scheme of finite type and nonsingular over the field k, and suppose that X_0 is affine. Let A be a local Artinian ring with residue field k. Then there exists an A-scheme X, flat over A, such that $X \otimes_A k = X_0$, and any two such schemes are necessarily isomorphic.

The isomorphism in question is not canonical, since in general X will have nontrivial A-automorphisms inducing the identity on X_0. Moreover, in general there is no "canonical" choice of an X satisfying the stated conditions, except when A is a k-algebra (the equal-characteristic case), where one may take $X = X_0 \otimes_k A$, i.e. $\underline{O}_X = \underline{O}_{X_0} \otimes_k A$ (whether or not X_0 is affine). In

unequal characteristic, when X_0 is not affine, I do not know in general whether X_0 can be "lifted" to an X defined over A. Nevertheless, let $n > 0$ be an integer, put $A_{n-1} = A/\mathfrak{m}^n$, and suppose that X_0 has been lifted to a flat A_{n-1}-scheme X_{n-1}; we seek to lift X_{n-1} to a flat A_n-scheme X_n. Theorem 8 already shows that this is possible locally. Moreover, one checks easily that if U_n lifts an open subset U_{n-1} of X_{n-1}, then the sheaf of groups of automorphisms of U_n (inducing the identity on U_{n-1}) is canonically isomorphic to

$$\Theta_{X_0/k} \otimes_k (\mathfrak{m}^n/\mathfrak{m}^{n+1})$$

restricted to U_{n-1}, and in particular is commutative ($\Theta_{X_0/k}$ denotes the sheaf of germs of k-derivations on X_0). It follows easily that there is an obstruction to constructing X_n lifting X_{n-1}, lying in

$$H^2(X_0, \Theta_{X_0/k}) \otimes_k \mathfrak{m}^n/\mathfrak{m}^{n+1}.$$

Consequently:

COROLLARY 1. — Let X_0 be a scheme of finite type and nonsingular over k, and suppose that $H^2(X_0, \Theta_{X_0/k}) = 0$. Then, for every local Artinian ring A with residue field k, there exists an A-scheme X, flat over A, such that $X \otimes_A k = X_0$.

Moreover, if one has found an X flat over A lifting X_0, then by Theorem 8 the set of classes (up to isomorphism) of flat A-schemes lifting X_0 identifies with

$$H^1(X_0, \underline{\mathrm{Aut}}(X)),$$

where $\underline{\mathrm{Aut}}(X)$ denotes, of course, the sheaf of germs of automorphisms of the sheaf of A-algebras $\underline{O}_X$ that are compatible with the augmentation. The filtration of $\underline{O}_X$ defines a filtration of $\underline{\mathrm{Aut}}(X)$, the quotient of this sheaf by the nth subgroup of the filtration being $\underline{\mathrm{Aut}}(X_n)$. The graded object associated with this filtration is commutative and identifies with $\Theta_{X_0/k} \otimes_k \mathrm{gr}(A)$. In particular, if $\mathfrak{m}^{n+1}$ is the first power of $\mathfrak{m}$ that vanishes, then $F^n(\underline{\mathrm{Aut}}(X))$ (the last step of the filtration) is contained in the center of $\underline{\mathrm{Aut}}(X)$ and is isomorphic to $\Theta_{X_0/k} \otimes_k \mathfrak{m}^n$; it is also the sheaf of germs of automorphisms of X that induce the identity on $X_{n-1} = X \otimes_A A/\mathfrak{m}^n$. These results immediately give the following statements. 205

COROLLARY 2. — Let X_0 be a scheme of finite type and nonsingular over k. Let A be a local Artinian ring with residue field k and maximal ideal $\mathfrak{m}$, suppose that $\mathfrak{m}^{n+1} = 0$, put $A_{n-1} = A/\mathfrak{m}^n$, and let X_{n-1} be a flat A_{n-1}-scheme such that $X_{n-1} \otimes_A k = X_0$. Then the set of classes (up to isomorphism inducing the identity on X_{n-1}) of flat A-schemes X_n such that $X_n \otimes_A A_{n-1} = X_{n-1}$ is either empty or a principal homogeneous space under

$$H^1(X_0, \Theta_{X_0/k}) \otimes_k \mathfrak{m}^n.$$

(Note that in general there is no privileged origin in this latter space, since there is no privileged way to lift X_{n-1} to X_n.)

COROLLARY 3. — Let X_0 be a scheme of finite type and nonsingular over k, and suppose that $H^1(X_0, \Theta_{X_0/k}) = 0$. Then, for every local Artinian ring A with residue field k, there exists at most one (up to isomorphism) flat A-scheme X such that $X \otimes_A k = X_0$.

Corollaries 1 and 3 immediately imply the apparently more general statements obtained by assuming only that A is a complete Noetherian local ring with residue field k, provided that X is introduced as a formal scheme over A:

THEOREM 9. — Let k be a field and let X_0 be a scheme of finite type and nonsingular over k. For every complete Noetherian local ring A with residue field k, let $F(A)$ be the set of classes (up to isomorphism inducing the identity on X_0) of formal schemes $\mathfrak{X}$ over A, of finite type and flat over A, such that $\mathfrak{X} \otimes_A k = X_0$. With this notation:

i. If $H^1(X_0, \Theta_{X_0/k}) = 0$, then for every A, $F(A)$ has at most one element.

ii. If $H^2(X_0, \Theta_{X_0/k}) = 0$, then for every A, $F(A)$ has at least one element.

206

COROLLARY 1. — Suppose that X_0 is proper over k. Under condition (i), for every A there exists at most one (up to isomorphism inducing the identity on X_0) scheme X, proper and flat over A, such that $X \otimes_A k = X_0$.

Here one uses Theorem 3, Corollary 2. For example:

COROLLARY 2. — If X is a scheme proper and flat over A such that $X \otimes_A k$ is isomorphic to the standard projective scheme $\mathbf{P}^r_k$ of dimension r over k, then X is isomorphic to $\mathbf{P}^r_A$.

(This result may also be deduced from Proposition 3, Corollary 1.)

COROLLARY 3. — Let X_0 be a nonsingular projective scheme over k, and suppose that

$$H^2(X_0, \underline{O}_{X_0}) = H^2(X_0, \Theta_{X_0/k}) = 0.$$

Then for every A there exists a projective scheme X, flat over A, such that $X \otimes_A k = X_0$.

Combine Theorem 9(ii) with Theorem 4. In particular:

COROLLARY 4. — Let X_0 be the scheme of a complete nonsingular algebraic curve over k. Then, for every complete Noetherian local ring A with residue field k, there exists a "smooth-curve scheme" X over A such that $X \otimes_A k = X_0$.

REMARKS. —

1) Corollaries 3 and 4 are chiefly interesting when k has characteristic $p \neq 0$, by taking for A a discrete valuation ring of characteristic 0 with residue field k; for example, the "smallest possible A," namely one whose maximal ideal is generated by p. (In fact, because of Cohen's theorems, it is enough to know Corollaries 3 and 4 for one such ring A.) It may be noted in this connection that, according to the specialists, it is not known whether there exist schemes over a field k that are not the reduction modulo p of a flat scheme defined over such a ring A. At least the results of this section provide a method for systematically investigating this question. One should begin by examining whether the first obstruction in $H^2(X_0, \Theta_{X_0/k})$ is necessarily zero.

1962 supplement. — J.-P. Serre constructed a nonsingular projective threefold over an algebraically closed field k of characteristic $p > 0$ that is not obtained by reduction from a proper scheme over an integral local ring with residue field k and fraction field of characteristic zero. Mumford is said to have found an analogous example with a nonsingular projective surface.

2) It should be noted that Theorem 3, and the corresponding technique, applies only when the base ring (local, to fix ideas) is complete.

To pass from results known for the completion of a local ring to corresponding results for the local ring itself, one would need a fourth "fundamental theorem," whose definitive statement re-
207 mains to be found.

3) The results of this section (especially the preceding Corollaries 1 and 2) and of the next should be compared with the results of Kodaira–Spencer on the variation of complex structures. Using the conjectural theorem just mentioned, one should be able to conclude, under the hypotheses of Corollary 1 but without assuming A complete, that there exists a ring A' containing A, finite and unramified over A, such that $X \otimes_A A'$ and $X' \otimes_A A'$ are A'-isomorphic (where X, X' are two given proper flat A-schemes satisfying $X \otimes_A k = X' \otimes_A k = X_0$). This can at least be proved when $X_0 = \mathbf{P}^r_k$, using the corollary to Proposition 2. In any case, when $H^1(X_0, \Theta_{X_0/k}) = 0$, one can prove that the fibers of X, X' over every point y of $Y = \mathrm{Spec}(A)$ are isomorphic, at least after passing to the algebraic closure of the residue field $\kappa(y)$. (There is a local result, apparently stronger, in which A need no longer be assumed local.) Concerning "variations of structure" for projective space, let us also mention the following question, suggested by a corresponding problem

of Kodaira–Spencer. Let X be a scheme proper and flat over an integral local ring A with fraction field K and residue field k. Suppose that $X \otimes_A K$ is isomorphic to $\mathbf{P}^r_K$. Is it true that $X \otimes_A k = X_0$ is isomorphic to $\mathbf{P}^r_k$ (at least over the algebraic closure of k)? In this problem one may assume that A is a complete discrete valuation ring. There is an analogous problem when X_0 is an abelian variety.

1962 supplement. — The speaker is by then less optimistic about the results conjectured above. The problem concerning variations of structure of projective space was nevertheless solved affirmatively by Hironaka, and the analogous problem for abelian varieties by Koizumi.

7. Application to the theory of moduli

Since the speaker has only just begun to study this theory himself, we must limit ourselves to a few indications. For simplicity we work over a field k, i.e. in equal characteristic, although Theorem 8 also makes it possible to treat the general case, apparently without essential change. At present we have not gone beyond the "formal" stage; nevertheless, the speaker hopes to be able to construct genuine moduli schemes in certain cases and, in particular, for every integer g, to construct a scheme over the integers playing the role of a universal moduli scheme for nonsingular curves of genus g.

1962 supplement. — Mumford constructed the moduli schemes for curves of genus g (cf. SHMI). Theorem 10 also proves that the "Jacobi schemes of level n" in the theory of moduli are nonsingular, and indeed smooth over $\mathbf{Z}$.

208

Return to the situation and notation of Theorem 9, now supposing that A is a finite-dimensional local algebra over k, which for simplicity is assumed algebraically closed. Then $F(A)$ may be regarded as a covariant functor of A with values in the category of sets: a homomorphism of k-algebras $A \longrightarrow B$ defines a map $F(A) \longrightarrow F(B)$, since every flat A-scheme X such that $X \otimes_A k = X_0$ gives rise to a B-scheme $X \otimes_A B$ with the same properties. Suppose that one can find a complete Noetherian local k-algebra $\mathcal{O}$ and a functorial isomorphism

$$\mathrm{Hom}(\mathcal{O}, A) \xrightarrow{\sim} F(A) \qquad (*)$$

(where the first member denotes homomorphisms of k-algebras). It is easy to see that such an $\mathcal{O}$ is determined up to canonical isomorphism. The formal spectrum $\mathfrak{Y}$ of $\mathcal{O}$ (i.e. the one-point topological space equipped with the sheaf of topological rings reduced to $\mathcal{O}$) is then called the *formal moduli scheme* for X_0. (It need not exist.) Let $\mathfrak{r}$ be the maximal ideal of $\mathcal{O}$, and for every n put $\mathcal{O}_n = \mathcal{O}/\mathfrak{r}^{n+1}$ (so $\mathcal{O}_0 = k$). The canonical homomorphism $\mathcal{O} \longrightarrow \mathcal{O}_n$ is an element of $\mathrm{Hom}(\mathcal{O}, \mathcal{O}_n)$, and hence defines an element of $F(\mathcal{O}_n)$, i.e. a flat $\mathcal{O}_n$-scheme X_n whose reduction modulo $\mathfrak{r}$ is X_0. These X_n are obtained from one another by extension of scalars (here, by reduction), and consequently arise from a well-determined formal scheme $\mathfrak{X}$ over the formal moduli scheme $\mathfrak{Y}$; $\mathfrak{X}$ is flat over $\mathfrak{Y}$ and $\mathfrak{X}_0 = X_0$. The isomorphism (*) is then given, as is immediately seen, by associating to every homomorphism of k-algebras $\mathcal{O} \longrightarrow A$ the class of the A-scheme $\mathfrak{X} \otimes_{\mathcal{O}} A$ (i.e. to every morphism of k-schemes $Y' = \mathrm{Spec}(A) \longrightarrow \mathfrak{Y}$, one associates the Y'-scheme $\mathfrak{X} \otimes_{\mathfrak{Y}} Y'$ obtained by base change). Moreover, the isomorphism (*) and its preceding description remain valid if one assumes only that A is a complete Noetherian local k-algebra (not necessarily Artinian). Of course, as usual, $\mathcal{O}$ may very well have nilpotent elements *a priori*, and it seems probable that cases should exist in which $\mathcal{O}$ is itself Artinian without being equal to k. This shows how inadequate the Kodaira–Spencer point of view (which restricts to A that are regular rings) is *a priori* in the general case.

209

It remains to give sufficient conditions for a formal moduli scheme to exist for X_0, assumed proper over k. In general, it is easy to give simple necessary and sufficient conditions on a functor $A \longmapsto F(A)$ (from finite-dimensional local k-algebras to sets) for it to be expressible in the form

$\mathrm{Hom}(\mathcal{O}, A)$ for a suitable $\mathcal{O}$. We shall not detail them here. Let us merely point out that, in the case at hand, these conditions impose nontrivial cohomological conditions on X_0, and it seems unlikely that they always hold, although the speaker has not constructed a counterexample. On the other hand, it seems plausible that $H^0(X_0, \Theta_{X_0/k}) = 0$ is a sufficient condition (though by no means a necessary one) for the existence of a formal moduli scheme. We confine ourselves here to stating a theorem in a particularly simple case (whose analogue in the theory of analytic spaces is well known; cf. Kodaira–Spencer), which follows without difficulty from the results of the preceding section.

THEOREM 10. — Let X_0 be a scheme proper and nonsingular over the field k, such that

$$H^0(X_0, \Theta_{X_0/k}) = H^2(X_0, \Theta_{X_0/k}) = 0.$$

Then a formal moduli scheme exists for X_0, corresponding to a regular local ring $\mathcal{O}$ (i.e. a formal power-series algebra over k).

As already noted, it is not true in general that the formal scheme $\mathfrak{X}$ over $\mathcal{O}$ is algebraizable; it is nevertheless known to be so when X_0 is projective and $H^2(X_0, \underline{O}_{X_0}) = 0$ (Theorem 4), for example when X_0 has dimension 1. This lends some support to the hope of constructing a moduli scheme over the integers for curves of a given genus...

Methods such as those presented in this section can also be applied to the construction and study of Picard varieties, as well as to many other constructions. We shall return to this soon.

8. Application to the fundamental group

The techniques presented above make it possible to undertake a systematic study of the fundamental group, modeled on topological theory. The first two theorems stated in this section generalize results of a recent paper by Lang–Serre.

Let X be a scheme. An X-scheme X' is called an unramified covering of X if

i. X' is finite over X, i.e. it is defined by a coherent sheaf of algebras $\mathcal{A} = \mathcal{A}(X')$ on X;

ii. $\mathcal{A}$ is a locally free sheaf on X;

iii. for every $x \in X$, the quotient

$$\mathcal{A}_x/\mathfrak{m}_x\mathcal{A}_x = \mathcal{A}_x \otimes_{\underline{O}_x} \kappa(x)$$

is a separable algebra over $\kappa(x)$.

210

This notion of unramified covering (due to Serre and the speaker) has all the elementary properties one can reasonably expect, which we shall not list. Let us merely say that it gives rise to a Galois theory modeled on classical Galois theory (and containing it, with proofs that are rather simpler than the usual proofs of the latter) and on the Galois theory of topological coverings. More precisely, call a *geometric point* of a scheme X a morphism a from the spectrum ξ of an algebraically closed field Ω into X, i.e. the datum of an algebraically closed extension of the residue field $\kappa(x)$ of a point $x = |a|$ of X (called the *locality* of the geometric point a). If X' is an unramified covering of X, associate to it the set $E_a(X')$ of "geometric points of X' above a," i.e. the set of pairs consisting of a point $x' \in X'$ above x and a $\kappa(x)$-homomorphism $\kappa(x') \longrightarrow \Omega$. Thus, for fixed (X, a), one obtains a functor $F(X, a)$ from the category $R(X)$ of unramified coverings X' of X to the category of finite sets. If X is connected, the pair formed by $R(X)$ and $F(X, a)$ has the formal properties needed to be isomorphic to the analogous pair defined by a suitable compact totally disconnected topological group π (i.e. an inverse limit of finite groups): one takes the category $C(\pi)$ of finite

sets E on which π acts continuously, together with the forgetful functor $F(\pi)(E) = E$ from this category to the category of finite sets. Moreover, π is determined up to canonical isomorphism by the condition that $(C(\pi), F(\pi))$ be isomorphic to a given pair. Here π is called the *fundamental group* of the connected scheme X at the geometric point a, and is denoted $\pi_1(X, a)$. If X is not connected, it is replaced by the connected component of $x = |a|$. If X is connected, however, then the groups $\pi_1(X, a)$ for two geometric points a', a'' of X are isomorphic (the isomorphism being determined only up to inner automorphisms), so that, as usual, one may choose a to suit the problem, for example at the generic point of X, assumed irreducible. Of course, $\pi_1(X, a)$ is a covariant functor of the pointed scheme (X, a). Every statement concerning the classification of unramified coverings[2] can then be translated into group-theoretic language according to the familiar dictionary (except that here one must take account of the fact that the groups are topological). 211

Our aim is to develop the analogue of the homotopy exact sequence for fiber spaces, relative to a proper morphism $f : X \longrightarrow Y$. Since we do not know what the higher homotopy groups should be, the results will necessarily be incomplete. To apply the fundamental theorems of Section 3, we must first make explicit a few elementary lemmas concerning schemes over fields or Artinian rings (in accordance with the general procedure!).

LEMMA 1. — Let (X', a') be a pointed unramified covering associated to a pointed representation of $\pi_1(X, a)$ on a finite set E (with marked point e). Then the canonical morphism $\pi_1(X', a') \longrightarrow \pi_1(X, a)$ identifies the former group with the stabilizer of e in $\pi_1(X, a)$ (and is therefore injective).

LEMMA 2. — Let X be an algebraic scheme over a field k, and let k' be a radicial extension of k. Then every unramified covering of $X \otimes_k k'$ is obtained by inverse image (i.e. extension of scalars) from an unramified covering of X, determined up to canonical isomorphism.

It follows in particular from these two lemmas that for every algebraic extension K of k, and every geometric point a' of $X' = X \otimes_k K$ mapping to the geometric point a of X, the functorial homomorphism $\pi_1(X', a') \longrightarrow \pi_1(X, a)$ is injective.

LEMMA 3. — Let X be a complete scheme over a local Artinian ring A such that $H^0(X, \underline{O}_X) = A$. Let X' be an unramified covering of X, and put $A' = H^0(X', \underline{O}_{X'})$, which is therefore a finite ring over A (and may *a priori* be ramified over A). Let X_0, X_0' be the associated reduced subschemes of X, X' (obtained by quotienting by the sheaves of nilpotent elements in $\underline{O}_X, \underline{O}_{X'}$), and let k be a subfield of $A/\mathfrak{r}(A)$ over which $A/\mathfrak{r}(A)$ is finite (so X_0 is a complete algebraic scheme over k, and X_0' is an unramified covering of it). Finally, let Ω be an algebraically closed extension of k, and consider the unramified covering $X_0' \otimes_k \Omega$ of $X_0 \otimes_k \Omega$. Then the following two conditions are equivalent:

i. $X_0' \otimes_k \Omega$ is completely split over $X_0 \otimes_k \Omega$.

ii. The natural morphism $X' \longrightarrow X \otimes_A A'$ is an isomorphism.

Under these conditions, A' is an unramified extension of A. Finally, if X' is connected, condition (i) is equivalent to the following apparently weaker condition:

i bis. $X_0' \otimes_k \Omega$ admits a regular section over $X_0 \otimes_k \Omega$.

When condition (ii) holds, we shall say that the unramified covering X' of X is *geometrically trivial*. 212

LEMMA 4. — Let $f : X \longrightarrow Y$ be a proper morphism such that $f_*(\underline{O}_X) = \underline{O}_Y$. Let a be a geometric point of X, and let b be its image in Y. Then $\pi_1(X, a) \longrightarrow \pi_1(Y, b)$ is surjective.

[2]The printed text says "inseparable coverings." The paragraph concerns the category $R(X)$ of unramified coverings and its equivalence with finite sets carrying a continuous $\pi_1(X, a)$-action.

Indeed, it is enough to prove the following: if an unramified covering Y' of Y (corresponding to a locally free sheaf of algebras $\mathcal{A}$) is such that $X \otimes_Y Y'$ is disconnected, then Y' is disconnected as well. Indeed, $f^*(\mathcal{A})$ is then a direct sum of two nonzero sheaves of rings, and hence so is its direct image; but the latter is precisely $\mathcal{A} \otimes f_*(\underline{O}_X) = \mathcal{A}$.

LEMMA 5. — Let X be a complete scheme over a field k. Suppose that $H^0(X, \underline{O}_X)$ is a local ring A and that $A/\mathfrak{r}(A)$ is radicial over k. Let Ω be an algebraic closure of k, and put $\overline{X} = X \otimes_k \Omega$ (which is connected). Choose a geometric point $\overline{a}$ of $\overline{X}$ mapping to the geometric point a of X. Then there is an exact sequence

$$e \longrightarrow \pi_1(\overline{X}, \overline{a}) \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(k, b) \longrightarrow e$$

(where $\pi_1(k, b)$ is the Galois group of Ω over k).

The injectivity of the first homomorphism was seen in Lemmas 1 and 2; exactness in the middle follows from Lemma 3; and the surjectivity of the last homomorphism (the only part that uses the fact that $A/\mathfrak{r}(A)$ is radicial) follows from Lemma 4.

PROPOSITION 4. — Let $f : X \longrightarrow Y$ be a proper flat morphism such that for every $y \in Y$ the algebra $H^0(f^{-1}(y), \underline{O}_{f^{-1}(y)})$ is separable over the residue field $\kappa(y)$ (as is the case, for example, if $f^{-1}(y)$ is a separable scheme over $\kappa(y)$, i.e. reduced and such that the fields corresponding to its irreducible components are separable extensions of $\kappa(y)$). Then the covering Y' of Y associated with $f_*(\underline{O}_X)$ is unramified.

213 The proof is easy from Theorem 2.

Together with Lemma 1, this proposition reduces the homotopy-theoretic study of proper flat morphisms (with separable fibers), for all practical purposes, to the case $f_*(\underline{O}_X) = \underline{O}_Y$ (for, using Stein factorization, one replaces Y by Y').

REMARK. — A flat morphism of finite type whose fibers are separable (respectively nonsingular) schemes is called *separable* (respectively *smooth*). One proves that if f is flat and $f^{-1}(y)$ is separable (respectively nonsingular), then there is a neighborhood of $f^{-1}(y)$ on which f is separable (respectively smooth). The same result holds for "absolutely normal" ("Bertini's theorem").

Let $f : X \longrightarrow Y$ be a proper morphism such that

$$f_*(\underline{O}_X) = \underline{O}_Y, \tag{i}$$

and let X' be a scheme finite over X. Let Y' be the covering of Y corresponding to the Stein factorization of $X' \longrightarrow Y$ (cf. Theorem 5). Let $y \in Y$. The set of connected components of the fiber F' of X' over y therefore identifies with the set of points y' of Y' above y (Theorem 5). Consider the evident morphism

$$X' \longrightarrow X \times_Y Y' \tag{*}$$

induced by the natural morphisms $X' \longrightarrow X$ and $X' \longrightarrow Y'$.[3] This is an isomorphism whenever X' has the form $X \times_Y Y''$, where Y'' is an unramified covering of Y; then Y' is simply Y'' and (*) is the identity. We seek conditions under which X' has the form just described, i.e. under which Y' is unramified and (*) is an isomorphism. To this end, let F be the fiber of X over y; it is a proper scheme over $\kappa(y)$, and F' is a covering of it (unramified if X' is). Let F'_1 be a connected component of F' corresponding to a point y'_1 of Y' above y. Suppose in addition that

(*ii*) X' is unramified over X at the points of F'_1 (hence F'_1 is an unramified covering of F), and

(*iii*) F'_1 is a geometrically trivial covering of F (cf. Lemma 3).

[3] The printed text omits the prime on the second X'; the morphism to Y' is the one in the Stein factorization of $X' \to Y$.

THEOREM 11. — Under these conditions, there exists an open neighborhood U' of y'_1 in Y' such that (*) is an isomorphism above U'. Moreover, Y' is unramified at y'_1 over Y (although it may be ramified at the other points y' of Y' above y).

The conditions (*ii*) and (*iii*) are of course also necessary for the conclusion of the theorem to hold. The proof is easy, using Lemma 3 and Theorem 2. 214

COROLLARY 1. — Continue to assume (*i*). An unramified covering X' of X is isomorphic to the inverse image of an unramified covering Y' of Y if and only if X' induces a geometrically trivial covering on every fiber $f^{-1}(y)$.

By Theorem 11, the set of points of Y for which this condition holds is open, so it is enough to verify it at the closed points y... The following statement is equivalent to Corollary 1: the kernel of the homomorphism $\pi_1(X) \longrightarrow \pi_1(Y)$ (surjective by Lemma 4) is the closed normal subgroup generated by the images in $\pi_1(X)$ of the groups $\pi_1(\overline{f^{-1}(y)})$, where

$$\overline{f^{-1}(y)} = f^{-1}(y) \otimes_{\kappa(y)} \overline{\kappa(y)},$$

and $\overline{\kappa(y)}$ denotes an algebraic closure of $\kappa(y)$. Since one cannot choose the same base point for all the fibers, the homomorphisms $\pi_1(\overline{f^{-1}(y)}) \longrightarrow \pi_1(X)$ are in any event determined (once a base point for X, and hence for Y, has been chosen) only up to composition with an inner automorphism of $\pi_1(X)$.

COROLLARY 2. — Under the general hypotheses of Theorem 11, suppose in addition that Y, X, X' are integral, and let K, L, L' be their function fields. Then there exists a separable intermediate extension K' of K in L', linearly disjoint from L, such that $L' = LK'$ (whence $L' = L \otimes_K K'$).

(Apply the last part of Lemma 3 to the geometric fiber of X.) The most interesting application of Theorem 11 occurs when f is separable. Then X' is also separable over Y, so by Proposition 4, Y' is unramified over Y; hence the second member $X \times_Y Y'$ in (*) is unramified over X. It follows easily that:

COROLLARY 3. — Suppose, in addition to (*i*), that f is separable. Let X' be a connected unramified covering of X. Then X'[4] is the inverse image of an unramified covering Y' of Y if and only if the covering $\overline{F}'$ induced on one geometric fiber $\overline{F} = f^{-1}(y)$ admits a regular section.

Note that it was not necessary to suppose that $\overline{F}'$ was geometrically trivial over $\overline{F}$ (this follows *a posteriori*, although *a priori* it is a much stronger condition). Corollary 3 is in fact equivalent to the following statement. 215

COROLLARY 4. — Let $f : X \longrightarrow Y$ be a proper separable morphism such that $f_*(\underline{O}_X) = \underline{O}_Y$. Let $\overline{F}$ be the geometric fiber of a point $y \in Y$, and choose a geometric point in $\overline{F}$; the morphisms $\overline{F} \longrightarrow X \longrightarrow Y$ then give geometric points in X, Y, which are taken as base points for the fundamental groups of $\overline{F}, X, Y$. Under these conditions there is an exact sequence

$$\boxed{\pi_1(\overline{F}) \longrightarrow \pi_1(X) \longrightarrow \pi_1(Y) \longrightarrow e}.$$

The following two statements of Serre–Lang, now freed of any normality hypothesis, follow easily.

COROLLARY 5. — Let X, Y be connected schemes over a field k. Suppose that the reduced scheme X_{red} is separable over k (as is automatically the case if k is perfect), and that X or Y is proper over k.[5] Choose a geometric point a (respectively b) in X (respectively Y), hence a geometric point $c = (a, b)$ in $X \times_k Y$ and a natural morphism

$$\pi_1(X \times_k Y, c) \longrightarrow \pi_1(X, a) \times \pi_1(Y, b)$$

[4]The printed text has "X" here; the subject of the criterion is the covering X' introduced in the preceding sentence.
[5]Condition added by the 1962 erratum.

(induced by the functorial morphisms from $\pi_1(X \times Y, c)$ to $\pi_1(X, a)$ and $\pi_1(Y, b)$). This morphism is injective, and is even bijective if k is algebraically closed.

(Surjectivity in the latter case is more or less trivial.) One deduces, with Serre–Lang:

COROLLARY 6. — Let X be a connected algebraic scheme over an algebraically closed field k, and let K be an algebraically closed extension of k. Then the fundamental groups of X and $X \otimes_k K$ are the same; i.e. every unramified covering of the latter scheme is obtained by extension of scalars from an unramified covering of X, unique up to isomorphism.

REMARKS. —

1) Using Proposition 4, one sees that the hypothesis $f_*(\underline{O}_X) = \underline{O}_Y$ in Corollary 4 is not essential. In the general case, instead of placing the unit group e after $\pi_1(Y)$, one should continue with

$$\pi_0(\overline{F}) \longrightarrow \pi_0(X) \longrightarrow \pi_0(Y) \longrightarrow e.$$

216 as in algebraic topology.

2) In general, nothing can yet be said about the kernel of $\pi_1(\overline{F}) \longrightarrow \pi_1(X)$, which ought to involve a $\pi_2(Y)$. It nevertheless seems that one should be able to prove that $\pi_1(\overline{F}) \longrightarrow \pi_1(X)$ is injective when Y is the spectrum of a local ring A, using Theorem 12 below (which asserts this when A is complete).

Theorem 11 used only Theorems 1 and 2. We now apply Theorem 3, using the following elementary lemma.

LEMMA 6. — Let X be a scheme and let X_0 be the corresponding reduced scheme (i.e. with the nilpotent elements killed). Then every unramified covering X_0' of X_0 is induced by an unramified covering X' of X, determined up to canonical isomorphism.

This purely local lemma plays a role here analogous to that of Theorem 8 in the theory of moduli. Combining it with the existence theorem (Theorem 3) gives:

THEOREM 12. — Let A be a complete Noetherian local ring with residue field k, and let X be a scheme proper over A. Then every unramified covering X_0' of $X_0 = X \otimes_A k$ is induced by an unramified covering X' of X, unique up to isomorphism.

In other words:

COROLLARY 1. — Choose a geometric point in X_0 as base point for the fundamental groups of X_0 and X. Then the canonical homomorphism $\pi_1(X_0) \longrightarrow \pi_1(X)$ is an isomorphism.

Now apply Lemma 5 to X_0 (assuming for simplicity that $H^0(X_0, \underline{O}_{X_0}) = k$), and observe that since A is complete, its unramified extensions correspond to the unramified extensions of its residue field, i.e. $\pi_1(Y) = \pi_1(k)$ (where $Y = \mathrm{Spec}(A)$). This gives the exact sequence

$$e \longrightarrow \pi_1(\overline{X}_0) \longrightarrow \pi_1(X) \longrightarrow \pi_1(Y) \longrightarrow e.$$

COROLLARY 2. — Let $f : X \longrightarrow Y$ be a proper flat morphism, let y_1 be a point of Y, and let y_0 be a specialization of y_1. Consider the corresponding "geometric" fibers $\overline{X}_1$ and $\overline{X}_0$, and suppose that $\overline{X}_0$ is separable and connected (which implies that $\overline{X}_1$ satisfies the same conditions). Then one can find a group homomorphism $\pi_1(\overline{X}_1) \longrightarrow \pi_1(\overline{X}_0)$, defined up to an inner automorphism,
217 and this homomorphism is surjective.

One might hope that this homomorphism is always bijective. Unfortunately this is not so in general when $\kappa(y_0)$ has characteristic > 0. Below we shall nevertheless obtain a bound on the kernel of this homomorphism (at least when $\overline{X}_0$ is nonsingular), implying that if $\kappa(y_0)$ has characteristic 0, then the homomorphism above is bijective (a result that could also be proved by transcendental methods). In any case, we already obtain a bound on the π_1 of a special fiber in terms of that of a

generic fiber. For example, using the fact that an algebraic curve in characteristic p lifts to a curve in characteristic 0 (Theorem 9, Corollary 4), one finds by transcendental methods:

COROLLARY 3. — Let X_0 be the scheme of a complete nonsingular curve over an algebraically closed field of arbitrary characteristic, and let g be the genus of X_0. Then $\pi_1(X_0)$ admits $2g$ topological generators, subject to the familiar relation.

By a familiar hyperplane-section technique one deduces:

COROLLARY 4. — Let X be a nonsingular projective scheme over an algebraically closed field of arbitrary characteristic. Then $\pi_1(X)$ admits a finite number of topological generators.

Let us seek the kernel of the homomorphism $\pi_1(\overline{X}_1) \longrightarrow \pi_1(\overline{X}_0)$. For this purpose we may suppose that Y is the spectrum of a complete discrete valuation ring $V = \underline{O}_y$ $(y = y_0)$. The question is equivalent to the following: given an unramified covering $\overline{X}_1'$ of $\overline{X}_1$ (which may, if desired, be assumed Galois), determine under what conditions it comes from an unramified covering of X_0. *A priori*, the given covering is obtained by extension of scalars from an unramified covering X_1' of $X_1 \otimes_K K'$, where K' is a finite subextension of the algebraic closure $\overline{K}$ of the fraction field K of V; if $\overline{X}_1'$ is Galois with group G, one may choose X_1' Galois with group G. One then has: for $X_1' \otimes_{K'} \overline{K} = \overline{X}_1'$ to come from an unramified covering of X_0, it is necessary and sufficient that there exist a finite extension K'' of K' in $\overline{K}$ such that $X_1'' = X_1' \otimes_{K'} K''$ has the form $X'' \otimes_{V''} K''$, where V'' is the integral closure of V in K'' and X'' is an unramified covering of $X \otimes_V V''$. Suppose, for example, that X_0 is absolutely normal. It follows that $X \otimes_V V''$ is normal (being flat over V'' with normal special fiber), and its function field is the same field $K''(X_1)$ as that of

$$X_1 \otimes_K K'' = (X \otimes_V K) \otimes_K K'' = X \otimes_V K'' = (X \otimes_V V'') \otimes_{V''} K''.$$

Let $L'' = K''(X_1')$ be the function field of $X_1' \otimes_{K'} K''$. It is a finite separable extension of $K''(X_1)$, and the preceding condition also means that L'' is an unramified extension of the function field of $X \otimes_V V''$ (i.e. the normalization of $X \otimes_V V''$ in L'' is unramified over $X \otimes_V V''$). It is enough to check that L'' is unramified at the points of the special fiber of $X \otimes_V V''$ (since it is unramified over the generic fiber $X_1 \otimes_K K''$). If X_0 is nonsingular, the "purity theorem" of Nagata–Zariski shows that it is enough to check that L'' is unramified over the local ring O'' at the generic point of the special fiber of $X \otimes_V V''$. This is a discrete valuation ring, the normalization in $K''(X_1)$ of the local ring $O \subset K(X_1)$ at the generic point of the special fiber of X. Thus we have obtained: 218

COROLLARY 5. — Under the preceding hypotheses and notation, the unramified covering $X_1' \otimes_{K'} \overline{K}$ of $\overline{X}_1 = X_1 \otimes_K \overline{K}$ comes from an unramified covering of $\overline{X}_0$ if and only if there exists a finite subextension K'' of $\overline{K}/K'$ such that $K''(X_1')$ is unramified over the discrete valuation ring $O'' \subset K''(X_1)$.

Now observe that O'' is the normalization in $K''(X_1)$ of the discrete valuation ring $O' \subset K'(X_1)$ (the normalization of O in $K'(X_1)$), and that O' contains the normalization V' of V in K', a uniformizer u of V' being also a uniformizer of O'. Suppose now that X_1' is Galois with Galois group G of order n prime to the characteristic p of $\kappa(y_0)$ (which is also the characteristic of the residue field of O'). Thus $K'(X_1')$ is tamely ramified over O', and it follows easily ("Abhyankar's lemma") that after adjoining an nth root v of a uniformizer of O', it becomes unramified over the normalization of O' in $K'(X_1)(v)$. But one may take for v an nth root of a uniformizer of V', which proves that the condition of Corollary 5 is satisfied. (The possibility of using Abhyankar's lemma and the purity theorem was sold to me by Serre.) To express the result obtained, for every compact totally disconnected group π let $\tilde{\pi}$ denote the quotient of π by the closed subgroup generated by its Sylow p-subgroups, i.e. the inverse limit of the discrete quotient groups of π whose orders are prime to p. With this notation one obtains:

THEOREM 13. — Let $f : X \longrightarrow Y$ be a proper flat morphism, let y_1 be a point of Y, and let y_0 be a specialization of y_1. Suppose that $\overline{X}_0$ is connected and nonsingular. Then the homomorphism

$\tilde{\pi}_1(\overline{X}_1) \longrightarrow \tilde{\pi}_1(\overline{X}_0)$ induced by the homomorphism that is surjective by Theorem 12, Corollary 2, is an isomorphism.

219 In other words:

COROLLARY 1. — The classification of unramified Galois coverings whose Galois group has order prime to the characteristic p of $\kappa(y_0)$ is the same for $\overline{X}_0$ and $\overline{X}_1$.

In particular, if $\kappa(y_0)$ has characteristic zero, one obtains algebraically that $\pi_1(\overline{X}_1) \longrightarrow \pi_1(\overline{X}_0)$ is bijective.

Finally, the techniques used also give the following result, which is more general than Theorem 13.

THEOREM 14. — Let $f : X \longrightarrow Y$ be a proper smooth morphism, and let D be a closed subscheme of X, smooth over Y and of codimension 1 at all its points. For a fiber $Z = f^{-1}(z)$ of f, put $Z' = Z - Z \cap D$ and let $\pi_1^t(Z')$ be the quotient of the fundamental group $\pi_1(Z')$ that classifies the unramified coverings of Z' which are tamely ramified above $\overline{Z \cap D}$. Let y_0, y_1 be as in Theorem 13. Then there is a surjective homomorphism (defined up to an inner automorphism)

$$\pi_1^t(\overline{X}_1') \longrightarrow \pi_1^t(\overline{X}_0'),$$

and the corresponding homomorphism

$$\tilde{\pi}_1^t(\overline{X}_1') \longrightarrow \tilde{\pi}_1^t(\overline{X}_0')$$

is an isomorphism.

Corresponding variants of the corollaries to Theorem 13 and of Corollary 4 to Theorem 12 follow. Likewise, using Theorem 9, Corollary 3, one finds by transcendental methods:

COROLLARY. — Let X_0 be the scheme of a complete nonsingular curve over an algebraically closed field of arbitrary characteristic, and let $S = (s_i)_{1 \leq i \leq n}$ be a finite subset of X_0 with n elements. Then $\pi_1^t(X_0 - S)$ admits $2g + n$ topological generators x_i, y_i $(1 \leq i \leq g)$ and σ_j $(1 \leq j \leq n)$, subject to the relation

$$\left(\prod_i x_i y_i x_i^{-1} y_i^{-1} \right) \sigma_1 \cdots \sigma_n = 1,$$

where the σ_j are generators of the inertia groups corresponding to the s_j. For every finite group G of order prime to the characteristic, generated by elements x_i, y_i, σ_j satisfying the preceding relation, there exists an unramified Galois covering of $X_0 - S$ with group G, whose inertia groups at the points s_j are generated by the σ_j.

When X_0 has genus 0 and $n = 3$, this gives a solution of the "Problem of the three points," at least for Galois coverings of order prime to the characteristic. (Here Theorem 9 is unnecessary, and it seems moreover that the preceding corollary can be deduced from the special case considered in
220 the problem of the three points.)

REMARKS. —

1) A more complete study, no doubt involving generalized Galois coverings of X, X_0, X_1 (with possibly infinitesimal Galois group), should make it possible to recover the kernel in Theorem 12, Corollary 2. By contrast, a study of coverings admitting ramification that is not tame seems likely to be much more difficult.

2) Lemma 6, together with a result of Grauert concerning the formal completion of a nonsingular projective scheme along a hyperplane section (or with the theorem, not yet proved, mentioned in Remark 2 after Theorem 11), also makes it possible to prove in "abstract" algebraic geometry the classical Lefschetz theorem on the fundamental group.

BIBLIOGRAPHY

[1] Dieudonné (J.) and Grothendieck (A.). — *Éléments de géométrie algébrique*, Publications mathématiques de l'Institut des Hautes Études Scientifiques (forthcoming).

[2] Grothendieck (Alexander). — *The cohomology theory of abstract algebraic varieties*. International Congress of Mathematicians [1958. Edinburgh] (forthcoming).

[3] Serre (Jean-Pierre). — *Faisceaux algébriques cohérents*, Annals of Math., Series 2, vol. 61, 1955, pp. 197–278.

[4] Serre (Jean-Pierre). — *Géométrie algébrique et géométrie analytique*, Ann. Institut Fourier Grenoble, vol. 6, 1955–1956, pp. 1–42.

Erratum to Exposé no. 182

298 **2. Formal geometry and algebraic geometry.**

[Bourbaki Seminar, vol. 11, 1958/59, no. 182, 28 pp.]

Page 8, Theorem 6. — The hypothesis "f is quasi-projective" may be replaced by the weaker hypothesis "f is separated," by virtue of the following result (cf. SGA VIII, 6.2): every quasi-finite
299 separated morphism $f : X \longrightarrow Y$ is quasi-projective.

Page 12, and page 14, Remark 1. — J.-P. Serre constructed a nonsingular projective variety of dimension 3 over an algebraically closed field k of characteristic $p > 0$ which is not obtained by reduction from a proper scheme over an integral local ring with residue field k and fraction field of characteristic zero.[1] Mumford is said to have found an analogous example involving a nonsingular projective surface.

Page 15, Remarks 2 and 3. — I am now less optimistic about the results conjectured here. Nevertheless, the problem concerning variations of structure for projective space, mentioned at the end of Remark 3, has been solved affirmatively by Hironaka, and the analogous problem for abelian varieties has been solved by Koizumi.

Page 16, line 3. — Mumford recently constructed the moduli schemes for curves of genus g (cf. SHMI). Theorem 10 also proves that the "Jacobi schemes of level n" in the theory of moduli are nonsingular (and in fact smooth over $\mathbf{Z}$).

Page 23, line 12, first line of Corollary 5. — Add: "X or Y being proper over k."

The substance of §§ 1 through 5 is contained in the published part of EGA III, and that of §§ 6 and 7 is contained in SGA III. For the study of the fundamental group, see SGA V, IX, X, XI, as well as SGA 1962 (Exposés X, XII, and XIII) for Lefschetz-type theorems and numerous open questions. Only the theory of tamely ramified coverings (cf. Theorem 14) has not yet received a formal exposition. The corollary to Theorem 14, which gives the complete determination of Galois coverings of order prime to the characteristic of an algebraic curve over an algebraically closed field, has been used essentially on three occasions:

1° in Igusa's proof of the Picard inequality for nonsingular projective surfaces in arbitrary characteristic;

2° in the study (developed independently by Ogg and Šafarevič) of the group of homogeneous principal bundles under an abelian variety defined over a one-variable function field, in arbitrary characteristic;

3° in Artin's recent proof of certain key theorems on the "Weil cohomology" of algebraic varieties.

[1] Serre (Jean-Pierre). — Exemples de variétés projectives en caractéristique p non relevables en caractéristique zéro, *Proc. Nat. Acad. Sc. U.S.A.*, vol. 47, 1961, pp. 108–109.

Chapter 3

Exposé 190 – Descent Technique I

299

BOURBAKI SEMINAR
12th year, 1959/60, no. 190
December 1959

DESCENT TECHNIQUE AND EXISTENCE THEOREMS IN ALGEBRAIC GEOMETRY
I. GENERALITIES. DESCENT BY FAITHFULLY FLAT MORPHISMS

by Alexander GROTHENDIECK

From a technical point of view, the present exposé and those that follow it may be regarded as variations on Hilbert's celebrated "Theorem 90." The introduction of the descent method into algebraic geometry seems to be due to A. Weil, under the name "descent of the base field." Weil restricts himself to finite separable field extensions. The case of radicial extensions of height 1 was subsequently treated by P. Cartier. For lack of the language of schemes, and more particularly for lack of nilpotent elements in the rings one allowed oneself to consider, Cartier had been unable to formulate the essential identity between these two cases.

At present, it seems that the general descent technique to be presented (combined, where appropriate, with the fundamental theorems of "formal geometry," cf. [3]) underlies most existence theorems in algebraic geometry.[1] It should moreover be noted that this technique can certainly be carried over to "analytic geometry," and one may hope that in the near future specialists will be able to prove the "analytic" analogues of the existence theorems in formal geometry that will be given in Exposé II. In any case, the recent work of Kodaira–Spencer, whose methods seem ill-suited to the definition and study of "moduli varieties" near their singular points, shows quite clearly the need for methods closer to scheme theory (which must of course be supplemented by transcendental techniques).

In the present Exposé I we shall treat the most elementary case of descent (indicated in the title). The applications of Theorems 1, 2, and 3 below are nevertheless already very numerous. We have confined ourselves to giving a few of them as examples, without attempting to state them here in the greatest possible generality.

We shall freely use the language of schemes, for which we refer to the exposé already cited, as well as to [2]. We note explicitly, however, that the preschemes considered in the present exposé are not necessarily Noetherian, and the

300

[1] The 1962 erratum calls this formulation excessive: it is true to a large extent for the nonprojective techniques treated in the first two exposés of the TDTE I–VI series, but not for the projective techniques (TDTE IV, V, and VI).

morphisms are not necessarily of finite type. Thus, if A is a Noetherian local ring with completion $\widehat{A}$, we shall have to consider the non-Noetherian ring $\widehat{A} \otimes_A \widehat{A}$ and the morphisms of affine schemes corresponding to the inclusions among the rings under consideration.

A. Preliminaries on categories

1. Fibered categories, descent data, and $\mathcal{F}$-descent morphisms

a. DEFINITION 1.1. — A *fibered category* $\mathcal{F}$ over a base $\mathcal{C}$ consists of a category $\mathcal{C}$, a category $\mathcal{F}_X$ for every $X \in \mathcal{C}$, a functor $f^* : \mathcal{F}_Y \longrightarrow \mathcal{F}_X$ for every $\mathcal{C}$-morphism $f : X \longrightarrow Y$, also denoted by

$$f^*(\xi) = \xi \times_Y X$$

for $\xi \in \mathcal{F}_Y$ (with X regarded as an "object of $\mathcal{C}$ over Y," i.e. equipped with the morphism f), and finally, for two composable morphisms $X \overset{f}{\to} Y \overset{g}{\to} Z$, an isomorphism of functors

$$c_{f,g} : (gf)^* \longrightarrow f^* g^*,$$

these data being subject to the following conditions:

(i) $\mathrm{id}_X^* = \mathrm{id}$;

(ii) $c_{f,g}$ is the identity isomorphism if either f or g is an identity isomorphism;

(iii) given three composable morphisms $X \overset{f}{\to} Y \overset{g}{\to} Z \overset{h}{\to} T$, the following diagram of isomorphisms, formed using isomorphisms of the form $c_{u,v}$, is commutative:

$$\begin{array}{ccc} (h(gf))^* = ((hg)f)^* & \longrightarrow & f^*(hg)^* \\ \downarrow & & \downarrow \\ (gf)^* h^* & \longrightarrow & (f^*g^*)h^* = f^*(g^*h^*). \end{array}$$

EXAMPLE 1. — Let $\mathcal{C}$ be a category in which fiber products exist. A fibered category $\mathcal{F}$ over $\mathcal{C}$ is then defined by letting $\mathcal{F}_X$ be the category of objects of $\mathcal{C}$ over X; the functor $f^* : \mathcal{F}_Y \longrightarrow \mathcal{F}_X$ corresponding to a morphism $f : X \longrightarrow Y$ is defined by the fiber product $Z \longmapsto Z \times_Y X$.

EXAMPLE 2. — Let $\mathcal{C}$ be the category of preschemes, and for $X \in \mathcal{C}$ let $\mathcal{F}_X$ be the category of quasi-coherent sheaves of modules on X. If $f : X \longrightarrow Y$ is a morphism of preschemes, then
301 $f^* : \mathcal{F}_Y \longrightarrow \mathcal{F}_X$ is the inverse-image functor
on sheaves of modules. This gives a fibered category over $\mathcal{C}$.

b. DEFINITION 1.2. — A diagram

$$E \overset{u}{\to} E' \underset{v_2}{\overset{v_1}{\rightrightarrows}} E''$$

of maps of sets is called *exact* if u is a bijection from E onto the part of E' consisting of the $x' \in E'$ such that $v_1(x') = v_2(x')$.

DEFINITION 1.3. — Let $\mathcal{F}$ be a fibered category over $\mathcal{C}$, and consider a diagram

$$S \overset{\alpha}{\leftarrow} S' \underset{\beta_2}{\overset{\beta_1}{\rightrightarrows}} S'' \qquad (+)$$

of morphisms in $\mathcal{C}$ such that $\alpha\beta_1 = \alpha\beta_2$. This diagram is called $\mathcal{F}$-*exact* if, for every pair ξ, η of objects of $\mathcal{F}_S$, the following diagram of maps of sets is exact:

$$\mathrm{Hom}(\xi,\eta) \xrightarrow{\alpha^*} \mathrm{Hom}(\alpha^*\xi,\alpha^*\eta) \underset{\beta_2^*}{\overset{\beta_1^*}{\rightrightarrows}} \mathrm{Hom}(\gamma^*\xi,\gamma^*\eta), \qquad (+')$$

where $\gamma = \alpha\beta_1 = \alpha\beta_2$. In the latter diagram we have identified $\beta_i^*\alpha^*$ with $(\alpha\beta_i)^* = \gamma^*$ by means of $c_{\beta_i,\alpha}$, for simplicity.

DEFINITION 1.4. — Let $\mathcal{F}$ be a fibered category over $\mathcal{C}$, and consider two morphisms $\beta_1, \beta_2 : S'' \longrightarrow S'$ in $\mathcal{C}$. Let $\xi' \in \mathcal{F}_{S'}$. A *gluing datum* on ξ' (relative to the pair (β_1, β_2)) is an isomorphism from $\beta_1^*(\xi')$ to $\beta_2^*(\xi')$. If $\xi', \eta' \in \mathcal{F}_{S'}$ are each equipped with a gluing datum, a morphism $u : \xi' \longrightarrow \eta'$ in $\mathcal{F}_{S'}$ is said to be *compatible with the gluing data* if the following diagram is commutative:

$$\begin{array}{ccc} \beta_1^*(\xi') & \longrightarrow & \beta_2^*(\xi') \\ \downarrow & & \downarrow \\ \beta_1^*(\eta') & \longrightarrow & \beta_2^*(\eta'). \end{array}$$

In this way the objects of $\mathcal{F}_{S'}$ equipped with a gluing datum (relative to β_1, β_2) form a category. If $\alpha : S' \longrightarrow S$ is a morphism such that $\alpha\beta_1 = \alpha\beta_2$, then for every $\xi \in \mathcal{F}_S$ the object $\xi' = \alpha^*(\xi)$ 302
of $\mathcal{F}_{S'}$ carries a canonical gluing datum, since

$$\beta_i^*\alpha^*(\xi) \simeq (\alpha\beta_i)^*(\xi) = \gamma^*(\xi),$$

where again $\gamma = \alpha\beta_1 = \alpha\beta_2$; moreover, if $u : \xi \longrightarrow \eta$ is a morphism in $\mathcal{F}_S$, then

$$\alpha^*(u) : \alpha^*(\xi) \longrightarrow \alpha^*(\eta)$$

is a morphism in $\mathcal{F}_{S'}$ compatible with the canonical gluing data. Thus one obtains a canonical functor from $\mathcal{F}_S$ to the category of objects of $\mathcal{F}_{S'}$ equipped with a gluing datum relative to the pair (β_1, β_2). Definition 1.3 may therefore also be expressed by saying that the diagram $(+)$ there is $\mathcal{F}$-exact if the preceding functor is "fully faithful," i.e. defines an equivalence of $\mathcal{F}_S$ with a full subcategory of the category of objects of $\mathcal{F}_{S'}$ equipped with a gluing datum relative to (β_1, β_2).

DEFINITION 1.5. — A gluing datum on $\xi' \in \mathcal{F}_{S'}$ is called *effective* (relative to α) if ξ', equipped with this datum, is isomorphic to some $\alpha^*(\xi)$ with $\xi \in \mathcal{F}_S$.

When the diagram $(+)$ is $\mathcal{F}$-exact, the object ξ in the preceding definition is therefore determined up to unique isomorphism, and the category $\mathcal{F}_S$ is equivalent to the category of objects of $\mathcal{F}_{S'}$ equipped with an effective gluing datum.

c. The most important case is that in which

$$S'' = S' \times_S S',$$

the β_i being the two projections p_1 and p_2 from $S' \times_S S'$ onto its two factors (we now assume that fiber products exist in $\mathcal{C}$). There are then two immediately necessary conditions for a gluing datum $\varphi : p_1^*(\xi') \longrightarrow p_2^*(\xi')$ on an object $\xi' \in \mathcal{F}_{S'}$ to be effective. One has

$$\Delta^*(\varphi) = \mathrm{id}_{\xi'}, \qquad (\mathrm{i})$$

where $\Delta : S' \longrightarrow S' \times_S S'$ denotes the diagonal morphism, and where $\Delta^* p_i^*(\xi')$ has been identified with $(p_i\Delta)^*(\xi') = \xi'$. One also has

$$p_{31}^*(\varphi) = p_{32}^*(\varphi)\, p_{21}^*(\varphi), \qquad (\mathrm{ii})$$

303

where p_{ij} denotes the canonical projection from $S' \times_S S' \times_S S'$ onto the partial product of its ith and jth factors.

DEFINITION 1.6. — A *descent datum* on $\xi' \in \mathcal{F}_{S'}$, relative to the morphism $\alpha : S' \longrightarrow S$, is a gluing datum on ξ' relative to the pair (p_1, p_2) of canonical projections $S' \times_S S' \longrightarrow S'$ satisfying conditions (i) and (ii) above.

DEFINITION 1.7. — A morphism $\alpha : S' \longrightarrow S$ is called an *$\mathcal{F}$-descent morphism* if the diagram

$$S \xleftarrow{\alpha} S' \underset{p_2}{\overset{p_1}{\rightrightarrows}} S' \times_S S'$$

is $\mathcal{F}$-exact (Definition 1.3); one says that α is a *strict $\mathcal{F}$-descent morphism* if, in addition, every descent datum (Definition 1.6) on an object of $\mathcal{F}_{S'}$ is effective.

The latter condition may also be stated more pictorially as follows: "it amounts to the same thing" to specify an object of $\mathcal{F}_S$ or an object of $\mathcal{F}_{S'}$ equipped with a descent datum.

We note that if a morphism $\alpha : S' \longrightarrow S$ admits a "section" $s : S \longrightarrow S'$ (i.e. a morphism s such that $\alpha s = \mathrm{id}_S$), then it is a strict $\mathcal{F}$-descent morphism.[2] Indeed, if $\xi' \in \mathcal{F}_{S'}$ is equipped with a descent datum relative to α, it "comes from" $\xi = s^*(\xi')$.

d. The preceding notions may be presented more intuitively by introducing, for a parameter object T of $\mathcal{C}$ over S, the set

$$\mathrm{Hom}_S(T, S') = S'(T),$$

whose elements will be denoted by t, t', and so forth. Given an object $\xi' \in \mathcal{F}_{S'}$, each $t \in S'(T)$ then determines an object $t^*(\xi')$ of $\mathcal{F}_T$, also denoted by ξ'_t. A gluing datum on ξ' (relative to p_1, p_2) is then specified, for every T over S and every pair of points $t, t' \in S'(T)$, by an isomorphism

$$\varphi_{t',t} : \xi'_t \longrightarrow \xi'_{t'}$$

(satisfying the evident functoriality conditions as T varies). Conditions (i) and (ii) of 1.c then become

$$\varphi_{t,t} = \mathrm{id}_{\xi'_t} \qquad \text{for every } T \text{ and every } t \in S'(T), \tag{i bis}$$

304

$$\varphi_{t,t''} = \varphi_{t,t'}\varphi_{t',t''} \qquad \text{for every } T \text{ and all } t, t', t'' \in S'(T). \tag{ii bis}$$

It is clear moreover that (ii bis), on setting $t = t' = t''$, implies $\varphi_{t,t} = \varphi_{t,t}^2$; since $\varphi_{t,t}$ is an isomorphism by hypothesis, this gives (i bis), which is therefore actually a consequence of (ii bis) (and hence (i) is a consequence of (ii)). But if the $\varphi_{t,t'}$ are no longer assumed a priori to be isomorphisms (i.e. if $\varphi : p_1^*(\xi') \longrightarrow p_2^*(\xi')$ is not assumed to be an isomorphism), then (ii bis) no longer necessarily implies (i bis); the conjunction of (ii bis) and (i bis) nevertheless implies that the $\varphi_{t,t'}$ are isomorphisms (for then $\varphi_{t,t'}\varphi_{t',t} = \varphi_{t,t} = \mathrm{id}_{\xi'_t}$).

2. Exact diagrams and strict epimorphisms, descent morphisms. Examples

a. DEFINITION 2.1. — Let $\mathcal{C}$ be a category. A diagram

$$T \xrightarrow{\alpha} T' \underset{\beta_2}{\overset{\beta_1}{\rightrightarrows}} T''$$

[2] The 1962 erratum specifies that it is unnecessary to assume beforehand that α is an $\mathcal{F}$-descent morphism.

of morphisms in $\mathcal{C}$ is called *exact* if, for every $Z \in \mathcal{C}$, the corresponding diagram of maps of sets

$$\mathrm{Hom}(Z,T) \longrightarrow \mathrm{Hom}(Z,T') \rightrightarrows \mathrm{Hom}(Z,T'')$$

is exact (Definition 1.2). One then says that (T,α) (or, by abuse of language, that T) is a *kernel* of the pair of morphisms (β_1,β_2).

This kernel is plainly determined up to unique isomorphism. If $\mathcal{C}$ is the category of sets, the preceding definition is compatible with Definition 1.2. Dually, one defines exactness of a diagram

$$S \xleftarrow{\alpha} S' \underset{\beta_2}{\overset{\beta_1}{\rightrightarrows}} S''$$

of morphisms in $\mathcal{C}$; one then says that (S,α) is a *cokernel* of the pair of morphisms (β_1,β_2).

DEFINITION 2.2. — A morphism $\alpha : S' \longrightarrow S$ is called a *strict epimorphism* if it is an epimorphism and if, for every morphism $u : S' \longrightarrow Z$, the following necessary condition is also sufficient for u to factor as $S' \longrightarrow S \longrightarrow Z$: for every $S'' \in \mathcal{C}$ and every pair of morphisms $\beta_1,\beta_2 : S'' \longrightarrow S'$ such that $\alpha\beta_1 = \alpha\beta_2$, one also has $u\beta_1 = u\beta_2$.

If the fiber product $S' \times_S S'$ exists, this is equivalent to saying that the diagram

$$S \xleftarrow{\alpha} S' \underset{p_2}{\overset{p_1}{\rightrightarrows}} S' \times_S S'$$

is exact, i.e. S is a cokernel of the pair (p_1,p_2). In any case, a cokernel morphism is a strict epimorphism. Note also that a strict epimorphism which is a monomorphism is an isomorphism. We leave it to the reader to develop the dual notion of a *strict monomorphism*.

305

To make precise the relations between the notion of an $\mathcal{F}$-descent morphism and the notion of a strict epimorphism, we introduce the following further definitions.

DEFINITION 2.3. — A morphism $\alpha : S' \longrightarrow S$ is called a *universal epimorphism* (resp. a *universal strict epimorphism*) if, for every T over S, the fiber product $T' = S' \times_S T$ exists and the projection $T' \longrightarrow T$ is an epimorphism (resp. a strict epimorphism).

In very good categories (such as the category of sets, the category of sheaves of sets on a topological space, abelian categories, and so forth), the four notions of epimorphism just introduced coincide. By contrast, they are all distinct in a category such as the category of preschemes, or the category of preschemes over a given nonempty prescheme S, even if one restricts to S-schemes finite over S.

DEFINITION 2.4. — A morphism $\alpha : S' \longrightarrow S$ is called a *descent morphism* (resp. a *strict descent morphism*) if it is an $\mathcal{F}$-descent morphism (resp. a strict $\mathcal{F}$-descent morphism) (cf. Definition 1.7), where $\mathcal{F}$ denotes the fibered category over $\mathcal{C}$ of objects of $\mathcal{C}$ over objects of $\mathcal{C}$ (no. 1, Example 1).

PROPOSITION 2.1. — If finite products and (finite) fiber products exist in $\mathcal{C}$, then in $\mathcal{C}$ descent morphisms and universal strict epimorphisms are the same.

b. EXAMPLES. — Let $\mathcal{C}$ be the category of preschemes. Let $S \in \mathcal{C}$, and let S' and S'' be two preschemes finite over S, i.e. corresponding to sheaves of algebras $\mathcal{A}'$, $\mathcal{A}''$ on S which, as sheaves of modules, are quasi-coherent and of finite type (i.e. coherent if S is locally Noetherian). Let $\alpha : S' \longrightarrow S$ be the structure morphism of S', and let β_1,β_2 be two S-morphisms from S'' to S', defined by homomorphisms of algebras $\mathcal{A}' \longrightarrow \mathcal{A}''$, again denoted by β_1,β_2. Using the fact that a finite morphism is closed (the first Cohen–Seidenberg theorem), one proves easily that the diagram

$$S \xleftarrow{\alpha} S' \underset{\beta_2}{\overset{\beta_1}{\rightrightarrows}} S'' \qquad (+)$$

in $\mathcal{C}$ is exact if and only if the diagram of sheaves 306

$$\mathcal{O}_S = \mathcal{A} \longrightarrow \mathcal{A}' \overset{\beta_1}{\underset{\beta_2}{\rightrightarrows}} \mathcal{A}''$$

on S is exact. In particular, if $\alpha : S' \longrightarrow S$ is a finite morphism corresponding to a sheaf $\mathcal{A}'$ of algebras on S, then α is a strict epimorphism if and only if the diagram of sheaves

$$\mathcal{O}_S = \mathcal{A} \longrightarrow \mathcal{A}' \overset{p_1}{\underset{p_2}{\rightrightarrows}} \mathcal{A}' \otimes_{\mathcal{A}} \mathcal{A}'$$

is exact (it is an epimorphism if and only if $\mathcal{A} \longrightarrow \mathcal{A}'$ is injective). If S is affine with ring A, and hence S' is affine with ring A' finite over A, then $S' \longrightarrow S$ is a strict epimorphism if and only if $A \longrightarrow A'$ is an isomorphism from A onto the subring of A' formed by those $x' \in A'$ such that

$$1_{A'} \otimes_A x' - x' \otimes_A 1_{A'} = 0$$

(it is an epimorphism if and only if $A \longrightarrow A'$ is injective). As we have already pointed out, even if S is the scheme of a local Artinian ring, a finite morphism $S' \longrightarrow S$ which is an epimorphism need not be a strict epimorphism. It can nevertheless be proved that, if S is a Noetherian prescheme, every finite morphism $S' \longrightarrow S$ which is an epimorphism is the composite of a finite sequence of strict epimorphisms (also finite). This also shows that the composite of two strict epimorphisms need not be a strict epimorphism.

c. — If $(+)$ is an exact diagram of finite morphisms, then, for every flat morphism $T \longrightarrow S$ of preschemes, the diagram obtained from $(+)$ by the base change $T \longrightarrow S$ is again exact. It follows that, if X, Y are two S-preschemes, with X flat over S, then the following diagram of maps of sets (where X', Y' are the inverse images of X, Y over S', and X'', Y'' their inverse images over S'')

$$\mathrm{Hom}_S(X, Y) \longrightarrow \mathrm{Hom}_{S'}(X', Y') \rightrightarrows \mathrm{Hom}_{S''}(X'', Y'')$$

is exact. In particular, if $\mathcal{F}$ denotes the fibered category over the category $\mathcal{C}$ of preschemes such that, for $X \in \mathcal{C}$, $\mathcal{F}_X$ is the category of flat X-preschemes, then the diagram $(+)$ is $\mathcal{F}$-exact. (This result becomes false if no flatness hypothesis is made; in particular, a finite strict epimorphism need not be a descent morphism.) Likewise, $(+)$ is $\mathcal{F}$-exact if $\mathcal{F}$ denotes the fibered category for which $\mathcal{F}_X$ is the category of quasi-coherent flat sheaves on the prescheme X (here again the flatness 307 hypothesis

is essential). In either case, the question of the effectivity of a gluing datum (and more particularly of a descent datum when $S'' = S' \times_S S'$) on an object flat over S' is delicate (and its answer in various special cases is one of the main subjects of the present talks). The speaker does not know whether, for every finite strict epimorphism $S' \longrightarrow S$, every descent datum on a flat quasi-coherent sheaf on S' is effective (even if S is assumed to be the spectrum of a local Artinian ring and one restricts to locally free sheaves of rank 1). More generally, let A be a ring and A' an A-algebra (all rings are commutative) such that the diagram of maps

$$A \longrightarrow A' \rightrightarrows A' \otimes_A A'$$

is exact; equivalently, the corresponding morphism $S' \longrightarrow S$ on the spectra of A, A' is an $\mathcal{F}$-descent morphism, where $\mathcal{F}$ is the fibered category of flat quasi-coherent sheaves. Let M' be a flat A'-module equipped with a descent datum to A, i.e. an isomorphism

$$\varphi : M' \otimes_A A' \xrightarrow{\sim} A' \otimes_A M'$$

of $A' \otimes_A A'$-modules satisfying conditions (i) and (ii) of 1.c (whose formulation in terms of modules we leave to the reader). Is this datum effective (relative to the fibered category of flat quasi-coherent sheaves)? Let M be the subset of M' formed by the $x' \in M'$ such that

$$\varphi(x' \otimes_A 1_{A'}) = 1_{A'} \otimes_A x'.$$

It is an A-submodule of M'. The canonical injection $M \longrightarrow M'$ defines a homomorphism of A'-modules $M \otimes_A A' \longrightarrow M'$. The effectivity of φ then means the following: M is a flat A-module, and the preceding homomorphism is an isomorphism.

REMARK. — In the preceding considerations we made no flatness hypothesis on the morphisms in the diagram $(+)$; to obtain a descent technique, this compelled us to make flatness hypotheses on the objects over S, S' under consideration. In Section 2 we shall make a flatness hypothesis on $\alpha : S' \longrightarrow S$, which will allow us to have a descent technique for objects over S, S' that are no longer subject to any flatness condition. In every case, a flatness hypothesis enters. This is one of the principal reasons for the importance of the notion of flatness in algebraic geometry (a role which could not become apparent so long as one restricted to base fields, over which anything whatever is, indeed, flat!). 308

3. Application to étale morphisms

Let A be a local ring and B a local algebra over A whose maximal ideal induces that of A. We shall say that B is *étale* over A (instead of "unramified," used elsewhere) if it satisfies the following conditions:

(i) B is flat over A;

(ii) $B/\mathfrak{m}B$ is a finite separable extension of $A/\mathfrak{m} = k$ (where $\mathfrak{m}$ denotes the maximal ideal of A).

When A and B are Noetherian and k is algebraically closed, this means that the homomorphism $\widehat{A} \longrightarrow \widehat{B}$ on completions extending $A \longrightarrow B$ is an isomorphism. A morphism of finite type $f : T \longrightarrow S$ is said to be *étale at* $x \in T$, or T is said to be *étale over* S *at* x, if $\mathcal{O}_x$ is étale over $\mathcal{O}_{f(x)}$; and f is said to be *étale*, or f is called an *étale morphism*, or T is said to be *étale over* S, if f is étale at every $x \in T$. Observe moreover that, if S is locally Noetherian, the set of points of T where f is étale is open, and the use of Zariski's "main theorem" makes it possible to specify the structure of T/S in a neighborhood of such a point (by an equation of a well-known type).

When S is a scheme of finite type over the field of complex numbers, there corresponds to it an analytic space $\overline{S}$ in the sense of Serre [5], except that $\overline{S}$ may have nilpotent elements in its structure sheaf, which changes nothing essential in [5]. It is then easy to see that f is étale if and only if $\overline{f} : \overline{T} \longrightarrow \overline{S}$ is étale, i.e. if every point of $\overline{T}$ has a neighborhood on which $\overline{f}$ induces an isomorphism onto an open subset of $\overline{S}$. In particular, to every *étale covering* T of S (i.e. a finite étale morphism $f : T \longrightarrow S$) there corresponds an étale covering $\overline{T}$ of $\overline{S}$, which is connected if and only if T is [5]. It is also easy to see that, if T, T' are two schemes étale over S, then the natural map

$$\mathrm{Hom}_S(T, T') \longrightarrow \mathrm{Hom}_{\overline{S}}(\overline{T}, \overline{T'})$$

is bijective, i.e. the functor $T \longmapsto \overline{T}$ from the category of schemes étale over S to the category of analytic spaces étale over $\overline{S}$ is "fully faithful," and therefore defines an equivalence of the first category with a subcategory of the second. A theorem of Grauert–Remmert [2] implies that, if S is normal, one thereby obtains an equivalence between the category of *étale coverings* of S and the category of (finite) étale coverings of $\overline{S}$, i.e. every étale covering of $\overline{S}$ is $\overline{S}$-isomorphic to some $\overline{T}$,

where T is an étale covering of S. Let us show that the *Grauert–Remmert theorem remains valid without a normality hypothesis on* S. First let $S' \longrightarrow S$ be a finite strict epimorphism, assume the 309

theorem proved for S', and let us show that it is true for S. Indeed, let $\mathfrak{T}$ be an étale covering of $\overline{S}$, and consider its inverse image $\mathfrak{T}'$ over $\overline{S'}$, which corresponds to a coherent analytic sheaf $\mathcal{A}'$ of algebras on $\overline{S'}$, the inverse image of the sheaf of algebras $\mathcal{A}$ on $\overline{S}$ defining $\mathfrak{T}$. By hypothesis, over $\overline{S'}$, $\mathfrak{T}'$ comes from an étale covering T' of S', i.e. $\mathcal{A}'$ comes from a coherent sheaf of algebras A' on S'. On the other hand, $\mathcal{A}'$ is equipped with a canonical descent datum relative to $\overline{S'} \longrightarrow \overline{S}$, i.e. an isomorphism between its two inverse images on

$$\overline{S'} \times_{\overline{S}} \overline{S'} = \overline{(S' \times_S S')},$$

(satisfying conditions (i), (ii)); and, by what was said above, this isomorphism comes from an isomorphism on the corresponding algebraic sheaves, i.e. from a descent datum on A' relative to $S' \longrightarrow S$. One checks easily that the latter is effective (for the datum on $\mathcal{A}'$ is effective, and the effectivity of a descent datum, as made explicit in the preceding subsection, can be recognized locally on the *completions* of the modules involved). Hence there is a coherent sheaf of algebras A on S defining a covering T of S, which is the desired covering. The preceding result plainly remains valid if $S' \longrightarrow S$ is merely a composite of a finite number of finite strict epimorphisms, i.e. an arbitrary finite epimorphism (by the factorization result noted in Section 2). It follows that the Grauert–Remmert theorem remains valid if S is a *reduced* scheme, i.e. if $\mathcal{O}_S$ has no nilpotent elements, as one sees by introducing its normalization S'. The general case follows easily.

A wholly analogous proof, again using the factorization result for finite strict epimorphisms and the "formal" nature of the effectivity of descent data, proves the following result: let S be a locally Noetherian prescheme and let $S' \longrightarrow S$ be a morphism that is *finite, surjective, and radicial* (or, equivalently, a morphism of finite type such that, for every T over S, the morphism $T' = S' \times_S T \longrightarrow T$ is a homeomorphism, which is also expressed by saying that $S' \longrightarrow S$ is a *universal homeomorphism*). For every T étale over S, consider its inverse image $T' = T \times_S S'$, which is étale over S'. Then the functor $T \longmapsto T'$ is an *equivalence of categories from the category of preschemes T étale over S to the category of preschemes T' étale over S'*. (One uses the bijectivity of

$$\mathrm{Hom}_S(T_1, T_2) \longrightarrow \mathrm{Hom}_{S'}(T_1', T_2')$$

for two preschemes T_1, T_2 étale over S, a fact whose direct verification is easy, and the fact that the 310 stated theorem is true if $S' = (S, \mathcal{O}_S/\mathcal{J})$,

where $\mathcal{J}$ is a coherent nilpotent sheaf of ideals of $\mathcal{O}_S$ ([4], Lemma 6)). Note moreover that here we do not assume the T under consideration to be finite over S. This result implies in particular that the morphism $S' \longrightarrow S$ induces an isomorphism from the fundamental group $[\,]$ of S' onto that of S ("*topological invariance of the fundamental group of a prescheme*").

4. Relations with 1-cohomology

a. Let $\mathcal{C}$ be a category in which the product of any two objects always exists, and let $T \in \mathcal{C}$. For every finite set $I \neq \emptyset$, one can consider T^I; as I varies, this yields a covariant functor from the category of nonempty finite sets to $\mathcal{C}$, i.e. what may be called a *simplicial object* of $\mathcal{C}$, denoted by K_T. The latter depends covariantly on T; moreover, if u, v are two morphisms $T \longrightarrow T'$, then the corresponding morphisms $K_T \longrightarrow K_{T'}$ are homotopic. Say that T *dominates* T' if $\mathrm{Hom}(T, T') \neq \emptyset$; this is an increasing filtered preorder relation in $\mathcal{C}$. It follows from the foregoing that, if T dominates T', there is a canonical homotopy class of homomorphisms of simplicial objects $K_T \longrightarrow K_{T'}$; in particular, if K_T and $K_{T'}$ are such that each dominates the other, then K_T and $K_{T'}$ are homotopy equivalent. Now let F be a functor (contravariant, to fix ideas) from $\mathcal{C}$ to an abelian category $\mathcal{C}'$. Then

$$C^*(T, F) = F(K_T)$$

is a cosimplicial object of $\mathcal{C}'$, and hence defines in the usual way a (cochain) complex in $\mathcal{C}'$, whose cohomology may be taken:

$$H^*(T, F) = H^*(C^*(T, F)) = H^*(F(K_T)).$$

(One may put a $\mathcal{C}$ as a subscript on H^* if there is a possibility of confusion.) This is a cohomological functor in F, whose variance in the variable T follows from what was said about the K_T; more precisely, for F fixed and T varying in $\mathcal{C}$ (preordered by the domination relation), the $H^*(T, F)$ form an inductive system of graded objects of $\mathcal{C}'$. In particular, if T and T' are such that each dominates the other, then $H^*(T, F)$ and $H^*(T', F)$ are canonically isomorphic.

Suppose that fiber products exist in $\mathcal{C}$. Then, for a fixed $S \in \mathcal{C}$, one may apply the foregoing to the category $\mathcal{C}_S$ of objects of $\mathcal{C}$ over S; we shall write $C^*(T/S, F)$ and $H^*(T/S, F)$ instead of $C^*(T, F)$ and $H^*(T, F)$ if we wish to specify that we are working in the category $\mathcal{C}_S$; thus, 311

$C^*(T/S, F)$ is a cochain complex in $\mathcal{C}'$ whose term in degree n is

$$F(T \times_S T \times_S \cdots \times_S T)$$

(where the parenthesis contains $n + 1$ factors).

As usual, note that one may define $H^0(T/S, F)$ without assuming that $\mathcal{C}'$ is abelian: it is the kernel (Definition 2.1), if it exists, of the pair of morphisms $F(p_i)$ $(i = 1, 2)$

$$F(T) \rightrightarrows F(T \times_S T)$$

corresponding to the two projections $p_1, p_2 : T \times_S T \rightrightarrows T$. In particular, there is a natural morphism (called the *augmentation*)

$$F(S) \longrightarrow H^0(T/S, F),$$

which is an isomorphism in favorable cases (in particular if $T \longrightarrow S$ is a strict epimorphism and F is "left exact"). Likewise, when F takes values in the category of groups in a category $\mathcal{C}''$, one may also define $H^1(T/S, F)$. When $\mathcal{C}''$ is the category of sets (i.e. F takes values in the category of ordinary, not necessarily commutative, groups), $H^1(T/S, F)$ is the quotient of the subgroup $Z^1(T/S, F)$ of $C^1(T/S, F) = F(T \times_S T)$ formed by the g such that

$$F(p_{31})(g) = F(p_{32})(g)\, F(p_{21})(g)$$

by the operator group $F(T)$, acting on $C^1(T/S, F)$ and in particular on the subset $Z^1(T/S, F)$ by

$$\rho(g') \cdot g = F(p_2)(g')\, g\, F(p_1)(g')^{-1}.$$

b. For example, let $\mathcal{F}$ be a fibered category over $\mathcal{C}$. Let $\xi, \eta \in \mathcal{F}_S$, and for every S' over S, set

$$F_{\xi,\eta}(S') = \mathrm{Hom}(\xi \times_S S', \eta \times_S S').$$

Thus $F_{\xi,\eta}$ is a contravariant functor from $\mathcal{C}_S$ to the category of sets. With this notation, *to say that the augmentation morphism*

$$F_{\xi,\eta}(S) \longrightarrow H^0(S'/S, F_{\xi,\eta})$$

is an isomorphism for every pair of elements $\xi, \eta \in \mathcal{F}_S$ means that $\alpha : S' \longrightarrow S$ is an $\mathcal{F}$-descent morphism (Definition 1.7).

c. Similarly, for $\xi \in \mathcal{F}_S$ and every object S' of $\mathcal{C}$ over S, set

$$G_\xi(S') = \mathrm{Aut}(\xi \times_S S').$$

This defines a contravariant functor G_ξ from $\mathcal{C}_S$ to the category of 312

groups. One then sees that $Z^1(S'/S, G_\xi)$ *is canonically identified with the set of descent data on* $\xi' = \xi \times_S S'$ *relative to* $S' \longrightarrow S$ (Definition 1.6), and $H^1(S'/S, G_\xi)$ *is identified with the set of classes, up to isomorphism, of objects of* $\mathcal{F}_{S'}$ *equipped with a descent datum relative to* $\alpha : S' \longrightarrow S$ *which, as objects of* $\mathcal{F}_{S'}$*, are isomorphic to* $\xi' = \xi \times_S S'$. Thus, if $\alpha : S' \longrightarrow S$ is an $\mathcal{F}$-descent morphism (cf. (b)), then $H^1(S'/S, G_\xi)$ *contains as a subset the set of classes, up to isomorphism, of objects* η *of* $\mathcal{F}_S$ *such that* $\eta \times_S S'$ *is isomorphic in* $\mathcal{F}_{S'}$ *to* $\xi \times_S S'$*; and this inclusion is an identity if and only if every descent datum on* $\xi' = \xi \times_S S'$ *relative to* $\alpha : S' \longrightarrow S$ *is effective*. (This will hold in particular if $\alpha : S' \longrightarrow S$ is a strict $\mathcal{F}$-descent morphism.)

REMARK. — Cochain complexes of the form $C^*(T/S, F)$ contain as special cases most of the familiar standard complexes (Čech cohomology, group cohomology, and so forth), and play an important role in algebraic geometry, notably in the "Weil cohomology" of preschemes.

d.

EXAMPLE 1. — Let S' be an object over $S \in \mathcal{C}$, and let Γ be a group of automorphisms of S' such that S' is "formally principal over S, with group Γ," i.e. such that the natural morphism

$$\Gamma \times S' \longrightarrow S' \times_S S',$$

where $\Gamma \times S'$ denotes the coproduct of Γ copies of S', is an isomorphism. (We assume that the coproducts occurring here exist in $\mathcal{C}$.) Let F be a contravariant functor from $\mathcal{C}$ to the category of abelian groups. Then $C^*(S'/S, F)$ is canonically isomorphic to the simplicial group of standard homogeneous cochains $C^*(\Gamma, F(S'))$; hence $H^*(S'/S, F)$ is canonically isomorphic to $H^*(\Gamma, F(S'))$.

e.

EXAMPLE 2. — Let $\mathcal{C}$ be the category of preschemes. Denote by G_a ("additive group") the contravariant functor from $\mathcal{C}$ to the category of abelian groups defined by

$$G_a(X) = H^0(X, \mathcal{O}_X).$$

Similarly, define the functor G_m ("multiplicative group") by

$$G_m(X) = H^0(X, \mathcal{O}_X)^*$$

(the group of invertible elements of the ring $H^0(X, \mathcal{O}_X)$), and more generally the functor $\mathrm{Gl}(n)$
313 ("linear group of order n") by

$$\mathrm{Gl}(n)(X) = \mathrm{Gl}(n, H^0(X, \mathcal{O}_X)).$$

This is a functor from $\mathcal{C}$ to the category of groups (not necessarily commutative if $n > 1$; for $n = 1$ it is G_m). Moreover, $\mathrm{Gl}(n)$ may be interpreted as an automorphism functor (cf. (c)) by considering the fibered category $\mathcal{F}$ over $\mathcal{C}$ such that, for $X \in \mathcal{C}$, $\mathcal{F}_X$ is the category of locally free sheaves on X: indeed,

$$\mathrm{Gl}(n)(X) = \mathrm{Aut}_{\mathcal{F}_X}(\mathcal{O}_X^n).$$

By (b), if $\alpha : S' \longrightarrow S$ is an $\mathcal{F}$-descent morphism (cf. Section 2.c), $H^1(S'/S, \mathrm{Gl}(n))$ contains the set of classes, up to isomorphism, of locally free sheaves on S whose inverse image on S' is isomorphic to $\mathcal{O}_{S'}^n$; and this inclusion is an equality if and only if every descent datum on $\mathcal{O}_{S'}^n$ (relative to $\alpha : S' \longrightarrow S$) is effective. When S is the spectrum of a local ring, this therefore means that $H^1(S'/S, \mathrm{Gl}(n)) = (e)$, since every locally free sheaf on S is then trivial.

Note the equivalence of the following conditions on a morphism $\alpha : S' \longrightarrow S$:

(i) the augmentation homomorphism

$$H^0(S, \mathcal{O}_S) = G_a(S) \longrightarrow H^0(S'/S, G_a)$$

is an isomorphism;

(ii) $S' \longrightarrow S$ is an $\mathcal{F}$-descent morphism ($\mathcal{F}$ being the fibered category over $\mathcal{C}$ considered above).

If $S' \longrightarrow S$ is finite, these conditions are also equivalent to:

(iii) $S' \longrightarrow S$ is a strict epimorphism (cf. Section 2.c).

Now suppose that $S = \mathrm{Spec}(A)$ and $S' = \mathrm{Spec}(A')$. Then

$$C^n(S'/S, G_a) = C^n(A'/A, G_a) = \bigotimes_A^{n+1} A',$$

the coboundary operator $C^n(A'/A, G_a) \longrightarrow C^{n+1}(A'/A, G_a)$ being the alternating sum of the face operators

$$\partial_i(x_0 \otimes x_1 \otimes \cdots \otimes x_n) = x_0 \otimes \cdots \otimes x_{i-1} \otimes 1_{A'} \otimes x_i \otimes \cdots \otimes x_n.$$

Likewise,

$$C^n(S'/S, G_m) = C^n(A'/A, G_m) \simeq \left(\bigotimes_A^{n+1} A' \right)^*,$$

the simplicial operations in $C^*(A'/A, G_m)$ being induced by those of $C^*(S'/S, G_a)$. The simplicial operations in $C^*(A'/A, \mathrm{Gl}(n))$ are made explicit in the same way. *In all cases known to the speaker,*

$$H^i(A'/A, G_a) = 0 \qquad \text{for } i > 0,$$

and if A is local,

$$H^1(A'/A, G_m) = 0 \quad \text{and more generally} \quad H^1(A'/A, \mathrm{Gl}(n)) = (e)$$

(when $S' \longrightarrow S$ is an $\mathcal{F}$-descent morphism, i.e. the diagram $A \longrightarrow A' \rightrightarrows A' \otimes_A A'$ is exact; compare Section 2.c). Note that *Hilbert's "Theorem 90" is nothing other than the relation* 314

$$H^1(S'/S, G_m) = 0$$

when A is a field and A' is a finite Galois extension of A (cf. Example 1), and may also be expressed by saying that, in the case under consideration, $S' \longrightarrow S$ is a strict descent morphism for the fibered category of locally free sheaves of rank 1. It is in this latter form that Hilbert's theorem should be generalized, by varying the hypotheses both on the morphism $S' \longrightarrow S$ and on the quasi-coherent sheaves under consideration.

Finally, note the equivalence of the following properties when A is an Artinian local ring with maximal ideal $\mathfrak{m}$ and A' is an A-algebra (where, for every integer $k > 0$, A_k (respectively A'_k) denotes the ring $A/\mathfrak{m}^{k+1}$ (respectively $A'/\mathfrak{m}^{k+1}A'$)):

(i) $H^1(A'_k/A_k, G_a) = 0$ for every k.

(ii) $H^1(A'_k/A_k, G_m) = 0$ for every k.

(iii) $H^1(A'_k/A_k, \mathrm{Gl}(n)) = (e)$ for every k and every n.

If $S' \longrightarrow S$ is a strict epimorphism, the preceding conditions even imply that it is a strict descent morphism for free modules (of finite type or not) over A'.

REMARK. — The definition of the groups $H^i(S'/S, G_m)$, when S and S' are the schemes of fields A and A', is due to Amitsur. The group $H^2(S'/S, G_m)$ is especially interesting as a "global" variant of the Brauer group; for this variant one may refer to [1], Chapter VII.

B. Descent by faithfully flat morphisms

1. Statement of the descent theorems

DEFINITION 1.1. — A morphism $\alpha : S' \longrightarrow S$ of preschemes is called *flat* if, for every $x' \in S'$, $\mathcal{O}_{x'}$ is a flat module over the ring $\mathcal{O}_{\alpha(x')}$ (i.e. $\mathcal{O}_{x'} \otimes_{\mathcal{O}_{\alpha(x')}} M$ is an exact functor of the $\mathcal{O}_{\alpha(x')}$-module M). A morphism is called *faithfully flat* if it is flat and surjective.

For example, if $S = \mathrm{Spec}(A)$ and $S' = \mathrm{Spec}(A')$, then S' is flat over S if and only if A' is a flat A-module, and S' is faithfully flat over S if and only if A' is a faithfully flat A-module (i.e. the functor $A' \otimes_A M$ of the A-module M is exact and faithful); in Serre's terminology [5], this also means that the pair (A, A') is flat. If S' is faithfully flat over S, then the inverse-image functor on quasi-coherent sheaves over S is exact and faithful. In other words, for a sequence of
315 homomorphisms of quasi-coherent sheaves

over S to be exact, it is necessary and sufficient that its inverse image over S' be exact (in particular, for a homomorphism of quasi-coherent sheaves over S to be a monomorphism, respectively an epimorphism, respectively an isomorphism, it is necessary and sufficient that its inverse image over S' be so). This property remains true after restriction over any open subset of S, and in this form characterizes faithfully flat morphisms.

DEFINITION 1.2. — A morphism $\alpha : S' \longrightarrow S$ is called *quasi-compact* if the inverse image of every quasi-compact open subset U of S is quasi-compact (i.e. a finite union of affine open subsets).

It plainly suffices to check this property for the affine open subsets of S. For example, an affine morphism (i.e. one for which the inverse image of an affine open subset is affine) is quasi-compact.

The class of flat, respectively faithfully flat, respectively quasi-compact morphisms is stable under composition and "base extension," and contains the isomorphisms.

THEOREM 1. — Let $\alpha : S' \longrightarrow S$ be a faithfully flat and quasi-compact morphism of preschemes. Then α is a *strict descent morphism* (A, Definition 1.7) for the fibered category $\mathcal{F}$ of quasi-coherent sheaves (A, Section 1, Example 2).

This statement means two things:

(i) If F and G are two quasi-coherent sheaves over S, and F' and G' are their inverse images over S', then the natural homomorphism

$$\mathrm{Hom}(F, G) \longrightarrow \mathrm{Hom}(F', G')$$

is a bijection from the first term onto the subgroup of the second formed by the homomorphisms $F' \longrightarrow G'$ compatible with the canonical descent data on these sheaves, i.e. those whose inverse images under the two projections from $S'' = S' \times_S S'$ to S' give the same homomorphism $F'' \longrightarrow G''$.

(ii) Every quasi-coherent sheaf F' over S' equipped with a descent datum relative to the morphism $\alpha : S' \longrightarrow S$ (A, Definition 1.6) is isomorphic (together with this datum) to the inverse image of a quasi-coherent sheaf F over S.

Taking $F = \mathcal{O}_S$ in (i), one obtains:

COROLLARY 1. — Let G be a quasi-coherent sheaf over S, let G' and G'' be its inverse images over S' and over $S'' = S' \times_S S'$, and let p_1, p_2 be the two projections from S'' to S'. Then the
316 following diagram of maps of sets

$$\Gamma(G) \xrightarrow{\alpha^*} \Gamma(G') \underset{p_2^*}{\overset{p_1^*}{\rightrightarrows}} \Gamma(G'')$$

is exact (A, Definition 1.(a)).

On the other hand, combining (i), (ii), and Definition 1.1 gives:

COROLLARY 2. — Let G be as in Corollary 1. Then there is a one-to-one correspondence between the quasi-coherent subsheaves of G and the quasi-coherent subsheaves of G' whose inverse images over S'' under the two projections p_1, p_2 give the same subsheaf of G''.

There is, of course, an equivalent statement in terms of quotient sheaves. As seen in A, Section 4(e), Theorem 1 should be regarded as a generalization of Hilbert's "Theorem 90," and includes as special cases various formulations in terms of 1-cohomology. For the proof, one readily reduces to the case $S = \mathrm{Spec}(A)$, $S' = \mathrm{Spec}(A')$; and for (i) one readily reduces to proving Corollary 1, i.e. the exactness of the diagram

$$M = A \otimes_A M \longrightarrow A' \otimes_A M \rightrightarrows A' \otimes_A A' \otimes_A M$$

for every A-module M. This follows from the more general lemma:

LEMMA 1.1. — Let A' be a faithfully flat A-algebra. Then, for every A-module M, the M-augmented complex $C^*(A'/A, G_a) \otimes_A M$ (cf. A, Section 4(e)) is a *resolution* of M.

It suffices to prove that the augmented complex obtained from the preceding one by base extension from A to A' satisfies the same conclusions. This amounts to checking the statement after replacing A by A' and A' by $A' \otimes_A A'$, and hence reduces us to a case in which there is an A-algebra homomorphism $A' \longrightarrow A$ (or, in geometric terms, to the case in which S' over S admits a section). In that case the assertion follows from the general facts of A, Section 4(a). We note in passing the following corollary, which generalizes a familiar statement in Galois cohomology (compare A, Section 4(e)):

COROLLARY. — If A' is faithfully flat over A, then $H^0(A'/A, G_a) = A$ and

$$H^i(A'/A, G_a) = 0 \qquad \text{for } i \geq 1.$$

To prove part (ii) of Theorem 1, one proceeds as for (i), reducing to the case where S' over S admits a section; there it follows from (i) (cf. A, Section 1(c)).

Theorem 1 and its corollaries may plainly be varied ad libitum by introducing various additional structures on the quasi-coherent sheaves (or systems of sheaves) under consideration. For example, the datum over S of a sheaf of commutative quasi-coherent algebras "is equivalent" to the 317
datum over S' of such a sheaf equipped with a descent datum relative to $\alpha : S' \longrightarrow S$. In view of the functorial correspondence between such quasi-coherent sheaves over S and affine preschemes over S, one obtains the second assertion of the following theorem:

THEOREM 2. — Let $\alpha : S' \longrightarrow S$ be as in Theorem 1. Then α is a *descent morphism* (not strict in general) (A, Definition 2.4), and it is a *strict descent morphism* for the fibered category of schemes affine over preschemes (A, Definition 1.7).

The first assertion of the theorem means the following. Let X, Y be two preschemes over S, let X', Y' be their inverse images over S', and let X'', Y'' be their inverse images over $S'' = S' \times_S S'$. Then the following diagram of natural maps

$$\mathrm{Hom}_S(X, Y) \xrightarrow{\alpha^*} \mathrm{Hom}_{S'}(X', Y') \underset{p_2^*}{\overset{p_1^*}{\rightrightarrows}} \mathrm{Hom}_{S''}(X'', Y'')$$

is exact; i.e. α^* is a bijection from $\mathrm{Hom}_S(X, Y)$ onto the subset of $\mathrm{Hom}_{S'}(X', Y')$ formed by the homomorphisms compatible with the canonical descent data on X', Y' (i.e. those whose inverse images under the two projections from S'' to S' are equal). This follows readily from Theorem 1, Corollary 1, when Y is restricted to be affine over S; in the general case Theorem 1 must be combined with the following result:

LEMMA 1.2. — Let $\alpha : S' \longrightarrow S$ be a faithfully flat and quasi-compact morphism. Then S identifies with a *quotient topological space* of S'; i.e. every subset U of S such that $\alpha^{-1}(U)$ is open is itself open.

To complete Theorem 2, one must give effectiveness criteria for a descent datum on an S'-prescheme X' (when X' is not assumed affine over S'). First observe that *such a descent datum need not be effective*, even if S is the spectrum of a field k, S' is the spectrum of a quadratic extension k' of k, and X' is a proper algebraic scheme of dimension 2 over S' (as one can see, following Serre, by using Nagata's nonprojective surface). *For a descent datum on X'/S' relative to $\alpha : S' \longrightarrow S$ (faithfully flat and quasi-compact) to be effective, it is necessary and sufficient that X' be a union of open subsets X'_i, affine over S', that are "stable" under the descent datum on X'.* This certainly holds (whatever X'/S' and its descent datum may be) if the morphism $\alpha : S' \longrightarrow S$ is *radicial* (i.e. injective, with residue field extensions
318 that

are radicial). One can also show that it still holds if $\alpha : S' \longrightarrow S$ is *finite*, and every finite subset of X' contained in a fiber of X' over S is contained in an open subset of X' affine over S (this is *Weil's criterion*). In particular, this holds if X'/S' is quasi-projective; in that case one can show that the "descended" prescheme X/S is also quasi-projective (and projective if X'/S' is projective). In summary:

THEOREM 3. — Let $\alpha : S' \longrightarrow S$ be a faithfully flat and quasi-compact morphism of preschemes. If α is *radicial*, it is a *strict descent morphism*. If α is finite, it is a strict descent morphism relative to the fibered category of quasi-projective (or projective) preschemes over preschemes.

REMARKS. — I do not know whether, in the second assertion above, the hypothesis that α be a *finite* morphism is really necessary; in any case one checks "formally" that it may be replaced by the following, apparently weaker, hypothesis: *for every point of S there exist an open neighborhood U, a U' finite and faithfully flat over U, and an S-morphism from U' to S'.* A typical case not covered by the preceding condition is that in which $S = \mathrm{Spec}(A)$, $S' = \mathrm{Spec}(\widehat{A})$, where A is a Noetherian local ring and $\widehat{A}$ its completion; another is that in which S' is quasi-finite over S (i.e. locally isomorphic to an open subset of a finite S-scheme) but not finite. In these two cases, the speaker also does not know the answer to the following question: if X is an S-scheme such that $X' = X \times_S S'$ is projective over S', is X projective over S?[3]

2. Application to the descent of certain properties of morphisms

Let P be a class of morphisms of preschemes. Let $\alpha : S' \longrightarrow S$ be a morphism of preschemes, let $f : X \longrightarrow Y$ be a morphism of S-preschemes, and let $f' : X' \longrightarrow Y'$ be the inverse image of f under α. One may then ask whether "$f' \in P$" implies "$f \in P$." It appears that the answer is affirmative in many important cases when α is assumed *faithfully flat and quasi-compact* (the latter hypothesis sometimes being superfluous). This follows directly and without difficulty if P is the class of surjective morphisms, respectively radicial morphisms (these cases following from the surjectivity of α), respectively flat morphisms, faithfully flat morphisms, smooth morphisms (these cases following from the faithful flatness of α), or morphisms of finite type. Using Theorems 1 and 2 and Lemma 1.2, one also sees that the same holds when P is one of the following classes: isomorphisms, open immersions, closed immersions, immersions (if f is of finite type and Y is locally Noetherian), affine morphisms, finite morphisms, quasi-finite morphisms, open morphisms,
319 closed morphisms, homeomorphisms, separated morphisms,

[3]The 1962 erratum answers in the negative: a quasi-finite, étale, surjective morphism $S' \longrightarrow S$, or a morphism $\mathrm{Spec}(\widehat{A}) \longrightarrow \mathrm{Spec}(A)$, need not be a strict descent morphism, even if A is the local ring of an algebraic curve over an algebraically closed field k and $S = \mathrm{Spec}(A)$. One may thus find a proper smooth morphism $f : X \longrightarrow S$ making X a principal bundle over S under an elliptic curve E, such that $f' : X' \longrightarrow S'$ is projective but f is not. This is therefore also an example of a non-isotrivial homogeneous principal bundle under an abelian scheme.

and proper morphisms. The only important unresolved case is that of projective and quasi-projective morphisms, already noted in the remark of Section 1.

3. Descent by finite faithfully flat morphisms

Let $\alpha : S' \longrightarrow S$ be a finite morphism corresponding to a sheaf of algebras $\mathcal{A}'$ over S that is locally free of finite type as a sheaf of modules and is everywhere nonzero. Then α is faithfully flat and quasi-compact, so the preceding results apply. The datum of a quasi-coherent sheaf F' over S' is equivalent to that of the quasi-coherent sheaf $\alpha_*(F')$ over S equipped with its $\mathcal{A}'$-module structure (noting that $\mathcal{A}' = \alpha_*(\mathcal{O}_{S'})$). For simplicity, this sheaf over S will also be denoted by F'. The two inverse images $p_i^*(F')$ of F' over $S' \times_S S'$ likewise correspond to the quasi-coherent sheaves of $(\mathcal{A}' \otimes_{\mathcal{O}_S} \mathcal{A}')$-modules

$$F' \otimes_{\mathcal{O}_S} \mathcal{A}' \quad \text{and} \quad \mathcal{A}' \otimes_{\mathcal{O}_S} F'.$$

Giving a homomorphism over $S' \times_S S'$ from the first to the second is equivalent to giving a homomorphism of $(\mathcal{A}' \otimes_{\mathcal{O}_S} \mathcal{A}')$-modules and, since $\mathcal{A}'$ is locally free, this is equivalent to giving a homomorphism of $(\mathcal{A}' \otimes \mathcal{A}')$-modules:

$$\mathcal{U} = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{A}', \mathcal{A}') = \mathcal{A}' \otimes \check{\mathcal{A}}' \longrightarrow \mathrm{Hom}_{\mathcal{O}_S}(F', F'),$$

i.e. to giving, for every section ξ of $\mathcal{U}$ over an open subset V, a homomorphism of $\mathcal{O}_S$-modules $T_\xi : F'|_V \longrightarrow F'|_V$ satisfying

$$T_{f\xi}(x) = fT_\xi(x), \qquad T_{\xi f}(x) = T_\xi(fx), \tag{3.1}$$

where f, x are respectively sections of $\mathcal{A}', F'$ over an open subset of S contained in V. Conditions (i) and (ii) for a descent datum (A, Section 1(c)) then take the respective forms

$$T_{1_{\mathcal{U}}} x = x, \qquad \text{i.e.} \qquad T_{1_{\mathcal{U}}} = \mathrm{id}_{F'}, \tag{3.2}$$

$$T_{\xi\eta} = T_\xi T_\eta. \tag{3.3}$$

In other words, *a descent datum on F' is equivalent to a representation of the sheaf of $\mathcal{O}_S$-algebras $\mathcal{U} = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{A}', \mathcal{A}')$ in the sheaf of $\mathcal{O}_S$-algebras $\mathrm{Hom}_{\mathcal{O}_S}(F', F')$, satisfying the two linearity conditions (3.1).* If there is a pairing of quasi-coherent sheaves over S',

$$F_1' \times F_2' \longrightarrow F_3',$$

(which may be interpreted as a pairing of sheaves of $\mathcal{A}'$-modules over S), 320

and gluing data on the F_i', defined by homomorphisms

$$T_i : \mathcal{U} \longrightarrow \mathrm{Hom}_{\mathcal{O}_S}(F_i', F_i') \qquad (i = 1, 2, 3),$$

then these data are compatible with the given pairing, in the evident sense, if and only if the following condition is satisfied.

For every section ξ of $\mathcal{U}$ over an open subset, write

$$\Delta\xi = \sum_i \xi_i' \otimes_{\mathcal{A}'} \xi_i''$$

for the section of $\mathcal{U} \otimes_{\mathcal{A}'} \mathcal{U}$ (where $\mathcal{U}$ is regarded as an $\mathcal{A}'$-module through its left module structure) defined by

$$\xi \cdot (fg) = \sum_i \xi_i'(f)\xi_i''(g)$$

(where f and g are two sections of $\mathcal{A}'$ over a smaller open subset). Then

$$T_\xi^{(3)}(x \cdot y) = \sum_i T_{\xi_i'}^{(1)} x \cdot T_{\xi_i''}^{(2)} y \tag{3.4}$$

for every pair of sections x, y of F_1', F_2', respectively, over a smaller open subset.[4] (This property may be expressed by saying that the homomorphisms $T^{(i)}$ are compatible with the diagonal map of $\mathcal{U}$ relative to the given pairing.) In particular, formulas (3.1)–(3.4) allow descent data on a quasi-coherent sheaf of algebras over S' to be interpreted in terms of representations of algebras with diagonal maps, and hence also (upon restricting to commutative algebras) descent data on an affine S'-scheme.

From this one passes to an analogous interpretation of descent data on an arbitrary S'-prescheme X'. Giving such an X' is equivalent to giving a prescheme X' over S equipped with a homomorphism of $\mathcal{O}_S$-algebras

$$\mathcal{A}' \longrightarrow \mathcal{O}_{X'},$$

and a descent datum on X' is equivalent to giving a homomorphism of sheaves (compatible with the morphism $h : X' \longrightarrow S'$)

$$\mathcal{U} \longrightarrow \mathrm{Hom}_{h^{-1}(\mathcal{O}_S)}(\mathcal{O}_{X'}, \mathcal{O}_{X'}),$$

satisfying conditions analogous to (3.1)–(3.4) above.

Example 1 ("Weil"). – Suppose that S'/S is a Galois étale covering with Galois group Γ (cf. A, Sections 3 and 4(d)). Then a descent datum on a quasi-coherent sheaf F' over S' (respectively, on an S'-prescheme X') is equivalent to a representation of Γ by automorphisms of (S', F') (re-
321 spectively, of (S', X')) compatible with the action of Γ on S'. This result

is "formal," i.e. it can be proved in categorical terms, but from the viewpoint of this section it also follows from the explicit structure of $\mathcal{U}$ (equipped with its ring structure, the ring homomorphism $\mathcal{A}' \longrightarrow \mathcal{U}$, and the diagonal map), which is completely known by virtue of the following result: as a left $\mathcal{A}'$-module, $\mathcal{U}$ admits a basis consisting of those sections of $\mathcal{U}$ that correspond to the elements of Γ.

Example 2 ("Cartier"). – Let p be a prime number, suppose that $p\mathcal{O}_S = 0$ (i.e. $\mathcal{O}_S$ has characteristic p), that $(\mathcal{A}')^p \subset \mathcal{O}_S = \mathcal{A}$ (i.e. S'/S is radicial of height 1), and that the sheaf of algebras $\mathcal{A}'$ over $\mathcal{A}$ locally admits a p-basis, i.e. a family (x_i) of sections such that $\mathcal{A}'$ is generated as an algebra by the x_i, subject only to the relations $x_i^p = 0$. We suppose the set of indices i to be finite, of cardinality n. Let $\mathfrak{g}$ be the sheaf of $\mathcal{A}$-derivations of $\mathcal{A}'$. It is a locally free sheaf of $\mathcal{A}'$-modules of rank n; moreover, it is a sheaf of p-Lie algebras over $\mathcal{A}$ (but not over $\mathcal{A}'$), satisfying

$$[X, fY] = X(f)Y + f[X, Y]. \tag{3.5}$$

Lemma. – $\mathcal{U} = \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{A}', \mathcal{A}')$ is generated, as an $\mathcal{O}_S$-algebra equipped with an algebra homomorphism $\mathcal{A}' \longrightarrow \mathcal{U}$, by the left $\mathcal{A}'$-submodule $\mathfrak{g}$, subject to the additional relations

$$\begin{cases} Xf - fX = X(f), \\ XY - YX = [X, Y], \\ X^p = X^{(p)}. \end{cases} \tag{3.6}$$

[4] The printed text says "of $\mathcal{A}'$." The domains of $T^{(1)}$ and $T^{(2)}$, together with the pairing $F_1' \times F_2' \longrightarrow F_3'$, require x and y to be sections of F_1' and F_2', respectively.

It follows from the preceding lemma that a descent datum on the quasi-coherent sheaf F' over S' is equivalent to giving, for every $X \in \mathfrak{g}$, an $\mathcal{O}_S$-endomorphism $\overline{X}$ of F' satisfying

$$\overline{fX} = f\overline{X}, \tag{3.7}$$

$$\overline{X}(fx) = X(f)x + f\overline{X}(x), \tag{3.8}$$

$$\overline{[X,Y]} = [\overline{X},\overline{Y}], \tag{3.9}$$

$$\overline{X^{(p)}} = \overline{X}^p. \tag{3.10}$$

(This is what one might call a linear connection on F', without curvature and compatible with the p-th power.) The notion of a descent datum on an S'-prescheme X' is made explicit in the same way; here relation (3.4) is replaced by the condition that the $\overline{X}$ are derivations of $\mathcal{O}_{X'}$. Since the morphism $S' \longrightarrow S$ is radicial, Theorem 3 guarantees that every such descent datum is 322

effective and hence defines an S-prescheme X.

We note that no flatness, nonsingularity, or finiteness hypothesis of any kind has been imposed on F', respectively X'.

4. Application to rationality criteria

Let X be an S-prescheme such that the direct image of $\mathcal{O}_X$ on S is $\mathcal{O}_S$; this property then remains true under every flat extension $S' \longrightarrow S$ of the base S. If F is an invertible sheaf (i.e. locally free of rank 1) on X, the automorphisms of F, which are identified with the invertible sections of $\mathcal{O}_X$, correspond bijectively to the invertible sections of $\mathcal{O}_S$. Let s now be a section of X over S. We shall refer figuratively to a section of F over s, meaning a section of the invertible sheaf $s^*(F)$ on S. It follows from the foregoing that if F_i $(i = 1, 2)$ are two invertible sheaves on X, each equipped with a section over s, and if F_1 and F_2 are isomorphic, there exists exactly one isomorphism from F_1 to F_2 compatible with the sections in question (i.e. carrying the first to the second). Independently of the section s, let us agree to regard two invertible sheaves F_1 and F_2 on X as equivalent if every point of S has an open neighborhood U such that the restrictions of F_1 and F_2 to $X|U$ are isomorphic. Then every invertible sheaf F on X is equivalent to an invertible sheaf F_1 equipped with a marked section over s (take $F_1 = F\, s^*(F)^{-1}$), and F_1 is determined up to isomorphism. In other words, the classification of invertible sheaves on X up to equivalence is the same as the classification, up to isomorphism, of invertible sheaves equipped with a marked section.

Since these properties remain true under a flat extension $\alpha : S' \longrightarrow S$ of the base (with the section s replaced by its inverse image s' under α), Theorem 1 gives the following conclusion:

Let the prescheme X/S be as above and admit a section s, let $\alpha : S' \longrightarrow S$ be a faithfully flat and quasi-compact morphism, and let F' be an invertible sheaf on $X' = X \times_S S'$. Then F' is equivalent to the inverse image on X' of an invertible sheaf F on X if and only if its inverse images $p_1^*(F')$ and $p_2^*(F')$ on $X' \times_X X' = X \times_S (S' \times_S S')$ are equivalent. If so, F is determined up to equivalence. (In that case F' is said to be rational over S.)

Applying this principle when $\alpha : S' \longrightarrow S$ is as in Example 1 or Example 2 of the preceding section yields the rationality criteria of Weil or Cartier. (We note that these authors restrict themselves to the case in which S and S' 323

are spectra of fields; a fortiori, S is then the spectrum of a local ring, and the equivalence relation introduced above is simply isomorphism.) In the first case, F' is rational over S if and only if its transforms under Γ are equivalent to F'. To express the rationality criterion in the second case, consider, in general, the diagonal morphism

$$X' \longrightarrow X'' = X' \times_X X'$$

of X'/X, the corresponding sheaf of ideals $\mathcal{I}$ on $X' \times_X X'$, and the sheaf $\mathcal{I}/\mathcal{I}^2$, which is identified with its inverse image $\Omega^1_{X'/X}$ on X'. Since the restrictions to the diagonal of $F_i'' = p_i^*(F')$ $(i = 1, 2)$

are isomorphic (both being isomorphic to F'), i.e. $F''_1(F''_2)^{-1} = F''$ has a trivial restriction to the diagonal, it follows that the restriction of F'' to $(X'', \mathcal{O}_{X''}/\mathcal{I}^2)$ is given, up to isomorphism, by a well-determined element ξ of

$$H^1(X'', \mathcal{I}/\mathcal{I}^2) = H^1(X', \Omega^1_{X'/X}).$$

Moreover, in the present case,

$$\Omega^1_{X'/X} = \Omega^1_{S'/S} \otimes_{\mathcal{O}_S} \mathcal{O}_X,$$

and consequently, if $\Omega^1_{S'/S}$ is locally free over S (as in Cartier's case), ξ defines a section of

$$R^1 f'_*(\mathcal{O}_{X'}) \otimes \Omega^1_{S'/S}$$

over S', called the Atiyah–Cartier class of the invertible sheaf F' on X'/S. Its vanishing is necessary and sufficient for the inverse images of F' under the two projections from

$$(X'', \mathcal{O}_{X''}/\mathcal{I}^2) = X \times_S (S'', \mathcal{O}_{S''}/\mathcal{J}^2)$$

to X' to be equivalent (where $\mathcal{J}$ is the sheaf of ideals on $S'' = S' \times_S S'$ defined by the diagonal morphism $S' \longrightarrow S' \times_S S'$). This vanishing is therefore plainly necessary for the inverse images of F' on $X'' = X \times_S S''$ itself to be equivalent, and hence also for F' to be equivalent to the inverse image of an invertible sheaf F on X. The Atiyah–Cartier class may also be interpreted as the obstruction to the existence, locally over S', of a connection on F' relative to the derivations of X'/X; such a connection is further determined once one knows the derivations of F' corresponding to the natural prolongations to X' of the derivations of S'/S. From this and the developments in the preceding section, one readily concludes that in the case of Example 2 of that section, and when X/S admits a section, the vanishing of the Atiyah–Cartier class is also sufficient for F' to be rational over S.

5. Application to restricting the base in an abelian scheme

324 Let S be a prescheme. An abelian scheme over S is a scheme X smooth and proper over S whose fibers at the points $x \in S$ are schemes of abelian varieties over the fields $\kappa(x)$. Suppose that S is Noetherian and regular (i.e. its local rings are regular). Using Murre's connectedness theorem [4], one can then show (at least in the equal-characteristic case, in which the cited theorem is presently proved) that every rational section of X over S is defined everywhere (i.e. is a section), generalizing a classical theorem of Weil. More generally, it follows that if X' is smooth over S, then every rational S-map from X' to X is defined everywhere. This gives the following result, which generalizes a result of Chow–Iang:[5] Let S be Noetherian and regular, and let K denote its ring of rational functions (a direct product of fields). Let X be an abelian scheme over K. If X is isomorphic to a K-scheme of the form $X_0 \times_S \mathrm{Spec}(K)$, where X_0 is an abelian scheme over S, then X_0 is determined up to a unique isomorphism.

Using the preceding uniqueness result, one sees that the problem of restricting the base in X is local on S (and hence that it suffices to know how to carry out the restriction over the $\mathrm{Spec}(\mathcal{O}_x)$, with $x \in S$). In the same way, if $\alpha : S' \longrightarrow S$ is a smooth morphism of finite type, if K' is the ring of rational functions of S', and if $X \otimes_K K'$ is of the form $X'_0 \times_{S'} \mathrm{Spec}(K')$, then X'_0 carries a canonical descent datum relative to α. In view of Theorem 3, it follows that:

PROPOSITION 5.1. — Let S be an irreducible Noetherian regular prescheme with rational function field K, let K' be a finite extension of K unramified over S, let S' be the normalization of S in K' (thus an étale covering of S), and let X be an abelian scheme over K such that $X \otimes_K K'$ is

[5] The printed text says "Igusa–Lang"; the 1962 erratum prescribes "Chow–Iang."

of the form $X'_0 \times_{S'} \mathrm{Spec}(K')$, where X'_0 is a projective abelian scheme over S'. Then X is of the form $X_0 \times_S \mathrm{Spec}(K)$, where X_0 is a projective abelian scheme over S.

REMARKS. — The speaker does not know whether the hypothesis that $S' \longrightarrow S$ is a surjective étale covering (which permits the use of Theorem 3) can be replaced by the hypothesis that it is a surjective smooth morphism of finite type (even if it is assumed to be étale), or whether the proposition remains valid without assuming X'_0 projective over S' (a condition that may perhaps hold automatically).

6. Application to criteria for local triviality and isotriviality

Let S be a prescheme, let G be a "prescheme in groups" over S, and let P be a prescheme over S on which "G acts" (on the right). We say that P is formally principal homogeneous under G if the familiar morphism

$$G \times_S P \longrightarrow P \times_S P$$

325

induced by the action of G on P is an isomorphism. Henceforth we assume that G is flat over S (and therefore faithfully flat over S), and reserve the name principal homogeneous bundle under G for a formally principal homogeneous bundle P that is faithfully flat and quasi-compact over S. Equivalently, there is a faithfully flat and quasi-compact extension $S' \longrightarrow S$ of the base S such that the formally principal homogeneous bundle

$$P' = P \times_S S'$$

under $G' = G \times_S S'$ is trivial, i.e. isomorphic to G' (i.e. admits a section); one may in particular take $S' = P$. Note also that if S is locally Noetherian, then the hypothesis that P be faithfully flat is equivalent to the hypothesis that

$$\widehat{P}_s = P \times_S \mathrm{Spec}(\widehat{\mathcal{O}}_s)$$

be faithfully flat over $\widehat{\mathcal{O}}_s$ for every $s \in S$ (where $\widehat{\mathcal{O}}_s$ denotes the completion of the local ring $\mathcal{O}_s$), since $\widehat{\mathcal{O}}_s$ is faithfully flat over $\mathcal{O}_s$. Moreover, if P is of finite type over the locally Noetherian prescheme S, the set of points s satisfying the preceding condition is constructible. Thus, if S is a "Jacobson prescheme" (for example, a scheme of finite type over a field, or more generally the spectrum of a Jacobson ring), it suffices to check the condition at the closed points of S. This reduces us to the case in which the base is the spectrum of a complete local ring A. When $S = \mathrm{Spec}(A)$, where A is a complete Noetherian local ring, and P is of finite type over S, the faithful flatness of P/S is also equivalent to the existence of an S' finite and flat over S such that P' is trivial; if in addition G is smooth over S, one may take S' étale over S. Consequently, if the residue field of A is moreover algebraically closed (the "geometric case"), P is faithfully flat over A if and only if it is trivial. Thus, if S is an algebraic prescheme over an algebraically closed field and G is smooth and of finite type over S, the faithful flatness condition over S is equivalent to Serre's analytic triviality condition (SLF) ([6], pp. 1–12).

One may introduce other, stronger types of conditions on P having the nature of a "local triviality." We say in particular that P is isotrivial (respectively, strictly isotrivial) if, for every $s \in S$, there exist an open neighborhood U of s and a finite faithfully flat morphism (respectively, a surjective étale covering) $U' \longrightarrow U$ such that $P' = P \times_S U'$ is trivial. (Our terminology differs from Serre's [6], who calls locally isotrivial what we call strictly isotrivial.) Strict isotriviality is especially useful when G is smooth over S, but is an inadequate notion in the other cases.

If G is affine over S, every principal homogeneous bundle P under G is affine by Section 2, and Theorem 2 therefore makes it possible to "descend" 326

such bundles along faithfully flat and quasi-compact morphisms. In particular, take $G = \mathrm{Gl}(n)_S$, defined by the requirement that the functor on S-preschemes

$$S' \longmapsto \mathrm{Hom}_S(S', G)$$

(with values in the category of groups) be identified with the functor

$$\mathrm{Gl}(n)(S') = \mathrm{Gl}(n, H^0(S', \mathcal{O}_{S'}))$$

of A, Section 4(e). Use the facts (i) that every principal homogeneous bundle under G (respectively, every locally free sheaf of rank n over S) becomes isomorphic to the "trivial" object G (respectively, $\mathcal{O}_S^n$) after a suitable faithfully flat and quasi-compact extension of S; (ii) that objects of the type in question (principal homogeneous bundles under G, respectively, locally free sheaves of rank n) can be descended along such morphisms; and (iii) that the automorphism group of the trivial bundle over an S'/S is functorially isomorphic to the automorphism group of the trivial locally free sheaf of rank n over S'. One concludes "formally" that giving a principal homogeneous bundle with group G over S (or over an S'/S) is "the same thing" as giving a locally free sheaf of rank n there. (More precisely, there is an equivalence of fibered categories.) In particular:

PROPOSITION 6.1. — Every principal homogeneous bundle with group $\mathrm{Gl}(n)_S$ is locally trivial.

Known arguments give the same result for structure groups such as $\mathrm{Sl}(n)_S$, $\mathrm{Sp}(n)_S$, and products of such groups. It also follows that if F is a closed subgroup of $G = \mathrm{Gl}(n)_S$, flat over S, such that the quotient G/F exists, and if G is an isotrivial (respectively, strictly isotrivial) principal homogeneous bundle over G/F with group $F \times_S (G/F)$, then every principal homogeneous bundle with group F is isotrivial (respectively, strictly isotrivial). This applies to all the "linear groups" over S used so far, and in particular when $G = S \times_k \Gamma$, where S is a prescheme over the field k and Γ is a linear group in the classical sense, and in particular a smooth group, over k. For such groups this therefore resolves a question of Serre (loc. cit.).

We also note that for most known smooth groups over S, whether linear or not, and in any case for those of the form $S \times_k \Gamma$ above, one can show that every isotrivial principal homogeneous bundle is strictly isotrivial. This resolves in particular another question of Serre (loc. cit., pp. 1–14), since a principal homogeneous bundle obtained by descent à la Cartier (cf. Section 3, Example 2) is manifestly isotrivial.

REMARK. — One of the essential difficulties in these questions (apart from the question of the existence of the quotient schemes G/F) is the lack of effectivity criteria for a descent datum along
327 a faithfully flat nonfinite morphism.

BIBLIOGRAPHY

[1] Dieudonné (J.) and Grothendieck (Alexander). — *Éléments de géométrie algébrique*, forthcoming in the Publications mathématiques de l'Institut des Hautes Études Scientifiques.

[2] Grauert (H.) und Remmert (R.). — *Komplexe Räume*, Math. Annalen, vol. 136, 1958, pp. 245–318.

[3] Grothendieck (Alexander). — *Géométrie formelle et géométrie algébrique*. Séminaire Bourbaki, vol. 11, 1958/59, no. 182.

[4] Murre (J. P.). — *On a connectedness theorem for a birational transformation at a simple point*, Amer. J. Math., vol. 80, 1958, pp. 3–15.

[5] Serre (Jean-Pierre). — *Géométrie algébrique et géométrie analytique*, Ann. Inst. Fourier Grenoble, vol. 6, 1955–56, pp. 1–42.

[6] Serre (Jean-Pierre). — *Espaces fibrés algébriques*, Séminaire Chevalley, vol. 2, 1958: Anneaux de Chow et applications, no. 1.

Erratum to Exposé no. 190

300

3. Descent technique and existence theorems in algebraic geometry [TDTE I]. I: Generalities. Descent along faithfully flat morphisms.

[Bourbaki Seminar, vol. 12, 1959/60, no. 190, 29 pp.]

Page 1, line 10. — It now seems excessive to say that the descent technique is "the basis of most existence theorems in algebraic geometry." This is true to a large extent for the nonprojective techniques treated in the first two exposés of the TDTE I–VI series, but not for the projective techniques (TDTE IV, V, VI).

Page 5, line 16. — It is unnecessary to assume that α is an $\mathcal{F}$-descent morphism.

Page 20, remark. — A quasi-finite, étale, surjective morphism $S' \longrightarrow S$, or a morphism $\mathrm{Spec}(\widehat{A}) \longrightarrow \mathrm{Spec}(A)$, need not be a strict descent morphism, even if A is the local ring of an algebraic curve over an algebraically closed field k and $S = \mathrm{Spec}(A)$. Thus one can find a proper smooth morphism $f : X \longrightarrow S$ making X a principal bundle over S under an elliptic curve E, such that $f' : X' \longrightarrow S'$ is projective while f is not projective. This is therefore also an example of a non-isotrivial homogeneous principal bundle under an abelian scheme.

Page 26, line 9. — Read "Chow–Iang" instead of "Igusa–Lang."

For various details concerning descent theory, see SGA VI, VII, VIII.

Chapter 4

Exposé 195 – Descent Technique II

369
BOURBAKI SEMINAR
12th year, 1959/60, no. 195
February 1960

DESCENT TECHNIQUE AND EXISTENCE THEOREMS IN ALGEBRAIC GEOMETRY
II. THE EXISTENCE THEOREM IN FORMAL MODULI THEORY

by Alexander GROTHENDIECK

A. Representable and pro-representable functors

1. Representable functors

Let $\mathcal{C}$ be a category. For every $X \in \mathcal{C}$, let h_X be the contravariant functor from $\mathcal{C}$ to the category (Ens) of sets,

$$h_X : \mathcal{C} \longrightarrow (\mathrm{Ens}),$$

defined by the formula

$$h_X(Y) = \mathrm{Hom}(Y, X).$$

A morphism $X \longrightarrow X'$ in $\mathcal{C}$ gives, in the evident way, a homomorphism of functors $h_X \longrightarrow h_{X'}$: h_X is a covariant functor of X. In other words, we have defined a canonical covariant functor

$$h : \mathcal{C} \longrightarrow \mathrm{Hom}(\mathcal{C}^o, (\mathrm{Ens}))$$

from $\mathcal{C}$ to the category of covariant functors from the dual $\mathcal{C}^o$ of $\mathcal{C}$ to the category of sets. Recall:

PROPOSITION 1.1. — The functor h is *fully faithful*; in other words, for every pair X, X' of objects of $\mathcal{C}$, the natural map

$$\mathrm{Hom}(X, X') \longrightarrow \mathrm{Hom}(h_X, h_{X'})$$

is bijective.

In particular, if a functor $F \in \mathrm{Hom}(\mathcal{C}^o, (\mathrm{Ens}))$ is isomorphic to a functor of the form h_X, then X is determined up to a unique isomorphism. One then says that F is *representable*. The preceding proposition says that the canonical functor h defines an equivalence of $\mathcal{C}$ with the full subcategory

of $\mathrm{Hom}(\mathcal{C}^o, (\mathrm{Ens}))$ formed by the representable functors. This fact underlies the notion of a "solution of a Universal Problem": such a problem always consists in asking whether a given functor
370 (contravariant as here, or covariant in the dual case) from $\mathcal{C}$ to (Ens) is representable.

Moreover, by the very definition of a product in a category [1], the functor $h : X \longmapsto h_X$ commutes with products whenever they exist (and, more generally, with finite or infinite projective limits, in particular with fiber products, with the formation of "kernels" [2], etc., whenever they exist): there is an isomorphism of functors

$$h_{X\times X'} \xrightarrow{\sim} h_X \times h_{X'}$$

whenever $X \times X'$ exists; that is, there are bijections, functorial in Y,

$$h_{X\times X'}(Y) \xrightarrow{\sim} h_X(Y) \times h_{X'}(Y).$$

In particular, giving a morphism

$$X \times X' \longrightarrow X''$$

in $\mathcal{C}$ (that is, a "law of composition" in $\mathcal{C}$ among X, X', X'') is equivalent to giving a morphism

$$h_{X\times X'} = h_X \times h_{X'} \longrightarrow h_{X''};$$

that is, to giving for every $Y \in \mathcal{C}$ a law of composition of sets

$$h_X(Y) \times h_{X'}(Y) \longrightarrow h_{X''}(Y)$$

such that, for every morphism $Y \longrightarrow Y'$ in $\mathcal{C}$, the system of set maps

$$h_{X^{(i)}}(Y) \longleftarrow h_{X^{(i)}}(Y') \qquad (\text{for } i = 0, 1, 2)$$

is a morphism for the two laws of composition belonging to Y and Y'. In this way one sees that a structure of "$\mathcal{C}$-group," "$\mathcal{C}$-ring," etc. on an object X of $\mathcal{C}$ is expressed most conveniently (in theory as well as in practice) by saying that, for every $Y \in \mathcal{C}$, the set $h_X(Y)$ carries a group law (respectively a ring law, etc.) in the usual sense, and that the maps $h_X(Y) \longleftarrow h_X(Y')$ corresponding to morphisms $Y \longrightarrow Y'$ are group homomorphisms (respectively ring homomorphisms, etc.). This is, for example, the most intuitive and convenient way to define the various classical groups G_a, G_m, $\mathrm{Gl}(n)$, etc. over an arbitrary base prescheme S and to write the classical relations among these groups; to put a "vector bundle" structure on the affine scheme $V(\mathcal{F})$ over S defined by a quasi-coherent sheaf $\mathcal{F}$; and to define and study the associated flag varieties (Grassmannians, projective bundles), etc. The general yoga is simply to identify, by means of the canonical functor h, the objects of $\mathcal{C}$ with particular contravariant functors, namely the representable functors, 371

from $\mathcal{C}$ to the category of sets.

The usual procedure of reversing arrows, which is required, for example, in the case of affine schemes in passing from geometric language to the language of commutative algebra, leads one to dualize the preceding considerations and, in particular, also to introduce representable covariant functors $\mathcal{C} \longrightarrow (\mathrm{Ens})$, that is, functors of the form

$$Y \longmapsto \mathrm{Hom}(X, Y) = h'_X(Y).$$

2. Pro-representable functors and pro-objects

Let $\widetilde{X} = (X_i)_{i\in I}$ be a projective system of objects of $\mathcal{C}$. It has an associated covariant functor

$$h'_{\widetilde{X}} = \varinjlim_i h'_{X_i},$$

or, more explicitly,

$$h'_{\widetilde{X}}(Y) = \varinjlim_i h'_{X_i}(Y) = \varinjlim_i \mathrm{Hom}(X_i, Y),$$

from $\mathcal{C}$ to (Ens). A functor from $\mathcal{C}$ to (Ens) that is isomorphic to a functor of this type with I filtered is called *pro-representable*. By the preceding subsection, these are exactly the functors isomorphic to filtered inductive limits of representable functors. Let $\widetilde{X}' = (X'_j)_{j\in J}$ be a second filtered projective system in $\mathcal{C}$ (constructed on a second filtered preordered index set J). One readily verifies the canonical bijection (generalizing Proposition 1.1)

$$\mathrm{Hom}(h'_{\widetilde{X}'}, h'_{\widetilde{X}}) = \varprojlim_j \varinjlim_i \mathrm{Hom}(X_i, X'_j).$$

This leads one to introduce the category $\mathrm{Pro}(\mathcal{C})$ of pro-objects of $\mathcal{C}$. Its objects are the projective systems of objects of $\mathcal{C}$ (on variable filtered preordered index sets), and if $\widetilde{X} = (X_i)_{i\in I}$ and $\widetilde{X}' = (X'_j)_{j\in J}$ are two such objects, one puts

$$\mathrm{ProHom}(\widetilde{X}, \widetilde{X}') = \varprojlim_j \varinjlim_i \mathrm{Hom}(X_i, X'_j),$$

the composition of pro-homomorphisms being evident. By construction, $\widetilde{X} \longmapsto h'_{\widetilde{X}}$ may be regarded as a contravariant functor of $\widetilde{X}$ and establishes an equivalence of the dual of the category $\mathrm{Pro}(\mathcal{C})$ of pro-objects of $\mathcal{C}$ with the category of pro-representable covariant functors from $\mathcal{C}$ to
372 (Ens). Of course, an object X of $\mathcal{C}$ canonically defines a pro-object, again denoted by X,

so that $\mathcal{C}$ is equivalent to a full subcategory of $\mathrm{Pro}(\mathcal{C})$. If $\widetilde{X} = (X_i)_{i\in I}$ is any pro-object of $\mathcal{C}$, then (with the preceding identification)

$$\widetilde{X} = \varprojlim_i X_i,$$

where the projective limit is taken in $\mathrm{Pro}(\mathcal{C})$ (because $h'_{\widetilde{X}} = \varinjlim_i h'_{X_i}$). Notice that even when the projective limit of the X_i exists in $\mathcal{C}$, it will generally not be isomorphic to the projective limit $\widetilde{X}$ in $\mathrm{Pro}(\mathcal{C})$, as is already evident when $\mathcal{C}$ is the category of sets. By definition, $\varprojlim_{\mathcal{C}} X_i = L$ is defined by the requirement that the functor of $Y \in \mathcal{C}$

$$\varprojlim_i \mathrm{Hom}_{\mathcal{C}}(Y, X_i) = \mathrm{Hom}_{\mathrm{Pro}(\mathcal{C})}(Y, \widetilde{X}),$$

with values in (Ens), be representable by L, that is, be isomorphic to $\mathrm{Hom}_{\mathcal{C}}(Y, L)$. Consequently, $\varprojlim_{\mathcal{C}} X_i$ is already defined in terms of the pro-object $\widetilde{X}$, and depends functorially on $\widetilde{X}$ whenever it is defined; there is therefore no inconvenience in denoting it by $\varprojlim_{\mathcal{C}}(\widetilde{X})$. If projective limits always exist in $\mathcal{C}$, then $\varprojlim_{\mathcal{C}}(\widetilde{X})$ is a functor from $\mathrm{Pro}(\mathcal{C})$ to $\mathcal{C}$, and there is a canonical homomorphism of functors $\varprojlim_{\mathcal{C}}(\widetilde{X}) \longrightarrow \widetilde{X}$.

Every functor, say covariant to fix ideas,

$$F : \mathcal{C} \longrightarrow \mathcal{C}',$$

extends in the evident way to a functor

$$\mathrm{Pro}(F) : \mathrm{Pro}(\mathcal{C}) \longrightarrow \mathrm{Pro}(\mathcal{C}').$$

It follows that, if projective limits always exist in $\mathcal{C}'$, then F also canonically defines the composite functor

$$\overline{F} : \mathrm{Pro}(\mathcal{C}) \longrightarrow \mathcal{C}', \qquad \overline{F} = \varprojlim_{\mathcal{C}'} \circ \mathrm{Pro}(F),$$

which sends $\widetilde{X} = (X_i)_{i\in I}$ to $\varprojlim_{\mathcal{C}'} F(X_i)$.

A pro-object $\widetilde{X}$ is called a *strict pro-object* if it is isomorphic to a pro-object $(X_i)_{i\in I}$ whose transition morphisms $X_i \longrightarrow X_j$ are epimorphisms; a functor defined by such an object is called *strictly pro-representable*. One may then further require I to be filtered and ordered and every epimorphism $X_i \longrightarrow X'$ to be equivalent to an epimorphism 373

$X_i \longrightarrow X_j$ for a suitable $j \in I$ (uniquely determined by this condition). Under these conditions the projective system $(X_i)_{i\in I}$ is determined up to a unique isomorphism (in the usual sense of isomorphisms of projective systems). It follows that in $\mathrm{Pro}(\mathcal{C})$ the projective limit of a projective system of strict pro-objects $\widetilde{X}^{(\alpha)}$ always exists and, with the notation above for F and $\overline{F}$, one has

$$\overline{F}\left(\varprojlim_{\alpha} \widetilde{X}^{(\alpha)}\right) = \varprojlim_{\alpha,\mathcal{C}'} \overline{F}\left(\widetilde{X}^{(\alpha)}\right).$$

In particular, if every pro-object of $\mathcal{C}$ is strict (cf. the next subsection), the extended functor $\overline{F}$ commutes with projective limits.

3. Characterization of pro-representable functors

Let $\mathcal{C}$ and $\mathcal{C}'$ be two categories in which finite projective limits (that is, limits relative to finite preordered sets, not necessarily filtered) exist, or, equivalently, in which finite products and finite fiber products exist. This implies in particular the existence of a "right unit object" e such that $\mathrm{Hom}(X, e)$ consists of one element for every X. Let F be a covariant functor from $\mathcal{C}$ to $\mathcal{C}'$. The following conditions are equivalent:

(i) F commutes with finite projective limits;

(ii) F commutes with finite products and finite fiber products;

(iii) F commutes with finite products, and for every exact diagram

$$X \longrightarrow X' \rightrightarrows X''$$

in $\mathcal{C}$ ([3], A, Definition 2.1), the diagram obtained by applying F,

$$F(X) \longrightarrow F(X') \rightrightarrows F(X''),$$

is exact.

One then says that F is *left exact*.

Henceforth we assume that finite projective limits exist in $\mathcal{C}$. It is then immediate from the definitions that a representable functor is left exact, and, on passing to the limit, that a pro-representable functor is left exact.

For a converse, let

$$F : \mathcal{C} \longrightarrow (\mathrm{Ens})$$

be a covariant functor, and let $X \in \mathcal{C}$ and $\xi \in F(X)$. We say that ξ (or the pair (X, ξ)) is *minimal* 374 if, for every $X' \in \mathcal{C}$, every $\xi' \in F(X')$, and every strict monomorphism ([3], A, 2) $u : X' \longrightarrow X$

such that $\xi = F(u)(\xi')$, the morphism u is an isomorphism. On the other hand, a pair (X, ξ) is said to *dominate* (X'', ξ'') ($\xi \in F(X)$, $\xi'' \in F(X'')$) if there is a morphism $v : X \longrightarrow X''$ such that $\xi'' = F(v)(\xi)$. If ξ is minimal and F is left exact, this morphism v is unique; if ξ'' is minimal, v is surjective. The following proposition follows readily:

PROPOSITION 3.1. — The functor F is strictly pro-representable if and only if it satisfies the following two conditions:

(i) F is left exact;

(ii) every pair (X, ξ), with $\xi \in F(X)$, is dominated by a minimal pair.

The latter condition is vacuous if every object of $\mathcal{C}$ is Artinian (take a subobject X' of X minimal among those for which there exists $\xi' \in F(X')$ whose image is ξ). Hence:

COROLLARY. — Let $\mathcal{C}$ be a category whose objects are Artinian and in which finite projective limits exist. Then the pro-representable functors from $\mathcal{C}$ to (Ens) are exactly the left exact functors, and they are in fact strictly pro-representable.

The latter fact also means that *every pro-object of $\mathcal{C}$ is strict*.

4. Example: groups of Galois type and pro-algebraic groups

If $\mathcal{C}$ is the category of ordinary finite groups, then Pro($\mathcal{C}$) is equivalent to the category of compact totally disconnected topological groups. Groups of this type, and their generalizations obtained by replacing ordinary finite groups with finite group schemes over a given base prescheme (for example, finite algebraic groups over a field k), will serve as absolute and relative fundamental, homotopy, and homology groups for preschemes. In all these examples, the corollary to Proposition 3.1 applies, and π_1 must indeed be defined by means of the associated functor [2]. The same is true when one starts from the category of algebraic or quasi-algebraic groups over a field (or, more generally, over a Noetherian prescheme): one obtains Serre's "pro-algebraic groups" [4].

5. Example: "formal varieties"

Let Λ be a Noetherian ring, and let $\mathcal{C}$ be the category of Λ-algebras A that are Artinian finite-type
375 modules over Λ (or, more briefly,

Artinian Λ-algebras). The hypotheses of the corollary to Proposition 3.1 are satisfied. Here the category Pro($\mathcal{C}$) is equivalent to the category of topological algebras O over Λ that are isomorphic to topological projective limits

$$O = \varprojlim_i O_i$$

of algebras $O_i \in \mathcal{C}$; that is, their topology is linear, separated, and complete, and for every open ideal J_i of O, the quotient O/J_i is an Artinian algebra over Λ. The functor $\mathcal{C} \longrightarrow$ (Ens) associated with such an algebra is precisely

$$\begin{aligned} F(A) = h'_O(A) &= \{\text{continuous homomorphisms of topological } \Lambda\text{-algebras } O \longrightarrow A\} \\ &= \varinjlim_i \mathrm{Hom}_{\Lambda\text{-algebras}}(O_i, A). \end{aligned}$$

Notice also that $\mathcal{C}$ is essentially the product of the analogous categories corresponding to the completed local rings of the $\Lambda_{\mathfrak{m}}$ for the maximal ideals $\mathfrak{m}$ of Λ; if desired, one may therefore restrict to the case in which Λ is such a complete local ring. In every case, O decomposes canonically as the topological product of local components corresponding to the "points" of the formal scheme [2] defined by O. Such a point is defined by an object ξ of some $F(K)$, where $K \in \mathcal{C}$ is a field

(for example, the residue field of the local component in question); two pairs (ξ, K) and (ξ', K') define the same point if and only if they are dominated by a common (ξ'', K''), or, equivalently, if they dominate a common (ξ''', K'''). (If the $\Lambda/\mathfrak{m}$ are algebraically closed, it suffices to take the disjoint union of the $F(\Lambda/\mathfrak{m})$.)

It is important to give conditions under which the local component O_ξ of O corresponding to $\xi \in F(K)$ is a Noetherian ring. When Λ is a complete local ring (Noetherian, as we recall), this is equivalent to saying that O_ξ is isomorphic to a quotient of a formal power series ring $\Lambda[[t_1, \dots, t_n]]$ over Λ. To give such a criterion, for every ring A introduce the A-algebra I_A of the "dual numbers" of A by

$$I_A = A[t]/t^2A[t].$$

Let $\varepsilon : I_A \longrightarrow A$ be the augmentation homomorphism. If $A \in \mathcal{C}$, it defines a map

$$F(\varepsilon) : F(I_A) \longrightarrow F(A),$$

376

and, using the fact that F is left exact, one intrinsically defines an A-module structure on the subset

$$F(I_A, \xi) = F(\varepsilon)^{-1}(\xi)$$

of $F(I_A)$ formed by the $\xi' \in F(I_A)$ that "reduce to ξ." Using the explicit form of F in terms of O, one finds that this K-module is identified with

$$\mathrm{Hom}_\Lambda(\mathfrak{m}_\xi/\mathfrak{m}_\xi^2, A),$$

where $\mathfrak{m}_\xi$ is the kernel of the homomorphism $\xi : O \longrightarrow A$; if A is a field, this is the maximal ideal of the local component O_ξ of O. The following proposition is immediate:

PROPOSITION 5.1. — Let $\xi \in F(K)$, where $K \in \mathcal{C}$ is a field. The corresponding local component O_ξ of O is Noetherian if and only if the set $F(I_K, \xi)$ of elements of $F(I_K)$ reducing to ξ is a finite-dimensional vector space over K. Under these conditions there is a canonical isomorphism

$$F(I_K, \xi) = \mathrm{Hom}\left(\frac{\mathfrak{m}_\xi/\mathfrak{m}_\xi^2}{\mathrm{im}(\mathfrak{n}_\xi/\mathfrak{n}_\xi^2)}, K\right),$$

where $\mathfrak{n}_\xi$ is the maximal ideal of Λ, the kernel of $\Lambda \longrightarrow K$. [1] In particular, the dimension of the K-vector space $F(I_K, \xi)$ equals the dimension of this corrected vector space over the field $O_\xi/\mathfrak{m}_\xi = K(\xi)$.

Suppose that O_ξ is Noetherian, and, to simplify the notation, suppose that Λ is complete local and $O = O_\xi$. We say that O is *smooth over* Λ if O is finite and étale over the completion of the localization of $\Lambda[t_1, \dots, t_n]$ at a maximal ideal inducing the maximal ideal of Λ. When the residue extension of O over Λ is trivial (for example, when the residue field of Λ is algebraically closed), this is equivalent to saying that O itself is isomorphic to such a formal power series algebra. [2] Finally, if O is no longer assumed Noetherian, we still say that O is smooth over Λ if O is isomorphic to a topological projective limit of quotient Λ-algebras that are Noetherian and smooth over Λ in the preceding sense. This generalizes at once to the case in which Λ and O are no longer assumed local. We then have:

PROPOSITION 5.2. — The algebra O is smooth over Λ if and only if the associated functor F sends epimorphisms to epimorphisms.

[1] The printed formula is correct only when Λ is a field. The 1962 erratum directs that in general $\mathfrak{m}_\xi/\mathfrak{m}_\xi^2$ be replaced by its quotient by the image of $\mathfrak{n}_\xi/\mathfrak{n}_\xi^2$.

[2] The 1962 erratum specifies that this definition is correct only when the residue extension k'/k is separable; for the general case it refers to SGA III, 1.1.

Thus, for every surjective homomorphism $A \longrightarrow A'$ in $\mathcal{C}$, the map $F(A) \longrightarrow F(A')$ is surjective. It is of course enough to verify this condition when A is local and, proceeding step by step, when the ideal of A that is the kernel of $A \longrightarrow A'$ is annihilated by the maximal ideal of A. In practice, this reduces matters to checking that a certain obstruction, related to infinitesimal
377 invariants

of the situation giving rise to the functor F, vanishes, a problem of a cohomological nature.

To conclude, let us say a few words in the preceding context about *rings of definition*. Let F still be a functor from $\mathcal{C}$ to (Ens), pro-represented by a topological Λ-algebra O. Then for every $A \in \mathcal{C}$ and every $\xi \in F(A)$, there is a smallest subring A' of A such that ξ is the image of an element ξ' of $F(A')$ (which is then uniquely determined). Indeed, it suffices to interpret ξ by means of a homomorphism from O to A and to take for A' the image of O under this homomorphism. One then says that A' is the ring of definition of the object $\xi \in F(A)$. If $u : A \longrightarrow B$ is an algebra homomorphism and $\eta = F(u)(\xi)$, then the ring of definition of η is the image under u of the ring of definition of ξ. When one starts from a functor F from $\mathcal{C}$ to (Ens), the existence of rings of definition and the properties just given are, up to minor qualifications, equivalent to F being pro-representable; thus they are usually far from trivial.

B. The two existence theorems

Retain the notation of A, Section 5, and start with a covariant functor

$$F : \mathcal{C} \longrightarrow (\text{Ens}).$$

We seek manageable criteria for F to be pro-representable, that is, expressible by means of a Λ-algebra O as above. By the corollary to A, Proposition 3.1, this is equivalent to F being left exact. In the present state of the descent technique (cf. the questions posed in [3], p. 9), this criterion cannot be directly verified in this form in the most important cases, and criteria that appear less demanding are needed.

THEOREM 1. — The functor F is pro-representable if and only if it satisfies the following two conditions:

(i) F commutes with finite products;

(ii) for every algebra $A \in \mathcal{C}$ and every homomorphism $A \longrightarrow A'$ in $\mathcal{C}$ such that the diagram

$$A \longrightarrow A' \rightrightarrows A' \otimes_A A'$$

is exact ([3], A, Definition 1.2), the transformed diagram

$$F(A) \longrightarrow F(A') \rightrightarrows F(A' \otimes_A A')$$

378

is exact.

Moreover, it is enough to verify (ii) when A is local and, in addition, one of the following two cases holds:

(a) A' is a free module over A;

(b) the quotient module A'/A is an A-module of length 1.

The proof of this theorem is rather delicate and cannot be sketched here. We merely note that it rests essentially on a study of equivalence relations (in the categorical sense) in the spectrum of an Artinian algebra, a study that still raises several problems whose solution seems indispensable for the further development of the theory.

In applications, condition (i) is always trivial to verify. The verification of (ii) splits in two: the case in which A' is a free A-module belongs to the descent technique along flat morphisms ([3], Theorems 1, 2, and 3) and presents no difficulty. To treat case (b), we use the following result:

THEOREM 2. — Let A be an Artinian local ring with maximal ideal $\mathfrak{m}$, and let A' be an A-algebra containing A such that $\mathfrak{m}A' \subset A$ and

$$A \longrightarrow A' \rightrightarrows A' \otimes_A A'$$

is exact (as is the case, in particular, if A'/A is an A-module of length 1). Let $\mathcal{F}$ be the fibered category ([3], A, Definition 1.1) of flat quasi-coherent sheaves on variable preschemes. Then the morphism $\mathrm{Spec}(A') \longrightarrow \mathrm{Spec}(A)$ is a strict $\mathcal{F}$-descent morphism ([3], A, Definition 1.7).

In other words, giving a flat A-module M is completely equivalent to giving a flat A'-module M' together with an $A' \otimes_A A'$-isomorphism from $M' \otimes_A A'$ onto $A' \otimes_A M'$ that satisfies the usual transitivity condition for a descent datum (loco citato).

Theorem 2 is proved by first showing that

$$H^i(A'/A, G_a) = 0 \qquad \text{for } i \geq 1$$

([3], A, 4(e)); the hypothesis $\mathfrak{m}A' \subset A$ allows one to reduce readily to the case in which A is a field k (namely $A/\mathfrak{m}$). One then applies the equivalences noted in [3], page 16.

C. Applications to some special cases

1. General remarks on functors represented by preschemes

Let S be a locally Noetherian prescheme. A prescheme X over S is called locally of finite type over 379
S if, for every $x \in X$ mapping to $y \in S$,[3] there is an affine neighborhood of y, with ring A, and an affine neighborhood of x over it, with ring B, such that B is an A-algebra of finite type. One finds many important examples of preschemes locally of finite type over S that are not of finite type over S, as solutions of classical universal problems. Thus it is important to be able to regard the Picard scheme of a curve as the union of infinitely many connected components (which must not be confused with the connected component of the identity element, i.e. the "Picard variety"). It is then sometimes convenient to work in the category $\mathcal{C}$ of preschemes locally of finite type over S, and there examine the representability of a contravariant functor F. The principal aim of these exposés is to develop a general technique for recognizing whether such a functor F is representable, and for studying the properties of the corresponding S-prescheme X by means of those of F. Let us mention in passing that this study produces nonpathological examples of preschemes over S that are not separated over S, notably as "Picard preschemes" of excellent S-schemes. Preschemes that are not schemes must therefore not be banished from algebraic geometry.

Let X be a prescheme locally of finite type over S, and let

$$F(Y) = \mathrm{Hom}_S(Y, X)$$

be the associated contravariant functor. We may consider the restriction F_0 of F to the subcategory $\mathcal{C}_0$ of $\mathcal{C}$ formed by the preschemes Y over S that are Artinian and finite over S: when $S = \mathrm{Spec}(\Lambda)$,

[3] The printed text has "$y \in Y$" here and uses Y again at the end of the paragraph. Since the morphism under consideration is $X \longrightarrow S$, both occurrences are read as S.

$\mathcal{C}_0$ is therefore the dual of the category of Artinian Λ-algebras considered in B. If $Y = \mathrm{Spec}(A)$, where A is an Artinian local ring, then Y consists of a single point y lying over a closed point s of S, and an S-homomorphism from Y to X (i.e. an element of $F(Y)$) is specified by a point $x \in X$ over s and an $\mathcal{O}_{s,S}$-homomorphism from $\mathcal{O}_{x,X}$ to A. If such a homomorphism exists, then x is necessarily a closed point of X (its residue field being algebraic over that of s). This shows that the restriction F_0 of F to the "Artinian Λ-algebras" is pro-representable, and is represented by the topological Λ-algebra whose local components are the completions $\widehat{\mathcal{O}}_{x,X}$ of the local rings of X at the closed points x of X lying over closed points of S. Thus knowledge of F_0 alone gives precise information about the structure of X, namely the structure of the completions of its local rings at
380 the indicated points. Note that even when S

is the spectrum of an algebraically closed field, it is only through the systematic consideration of "varieties" Y for which $\mathcal{O}_Y$ may have nilpotent elements, and in particular by working with spectra of Artinian local rings, that one arrives at the "correct formulation" of classical universal problems and grasps their "infinitesimal" aspect.

When one starts from a functor F given in advance and wants to decide whether it is representable, the study of F_0 (using Theorems 1 and 2) gives nearly complete indications. It may happen, as it frequently does (for example, by testing simply the nature of the sets $F(I_K, \xi)$ and their functorial behavior; cf. A), that F_0 is already found not to be pro-representable. This explains the failure of the attempts made thus far to define, in a more or less natural way, moduli varieties classifying vector bundles of rank > 1. Or one may verify that F_0 is representable but that the vector spaces $F(I_K, \xi)$ are not finite-dimensional, in which case one must be content with the "formal" solution. Finally, F_0 may be represented by a product of complete Noetherian local rings. This gives very strong grounds for believing that F itself is representable and, together with analogous but more global properties that we intend to develop later, will doubtless suffice to imply that it actually is. Lastly, one encounters interesting geometric problems (see paragraphs 4 and 5 below) for which only the functor F_0 is available (and does not arise from any "global" functor F), and one will count oneself fortunate if a "formal moduli scheme" can be associated with it.

To conclude these general remarks, let us indicate how scheme theory explains apparent anomalies such as the Igusa surface V, whose "Picard variety" P is a point even though

$$H^1(V, \mathcal{O}_V) \neq 0.$$

In this case P is a "purely infinitesimal" group not reduced to the identity group; that is, it is defined by a local algebra O of finite rank over the base field k, equipped with a diagonal map corresponding to the multiplicative structure of P. If $\mathfrak{m}$ is the maximal ideal of O, the dual of $\mathfrak{m}/\mathfrak{m}^2$ is canonically isomorphic to $H^1(V, \mathcal{O}_V)$ (cf. paragraph 3 below). Only when the Picard group is an algebraic group in the classical sense, i.e. smooth over the base field k, is the dimension of $H^1(V, \mathcal{O}_V)$ (always equal to that of $\mathfrak{m}/\mathfrak{m}^2$) equal to the dimension of the Picard group.

2. Schemes $\underline{\mathrm{Hom}}_S(X, Y)$, $\Pi_{X/S} Z$, $\underline{\mathrm{Aut}}_S(X)$, etc.

381 Let X, Y be two preschemes over S; for every prescheme T over S,

put $X_T = X \times_S T$, $Y_T = Y \times_S T$, and consider the set

$$F(T) = \mathrm{Hom}_T(X_T, Y_T) = \mathrm{Hom}_S(X_T, Y) = \mathrm{Hom}_S(X \times_S T, Y)$$

as a contravariant functor of T. If it is representable, the prescheme over S that represents it is denoted by $\underline{\mathrm{Hom}}_S(X, Y)$; thus there is a functorial isomorphism

$$\mathrm{Hom}_S(T, \underline{\mathrm{Hom}}_S(X, Y)) \xrightarrow{\sim} \mathrm{Hom}_S(T \times_S X, Y).$$

There are many variants of this universal problem whose solution reduces to it: the prescheme of S-automorphisms of an S-prescheme X (which will be a group prescheme), the prescheme of S-homomorphisms from one S-group prescheme to another (which will be a commutative group prescheme if the second group scheme is commutative), etc. One may also generalize the definition of $\underline{\mathrm{Hom}}_S(X, Y)$ by considering a prescheme Z over the prescheme X over S, and the functor

$$F(T) = \mathrm{Hom}_{X_T}(X_T, Z_T)$$

(the set of "sections" of Z_T over X_T). If this functor is representable, the S-prescheme representing it will be denoted $\Pi_{X/S}Z$; hence, by definition, there is a functorial isomorphism

$$\mathrm{Hom}_S(T, \Pi_{X/S}Z) = \mathrm{Hom}_{X_T}(X_T, Z_T).$$

Taking $Z = Y \times_S X$ recovers $\underline{\mathrm{Hom}}_S(X, Y)$. These definitions give a formulary for the newly introduced preschemes that is as trivial as it is useful; we shall not give it here, since it holds in every category in which products and fiber products exist. More serious is the question of the existence of schemes of the form $\underline{\mathrm{Hom}}_S(X, Y)$. One first observes that, for fixed X, $\underline{\mathrm{Hom}}_S(X, Y)$ can exist for every Y over S (or $\Pi_{X/S}Z$ for every Z over X) only if X is flat over S. Moreover, one is convinced that it is reasonable to expect a solution for sufficiently general Y only if X is also proper over S. It appears, furthermore, that these conditions suffice for the existence of $\underline{\mathrm{Hom}}_S(X, Y)$ and $\Pi_{X/S}Z$, provided that, when necessary, a "quasi-projective" hypothesis is imposed on Y/S or on Z/X, respectively. This can in any case be verified in a fair number of situations (for example, when Y is affine over S, or, by direct elementary constructions, when X is finite over S). Theorems 1 and 2 give the following: 382

PROPOSITION 2.1. — Let Λ be a Noetherian ring and let X, Y be arbitrary preschemes over Λ. Consider the functor

$$F(A) = \mathrm{Hom}_A(X_A, Y_A)$$

on the category $\mathcal{C}_0$ of Artinian Λ-algebras. If X is flat over Λ, this functor is pro-representable.

Moreover, for every $A \in \mathcal{C}_0$ and $\xi \in F(A)$, there is a canonical isomorphism

$$F(I_A, \xi) = H^1\Big(X_A, \underline{\mathrm{Hom}}_{\mathcal{O}_{X_A}}\big(\xi^*(\Omega^1_{Y_A/A}), \mathcal{O}_{X_A}\big)\Big),$$

where $\Omega^1_{Y_A/A}$ is the sheaf of Kähler 1-differentials of Y_A over A. Taking A to be a field and using A.5.1 and the finiteness theorem of [2], one obtains the following corollary:

COROLLARY. — Suppose that X is flat and proper over S and that Y is locally of finite type over S. Then F is pro-representable, and the local components of the corresponding topological Λ-algebra are Noetherian rings.

REMARKS. — The problems considered in this number, and many others, were commonly studied in classical algebraic geometry by means of the "Chow coordinates" of cycles in projective space, which allowed these cycles to be regarded as points of suitable projective varieties. This procedure, and the use of Chow coordinates in general, seems irredeemably inadequate from the scheme-theoretic viewpoint because it destroys nilpotent elements in parameter varieties and, in particular, is not suited to a satisfactory study of infinitesimal variations of cycles (quite apart from its nonintrinsic nature, which is tied to projective space). Unfortunately, Chow coordinates were the only language used by many algebraic geometers to study families of varieties or cycles. This seems to have seriously obstructed the clarification of these notions, despite the method's undoubted (and probably temporary) technical value. To obtain the analogue of Chow varieties in scheme theory, one is led to the following universal problem. Let X be a prescheme over S. For every prescheme T over S, let $F(T)$ be the set of closed subpreschemes of $X_T = X \times_S T$ that are

flat over T. One seeks to represent this functor of T by a prescheme over S. More generally, one may take a quasi-coherent sheaf G on X and let $F(T)$ be the set of quotient sheaves 383

of G_T that are flat over T. It appears that the problem has a solution, with a scheme C locally of finite type over S, when X is proper over the locally Noetherian prescheme S and G is coherent. In any case, assuming only that S is locally Noetherian, the restriction of F to the "Artinian S-algebras" is pro-representable; and if X is also proper over S and G is coherent, the local components of the corresponding topological ring O are Noetherian. Of course, even after the existence of the "Chow scheme" of X over S has been proved, one will still have to find a decomposition into disjoint open subsets C_i (corresponding to fixing continuous invariants such as the degree and dimension of the varying cycles) that are of finite type, not merely locally of finite type, over S; determine the relation between this scheme and the classical Chow varieties; and determine when a C_i is even projective, or at least quasi-projective, over S.[4]

3. Picard schemes

Let $f : X \longrightarrow S$ be an S-prescheme.[5] Consider the multiplicative sheaf $\mathcal{O}_X^*$ of units in the structure sheaf of X, and the group

$$P(X/S) = H^0(S, R^1 f_*(\mathcal{O}_X^*)),$$

called the relative Picard group of X/S. An element of this group is thus specified by an open covering (U_i) of S and an invertible sheaf L_i on each $f^{-1}(U_i)$, in such a way that for every pair (i, j) the restriction $L_i|f^{-1}(U_i \cap U_j)$ is isomorphic to $L_j|f^{-1}(U_i \cap U_j)$, at least locally over $U_i \cap U_j$ (that is, the two sheaves are "equivalent" in the sense of [3], B, 4). If X/S has a section, then $P(X/S)$ is simply the set of classes of invertible sheaves over X/S, up to "equivalence" (loco citato). Now put, for every T over S,

$$F(T) = P(X_T/T).$$

This gives a covariant functor of T, which may be called the Picard functor of X/S. If this functor is representable, the prescheme over S that represents it is called the Picard prescheme of X/S and is denoted $\underline{P}(X/S)$. Thus there is an isomorphism of functors

$$\mathrm{Hom}_S(T, \underline{P}(X/S)) \xrightarrow{\sim} P(X_T/T).$$

The formation of Picard preschemes is compatible with base extension. In particular, the Picard preschemes of the fibers of X over S (which are preschemes over the residue fields $K(s)$ of the points $s \in S$) are the fibers of $\underline{P}(X/S)$. Of course, since $P(X_T/T) = F(T)$ is a commutative group, Picard preschemes are group preschemes. Note also that the generalized Jacobians of Rosenlicht are simply the connected components of the identity in the Picard schemes of complete curves that may have singularities; this should make most of their properties evident once existence has been proved.

REMARK. — The definition adopted here is reasonable only when every point of S[6] has an open neighborhood U over which X has a section. In the general case the definition of the Picard functor must be modified slightly in order still to obtain an existence theorem.

Here the plausible conditions for the existence of a Picard prescheme are the following: X is proper and flat over S, $f_*(\mathcal{O}_X) = \mathcal{O}_S$, and X locally has a section over S. The last condition

[4] The 1962 erratum records that the projective case is completely solved by Hilbert schemes (TDTE IV), while examples of Nagata and Hironaka show that without a projectivity hypothesis these functors need not be representable, even for zero-dimensional subvarieties of a nonsingular complete threefold; this is connected with the possible nonexistence of the symmetric square.

[5] For a fuller study, the 1962 erratum refers to TDTE V.

[6] The printed text has "Y"; no base Y is defined here, and the condition concerns the base S.

enters naturally in the application of the descent technique: it eliminates the automorphisms of an invertible sheaf L on X by equipping it with a marked section over the section s ([3], B, 4). In particular, one obtains: 384

PROPOSITION 3.1. — Suppose that X is flat over $S = \mathrm{Spec}(\Lambda)$, where Λ is Noetherian, and that for every T of finite type over S one has

$$f_{T*}(\mathcal{O}_{X_T}) = \mathcal{O}_T$$

(if f is proper and separable,[7] or if S is the spectrum of a field, Künneth's theorem shows that the last condition is equivalent to $f_*(\mathcal{O}_X) = \mathcal{O}_S$). Then the Picard functor of X/S on the category of Artinian Λ-algebras is pro-representable.

Moreover, here one has

$$F(I_A, \xi) = H^1(X_A, \mathcal{O}_{X_A});$$

in particular:

COROLLARY. — If X is proper over S, then the local components of the topological Λ-algebra O corresponding to the Picard functor are Noetherian.

REMARKS. — The definitions and results of this number may be generalized to the classification of principal bundles on X whose structure group is a group scheme G over S that is affine, flat over S, and commutative. If G is not commutative, then the adjoint group bundle of a principal bundle (whose sections are the automorphisms of the principal bundle) is no longer trivial, and Proposition 3.1 is no longer valid as stated. The universal problem can nevertheless be modified so as still to obtain a solution (at least, for the moment, in formal geometry). The golden rule to retain, in the context of this number and those that follow, and whenever one seeks "moduli schemes" for 385 classes of objects defined only up to isomorphism, is always this: eliminate the possible automorphisms of the objects to be classified by introducing auxiliary structures when necessary (marked points or elements of sections, fixed differential forms, etc.), chosen innocuously enough not to alter the original problem substantially.

4. Formal moduli of a variety

Let Λ be a Noetherian local ring with residue field k (most often Λ will equal k, or will be a Cohen p-ring), and let X_0 be a prescheme over k. For every Artinian local Λ-algebra A, consider the set $F(A)$ of isomorphism classes of A-preschemes X flat over A, equipped with an isomorphism

$$(*) \qquad X \otimes_A \kappa(A) \xleftarrow{\sim} X_0 \otimes_k \kappa(A),$$

where $\kappa(A)$ is the residue field of A; of course, isomorphisms between such flat A-preschemes must respect the preceding isomorphism included in the structure. If A is an Artinian Λ-algebra not necessarily local, with local components A_i, define $F(A)$ to be the product of the $F(A_i)$. Thus F becomes a multiplicative functor of A, which may be called the moduli functor for X_0 (and Λ). If this functor is representable, it corresponds to a local topological Λ-algebra O with residue field k, and the formal spectrum of O is called the formal moduli scheme for X_0 (and Λ); cf. [2] for some details on this notion.

Here, for applying the descent technique, the "finite" automorphisms of X_0 are harmless, since they have no bearing on the existence of automorphisms (in the sense specified above) of the A-preschemes X. When A is not a field, the necessary and sufficient condition for X to have no nontrivial A-automorphism is

$$H^0(X_0, \Theta_{X_0/k}) = 0,$$

[7] The 1962 erratum explicitly prescribes the hypothesis that f be "proper and separable."

where $\Theta_{X_0/k}$ denotes the sheaf of k-derivations (the tangent sheaf) of X_0. One also finds readily (at least if X_0 is smooth over k) that

$$F(I_A, \xi) = H^1(X_A, \Theta_{X_A/A}).$$

386 Hence one concludes, as usual:

PROPOSITION 4.1. — Suppose that $H^0(X_0, \Theta_{X_0/k}) = 0$. Then the formal moduli scheme for X_0 exists. If X_0 is also proper over k, the formal moduli scheme is Noetherian.

REMARKS. — 1. When X_0 is not assumed smooth over k, $F(I_A, \xi)$ is identified with an A-submodule of

$$\mathrm{Ext}^1_{\mathcal{O}_{P_A}}(P_A; \mathcal{J}_{X_A}, \mathcal{O}_{X_A}),$$

where $P_A = X_A \times_\Lambda X_A$, where $\mathcal{O}_{X_A}$ is regarded as a coherent sheaf on P_A by means of the diagonal morphism $X_A \longrightarrow P_A$, and where $\mathcal{J}_{X_A}$ denotes the coherent ideal sheaf on P_A defined by the diagonal morphism. More precisely, a straightforward globalization of Hochschild theory shows that the above Ext^1 is identified with the set of isomorphism classes of flat sheaves of I_A-algebras $\mathcal{O}$ on X_A, equipped with an augmentation isomorphism

$$\mathcal{O} \otimes_{I_A} A \longrightarrow \mathcal{O}_{X_A}$$

(recall that $I_A = A[t]/(t^2)$). The submodule $F(I_A, \xi)$ is the one corresponding to sheaves of commutative algebras. The smoothness hypotheses are therefore not essential in moduli theory, as [2] had suggested.

2. Recall (loco citato) that, in particular, every algebraic curve X_0 that is smooth and proper over k has a formal moduli scheme, smooth over Λ, of relative dimension $3g - 3$ if its genus g is at least 2, and g if $g = 0$ or 1. The latter two cases no longer fall directly under Proposition 4.1. In the case of elliptic curves ($g = 1$), however, they could be reduced to it by the considerations that follow.

One may of course vary Proposition 4.1 ad libitum by considering systems of schemes over k equipped with various structures. Suppose, for example, that X_0 is an abelian scheme over k with a marked origin (i.e. X_0 is regarded as a group scheme over k), and let $F(A)$ be the set of isomorphism classes of abelian schemes over A (i.e. group schemes proper and smooth over A) equipped with an isomorphism of abelian schemes $(*)$. One verifies that imposing a multiplicative structure (or even only a "unit section") eliminates infinitesimal automorphisms, and consequently that a formal moduli scheme exists, corresponding to a complete Noetherian local ring O. Moreover, if X is a proper smooth scheme with "absolutely connected" fibers over a locally Noetherian prescheme S, every multiplicative structure on X having a unit section is necessarily associative and commutative (provided at least that it is so on one fiber and that S is connected), and it is
387 uniquely determined by the knowledge

of the unit section. Furthermore, suppose that S is the spectrum of an Artinian local ring A with residue field k, that X is proper over A and equipped with a section s, and that $X \otimes_A k$ is equipped with the structure of an abelian scheme over k whose identity element is the point of $X \otimes_A k$ corresponding to s. A straightforward obstruction calculation, together with an argument due to Serre, proves that X does in fact possess a multiplicative structure having s as its unit section. (Starting from this, one can use the "existence theorem" of [2] to pass to the case where A is complete Noetherian local, and then the descent technique of [3] for the general case, to prove the analogous statement for every connected locally Noetherian S.) This proves that the functor $F(A)$ considered here is isomorphic to the analogous functor defined at the beginning of this number by forgetting the multiplicative structure on X_0. In particular, if $\mathfrak{m}$ is the maximal ideal of O, then $\mathfrak{m}/\mathfrak{m}^2$ is canonically isomorphic to the dual of $H^1(X_0, \Theta_{X_0/k})$ and hence has dimension n^2,

where $n = \dim X_0$. It would be very interesting to determine whether O is smooth over Λ, i.e. isomorphic to a formal power-series algebra in n^2 indeterminates over Λ. Paragraph 1, Proposition 5.2, gives an equivalent formulation of this question as an existence problem for abelian schemes reducing to a given abelian scheme. Transcendental methods show in any case that the answer is affirmative if k has characteristic zero. In characteristic $p \neq 0$, one clearly need only consider the case in which Λ is the ring of Witt vectors constructed on an algebraically closed field k. Perhaps the time has come for the "Greenberg functor" to prove its worth...[8]

5. Extension of coverings

Let $\mathfrak{X}$ be a Noetherian formal prescheme [2], and let U be an open subset of $\mathfrak{X}$, locally defined by the "nonvanishing" of a section of $\mathcal{O}_{\mathfrak{X}}$ that is a non-zero-divisor not only on $\mathfrak{X}$ but on every X_n,[9] hence large enough that every section of $\mathcal{O}_{\mathfrak{X}}$ on an open subset V which vanishes on $U \cap V$ is zero. Let $\mathcal{J}$ be an "ideal of definition" for $\mathfrak{X}$, and put

$$X_n = (\mathfrak{X}, \mathcal{O}_{\mathfrak{X}}/\mathcal{J}^{n+1}),$$

an ordinary Noetherian prescheme. If $\mathfrak{X}'$ and $\mathfrak{X}''$ are two flat coverings of $\mathfrak{X}$ (i.e. two preschemes over $\mathfrak{X}$ defined by sheaves of algebras that are coherent and locally free as sheaves of modules), unramified over U, the evident map

$$\mathrm{Hom}_{\mathfrak{X}}(\mathfrak{X}', \mathfrak{X}'') \longrightarrow \mathrm{Hom}_{X_0}(X_0', X_0'')$$

is injective. In particular, an automorphism of $\mathfrak{X}'$ inducing the identity on X_0' is the identity. This permits the descent technique to be applied to the situation. In particular, start with a flat covering X_0' of X_0, unramified 388

over U_0, and let $G(\mathfrak{X})$ be the set of isomorphism classes (with the isomorphisms inducing the identity on X_0') of flat coverings $\mathfrak{X}'$ of $\mathfrak{X}$ that induce X_0' on X_0 (and hence are necessarily unramified over U). Define $G(V)$ similarly for every open subset V of $\mathfrak{X}$, and more generally define $G(\mathfrak{Y})$ for every formal prescheme $\mathfrak{Y}$ over $\mathfrak{X}$. The results of [2] and [3] first imply the following:
a. As V ranges over the open subsets of $\mathfrak{X}$, the $G(V)$ form a sheaf on $\mathfrak{X}$, denoted $G_{\mathfrak{X}} = G$. The restriction of this sheaf to U is the constant sheaf whose fibers each consist of one element.

More generally, determining the fibers of $G_{\mathfrak{X}}$ is a question about complete local rings. More precisely:
b. For every $x \in \mathfrak{X}$ one has

$$G_x = G(\mathrm{Spec}(\mathcal{O}_{\mathfrak{X},x})) \subset G(\mathrm{Spec}(\widehat{\mathcal{O}}_{\mathfrak{X},x})),$$

(the latter consists of isomorphism classes of finite free algebras B over $\widehat{\mathcal{O}}_{\mathfrak{X},x}$, equipped with an isomorphism of $B \otimes_{\widehat{\mathcal{O}}_{\mathfrak{X},x}} \mathcal{O}_{X_0,x}$ with $\widehat{\mathcal{O}}'_{0x}$, where $\mathcal{O}'_0$ is the sheaf of algebras on X_0 that defines X_0').[10]
c. There is a canonical isomorphism

$$G_{\mathfrak{X}} = \varprojlim_n G_{X_n};$$

[8] The 1962 erratum reports that the formal moduli scheme of an abelian variety over a field is smooth over the Witt ring: every abelian scheme over an Artinian local quotient lifts to the ring of which it is a quotient. The proof uses the variance properties of the lifting obstruction class introduced in Exposé 182, p. 12. It also recalls Mumford's constructions of moduli schemes for curves of genus g and for polarized abelian schemes (cf. SHMT).

[9] This qualification is imposed by the 1962 erratum.

[10] The printed formula identifies G_x directly with the completed term and omits the hat on $\mathcal{O}'_{0x}$. The inclusion and completion are prescribed by the 1962 erratum.

in other words, for every open subset V of $\mathfrak{X}$,

$$G(V) = \varprojlim_n G(V_n).$$

d. Suppose that $\mathfrak{X}$ is obtained from an ordinary proper scheme X over a complete Noetherian local ring Λ with an ideal of definition $\mathfrak{m}$, by taking the $\mathcal{J}$-adic completion of $\mathcal{O}_X$, where $\mathcal{J} = \mathfrak{m}\mathcal{O}_X$. Then $G(\mathfrak{X})$ is canonically isomorphic to the set of classes of flat coverings of the ordinary scheme X that "reduce to X'_0."

Figuratively, (a) and (b) establish the fundamental relations between the local and global aspects of the problem, (c) gives the relations between the "finite" and "infinitesimal" aspects, and finally (d) recalls, under the stated conditions, the identity between the "formal" and "algebraic" aspects.

Now suppose that $\mathfrak{X}$ is defined over a complete Noetherian local ring Λ, with $\mathcal{J} = \mathfrak{m}\mathcal{O}_{\mathfrak{X}}$ (so that X_0 is a prescheme over $\Lambda/\mathfrak{m}$). For every algebra A finite over Λ, put

$$F(A) = G(\mathfrak{X} \times_\Lambda A).$$

This is a covariant functor of A with values in the category of sets, and by (c) it is completely known if it is known for Artinian A. Saying that this functor is pro-representable is equivalent to saying that it has the form

$$F(A) = \mathrm{Hom}_{\text{topological } \Lambda\text{-algebras}}(O, A),$$

389

where O is a topological Λ-algebra of the type considered in A, paragraph 5; it is also equivalent to saying the same after restricting to Artinian A. The conjunction of Theorems 1 and 2 therefore gives:

PROPOSITION 5.1. — The preceding functor is pro-representable.

Of course, by (a), if $U = \mathfrak{X}$, then $G(\mathfrak{Y})$ consists of a single element for every $\mathfrak{Y}$ over $\mathfrak{X}$, and the functor F is then of little interest (one has $O = \Lambda$). In practically every other case, it appears that the topological local ring O is not Noetherian. Its existence nevertheless strikingly exhibits the "continuous" nature of the set $G(\mathfrak{X})$ of solutions (corresponding intuitively to the existence of a "continuous" choice in the way ramification propagates when X'_0 is extended). Compare this result with J.-P. Serre's viewpoint [4] in local class field theory, which likewise brings out the continuous character of the topological Galois group of the maximal abelian extension of a "geometric" local field. Its dual group (in the sense of Pontryagin) appears there as an inductive limit of algebraic, or at least quasi-algebraic, groups; here too, extensions are classified by infinite-dimensional "varieties." Moreover, in the preceding discussion one may take $\mathfrak{X}$ to be the formal spectrum of a complete local ring (of which Λ will be, for example, a Cohen subring), and one may hope that the results of this number can be used to study extensions of a complete local ring of dimension > 1. In both the local and global cases they may make it possible to formulate precise relations between higher ramification phenomena and phenomena in characteristic zero (which are accessible by transcendental methods). In any case, it is the analysis preceding Proposition 5.1 that makes it possible to extend the methods of [2] for studying the fundamental group to the "tamely ramified" case and to solve the "three-point problem" by transcendental methods.

Finally, note that the situation simplifies when X_0 has dimension one: by (a) and (b),

$$G(\mathfrak{X}) \simeq \prod_i G(\mathrm{Spec}(\widehat{\mathcal{O}}_{\mathfrak{X},x_i})),$$

where the x_i are the points of $X_0 - U$: the "local" extensions at the ramification points may be chosen arbitrarily. Moreover, when X_0 is normal, the formal moduli scheme guaranteed by 5.1 is

390 smooth over $\mathrm{Spec}(\Lambda)$.

BIBLIOGRAPHY

[1] Grothendieck (Alexander). — *On some points of homological algebra*, Tohoku Math. J., vol. 9, 1957, pp. 119–221.

[2] Grothendieck (Alexander). — *Formal geometry and algebraic geometry*, Bourbaki Seminar, vol. 11, 1958/59, fasc. 3, no. 182.

[3] Grothendieck (Alexander). — *Descent technique and existence theorems in algebraic geometry, I: Generalities. Descent along faithfully flat morphisms*, Bourbaki Seminar, vol. 12, 1959/60, fasc. 1, no. 190.

[4] Serre (Jean-Pierre). — *Local fields and isogenies*, Bourbaki Seminar, vol. 11, 1958/59, fasc. 3, no. 185.

Erratum to Exposé no. 195

300 **4. TDTE II: The existence theorem in formal moduli theory.**

[Bourbaki Seminar, vol. 12, 1959/60, no. 195, 22 pp.]

Page 8. — The formula written in Proposition 5.1 is correct only when Λ is a field. In general, $\mathfrak{m}_\xi/\mathfrak{m}_\xi^2$ must be replaced by the quotient of that space by the image of $\mathfrak{n}_\xi/\mathfrak{n}_\xi^2$, where $\mathfrak{n}$ is the maximal ideal of Λ. Moreover, the definition given for O smooth over Λ is correct only when the residue-field extension k'/k is separable. In the general case, cf. SGA III, 1.1.

Page 14, remark. — The problems raised here are completely solved in the projective case by "Hilbert schemes" (cf. TDTE IV). Examples of Nagata and Hironaka show, however, that the functors under consideration need not be representable without a projectivity hypothesis, even if one restricts to classifying zero-dimensional subvarieties of a nonsingular complete threefold. This is
301 connected with the fact that the symmetric square of such a variety may fail to exist.

Page 15, no. 3. — For a fuller study, see TDTE V.

Page 16, Proposition 3.1. — Read "if f is proper and separable" instead of "if f is proper."

Pages 18 and 19, Remarks, 2. — I have recently proved that the formal moduli scheme of an abelian variety over a field is indeed smooth over the Witt ring; in other words, every abelian scheme over an Artinian local ring that is a quotient of another ring is obtained by reduction from an abelian scheme over the latter. The proof uses only the variance properties of the obstruction class for lifting introduced in Exposé no. 182, page 12. Let us also recall that the moduli schemes for curves of genus g and for polarized abelian schemes have been constructed by Mumford (cf. SHMT).

Page 19, paragraph 5. — The section used to define U must be assumed to be a non-zero-divisor not only on $\mathfrak{X}$, but on every X_n.

Page 20, lines 12 and 13. — Instead of

$$G_x = G\big(\mathrm{Spec}(\widehat{\mathcal{O}}_{\mathfrak{X},x})\big),$$

read

$$G_x = G\big(\mathrm{Spec}(\mathcal{O}_{\mathfrak{X},x})\big) \subset G\big(\mathrm{Spec}(\widehat{\mathcal{O}}_{\mathfrak{X},x})\big),$$

and instead of "$\mathcal{O}'_{0x}$," read "$\widehat{\mathcal{O}}'_{0x}$."

Chapter 5

Exposé 212 – Quotient preschemes

Séminaire BOURBAKI
13th year, 1960/61, no. 212
February 1961

99

CONSTRUCTION TECHNIQUES AND EXISTENCE THEOREMS IN ALGEBRAIC GEOMETRY
III. QUOTIENT PRESCHEMES

by Alexander GROTHENDIECK

Introduction

The problems treated in the present exposé differ from those considered in the preceding two in that one seeks to represent certain covariant, rather than contravariant, functors of variable schemes. The procedure of passing to a quotient is nevertheless essential in many construction problems in algebraic geometry, including those of Exposés I and II ([1], [2]). Thus the question of the *effectivity of descent data* on a T-prescheme X, relative to a faithfully flat and quasi-compact morphism $T \longrightarrow S$, is equivalent to the question of the existence of a quotient of X (satisfying the reasonable properties examined below) by the flat equivalence relation in X defined by the descent data; the questions raised in [1], A, § 2 c will no doubt be resolved at the same time as the questions posed in paragraph 2 of the present exposé. Likewise, the *Picard scheme* of an S-scheme X (for the definition, see [2], C, § 3) may be defined in various ways as a quotient of certain other schemes (of positive divisors, or of immersions into a projective scheme) by flat equivalence relations, while the definition and construction of these auxiliary schemes is technically simpler: they are schemes of the form $\underline{\mathrm{Hom}}_S(X, Y)$ and variants defined in [2], C, § 2, whose construction will be the subject of the next exposé (under suitable projectivity hypotheses). Combining the results of the present exposé and the next therefore yields the construction of Picard schemes under suitable hypotheses.

The problem of passing to a quotient in the category of preschemes still presents several unresolved questions. The most important is mentioned in paragraph 8. It is at present the sole obstacle to the construction of *moduli schemes over the integers for curves of arbitrary genus, polarized abelian varieties*, etc. This shows that its solution merits the efforts of specialists in algebraic groups.

100

1. Equivalence relations and effective equivalence relations

Let $\mathcal{C}$ be a category and let X be an object of $\mathcal{C}$. A pair of morphisms

$$p_1, p_2 : R \rightrightarrows X$$

is called an *equivalence pair* in $\mathcal{C}$, with *target* X and *source* R, if for every object T of $\mathcal{C}$ the two corresponding maps

$$p_1(T), p_2(T) : R(T) \rightrightarrows X(T)$$

(where, for every object Y of $\mathcal{C}$, we put $Y(T) = \mathrm{Hom}(T, Y)$) define a map

$$R(T) \longrightarrow X(T) \times X(T)$$

that induces a bijection from $R(T)$ onto the graph of an equivalence relation on the set $X(T)$. There is an evident equivalence relation among the equivalence pairs with target X; an equivalence class for this latter relation is called a *$\mathcal{C}$-equivalence relation on* X, or simply an equivalence relation when no confusion can result.

If $X \times X$ exists, giving an equivalence relation on X is equivalent to giving a subobject R of $X \times X$ such that, for every object T of $\mathcal{C}$, the subset of

$$(X \times X)(T) = X(T) \times X(T)$$

corresponding to $R(T)$ is the graph of an equivalence relation on $X(T)$. Writing p_1 and p_2 for the morphisms from R to X induced by the projections pr_1 and pr_2, the preceding condition says that (p_1, p_2) is an equivalence pair. One may also express diagrammatically in $\mathcal{C}$ (provided that $X \times X$ and the fiber product $(R, p_2) \times_X (R, p_1)$ exist) the set-theoretic axioms of an equivalence relation for the $R(T)$ in the $X(T)$, in accordance with the general principle explained in [2], A, § 1. We shall not need this.

Whenever there is a pair of morphisms (p_1, p_2) with the same source R and the same target X, one may define the *cokernel* of the pair as the object Y of $\mathcal{C}$ that represents the functor covariant in Z

$$\mathrm{Hom}_{p_1,p_2}(X, Z),$$

the set of morphisms u from X to Z such that $up_1 = up_2$. If Y exists, it is determined up to a unique isomorphism. We denote it by $X/(p_1, p_2)$ or, by abuse of notation, by X/R, the latter
101 notation being used chiefly when (p_1, p_2) is an equivalence pair: it is then customary to identify in the notation the equivalence relation defined by the pair with R. Notice that if Y is regarded as a quotient of X, it in fact depends only on the equivalence relation defined by the pair (p_1, p_2).

Now start with a morphism

$$f : X \longrightarrow Y,$$

which allows us to regard X as an "object over Y", and suppose that the fiber product

$$R(f) = X \times_Y X$$

exists. Let p_1 and p_2 be its two projections. Then (p_1, p_2) is an equivalence pair, called the pair *associated* with the morphism f. It therefore defines an equivalence relation, called the relation associated with f.

A pair of morphisms (p_1, p_2) with target X and common source R is called an *effective equivalence pair* if

(i) the cokernel $X/(p_1, p_2) = Y$ exists;

(ii) the fiber product $X \times_Y X$ exists;

(iii) the morphism $R \longrightarrow X \times_Y X$ with components p_1 and p_2 is an isomorphism.

Then (p_1, p_2) is indeed an equivalence pair. The equivalence relation it defines is also called an *effective equivalence relation*.

A morphism $f : X \longrightarrow Y$ is called an *effective epimorphism* if

(i) the fiber product $X \times_Y X = R$ exists;

(ii) the quotient $X/(p_1, p_2)$ exists, where p_1 and p_2 are the two projections from R to X;

(iii) the morphism $X/(p_1, p_2) \longrightarrow Y$ induced by f is an isomorphism.

Then f is indeed an epimorphism, and even a strict epimorphism (see [1], A, § 2.3); the converse holds if the fiber product $X \times_Y X$ exists. The quotient object of X defined by the epimorphism f is also called an *effective quotient* of X.

The preceding definitions imply the following "Galois correspondence": 102

PROPOSITION 1.1. — There is a one-to-one correspondence, preserving the natural orders, between the set of effective equivalence relations R on X and the set of effective quotients Y of X: to R corresponds the effective quotient X/R, and to Y corresponds the effective equivalence relation defined by the canonical projection $X \longrightarrow Y$ (which is defined by the fiber product $X \times_Y X$ equipped with its two projections).

In very good categories (sets, sheaves of sets, etc.), every quotient is effective and every equivalence relation is effective. This is no longer so in categories such as the category of preschemes over a fixed prescheme S, even when S is the spectrum of a field, and even when one restricts to schemes finite over S. Questions of effectivity, and even (for preschemes not finite over S) questions of the existence of quotients, most often prove delicate.

2. Example: preschemes finite over S

Let $\mathcal{C}$ be the category of preschemes finite over S, where S is assumed locally Noetherian. It is equivalent to the opposite of the category of coherent sheaves of commutative algebras on S, or, when S is affine with ring A, to the opposite of the category of A-algebras finite over A (i.e. finitely generated as A-modules). It follows at once that finite projective limits and finite inductive limits exist in $\mathcal{C}$. This is well known (without any finiteness hypothesis...) for the former. Thus the fiber product of preschemes X, Y over S corresponds to the tensor product $B \otimes_A C$ of the corresponding algebras; the kernel of two morphisms $X \rightrightarrows Y$, defined by two homomorphisms of A-algebras $u, v : C \rightrightarrows B$, corresponds to the quotient of B by the ideal generated by the elements $u(c) - v(c)$, etc. For finite inductive limits it is enough to consider, on the one hand, finite sums, which correspond to the ordinary product of A-algebras, and, on the other hand, cokernels of pairs of morphisms $X \rightrightarrows Y$, which correspond (as one sees immediately) to the subring of C consisting of the elements on which the homomorphisms $u, v : C \rightrightarrows B$ coincide (this subring is finite over A by the Noetherian hypothesis). One may moreover show, using the Noetherian hypothesis, that the finite inductive limits, and in particular the quotients, so constructed in the category $\mathcal{C}$ of preschemes finite over S are in fact quotients in the category of *all* preschemes. 103

As we said in [1], there are non-effective epimorphisms in $\mathcal{C}$ (or non-strict epimorphisms[1]; this is the same thing since fiber products exist). *I do not know whether equivalence relations are always effective* when no flatness hypothesis is imposed. In this direction I have obtained only very partial positive results, indispensable for the proof of the fundamental theorem in the formal theory of modules (see [2], B, th. 1). Let us point out that in the problem posed it is easy to reduce to the case where S is the spectrum of a local Artinian ring A with algebraically closed residue field. But even if A is a field, the answer is not known.

One may also consider a prescheme X over S that is no longer assumed finite over S, but with an equivalence relation R on X such that $p_1 : R \longrightarrow X$ is a finite morphism. Then R is called a finite equivalence relation. Suppose for simplicity that S and X are affine (and hence R is affine, so

[1] The printed text omits "non" before "strict". In the category under consideration fiber products exist, and the preceding paragraph establishes precisely the equivalence between effectivity and strict epimorphy.

that the situation reduces to one of pure commutative algebra). *Even in this case it is not known whether a quotient $X/R = Y$ exists and whether the canonical morphism $X \longrightarrow Y$ is finite*. (The simplest case is that in which S is the spectrum of a field k and X is the spectrum of $k[t]$, i.e. the affine line.) Of course, if the two preceding problems have affirmative solutions, one may conclude in the present situation that R is effective. Notice that the problem of the *existence* of a quotient Y and the finiteness of $f : X \longrightarrow Y$ arises in exactly the same terms if, instead of the graph of an equivalence relation on X, one has only an equivalence pregraph on X in the sense of paragraph 4.

The question of passing to the quotient by a more or less arbitrary finite equivalence relation arises in constructing preschemes by "gluing" given preschemes X_i along certain closed subpreschemes; the gluing law is expressed precisely by a finite equivalence relation on the sum prescheme X of the X_i. One should also expect the solution of the problems posed here, and of various variants, to be a preliminary condition for developing a general technique of nonprojective constructions in the direction initiated in [2].

The only general positive fact known to the author is the following:

PROPOSITION 2.1. — Let S be a locally Noetherian prescheme, let s be a point of S, and let Ω
104 be an algebraically closed extension of $k(s)$. Consider the corresponding "fiber functor" F, which assigns to every S-scheme X finite over S the set of Ω-valued points of X_s. This functor (trivially left exact) is *right exact*, i.e. it commutes with finite inductive limits, and in particular with cokernels of pairs of morphisms.

Using this result for all the "geometric points" of S, one deduces that the "quotient" category $\mathcal{C}'$ of $\mathcal{C}$, obtained by reasoning "modulo surjective radicial morphisms" (i.e. by formally adjoining inverses to the morphisms in question), is a "geometric" category, i.e. it satisfies the same properties "of finite nature" as the category of sets. In particular, every equivalence relation in it is effective. This implies that if R is an equivalence relation on an X finite over S, then the canonical morphism

$$R \longrightarrow X \times_Y X \qquad (\text{where } Y = X/R)$$

is *radicial and surjective* (in fact a surjective closed immersion, since it is a monomorphism).

3. The case of an operator group

Assume again that $\mathcal{C}$ is an arbitrary category. Let G, X be objects of $\mathcal{C}$, and suppose that G is a $\mathcal{C}$-operator group on the object X of $\mathcal{C}$. This means (see [2], A, § 1) that for every object T of $\mathcal{C}$ a group structure has been given on $G(T)$ and a $G(T)$-set structure on $X(T)$, in such a way that as T varies the structures in question "vary functorially" with T. If the products $G \times G$ and $G \times X$ exist in $\mathcal{C}$, such a structure may also be defined as a pair of morphisms

$$G \times G \longrightarrow G, \qquad \pi : G \times X \longrightarrow X,$$

subject to the condition that for every object T of $\mathcal{C}$, the corresponding laws of composition on the sets $G(T)$ and $X(T)$ make $G(T)$ a group acting on $X(T)$. Translating this axiom into the commutativity of certain diagrams in $\mathcal{C}$ is easy but tedious, and in fact completely useless in every case known to me.

Suppose that $G \times X$ exists, and consider the two morphisms

$$p_1, p_2 : G \times X \rightrightarrows X$$

105 with

$$p_1 = \mathrm{pr}_1, \qquad p_2 = \pi.$$

One sees immediately that (p_1, p_2) is an equivalence pair if and only if, for every object T of $\mathcal{C}$, the map

$$G(T) \times X(T) \simeq (G \times X)(T) \longrightarrow X(T) \times X(T),$$

defined by this pair is injective, i.e. if the group $G(T)$ acts *freely* on the set $X(T)$, that is, if $g \in G(T)$, $x \in X(T)$, and $g \cdot x = x$ imply that g is the identity element of $G(T)$. One then says that G *acts freely on* X (or that X is a $\mathcal{C}$*-principal space under* G). The equivalence relation associated with the pair (p_1, p_2) is then called the *equivalence relation defined by the group* G *acting freely on* X. If $X \times X$ also exists and one considers the morphism

$$p : G \times X \longrightarrow X \times X$$

defined by the pair (p_1, p_2), the condition that G act freely means that p is a *monomorphism*.

Of course, even when G does not act freely on X, one wants criteria for the existence of a quotient of X by G, i.e. of the cokernel of the preceding pair (p_1, p_2).

The cokernel in question will often be denoted by X/G, or preferably by $G\backslash X$ if G acts on the left (the former notation being reserved for the case in which G acts on the right). Notice that even when the "image" of $G \times X$ under p exists (this image being defined, for example, as the smallest subobject of $X \times X$ through which p factors), say R, it is most often not an equivalence relation on X. If one then tries to pass directly to the quotient by R (or more precisely by the pair of morphisms from R to X induced by the two projections pr_i), one irretrievably loses the special features of the original pair (p_1, p_2). It is therefore important to find a generalization of the notion of equivalence relation that applies directly to the pair defined by a $\mathcal{C}$-operator group.

4. Pre-equivalence relations

Recall that a category in which every morphism is an isomorphism is called a *groupoid*. On the other hand, a category must be defined as consisting of two underlying sets (X, R), the set of *objects* and the set of *arrows*, equipped with the following structures: 106

(i) A pair of maps

$$p_1, p_2 : R \rightrightarrows X$$

called the *source map* and the *target map*,

(ii) A map

$$\pi : (R, p_2) \times_X (R, p_1) \longrightarrow R$$

called the *composition map*.

These data must satisfy familiar axioms, which we shall not repeat here and which could be expressed by the commutativity of certain diagrams, together with the existence of a map $D : X \to R$ (necessarily unique) making two other diagrams commute. The map D sends an object to its corresponding identity arrow and satisfies

$$p_1 D = p_2 D = \mathrm{id}_X .$$

To say that the category is a groupoid then means that there exists a map

$$s : R \longrightarrow R$$

(necessarily unique), called the *inversion* of R, which transforms each arrow into an inverse arrow. This could be expressed by the commutativity of four further diagrams formed from s, Δ, and the preceding data; the first two are

$$p_1 s = p_2, qquad p_2 s = p_1.$$

With these points recalled, the general definitions in ([2], A, § 1) show, in particular, what is meant by a structure of a $\mathcal{C}$*-category*, respectively a $\mathcal{C}$*-groupoid*, on a pair of objects X, R of an

arbitrary category $\mathcal{C}$: by definition, it is the datum, for every object T of $\mathcal{C}$, of a category structure, respectively a groupoid structure, in the set-theoretic sense whose set of objects is $X(T)$ and whose set of arrows is $R(T)$, these structures "varying functorially" with the variable T. This therefore gives two morphisms

$$p_1, p_2 : R \rightrightarrows X$$

called the *source morphism* and the *target morphism*, and, when the required fiber product exists, a mor-
107 phism

$$\pi : (R, p_2) \times_X (R, p_1) \longrightarrow R$$

called the *composition morphism*. These three data then suffice to determine the category structure on X, R, the axiom imposed on them being as follows: for every T, the three corresponding data for $X(T), R(T)$ define a category structure (respectively a groupoid structure) on this pair of sets. When appropriate, this may be expressed by the commutativity of certain diagrams involving a uniquely determined morphism

$$D : X \longrightarrow R$$

and, in the case of groupoids, a uniquely determined morphism

$$s : R \longrightarrow R;$$

the diagrams are written out just as in the "set-theoretic" case. This laborious interpretation of the axioms is fortunately useless in practice. The sole theoretical interest of being able to express the data and axioms in terms of morphisms and equalities of morphisms between certain fiber products is the following: if $F : \mathcal{C} \to \mathcal{C}'$ is a left exact functor (i.e. one commuting with finite products and fiber products), then it transforms a $\mathcal{C}$-category (respectively a $\mathcal{C}$-groupoid) into a $\mathcal{C}'$-category (respectively a $\mathcal{C}'$-groupoid), provided finite products and fiber products exist in $\mathcal{C}$.

In practice, it is important to know how to interpret the morphisms p_1, p_2, π, D, s as *simplicial operations* in a suitable semisimplicial or simplicial object of $\mathcal{C}$, at least when fiber products exist in $\mathcal{C}$. To fix terminology, let us introduce the *category $\mathcal{S}$ of standard simplices* as the category whose objects are the finite sets of the form

$$\Delta_n = [0, n], qquad \text{where } n \geq 0$$

(the interval of integers from 0 to n), and whose morphisms or arrows are *arbitrary maps* between these finite sets. Notice that the category $\mathcal{S}$ is equivalent to the category of *nonempty* finite sets, with maps of finite sets as morphisms. In $\mathcal{S}$, the coproduct of a finite *nonempty* family of objects clearly exists, as does the pushout of two objects under a third (the operation dual to the fiber product). Let $\mathcal{S}'$ denote the subcategory of $\mathcal{S}$ having the same objects but whose morphisms are the *increasing maps* between the Δ_n. This category is equivalent to the category of nonempty totally ordered finite
108 sets. In this category the coproduct of two objects never exists, and the pushout of two objects A, B under a third C does not exist in general (take, for example, $C = \Delta_0$, $A = B = \Delta_1$, with the two structure maps $u : C \to A$ and $v : C \to B$ equal). It does exist in certain cases, however; for example,

$$C = \Delta_0, qquad A = \Delta_m, qquad B = \Delta_n, qquad u(0) = m, qquad v(0) = 0,$$

and in this case

$$\Delta_m \amalg_{\Delta_0} \Delta_n = \Delta_{m+n}.$$

A *simplicial object* (respectively a *semisimplicial object*) in a category $\mathcal{C}$ is, by definition, a contravariant functor K from $\mathcal{S}$ (respectively from $\mathcal{S}'$) to $\mathcal{C}$. Thus a simplicial object defines a semisimplicial object by restriction, but the first notion differs from the second essentially through the presence

of *symmetry operations* on the $K_n = K(\Delta_n)$, corresponding to the images under the functor K of the elements of the symmetric group on $n+1$ letters (regarded as the automorphism group of Δ_n in $\mathcal{S}$).

For every n, let Δ'_n (respectively Δ''_n) be the finite category whose object set is Δ_n, defined by the chaotic order relation (respectively the natural total order) on Δ_n (the set of arrows being the graph of that order relation). It is clear that Δ'_n (respectively Δ''_n) depends functorially on the object Δ_n of $\mathcal{S}$ (respectively $\mathcal{S}'$). If Z is a category, then $\mathrm{Hom}(\Delta'_n, Z)$ (respectively $\mathrm{Hom}(\Delta''_n, Z)$), as Δ_n varies, is a functor from $\mathcal{S}$ (respectively $\mathcal{S}'$) to the category of sets, i.e. a simplicial set (respectively a semisimplicial set), said to be *associated with the category* Z; denote them by Z' and Z''. There is moreover an evident natural homomorphism from the semisimplicial set associated with Z' to Z'', and the latter is an isomorphism if and only if Z is a groupoid. With this understood:

PROPOSITION 4.1. — The functor $Z \longmapsto Z''$ from the category of categories to the category of semisimplicial sets is fully faithful, and defines an equivalence between the category of *categories* and the category of semisimplicial sets, i.e. contravariant functors K from $\mathcal{S}'$ to (Set), which transform pushouts $A \amalg_C B$ of the type specified above into fiber products of sets. Likewise, the functor $Z \longmapsto Z'$ from the category of groupoids to the category of simplicial sets is fully faithful, and defines an equivalence between the category of *groupoids* and the category of simplicial sets, i.e. 109
contravariant functors K from $\mathcal{S}$ to (Set), which transform pushouts into fiber products.

Thus categories may be regarded as particular semisimplicial sets, and groupoids as particular simplicial sets, provided of course that one reasons "up to isomorphism," as is required when interpreting certain structures in terms of others. The usual procedure of reduction to the set-theoretic case then implies:

COROLLARY 4.2. — The preceding statement remains valid when categories, groupoids, and simplicial sets are replaced by $\mathcal{C}$-categories, $\mathcal{C}$-groupoids, and simplicial objects in $\mathcal{C}$, provided that fiber products exist in $\mathcal{C}$.

The semisimplicial object K in $\mathcal{C}$ associated with a category $(X, R, ...)$ in $\mathcal{C}$ may be made explicit by interpreting the component $K_n = K(\Delta_n)$ of K as the n-fold fiber power of (R, p_1) over X[2], or better, by the recurrence formula

$$K_0 = X, qquadK_n = (K_{n-1}, p_n^{(n-1)}) \times_X (R, p_1),$$

where the $p_i^{(n-1)}$ $(0 \leq i \leq n-1)$ are the natural projections from K_{n-1} to X (which are themselves also defined recursively). In this way, p_1, p_2, π, D, s are interpreted as the simplicial operations corresponding respectively to the following morphisms in $\mathcal{S}$: face 0 of Δ_1, face 1 of Δ_1, face $(0,2)$ of Δ_2, the degeneracy $\Delta_1 \to \Delta_0$, and the symmetry of Δ_1. All other semisimplicial (respectively simplicial) operations are obtained formally from the preceding four (respectively five) by composition and fiber products.

A *pre-equivalence relation* on an object X of a category is now defined to be the datum of a groupoid whose object of objects (so to speak) is X. Such a datum therefore includes, among other things, an object R and two morphisms

$$p_1, p_2 : R \rightrightarrows X.$$

But notice that these data alone do not determine the structure in question, in contrast to what happens for equivalence pairs. In this exposé, we are interested in this notion in order to obtain criteria for the possibility of passing to the quotient, i.e. for forming the cokernel of the pair (p_1, p_2). Thus the statement of the problem does not invoke the additional data inherent in a groupoid. In proving the results that follow, we shall nevertheless use these additional data, and in particular 110

[2]FGA-CANON-ERR-0001. The printed text says "$(n+1)$-fold fiber power" and, in the following formula, $K_0 = R$. The object component of the nerve is $K_0 = X$ and its arrow component is $K_1 = R$; the two index corrections are therefore linked.

the simplicial operations (including symmetry operations) through dimension 3 (involving the fourfold fiber power of R over X).

An equivalence relation on an object X of $\mathcal{C}$ defines a pre-equivalence relation: it suffices to see this in the set-theoretic case, where one associates with an equivalence relation on a set X the groupoid whose set of objects is X and whose set of arrows is the graph of the equivalence relation.

A $\mathcal{C}$-monoid G acting on an object X of $\mathcal{C}$ defines a $\mathcal{C}$-category whose underlying objects are $R = G \times X$ and X (provided that $G \times X$ exists), and this is a $\mathcal{C}$-groupoid if and only if G is a group. Again, it suffices to verify this in the set-theoretic case. The composition of the arrows (g, a) and $(g', g \cdot a)$ is then defined by

$$(g', g \cdot a)(g, a) = (g'g, a),$$

i.e. if $a, b \in X$, then $\mathrm{Hom}(a, b)$ is by definition the transporter from a to b, and the morphisms compose by means of composition of the elements of G.

REMARK. — The logical difficulties raised by a statement such as (4.1) can be avoided by understanding that all the objects under consideration lie in a fixed "Universe" (which is itself a set).

5. Quotient by a finite flat equivalence relation

THEOREM 5.1. — Let $X = \mathrm{Spec}(B)$ be an affine scheme and R a pre-equivalence relation on X whose component R_1 is affine, say $R_1 = \mathrm{Spec}(C)$. Suppose that the first projection $p_1 : R_1 \to X$ is finite and locally free; in other words, suppose that the corresponding ring homomorphism $p_1^* : B \to C$ makes C a finitely generated projective B-module. Let A be the subring of B that is the kernel of the pair of homomorphisms

$$p_1^*, p_2^* : B \rightrightarrows C$$

(the set of elements b such that $p_1^*(b) = p_2^*(b)$). Let $Y = \mathrm{Spec}(A)$, and let $f : X \to Y$ be the morphism defined by the inclusion of A into B. Under these conditions:

(i) B is integral over A, i.e. f is an integral morphism.

111 (ii) The morphism f is surjective, and its fibers are the equivalence classes $p_2(p_1^{-1}(x))$ in the set X modulo R; moreover, the topology of Y is the quotient of that of X.

(iii) Y is the quotient of X by R in the category of preschemes.

(iv) If R comes from an *equivalence relation*, then the morphism $f : X \to Y$ is finite locally free (i.e. B is a finitely generated projective A-module), and the equivalence relation is effective, i.e. $R_1 \to X \times_Y X$ is an isomorphism.

This theorem generalizes the familiar theorem concerning a finite group G acting by automorphisms on the ring B and the invariant ring A, and its proof is analogous to the familiar proof. Statement (iii) can be sharpened as follows:

COROLLARY 5.2. — The canonical morphism $R_1 \to X \times_Y X$ is *surjective*.

Let R still be a "finite and locally free" pre-equivalence relation on X, but now let X be an arbitrary prescheme. Suppose that one can find a prescheme Y and a morphism $f : X \to Y$ such that $fp_1 = fp_2$ and, moreover, such that the sequence of homomorphisms of sheaves of rings on Y

$$\mathcal{O}_Y \longrightarrow f_*(\mathcal{O}_X) \rightrightarrows g_*(\mathcal{O}_R)$$

is exact (where $g = fp_i$). The theorem then gives conclusions analogous to (i)–(iv) above; in particular, by (iii), Y is the quotient of X by R and is therefore determined up to unique isomorphism. Under these conditions the pre-equivalence relation R on X is said to be *admissible*. With this definition:

THEOREM 5.3. — Let X be a prescheme and R a pre-equivalence relation on X such that $p_1 : R_1 \to X$ is finite and locally free. Then R is admissible if and only if every set-theoretic equivalence class $p_2(p_1^{-1}(x))$ in X modulo R is contained in an affine open subset (a condition that is always satisfied if every finite subset of X is contained in an affine open subset, for example if X is quasi-projective over an affine scheme).

Indeed, one shows without difficulty that every equivalence class modulo R in X is then contained in an affine open subset *stable under* R, and one constructs the quotient Y by gluing the pieces obtained from Theorem 5.1. 112

COROLLARY 5.4. — Suppose that this condition is satisfied and that, in addition, R comes from an equivalence relation. Then that relation is effective, i.e. $R_1 \to X \times_Y X$ is an isomorphism, and $f : X \to Y$ is finite and locally free.

One immediately concludes by "descent":

COROLLARY 5.5. — Under the hypotheses of (5.4), X has rank n everywhere over Y if and only if (R_1, p_1) has rank n everywhere over X. If X, R_1 are Z-preschemes and p_1, p_2 are Z-morphisms, so that Y is a Z-prescheme, then X is flat over Z if and only if Y is flat over Z.

In summary:

SCHOLIUM. — The datum of a finite, locally free, and surjective morphism $f : X \to Y$ of preschemes is equivalent to the datum of a prescheme X equipped with an equivalence relation R such that $p_1 : R \to X$ is finite and locally free and every class $p_2(p_1^{-1}(x))$ is contained in an affine open subset.

REMARKS 5.6.

1° No noetherian hypothesis was needed.

2° This notion of passage to the quotient includes, as a special case, CARTIER's "inseparable descent," which amounts to determining the finite locally free morphisms $f : X \to Y$ such that $f_*(\mathcal{O}_X)$ admits a p-basis relative to $\mathcal{O}_Y$ (where X is a given prescheme whose sheaf $\mathcal{O}_X$ is annihilated by the prime number $p > 0$). Notice that the result is readily expressed without any regularity hypothesis on the local rings and without assuming that X is an algebraic scheme over a field. The JACOBSON–BOURBAKI theory is obtained by taking X to be the spectrum of a field of characteristic p.

3° GABRIEL had previously obtained a special case of Theorem (5.3) in the theory of passage to the quotient for finite commutative groups over a field k. [Compare (7.4).]

6. Quotient by a proper flat equivalence relation

THEOREM 6.1. — Let S be a locally noetherian prescheme, X a quasi-projective S-scheme, and R a pre-equivalence relation on X such that: 113

(a) $p_1 : R_1 \to X$ is proper and flat;

(b) $R_1 \to X \times_S X$ is finite, or equivalently, by (a), has finite fibers (a condition automatically satisfied if R comes from an equivalence relation).

Under these conditions:

(i) $X/R = Y$ exists and, if S is noetherian and R comes from an equivalence relation, is quasi-projective over S.[3]

(ii) The canonical morphism $f : X \to Y$ is surjective, proper, and open; its fibers are the equivalence classes $p_2(p_1^{-1}(x))$ in X modulo R. Thus Y is identified with the quotient topological space of X by the set-theoretic equivalence relation defined by R. Finally, $R_1 \to X \times_Y X$ is surjective.

(iii) If R comes from an equivalence relation, then that relation is effective, i.e. $R_1 \to X \times_Y X$ is an isomorphism, and moreover $f : X \to Y$ is flat (hence faithfully flat).

For the proof, one reduces to (5.1) by means of suitable quasi-sections of X for R, the proof being analogous to the construction of quotient algebraic groups in the Chevalley Seminar.

In summary:

SCHOLIUM. — Let X be quasi-projective over a locally noetherian S. The datum of a proper, faithfully flat, and surjective morphism $f : X \to Y$ from X to an S-prescheme Y is equivalent to the datum of an equivalence relation R on X such that $p_1 : R \to X$ is proper and flat.

The same method gives the following result:

THEOREM 6.2. — Let S be a noetherian prescheme, X a prescheme of finite type over S, and R a pre-equivalence relation on the S-prescheme X. Suppose that:

(a) $p_1 : R_1 \to X$ is flat and of finite type;

(b) the morphism $R_1 \to X \times_S X$ is quasi-finite (i.e. has finite fibers).

Then there exists a dense open subset U of X that is *saturated* for R, such that:

(i) If R_U is the pre-equivalence relation induced by R on U, then U/R_U exists and is of finite type over S.

(ii) The canonical morphism $U \to U/R_U$ is surjective and open; its fibers are the set-theoretic equivalence classes for R_U (thus U/R_U is the quotient topological space of U by the set-theoretic equivalence relation defined by R_U). Finally, the morphism

$$(R_U)_1 \longrightarrow U \times_{U/R_U} U$$

114 is surjective.

(iii) If R comes from an equivalence relation, one may arrange for $U \to U/R_U$ to be faithfully flat and for R_U to be effective.

This is a result of an essentially "birational" nature.

REMARKS 6.3.

1° I do not know whether hypothesis (b) is unnecessary in (6.1) and (6.2). In taking quotients by groups, it forces us in practice to restrict ourselves to the case in which all stabilizers are finite groups.

2° One may ask whether there are results analogous to (6.1) and (6.2) without a flatness hypothesis. I have no counterexample in this direction. On the other hand, even retaining the flatness hypothesis and restricting to equivalence relations for which $p_1 : R \to X$ is flat and quasi-finite (but not finite), with X affine, it can happen that R is not effective: take the equivalence relations induced on affine open subsets covering Nagata's variety (on which a group of order two acts in a "non-admissible" manner).

[3]The printed text does not contain the second hypothesis. The linked erratum to Exposé 212 says that quasi-projectivity then appears to have been proved only when R comes from an equivalence relation.

7. Applications

As we said in the introduction, the most important application of (6.1) is the construction of Picard schemes, as well as solutions to various other moduli problems, to which we shall return later.

One obtains a simple proof of the following result of SHIMURA:

PROPOSITION 7.1. — Let A be an abelian scheme defined over a discrete valuation ring V with fraction field K. Then every abelian scheme B' over K that is isogenous to a quotient of $A \otimes_V K$ "has good reduction over V," i.e. is isomorphic to some $B \otimes_V K$,[4] where B is an abelian scheme over V (essentially unique, as we recall).

We may suppose that B' is the quotient of A_K by a subgroup scheme C'. (N.B.: C' will not in general be "reduced," i.e. its local rings will have nilpotent elements.) Consider the closed subscheme C of A that is the "closure" of C', i.e. the smallest closed subscheme of A such that C_K contains C'. We have $C_K = C'$, and the fact that V is a discrete valuation ring readily implies that C is a subgroup scheme of A over V. Since A is proper over $\mathrm{Spec}(V) = S$, so is C; moreover, A is even projective over S. We may therefore apply (6.1) to construct $A/C = B$, which is the desired B. 115

Finally, essentially familiar arguments allow one to deduce from (6.2) the following result:

THEOREM 7.2. — Let S be the spectrum of an artinian ring, let F and G be group schemes of finite type over S, and let $u : F \to G$ be a homomorphism of group schemes over S. Suppose that:

(i) F is flat over S;

(ii) the kernel of u is finite.

Under these conditions, the quotient scheme G/F exists; the canonical morphism $G \to G/F$ is surjective and open, and its fibers are the set-theoretic equivalence classes defined by the right actions of F on G. Finally, if u is a monomorphism, then $G \to G/F$ is flat and the morphism

$$G \times F \longrightarrow G \times_{G/F} G$$

is an isomorphism. In other words, G is a principal homogeneous space over G/F with structure group (acting on the right) F, or rather $F \times_S (G/F)$ regarded as a group scheme over G/F (see [1], B, § 6).

COROLLARY 7.3. — Under these conditions, G is flat over S if and only if G/F is flat over S. If this condition holds, formation of the quotient by F commutes with every base extension of S; and if F is a normal subgroup of G, then G/F can be endowed with the structure of the *quotient group* of G by F.

The situation is particularly simple when S is the spectrum of a field, for then every S-prescheme is automatically flat over S. One obtains:

COROLLARY 7.4. — Let F, G be group schemes of finite type over a field k, and let $u : F \to G$ be a homomorphism of k-groups. Then u factors as $F \to F' \to G$, where $F \to F'$ is the quotient homomorphism by the closed subgroup $\mathrm{Ker}\, u$ of F, and $F' \to G$ is a group homomorphism that is a closed immersion. The quotient $G/F = G/F'$ exists. The usual formalism (of the kind provided by Noether's theorems) is valid among algebraic groups over k.

This result makes it possible to treat uniformly both passage to the quotient among algebraic groups in the classical sense (i.e. those which are irreducible over k and smooth over k) and passage to the quotient by "infinitesimal" subgroups considered by CARTIER. It is advantageous to 116

[4] The printed text has $B \otimes_V A$. Since B is a V-scheme and the assertion concerns its generic fiber, the base factor must be K.

regard the "hyperalgebras" introduced by CARTIER, following DIEUDONNÉ's work on formal groups, as groups in the category of formal schemes over k, and, when applicable (if they correspond to hyperalgebras of finite rank over k), as finite algebraic groups over k.

8. A conjecture

It answers the need to form quotients by the projective group acting on certain subschemes of "Hilbert schemes" (the latter replacing Chow varieties in scheme theory).

Let S be a prescheme and n an integer. To every prescheme S' over S, associate the group $\mathrm{Gl}(n, \Gamma(S', \mathcal{O}_{S'}))$ of invertible n-by-n matrices with entries in the ring of sections of $\mathcal{O}_{S'}$. This gives a contravariant functor in S', which is readily shown to be representable; it therefore corresponds to a group scheme over S, in fact affine over S, denoted $\mathrm{Gl}(n)_S$. Its formation is compatible with base change, so in fact everything comes from a group scheme over $\mathbf{Z}$, denoted $\mathrm{Gl}(n)$. The group $\mathrm{Gl}(1)$, called the multiplicative group and often denoted $\mathbf{G}_m$, corresponds to the functor $S \longmapsto \Gamma(S, \mathcal{O}_S)^*$, the group of "units" over S. There is an evident homomorphism $\mathrm{Gl}(1) \to \mathrm{Gl}(n)$, and one readily constructs the quotient group, denoted $\mathrm{GP}(n-1)$ and called the *projective group of degree* $n-1$ *over* $\mathbf{Z}$. It represents the functor associating to S the group of sections of the sheaf $\widetilde{\mathrm{Gl}(n)}_S/\widetilde{\mathrm{Gl}(1)}_S$, where $\widetilde{\mathrm{Gl}(n)}_S$ denotes the sheaf of germs of sections of $\mathrm{Gl}(n)_S$ over S. (Care must be taken: sections of $\mathrm{GP}(n-1)_S$ over S do not in general come from sections of $\mathrm{Gl}(n)_S$ over S!) Notice that one can prove that $\mathrm{GP}(n-1)$ also represents the functor $S \longmapsto \mathrm{Aut}_S(\mathbf{P}_S^{n-1})$, where $\mathbf{P}_S^{n-1}$ is standard projective space of relative dimension $n-1$ over S, at least for noetherian S. This is how it enters moduli theory.

Let S be a noetherian scheme, which may if desired be assumed affine, and let X be a quasi-projective S-prescheme equipped with an invertible sheaf $\mathcal{L}$ that is very ample relative to S. Suppose that the group $G = \mathrm{GP}(n)_S$ acts on X and at the same time on $\mathcal{L}$ (compatibly with its action
117 on X), and that it acts *freely* on X.[5]

CONJECTURE 8.1. — [6] Under the preceding conditions:

1° *The equivalence relation defined by* G *is effective, the quotient* $Y = X/G$ *is of finite type over* S*, and the canonical morphism* $f : X \to Y$ *is flat and surjective* (thus X becomes a principal homogeneous bundle over Y with group $G \times_S Y = \mathrm{GP}(n)_Y$).

2° Let $\mathcal{L}'$ be the invertible sheaf on Y deduced from $\mathcal{L}$ by "faithfully flat descent" along f (see [1], B, § th. 1). Then $\mathcal{L}'$ is *pre-ample on* Y *relative to* S, i.e. there exist an integer m and a quasi-finite morphism from Y to a suitable standard projective scheme $\mathbf{P}_S^N$ such that $(\mathcal{L}')^{\otimes m}$ is isomorphic to the inverse image of $\mathcal{O}_{\mathbf{P}_S^N}(1)$.

Notice that even if X is separated over S, it may happen that Y is not separated over S (a situation occurring in entirely nonpathological moduli problems). If 1° holds, then Y is separated if and only if the equivalence relation defined by G has a closed graph, i.e. if $G \times X \to X \times X$ has closed image (it is then a closed immersion). If Y is separated, then $\mathcal{L}'$ is pre-ample on Y relative to S if and only if it is ample, i.e. if a suitable tensor power defines a projective immersion. In the moduli problems mentioned in the introduction, one can show that the resulting equivalence relation does indeed have a closed graph.

REMARKS 8.1. — We have assumed $G = \mathrm{GP}(n)_S$ to fix ideas and because this is presently the most important case in practice. A reasonable hypothesis on G would instead seem to be that G is one of the "forms" over S of one of the Tohokû groups (whose construction over the integers was also carried out by CHEVALLEY). The only positive fact known to me in the direction of the

[5] The printed text says "on S." The action under consideration and the quotient condition concern X.

[6] The linked erratum to Exposé 212 states that this conjecture is definitively false. It also reports that the positive fact in Remark 8.1 was subsequently proved and refers to Mumford's theorem and later refinements.

preceding conjecture is the following: let X be *an affine scheme over a field k of characteristic 0, on which the group $\mathrm{Gl}(n)_k$ or $\mathrm{GP}(n-1)_k$ acts freely. Then the equivalence relation defined by G is effective, the quotient X/G is also affine, and the morphism $X \to X/G$ is flat and surjective*. The proof uses the following fact (which at present has been proved only in characteristic 0): when G is made to act on the affine coordinate ring A of G, regarded as a vector space over k, the trivial representation of G occurs only once (in a composition series of a finite-dimensional G-stable vector subspace over k). It seems possible that a systematic use of the theory of linear representations of G will eventually provide a proof of the 118
conjecture, at least over a base field. When the base is no longer a field, the speaker knows nothing.

BIBLIOGRAPHY

[1] GROTHENDIECK (Alexander). — Techniques de descente et théorèmes d'existence en géométrie algébrique, I: Généralités, Séminaire Bourbaki, 1959/60, no. 190, 29 pp.

[2] GROTHENDIECK (Alexander). — Techniques de descente et théorèmes d'existence en géométrie algébrique, II: Le théorème d'existence en théorie formelle des modules, Séminaire Bourbaki, 1959/60, no. 195, 22 pp.

[3] GROTHENDIECK (A.) and DIEUDONNÉ (J.). — Éléments de géométrie algébrique, I: Le langage des schémas. — Paris, Presses universitaires de France, 1960 (Institut des hautes Études scientifiques, Publications mathématiques, 4).

Erratum to Exposé no. 212

301 **5. TDTE III: Quotient preschemes.**

[Bourbaki Seminar, vol. 13, 1960/61, no. 212, 20 pp.]

Page 15, assertion (i). — The fact that $Y = X/R$ is quasi-projective over S appears, for the present, to have been proved only when R arises from an equivalence relation.

Pages 18 and 19. — As we point out at the end of the following exposé, the conjecture under consideration is definitely false. The "positive fact" noted in Remark 8.1 appears to have been proved simultaneously by various authors (Nagata, Rosenlicht, Grothendieck, ...).

Let us point out that the application in TDTE V of the theory of passage to the quotient developed here to the construction of Picard schemes may also be replaced by an appropriate use of Hilbert schemes (cf. SHMT). As was stated in § 8, the most important gap in the theory presented
302 here is the lack of a criterion for the existence of quotients by a nonproper equivalence relation, such as equivalence relations arising from certain actions of the projective group. An important theorem in this direction has been obtained by Mumford[3]. For a refinement of his result and various applications to the theory, see SHMT.

[3]Mumford (David). — An elementary theorem in geometric invariant theory, *Bull. Amer. Math. Soc.*, vol. 67, 1961, pp. 483–486.

Chapter 6

Exposé 221 – Hilbert schemes

Séminaire BOURBAKI
13th year, 1960/61, no. 221
May 1961

CONSTRUCTION TECHNIQUES AND EXISTENCE THEOREMS IN ALGEBRAIC GEOMETRY
IV. HILBERT SCHEMES

by Alexander GROTHENDIECK

Introduction

The techniques presented in [2], I and II, were for the most part independent of any projectivity hypothesis on the schemes under consideration. Unfortunately, they do not at present make it possible to solve the existence problems posed in Exposé II of [2]. In the present exposé and the next, we shall solve these problems under projectivity hypotheses. The techniques employed are typically projective and make practically no use of the results of I and II of [2]. Here we shall construct the "Hilbert schemes" intended to replace the use of Chow coordinates, as stated in [2], II, no. 2. In the next exposé, the theory of passage to the quotient in schemes developed in III, together with the theory of Hilbert schemes, will enable us, for example, to construct Picard schemes (defined in [2], II, no. 3) under fairly general conditions.

In summary, one may say that a fairly satisfactory technique of projective constructions is now available, except that there is still lacking[1] a theorem on passage to the quotient by groups such as the projective group, acting "without fixed points" (cf. [2], III, no. 8). The situation even appears somewhat better in analytic geometry (when one restricts to studying analytic spaces "projective" over a given analytic space), since for analytic spaces the difficulty of passing to the quotient by a group acting nicely disappears. Moreover, in algebraic geometry as in analytic geometry, it remains to develop a construction technique valid without projectivity hypotheses.

1. Bounded families of sheaves: permanence properties

Let k be a field and X a k-prescheme, of finite type for simplicity. For every extension K/k one thus obtains a K-prescheme $X_K = X \otimes_k K$. If $\mathcal{F}$ is a coherent sheaf on X_K and K' is an extension

[1] See the addendum at the end of the exposé.

of K, then $\mathcal{F} \otimes_K K' = \mathcal{F}_{K'}$ is a quasi-coherent sheaf on $X_K \otimes_K K' = X_{K'}$. Now, if K and
250 K' are any two extensions of k, $\mathcal{F}$ is a quasi-coherent sheaf on X_K, and $\mathcal{F}'$ is a quasi-coherent sheaf on $X_{K'}$, we say that $\mathcal{F}$ and $\mathcal{F}'$ are *equivalent* if there exist k-homomorphisms from K and K' into a common extension K'' of k such that $\mathcal{F}_{K''}$ and $\mathcal{F}'_{K''}$ are isomorphic on $X_{K''}$. This is an equivalence relation, and we shall be interested in equivalence classes of sheaves under this relation and in sets of equivalence classes. Note that if X is of finite type over k,[2] every class of coherent sheaves can be defined by a coherent sheaf on some X_K, where K is an extension of finite type over k. In defining classes of coherent sheaves one may therefore restrict to algebraically closed extensions of k; one may also restrict to a fixed algebraically closed extension Ω of k of infinite transcendence degree. Two coherent sheaves $\mathcal{F}$ and $\mathcal{F}'$ on X_Ω are then equivalent if and only if there exists a k-automorphism σ of Ω such that $\mathcal{F} \otimes_\Omega (\Omega, \sigma)$ is isomorphic to $\mathcal{F}'$. There is thus a bijective correspondence between the classes of coherent sheaves under either definition.

Let E and E' be two sets of classes of coherent sheaves on X. Consider the classes of all sheaves of the form $\mathcal{F} \otimes \mathcal{F}'$, where $\mathcal{F}$ and $\mathcal{F}'$ are coherent sheaves on the same X_K, the class of $\mathcal{F}$ belonging to E and that of $\mathcal{F}'$ to E'. One thereby obtains a set of classes of coherent sheaves, denoted $E \otimes E'$. The sets $\mathrm{Tor}_i(E, E')$, etc., are defined in the same way. More generally, to every function U assigning to each sequence $\mathcal{F}_1, \dots, \mathcal{F}_n$ of n coherent sheaves on a common X_K a set $U(\mathcal{F}_1, \dots, \mathcal{F}_n)$ of coherent sheaves on X_K, and having the evident compatibility with sheaf isomorphisms and inverse images under base change, we associate a function, denoted by the same symbol U, which assigns to every sequence of n sets $E_1, \dots, E_n$ of classes of coherent sheaves a set $U(E_1, \dots, E_n)$ of classes of coherent sheaves.

Our purpose in this section is to define certain sets of classes of sheaves, called *bounded*, and to show that the most common operations U, when applied to bounded sets, again give bounded sets.

Let X be a prescheme of finite type over a noetherian S. For every $s \in S$, the fiber X_s is a prescheme of finite type over $k(s)$, and we shall consider classes of coherent sheaves on X_s in the preceding sense. This gives meaning to the phrase "class of coherent sheaves on a fiber of X/S," and to analogous phrases. Likewise, proceeding separately on each fiber, one may still consider
251 operations such as $E \otimes E'$, etc., which assign to systems of sets of classes of coherent sheaves on the fibers of X/S another set of classes of coherent sheaves on the fibers of X/S.

DEFINITION 1.1. — Let E be a set of classes of coherent sheaves on the fibers of X/S. We say that E is *bounded* if there exist a prescheme S' of finite type over S and a coherent sheaf $\mathcal{F}'$ on $X' = X \times_S S'$ such that E is contained in the set of classes of sheaves on the fibers of X/S defined by $\mathcal{F}'$.

By definition, the latter assigns to an $s \in S$ the classes of the sheaves $\mathcal{F}' \otimes_{S'} k(s')$, where s' ranges over the points of S' above s (so that $k(s')$ is an extension of $k(s)$ and $X_s \otimes_{k(s)} k(s')$ is identified with the fiber $X' \otimes_{S'} k(s') = X'_{s'}$ of X' at s'). Thus bounded families may be described as those contained in an *algebraic family* of coherent sheaves parametrized by an S' of finite type over S.

A finite union of bounded families is bounded (take the disjoint-union prescheme of the parameter preschemes S_i defining the dominating algebraic families). A base change $T \longrightarrow S$ transforms a bounded family relative to X/S into a bounded family relative to X_T/T, and the converse holds if $T \longrightarrow S$ is surjective (or, more generally, if its image contains the s that actually occur in the given family E for X/S). This reduces, in principle, the determination of bounded families to the case in which S is the spectrum of an algebra of finite type over the ring of integers $\mathbf{Z}$.

If E and E' are bounded families of classes of sheaves relative to X/S, then $E \otimes E'$ is also bounded. Indeed, suppose E and E' are dominated, respectively, by the algebraic families defined by $T \longrightarrow S$ and $\mathcal{F}$ on X_T, and by $T' \longrightarrow S$ and $\mathcal{F}'$ on $X_{T'}$. Then $E \otimes E'$ is dominated by the

[2] The printed text has X_0 here; only X is defined in this passage.

algebraic family defined by $T'' = T \times_S T' \longrightarrow S$ and the sheaf $\mathcal{F}''$ on $X_{T''}$ obtained as the tensor product of the inverse images of $\mathcal{F}$ and $\mathcal{F}'$ on $X_{T''}$. This argument is valid only because the functor $\mathcal{F} \otimes \mathcal{F}'$ in $\mathcal{F}, \mathcal{F}'$ is right exact and therefore commutes with base extension (and in particular with passage to the fibers). It is not directly applicable to local operations such as $\mathrm{Tor}_i(E, E')$, $\mathrm{Hom}(E, E')$, and $\mathrm{Ext}^i(E, E')$. One can nevertheless show that these operations again transform bounded sets[3] into bounded sets, proceeding as for $E \otimes E'$ but also using results of the following kind (all contained in [3], IV, 6.11): a bounded family E is always dominated by an algebraic family defined by a coherent sheaf $\mathcal{F}$ on an X_T (T of finite type over S) that is flat relative to T. (One "cuts into pieces"[4] the initial parameter space.) Such flatness properties for suitable 252
sheaves indeed ensure that operations such as $\mathrm{Tor}_i(\mathcal{F}, \mathcal{F}')$ commute with arbitrary base change. The same method applies to operations of a global nature: direct images and derived direct images of coherent sheaves under proper morphisms, global Ext relative to proper morphisms (cf. [5], III, § 6), etc. All these operations transform bounded families of sheaves into bounded families of sheaves. (N.B. – Here the preschemes on which the various sheaves are taken may change under operations of the kind considered.)

The following two statements are proved essentially by the same flatness technique; for primary decomposition on the fibers of a morphism of finite type, see in particular [5], IV.

PROPOSITION 1.2. — Let E and E' be bounded sets of classes of sheaves on the fibers of X/S, where X is assumed proper over S. Then:

(i) The family of kernels, cokernels, and images of homomorphisms $\mathcal{F} \longrightarrow \mathcal{F}'$, where the class of $\mathcal{F}$ belongs to E and that of $\mathcal{F}'$ belongs to E', is bounded.

(ii) The family of sheaves $\mathcal{F}''$ that are extensions of a $\mathcal{F}$ by a $\mathcal{F}'$, where the class of $\mathcal{F}$ belongs to E and that of $\mathcal{F}'$ belongs to E', is bounded.

After a suitable base change, one may suppose that E and E' are defined, respectively, by coherent sheaves $\mathcal{G}$ and $\mathcal{G}'$ on an X_T/T, where T is of finite type over S. One may further suppose that certain flatness hypotheses hold, implying that the formation of the sheaves $f_{T*}(\mathrm{Hom}_{X_T}(\mathcal{G}, \mathcal{G}'))$ and $\mathrm{Ext}^1_{f_T}(\mathcal{G}, \mathcal{G}')$ commutes with base change for an arbitrary morphism $T' \longrightarrow T$. One may moreover suppose that these coherent sheaves on T are locally free. Let T_0 and T_1 be the vector bundles over T whose sheaves of germs of sections are, respectively, the preceding sheaves. One then canonically defines a homomorphism $\mathcal{G}_{T_0} \longrightarrow \mathcal{G}'_{T_0}$ of coherent sheaves on X_{T_0}, and an extension[5]

$$0 \longrightarrow \mathcal{G}'_{T_1} \longrightarrow \mathcal{G}'' \longrightarrow \mathcal{G}_{T_1} \longrightarrow 0$$

of coherent sheaves on X_{T_1} having an evident universal property. The second sheaf defines an algebraic family dominating the family considered in (ii). The same is true of the kernel, cokernel, and image of the homomorphism above, whose consideration proves (i) (provided that the cokernel 253
is assumed flat relative to T_0, a case to which one again reduces by cutting T_0 into pieces...).

PROPOSITION 1.3. — Let E be a bounded family of classes[6] of sheaves on the fibers of X/S. Then the classes of the structure sheaves of $(\mathrm{supp}\,\mathcal{F})_{\mathrm{red}}$, where $\mathcal{F}$ is a coherent sheaf on an X_K, with K algebraically closed, whose class belongs to E, form a bounded family.

Here $(\mathrm{supp}\,\mathcal{F})_{\mathrm{red}}$ denotes the support of $\mathcal{F}$ endowed with the induced reduced structure; that is, its structure sheaf is the quotient of $\mathcal{O}_{X_K}$ by the largest sheaf of ideals defining $\mathrm{supp}\,\mathcal{F}$. One can prove the analogue of (1.3) for the sheaves canonically deduced from $\mathcal{F}$ by the theory of primary

[3] FGA-CANON-ERR-0002. The printed French omits the determiner *des*; the English wording was already idiomatic.

[4] FGA-CANON-ERR-0003. The printed French has the malformed phrase « coupe au morceaux »; the English wording already expresses the corrected « coupe en morceaux ».

[5] FGA-CANON-ERR-0004. The printed text reverses the two end terms of the universal extension; the class under consideration lies in $\mathrm{Ext}^1(\mathcal{G}, \mathcal{G}')$.

[6] FGA-CANON-ERR-0005. The printed French has the singular « classe »; the English was already plural.

decomposition, for example the $\mathcal{F}/\mathcal{F}_i$, where the $\mathcal{F}_i$ are the subsheaves of $\mathcal{F}$ primary for the components of the support of $\mathcal{F}$ and minimal with this property; or the $\mathcal{O}_{X_K}/\mathfrak{p}$, where $\mathfrak{p}$ is a sheaf of prime ideals associated with $\mathcal{F}$; or the $\mathcal{O}_{X_K}/\mathfrak{q}$, where $\mathfrak{q}$ is a sheaf of primary ideals associated with a component of the support of $\mathcal{F}$ (the reference field being algebraically closed).

2. Bounded families and the Hilbert polynomial

In what follows, we suppose that X is projective over S and is equipped with a very ample sheaf denoted $\mathcal{O}_X(1)$. For every extension K of a residue-field extension $k(s)$ of a point s of S, we consider on X_K the corresponding sheaf $\mathcal{O}_{X_K}(1)$, which is again very ample.

To every coherent sheaf $\mathcal{F}$ on X_K we associate the function

$$P_{\mathcal{F}}(n) = \text{the Euler–Poincaré characteristic of } \mathcal{F}(n) \text{ on } X_K,$$

which is a polynomial in the integer n, called the *Hilbert polynomial* of $\mathcal{F}$. For large values of n, $P(n)$ is simply the dimension of $H^0(X_K, \mathcal{F}(n))$ over K, since the $H^i(X_K, \mathcal{F}(n))$ vanish for $i > 0$ and n large.

Now suppose that $\mathcal{F}$ is a coherent sheaf on X, flat over S. Then the Hilbert polynomials of the sheaves $\mathcal{F}_s$ induced on fibers X_s belonging to the same connected component of S are all equal [5], III, § 7. It follows (without a flatness hypothesis) that the set of Hilbert polynomials of the
254 sheaves $\mathcal{F}_s$, $s \in S$, is finite for every coherent sheaf $\mathcal{F}$ on X. Recall also that if $\mathcal{F}$ is a coherent sheaf on X, it is isomorphic to a quotient sheaf of a sheaf $\mathcal{O}_X(-n)^N$, for n and N sufficiently large. Thus the sheaves $\mathcal{F}_s$ induced on the fibers are also quotients of the sheaf $\mathcal{O}(-n)$ on the fiber.

These two observations give the "only if" part of the following theorem:

THEOREM 2.1. — Let X be projective over a noetherian S, with $\mathcal{O}_X(1)$ very ample on X relative to S. Let E be a set of classes of sheaves on the fibers of X/S. For E to be bounded, it is necessary and sufficient[7] that it satisfy the following conditions:

(a) There exists a coherent sheaf $\mathcal{L}$ on X (which may be assumed to have the form $\mathcal{O}_X(-n)^N$) such that E is contained in the family of classes of coherent quotient sheaves of sheaves of the form $\mathcal{L}_K$.

(b) The Hilbert polynomials $P_{\mathcal{F}}$ of the sheaves $\mathcal{F}$ whose class belongs to E are members of one fixed finite set of polynomials.

It remains to prove the "if" part, which will be a special case of a more precise result. For every coherent Module $\mathcal{F}$ on a prescheme of finite type over a field K, and every integer r, let N_r be the sub-Module of $\mathcal{F}$ whose sections on an open set are those sections of $\mathcal{F}$ on that open set whose support has dimension $< r$. Thus $N_r = \mathcal{F}$ for $r > \dim \operatorname{supp} \mathcal{F}$, while $N_r = 0$ for $r \leq 0$, and one obtains a finite increasing filtration of $\mathcal{F}$ whose factors N_{r+1}/N_r[8] have as their associated prime cycles exactly the prime cycles associated with $\mathcal{F}$ that have dimension r. Put

$$\mathcal{F}_{(r)} = \mathcal{F}/N_r.$$

Then the associated prime cycles of $\mathcal{F}_{(r)}$ are exactly those prime cycles associated with $\mathcal{F}$ that have dimension $\geq r$; in particular, it equals $\mathcal{F}$ if and only if the prime cycles associated with $\mathcal{F}$ have dimension $\geq r$. With this notation:

THEOREM 2.2. — Under the hypotheses of (2.1), let s be an integer and suppose that E satisfies condition (a) and the following weakened form of (b):

[7] The printed exposé says only "necessary"; the 1962 erratum to Exposé 221 prescribes "necessary and sufficient."

[8] The printed text reverses the quotient as N_r/N_{r+1}, which is incompatible with the increasing filtration just defined.

(b_s) The Poincaré polynomials $P_{\mathcal{F}}$ of the sheaves $\mathcal{F}$ whose class belongs to E have bounded coefficients in degrees $\leq s-1$.

Under these conditions, the sheaves $\mathcal{F}_{(s)}$ (as the class of $\mathcal{F}$ ranges through E) form a bounded family. Moreover, the coefficients of degree $s-2$ of the $P_{\mathcal{F}}$ are bounded below. Thus: 255

COROLLARY 2.3. — Suppose that the sheaves $\mathcal{F}$ whose class belongs to E are such that all their associated prime cycles have dimension d satisfying $s \leq d \leq r$. Then in (2.1) (b), it is enough to consider the coefficients of the $P_{\mathcal{F}}$ in degrees from $s-1$ through r.

The remainder of this section is devoted to an outline of the proof of (2.2). The key lemmas are the following two, the first of which is well known (and summarizes the useful mathematical content of Chow coordinates):

LEMMA 2.4 (CHOW). — Consider the structure sheaves of subschemes Y of fibers X_K (K ranging over algebraically closed extensions of residue fields of S), where Y is reduced and all its components have the same dimension r (with $\mathcal{O}_Y$ regarded as a quotient sheaf of $\mathcal{O}_X$). If the degrees of the Y are bounded, the Y form a bounded family.

Here the degree a of Y can most conveniently be defined by the coefficient of the leading term of

$$P_{\mathcal{O}_Y} = a\frac{n^r}{r!} + \cdots.$$

LEMMA 2.5. — Let $\mathcal{L}$ be a coherent sheaf on X, and let E be a set of classes of quotient sheaves $\mathcal{F}$ of sheaves $\mathcal{L}_K$ (K a residue-field extension of S). Suppose that the fibers of X over S have dimension $\leq r$, and write

$$P_{\mathcal{F}}(n) = a_{\mathcal{F}}\frac{n^r}{r!} + b_{\mathcal{F}}\frac{n^{r-1}}{(r-1)!} + \text{terms of degree} < r-1.$$

Then the coefficient $a_{\mathcal{F}}$ is bounded, while $b_{\mathcal{F}}$ is bounded below. If $b_{\mathcal{F}}$ is bounded, then the family of the $\mathcal{F}_{(r)}$ is bounded.

Proof. – After replacing S by a union of subschemes of S covering S, one may suppose there is a finite morphism $f : X \longrightarrow \mathbf{P}^r_S$ such that $\mathcal{O}_X(1)$ is isomorphic to the inverse image of $\mathcal{O}_{\mathbf{P}^r_S}(1)$. Hence for every coherent sheaf $\mathcal{F}$ on X one has $P_{\mathcal{F}} = P_{f_*(\mathcal{F})}$. On the other hand, one checks readily (by the technique of the preceding section) that a set of sheaves $\mathcal{F}$ on X is bounded if and only if the set of $f_*(\mathcal{F})$ is bounded. Finally,

$$f_*(\mathcal{F})_{(r)} = f_*(\mathcal{F}_{(r)}).$$

This reduces us to the case $X = \mathbf{P}^r_S$. Moreover, one may suppose that $\mathcal{L} = \mathcal{O}_{\mathbf{P}^r_S}(k)^s$ for suitable k 256
and s. The coefficient $a_{\mathcal{F}}$ satisfies

$$0 \leq a_{\mathcal{F}} \leq s$$

and is therefore bounded. It is equivalent to say that the coefficient of n^{r-1} in the $P_{\mathcal{F}}(n)$ is bounded below (respectively, bounded), or to say the same for $P_{\mathcal{F}}(n-k) = P_{\mathcal{F}(-k)}(n)$. We are thus reduced to the case

$$\mathcal{L} = \mathcal{O}^s_{\mathbf{P}^r_S}.$$

Consider the exact sequence

$$0 \longrightarrow N_r \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}_{(r)} \longrightarrow 0,$$

whence

$$P_{\mathcal{F}} = P_{\mathcal{F}_{(r)}} + P_{N_r}.$$

Since the coefficient of n^{r-1} in P_{N_r} is positive (because $\dim \operatorname{supp} N_r \leq r-1$), we have

$$b_{\mathcal{F}_{(r)}} \leq b_{\mathcal{F}}.$$

Thus, in proving the assertions of the lemma, we may replace $\mathcal{F}$ by $\mathcal{F}_{(r)}$; that is, we may assume that the quotient sheaves $\mathcal{F}$ of $\mathcal{L}$ under consideration are torsion-free.

Since $\mathbf{P}^r_K$ is normal, it follows that $\mathcal{F}$ is locally free of rank $a = a_{\mathcal{F}}$ on an open set $U = \mathbf{P}^r_K - Y$, where Y has codimension ≥ 2. Thus $\bigwedge^a \mathcal{F}$ is a sheaf on $\mathbf{P}^r_K$ whose restriction to U is invertible; hence (since $\mathbf{P}^r_K$ is regular and Y has codimension ≥ 2) it is isomorphic to the restriction of an invertible sheaf on $\mathbf{P}^r_K$, determined up to isomorphism. The latter has the form $\mathcal{O}_{\mathbf{P}^r_K}(d)$ for a well-determined integer d. Since $\bigwedge^a \mathcal{F}$ is a quotient of $\bigwedge^a \mathcal{O}^s_{\mathbf{P}^r_K} \simeq \mathcal{O}^{N}_{\mathbf{P}^r_K}$[9], where $N = \binom{s}{a}$, it has
257 N canonical sections, and these therefore define sections of $\mathcal{O}_{\mathbf{P}^r_K}(d)$ on U that are restrictions of sections s_i $(1 \leq i \leq N)$ of $\mathcal{O}_{\mathbf{P}^r_K}(d)$ (since $\mathbf{P}^r_K$ is normal and Y has codimension ≥ 2). These sections generate $\mathcal{O}_{\mathbf{P}^r_K}(d)$ at the points of U, so they are not all zero; this implies $d \geq 0$. On the other hand, an easy calculation gives

$$b_{\mathcal{F}} = a_{\mathcal{F}}(r+1)/2 + d.$$

In particular, this shows that $b_{\mathcal{F}} > 0$, so $b_{\mathcal{F}}$ is bounded below. It is bounded if and only if d is bounded; let us show that in this case $\mathcal{F}$ belongs to a bounded family. We may suppose that $a_{\mathcal{F}}$ and $b_{\mathcal{F}}$ are fixed, say equal to a and b, so that d is fixed. Knowledge of the N sections s_i of $\mathcal{O}_{\mathbf{P}^r_K}(d)$, that is, of a homomorphism

$$s : \bigwedge^a \mathcal{L}_K \longrightarrow \mathcal{O}_{\mathbf{P}^r_K}(d),$$

allows one to recover $\mathcal{F}$ as the coimage of the corresponding composite homomorphism

$$\mathcal{L}_K \longrightarrow \operatorname{Hom}\left(\bigwedge^{a-1} \mathcal{L}_K, \bigwedge^a \mathcal{L}_K\right) \longrightarrow \operatorname{Hom}\left(\bigwedge^{a-1} \mathcal{L}_K, \mathcal{O}_{\mathbf{P}^r_K}(d)\right),$$

where the first is the canonical homomorphism arising from the exterior product and the second is deduced from s. The conclusion now follows from (1.2) (i).

Combining the preceding two lemmas proves:

LEMMA 2.6. — Under the preliminary hypotheses of (2.1), suppose that for every $\mathcal{F}$ one has

$$P_{\mathcal{F}}(n) = a_{\mathcal{F}} \frac{n^r}{r!} + b_{\mathcal{F}} \frac{n^{r-1}}{(r-1)!} + \text{terms of degree} < r-1,$$

and that the coefficients $a_{\mathcal{F}}$ are bounded. Then the coefficients $b_{\mathcal{F}}$ are bounded below. If the $b_{\mathcal{F}}$ are bounded, then the $\mathcal{F}_{(r)}$ form a bounded family.

We may suppose that the base fields K for the sheaves $\mathcal{F}$ are algebraically closed. Endow each $\operatorname{supp} \mathcal{F}_{(r)}$, the union of the components of $\operatorname{supp} \mathcal{F}$ of dimension r, with the induced reduced structure. Then the degrees of the $\operatorname{supp} \mathcal{F}_{(r)}$ are bounded above by a, so by (2.4) the $\operatorname{supp} \mathcal{F}_{(r)}$ form a bounded set. Moreover, for every component of $\operatorname{supp} \mathcal{F}_{(r)}$, the length of $\mathcal{F}_{(r)}$ along that
258 component is $\leq a$. Hence if $\mathcal{J}_{\mathcal{F}}$ is the Ideal defining $\operatorname{supp} \mathcal{F}_{(r)}$, then $\mathcal{F}_{(r)}$ may be regarded as a Module on the subscheme $Y_{\mathcal{F}}$ of X defined by $\mathcal{J}^a_{\mathcal{F}}$. As in the preceding lemma, we may also reduce to the case $\mathcal{F} = \mathcal{F}_{(r)}$, so that $\mathcal{F}$ comes from a Module on $Y_{\mathcal{F}}$. The $Y_{\mathcal{F}}$ correspond to a bounded family of quotient Modules of the $\mathcal{O}_{X_K}$, and hence arise from a closed subscheme Y of a scheme $X \times_S T$. We may then apply (2.5) to Y/T and $\mathcal{L} \otimes_X Y$, which gives the conclusion.

We can now prove (2.2) by induction on the upper bound r of the $\dim \operatorname{supp} \mathcal{F}$. The statement is trivial for $r < 0$; suppose then that $r \geq 0$ and that the statement has been proved for $r' < r$. By

[9] The printed text replaces the exponent s here by an undefined n; s is the rank of the free sheaf surjecting onto $\mathcal{F}$.

(2.6), the $\mathcal{F}_{(r)}$ form a bounded set; hence, by (1.2) (i), so do the kernels of the homomorphisms $\mathcal{L}_K \longrightarrow \mathcal{F}_{(r)}$. There is therefore a coherent Module $\mathcal{L}'$ on X such that the kernels in question, and consequently also the

$$N_r(\mathcal{F}) = \mathrm{Ker}(\mathcal{F} \longrightarrow \mathcal{F}_{(r)}),$$

are quotients of the Modules $\mathcal{L}'_K$. Since the $\mathcal{F}_{(r)}$ are bounded, the $P_{\mathcal{F}_{(r)}}$ remain bounded, and the formula

$$P_{\mathcal{F}} = P_{\mathcal{F}_{(r)}} + P_{N_r}$$

shows that the P_{N_r} satisfy the same condition (b_s) as the $P_{\mathcal{F}}$. Thus the N_r satisfy conditions (a) and (b_s), and by the induction hypothesis the $(N_r)_{(s)}$ remain bounded. Now $\mathcal{F}_{(s)}$ is an extension of $\mathcal{F}_{(r)}$ by $(N_r)_{(s)}$; hence by (1.2) (ii), the $\mathcal{F}_{(s)}$ remain bounded. For the final assertion of (2.2), observe that the kernels N_s of $\mathcal{F} \longrightarrow \mathcal{F}_{(s)}$ remain bounded by (1.2) (i), and that the coefficient of the term in n^{s-1} in P_{N_s} remains bounded; Lemma 2.6 then proves that the coefficient of the next term is bounded below. This completes the proof.

3. Hilbert schemes: definition and existence theorem

Let X be a prescheme over another prescheme S, and let $\mathcal{F}$ be a quasi-coherent Module on X. Denote by

$$\mathrm{Quot}(\mathcal{F}/X/S)$$

the set of quasi-coherent quotient Modules of $\mathcal{F}$ that are flat over S. Let now $S' \longrightarrow S$ be a base-change morphism, put $X' = X \times_S S'$ and $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$. Thus X' is a prescheme over S' equipped with a Module $\mathcal{F}'$, and we may consider $\mathrm{Quot}(\mathcal{F}'/X'/S')$. Put 259

$$\underline{\mathrm{Quot}}_{\mathcal{F}/X/S}(S') = \mathrm{Quot}(\mathcal{F}'/X'/S') \qquad (\text{where } X' = X \times_S S').$$

If now $S'' \longrightarrow S'$ is an S-morphism, then $X'' = X \times_S S''$ is isomorphic to $X' \times_{S'} S''$, and $\mathcal{F}''$ to $\mathcal{F}' \otimes_{\mathcal{O}_{S'}} \mathcal{O}_{S''}$. Since the inverse-image functor

$$\mathcal{G}' \longmapsto \mathcal{G}' \otimes_{\mathcal{O}_{S'}} \mathcal{O}_{S''}$$

from the category of quasi-coherent Modules on X' to the category of quasi-coherent Modules on X'' is right exact and transforms[10] S'-flat Modules into S''-flat Modules, there is a natural map

$$\mathrm{Quot}(\mathcal{F}'/X'/S') \longrightarrow \mathrm{Quot}(\mathcal{F}''/X''/S''),$$

so $\underline{\mathrm{Quot}}_{\mathcal{F}/X/S}(S')$ is a contravariant functor of S' (a prescheme over S), with values in the category of sets. From now on we shall suppose that X is projective over a noetherian S and that $\mathcal{F}$ is coherent; for simplicity we shall restrict to S' over S that are locally noetherian.

THEOREM 3.1. — Under these hypotheses, the contravariant functor $\underline{\mathrm{Quot}}_{\mathcal{F}/X/S}$ on the category of locally noetherian S-preschemes is representable by an S-prescheme $\mathrm{Quot}_{\mathcal{F}/X/S}$ that is a disjoint union of a sequence of projective S-schemes (in particular, $\mathrm{Quot}_{\mathcal{F}/X/S}$ is locally of finite type over S).

We shall obtain such a decomposition as follows. Let $\mathcal{O}_X(1)$ be an invertible sheaf on X that is very ample relative to S. For every polynomial $P(n)$ with rational coefficients, let $\mathrm{Quot}^P(\mathcal{F}/X/S)$

[10] FGA-CANON-ERR-0006. The printed French omits the article *les* before its direct object; the English was already grammatical.

be the subset of $\mathrm{Quot}(\mathcal{F}/X/S)$ consisting of the coherent quotients $\mathcal{G}$ of $\mathcal{F}$ that are flat over S and whose Hilbert polynomial at every $s \in S$ equals P. Put

$$\underline{\mathrm{Quot}}^P_{\mathcal{F}/X/S}(S') = \mathrm{Quot}^P(\mathcal{F}'/X'/S').$$

This gives a subfunctor of $\underline{\mathrm{Quot}}_{\mathcal{F}/X/S}$. The invariance property of Hilbert polynomials recalled in Section 2 implies the following: the functor $\underline{\mathrm{Quot}}_{\mathcal{F}/X/S}$ is representable if and only if the $\underline{\mathrm{Quot}}^P_{\mathcal{F}/X/S}$ are representable; in that case the representing S-prescheme $\mathrm{Quot}_{\mathcal{F}/X/S}$ is isomorphic to the disjoint union of the $\mathrm{Quot}^P_{\mathcal{F}/X/S}$ representing the functors $\underline{\mathrm{Quot}}^P_{\mathcal{F}/X/S}$. Thus (3.1)
260 follows from the following theorem:

THEOREM 3.2. — With the preceding notation, the functor $\underline{\mathrm{Quot}}^P_{\mathcal{F}/X/S}$ is represented by a projective S-prescheme $\mathrm{Quot}^P_{\mathcal{F}/X/S}$.

The remainder of this section is devoted to the proof of (3.2).

Let ν be an integer. For every S' over S, denote by $A_\nu(S')$ the set of quotients $\mathcal{G} = \mathcal{F}'/\mathcal{K}$ of $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ that are coherent and flat over S', and satisfy the following conditions:

a. $R^i f'_*(\mathcal{G}(n)) = 0$ for $i > 0$ and $n \geq \nu$,

b. $R^i f'_*(\mathcal{K}(n)) = 0$ for $i > 0$ and $n \geq \nu$,

c. $f'_*(\mathcal{K}(\nu + k)) = \mathcal{S}'_k f'_*(\mathcal{K}(\nu))$ for $k \geq 0$.

For the last relation, X is assumed to be written as the homogeneous prime spectrum of a quasi-coherent graded algebra $\mathcal{S}_*$ over S, positively graded and generated by $\mathcal{S}_1$, and we put $\mathcal{S}'_* = \mathcal{S}_* \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$, so that X' is the homogeneous prime spectrum of $\mathcal{S}'_*$. To prove (3.2), one readily reduces to the case $X = \mathbf{P}^r_S$ (since S is a union of open sets U such that $X|U$ is a closed subscheme of $\mathbf{P}^r_U$ and $\mathcal{O}_X(1)$ is induced by $\mathcal{O}_{\mathbf{P}^r_U}(1)$), and to the case in which $\mathcal{F}$ has the form $\mathcal{O}^{\oplus N}_{\mathbf{P}^r_S}$ and is therefore flat over S. In the preceding discussion the sheaves $\mathcal{K}$ are then also flat over S'. The Künneth relations ([5], III, § 7) show that conditions (a) and (b) are stable under base change and imply that, for $n \geq \nu$, the formation of $f'_*(\mathcal{G}(n))$ and $f'_*(\mathcal{K}(n))$ commutes with base extension (loc. cit.); hence (c) is also stable under base extension.

In other words, $A_\nu(S')$ is a contravariant functor of S', more precisely a subfunctor of $A(S') = \underline{\mathrm{Quot}}^P_{\mathcal{F}/X/S}(S')$. As ν varies, one obtains an increasing sequence of subsets $A_\nu(S')$ of $A(S')$ whose union is $A(S')$, by a well-known theorem of Serre [5], III, § 2. Now suppose that $\mathcal{G}$ is a coherent quotient of $\mathcal{F}'$, flat over S', and that s is a point of S' such that base change $\mathrm{Spec}(k(s)) \longrightarrow S'$ gives a quotient $\mathcal{G}_s$ of $\mathcal{F}_s$ satisfying conditions (a), (b), and (c), that is, belonging to $A_\nu(\mathrm{Spec}(k(s)))$. Then there exists an open neighborhood U of s on which the same conditions hold for $\mathcal{G}|f'^{-1}(U)$, that is, this quotient belongs to $A_\nu(U)$. For (a) and (b), this follows from the "theorem on holomorphic functions" [5], III, § 7; condition (c) follows from Nakayama's lemma and from the fact
261 that in any case one knows $f'_*(\mathcal{K}(n+k)) = \mathcal{S}'_k f'_*(\mathcal{K}(n))$ for n sufficiently large and $k \geq 0$ (cf. [5], III, § 2). From these observations one concludes the following (compare [4], IV). The functor A is representable if and only if the functors A_ν are representable; in that case the S-prescheme Q representing A is an increasing union of open subschemes Q_ν representing the A_ν.

Let

$$M_* = \bigoplus_{n \geq 0} f_*(\mathcal{F}(n)) = \mathcal{S}_*^{\oplus N},$$

so that

$$M'_* = M_* \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} = \bigoplus_{n \geq 0} f'_*(\mathcal{F}'(n)) = \mathcal{S}'^{\oplus N}_*.$$

It follows from (a) that

(a′) $f'_*(\mathcal{G}(n))$ is locally free of rank $P(n)$ for $n \geq \nu$, [5], III, § 7; and from (b), with $i = 1$, that (a″) $f'_*(\mathcal{G}(n))$ is a quotient Module of M'_n.

Moreover, knowledge of this quotient Module for $n = \nu$ determines, by (c), the sub-Modules $f'_*(\mathcal{K}(n))$ of M'_n for $n \geq \nu$, and therefore determines $\mathcal{K}$ and hence $\mathcal{G}$. Thus one obtains an injective map

$$A_\nu(S') \longrightarrow \mathrm{Grass}_{P(\nu)}(M'_\nu)$$

from $A_\nu(S')$ to the set of locally free quotient Modules of M'_ν of rank $P(\nu)$, and hence a functorial homomorphism

$$i_\nu : A_\nu(S') \longrightarrow \underline{\mathrm{Grass}}_{P(\nu)}(M_\nu)(S'),$$

where the functor on the right is represented by the Grassmann scheme $\mathrm{Grass}_{P(\nu)}(M_\nu)$ (compare [4], V), which is projective over S. I claim:

LEMMA 3.3. — The functor $A_\nu(S')$ is representable, and the morphism $Q_\nu \longrightarrow \mathrm{Grass}_{P(\nu)}(M_\nu)$ representing the homomorphism i_ν is an immersion (which implies that Q_ν is quasi-projective over S).

This assertion is equivalent to the following (compare [4], IV). Suppose given a quotient Module N of M'_ν, locally free of rank $P(\nu)$. Then there exists a subprescheme Z of S' such that for every locally noetherian prescheme T' over S', the inverse image of N on T' belongs to Im $A_\nu(T')$ if 262
and only if $T' \longrightarrow S'$ factors through the subprescheme Z. Changing notation, we may suppose $S' = S$; that is, we are given a quotient N_ν of M_ν by a sub-Module R_ν. It comes from an element of $A_\nu(S)$ if and only if it satisfies the following two conditions:

(i) $M_{\nu+k}/\mathcal{S}_k R_\nu$ is locally free of rank $P(\nu + k)$ for $k \geq 0$.

(ii) The subsheaf $\mathcal{K}$ of $\mathcal{F}$ defined by the graded sub-Module

$$R_* = \sum_{k \geq 0} \mathcal{S}_k R_\nu$$

of M_* ([5], II, § 3), and the quotient $\mathcal{G} = \mathcal{F}/\mathcal{K}$, satisfy conditions (a) and (b) above.

These conditions are manifestly necessary. Conversely, if they hold, then the sheaf $\mathcal{G}$ defined in (ii), being isomorphic to the sheaf associated with the graded $\mathcal{S}_*$-Module N_* that is the sum of the $M_{\nu+k}/\mathcal{S}_k R_\nu$, is flat over S (because its fibers are direct factors of localizations of ${N_*}^{11}$ at homogeneous prime ideals of $\mathcal{S}_*$), and has Hilbert polynomial P by (i). In view of (ii), (a′) and (a″) hold, so for $n \geq \nu$ the natural homomorphism $N_n \longrightarrow f_*(\mathcal{G}(n))$ is a surjective homomorphism between locally free Modules of the same rank, and hence an isomorphism. Therefore

$$f_*(\mathcal{K}(\nu + k)) = \mathcal{S}_k R_\nu \quad \text{for every } k \geq 0.$$

This proves that $\mathcal{G} \in A_\nu(S)$ and that R_ν is the element of $\mathrm{Grass}_{P(\nu)}(M_\nu)$ defined by $\mathcal{G}$.

Criterion (i)–(ii) also applies after a base change $S' \longrightarrow S$. We shall first prove that condition (i) holding after the base change $S' \longrightarrow S$ can be expressed by requiring $S' \longrightarrow S$ to factor through a certain subprescheme Z of S. Once this is done, replacing S by Z reduces us to the case in which (i) already holds on S; since it is stable under base change, it remains to express condition (ii). Let U be the set of $s \in S$ such that the cohomology in dimension > 0 of the sheaves induced on the fiber X_s by $\mathcal{G}(n)$ and $\mathcal{K}(n)$ vanishes for $n \geq \nu$. We have already observed that U is open, and condition (ii) holds after a base change $S' \longrightarrow S$ if and only if $S' \longrightarrow S$ factors through U. This proves Lemma 3.3. It remains to prove the following lemma:

LEMMA 3.4. — Let S be a locally noetherian prescheme equipped with a positively graded quasi-coherent algebra $\mathcal{S}_*$ generated by $\mathcal{S}_1$; let M_* be a quasi-coherent graded $\mathcal{S}_*$-Module of finite type, P a polynomial with rational coefficients, and ν an integer. Then there exists an obviously unique

subprescheme Z of S with the following property: for every prescheme S' over S, the Module $M_n \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ is locally free of rank $P(n)$ for every $n \geq \nu$ if and only if $S' \longrightarrow S$ factors through Z. 263

We may clearly suppose S affine, hence noetherian. Then:

LEMMA 3.5. — For every integer $N \geq \nu$, let U_N be the open subset of S consisting of those $s \in S$ such that

$$\operatorname{rank}_{k(s)} M_{ns} \otimes_{\mathcal{O}_{S,s}} k(s) \leq P(n) \quad \text{for every } \nu \leq n \leq N.$$

Then the decreasing sequence of open subsets U_N is stationary.

It is known [3], IV, that S has a finite partition into reduced subschemes S_i such that each $M \otimes_{\mathcal{O}_S} \mathcal{O}_{S_i}$ is flat over S_i. We may therefore suppose M flat, and hence the M_n flat. Finally, we may clearly suppose S connected. There then exist an integer n_0 and a polynomial Q such that

$$\operatorname{rank}_{k(s)} M_{ns} \otimes_{\mathcal{O}_{S,s}} k(s) = Q(n) \quad \text{for } n \geq n_0,$$

[5], III, § 7. First suppose $P < Q$[12], so that $P(n) < Q(n)$ for large n. Then plainly $U_N = \emptyset$ for N large, and the sequence U_N is stationary. In the complementary case, $P(n) \geq Q(n)$ for large n, so $U_N = U_{n_0}$ for $N \geq n_0$, and again the sequence U_N is stationary.

In particular, the set U_∞ of $s \in S$ such that

$$\operatorname{rank}_{k(s)} M_{ns} \otimes_{\mathcal{O}_{S,s}} k(s) \leq P(n) \quad \text{for every } n \geq \nu \tag{*}$$

is open, since it is the intersection of the U_N. For the proof of (3.4), we may therefore replace S by the open set U_∞[13], reducing to the case in which inequality (*) holds at every $s \in S$. Moreover:

LEMMA 3.6. — Let M be a Module on a locally noetherian prescheme S, and let r be an integer. Then there exists an obviously unique subprescheme Z of S with the following property: for every S' over S, the Module $M \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ is locally free of rank r if and only if $S' \longrightarrow S$ factors through Z. If

$$\operatorname{rank}_{k(s)} M_s \otimes_{\mathcal{O}_{S,s}} k(s) \leq r \quad \text{for every } s,$$

264 then Z is a closed subprescheme of S (with M assumed coherent). Indeed, the preceding argument reduces us to the case in which inequality (*) holds for every $s \in S$ (replacing S, if necessary, by the open subset of points where it holds). We may then suppose that M occurs in an exact sequence

$$\mathcal{O}_S^q \longrightarrow \mathcal{O}_S^r \longrightarrow M \longrightarrow 0.$$

The condition under consideration for S' over S also means that in the corresponding exact sequence $\mathcal{O}_{S'}^q \longrightarrow \mathcal{O}_{S'}^r \longrightarrow M' \longrightarrow 0$, the second arrow is an isomorphism; that is, the first is zero. Hence the closed subprescheme Z of S defined by the Ideal generated by the entries of the matrix defining $\mathcal{O}_S^q \longrightarrow \mathcal{O}_S^r$ has the required property.

Return now to the proof of (3.4). Let Z_n be the closed subscheme of S associated by (3.6) with the Module M_n and the integer $r = P(n)$, and let Z'_N be the infimum of the Z_n for $\nu \leq n \leq N$. The Z'_N form a decreasing sequence of closed subschemes of S[14], and hence are necessarily stationary. Let Z be their constant value for N large. This is the Z sought in (3.4). This completes the proof of (3.4), and hence also of (3.3).

We have therefore proved that $\underline{\mathrm{Quot}}^P_{\mathcal{F}/X/S}$ is represented by an S-prescheme Q that is an increasing union of open subschemes Q_ν quasi-projective over S. To go further, invoke Theorem 2.1,

[11] FGA-CANON-ERR-0007. The printed text has N; homogeneous localization applies to the graded $\mathcal{S}_*$-Module N_*.

[12] The printed text has $P \neq Q$; the 1962 erratum prescribes $P < Q$, replaces the "$P = Q$" case by the complementary case $P(n) \geq Q(n)$ for large n, and replaces the strict inequalities in (*) by non-strict ones.

[13] FGA-CANON-ERR-0008. The printed text has U; the open subset just defined is U_∞.

[14] FGA-CANON-ERR-0009. The printed text has Z, which is defined only in the following sentence; each Z'_N is constructed as a closed subscheme of S.

from which it follows readily that Q is quasi-compact (for it is the image of the prescheme S' of finite type over S parametrizing the family of quotient sheaves of the $\mathcal{F}_K$ with Hilbert polynomial P). Thus Q equals one of the Q_ν, and so is quasi-projective over S. To prove that it is projective over S, it remains to prove that it is proper over S. For this, it suffices to invoke the valuative criterion for properness in the form [5], II, 7.3.8. We need only verify:

LEMMA 3.7. — Let S be the spectrum of a discrete valuation ring, let s be its generic point, let X be a prescheme over S, let $\mathcal{F}$ be a quasi-coherent Module on X, and let $\mathcal{G}_s$ be a quasi-coherent quotient Module of $\mathcal{F}_s = \mathcal{F} \otimes_{\mathcal{O}_S} k(s)$ on X_s. Then there exists a unique quasi-coherent quotient Module $\mathcal{G}$ of $\mathcal{F}$, flat over S, whose restriction to X_s is $\mathcal{G}_s$. Indeed, if $\mathcal{G}_s = \mathcal{F}_s/\mathcal{K}_s$, it is enough 265
to take the largest subsheaf $\mathcal{K}$ of $\mathcal{F}$ inducing $\mathcal{K}_s$, [5], I, 9.4.2, and put $\mathcal{G} = \mathcal{F}/\mathcal{K}$. One checks readily that this sheaf has the required properties.

Theorem 3.2, and consequently (3.1), is now completely proved.

The proof establishes at the same time:

PROPOSITION 3.8. — Under the hypotheses of (3.2), let $Q = \mathrm{Quot}^P_{\mathcal{F}/X/S}$, $X_Q = X \times_S Q$, and $\mathcal{F}_Q = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_Q$; and let $\mathcal{G}$ be the coherent quotient of $\mathcal{F}_Q$, flat over Q, with relative Hilbert polynomial P, such that $(Q, \mathcal{G})$ represents the functor $\underline{\mathrm{Quot}}^P_{\mathcal{F}/X/S}$. There exists an integer ν such that for $n \geq \nu$, $(f_Q)_*(\mathcal{G}(n))$ is a Module on Q, locally free of rank $P(n)$ and very ample relative to S; that is, it defines an immersion of Q into a Grassmann scheme $\mathrm{Grass}_{P(n)}(M)$ over S. In particular, for $n \geq \nu$, the sheaf

$$\bigwedge^{P(n)} (f_Q)_*(\mathcal{G}(n))$$

on Q is invertible and very ample relative to S.

As in (3.2), we may reduce to the case in which $\mathcal{F}$ is flat over S; then it is enough to choose an integer ν such that $A_\nu = A$ (with the preceding notation).

The most important application of (3.2) is the case $\mathcal{F} = \mathcal{O}_X$. We then write

$$\mathrm{Quot}_{\mathcal{O}_X/X/S} = \mathrm{Hilb}_{X/S}, \qquad \mathrm{Quot}^P_{\mathcal{O}_X/X/S} = \mathrm{Hilb}^P_{X/S},$$

and obtain a decomposition

$$\mathrm{Hilb}_{X/S} = \coprod_P \mathrm{Hilb}^P_{X/S}\,.$$

By definition, $\mathrm{Hilb}_{X/S}$ represents the functor $\underline{\mathrm{Hilb}}_{X/S}(S')$, the set of closed subschemes of $X' = X \times_S S'$ that are flat over S'; and $\mathrm{Hilb}^P_{X/S}$ represents the corresponding subfunctor of closed subschemes with a given Hilbert polynomial P. These preschemes are also called the *Hilbert prescheme* of X over S and the *Hilbert prescheme of index* P, respectively. The terminology is justified by the role played in the theory by Hilbert polynomials. Their difference in nature from the classical Chow varieties (which parametrize cycles rather than varieties) is of the same order as that between the 266
Chow ring of cycle classes on a variety and the ring of sheaf classes on the variety (as it appears[15] in the Riemann–Roch theorem [1]). Indeed, when $X = \mathbf{P}^r_S$ and S is the spectrum of a field, knowledge of the Hilbert polynomial of a coherent Module $\mathcal{F}$ on X is also equivalent to knowledge of the Chern classes of $\mathcal{F}$, or of the class of $\mathcal{F}$ in the ring of classes of coherent sheaves on X.

REMARKS 3.9. — Note also that the construction of $\mathrm{Quot}_{\mathcal{F}/X/S}$ and $\mathrm{Quot}^P_{\mathcal{F}/X/S}$ was reduced to the case $X = \mathbf{P}^r_S$, $\mathcal{F} = \mathcal{O}_X^N$, with $\mathcal{O}_X(1)$ the usual very ample sheaf; more precisely, the general $\mathrm{Quot}_{\mathcal{F}/X/S}$ occur as closed subschemes of these. Since the formation of the $\mathrm{Quot}^P_{\mathcal{F}/X/S}$ is plainly

[15] FGA-CANON-ERR-0010. The printed French has feminine « telle », although its antecedent *anneau* is masculine; the English was already idiomatic.

compatible with base changes $S' \longrightarrow S$, we are reduced further to $S = \mathrm{Spec}(\mathbf{Z})$. Ultimately, then, one is reduced to studying the following projective schemes over $\mathbf{Z}$:

$$\mathfrak{Q}^P_{r,N} = \mathrm{Quot}^P_{(\mathcal{O}_{\mathbf{P}^r_{\mathbf{Z}}})^N/\mathbf{P}^r_{\mathbf{Z}}/\mathrm{Spec}(\mathbf{Z})},$$

and, more particularly, the absolute Hilbert schemes

$$\mathrm{Hilb}^P_r = \mathfrak{Q}^P_{r,1}.$$

A more detailed study of these schemes, beginning with the determination of their connected components (are they connected?),[16] and their irreducible components (by Serre [7], there may be irreducible components lying entirely over a nonzero prime p), would be very interesting. Recall Weil's question whether the irreducible components of the fibers of Hilb^P_r over $s \in \mathrm{Spec}(\mathbf{Z})$ correspond to "regular" extensions of the prime field, that is, whether they are "relatively connected." These questions may be more approachable for Hilbert schemes than for "Chow varieties."

4. Variants

a. Under the hypotheses of (3.1), let U be an open subset of X, and let A' be the subfunctor of $A = \underline{\mathrm{Quot}}_{\mathcal{F}/X/S}$ such that $A'(S')$ is the set of quotient Modules $\mathcal{G}$ of $\mathcal{F}'$, flat over S', whose support is contained in U'. Plainly A' is represented by an open subset of the prescheme $\mathrm{Quot}_{\mathcal{F}/X/S}$
267 representing A. It follows that Theorems (3.1) and (3.2) remain valid when X is assumed quasi-projective over S instead of projective over S, provided "projective" is also replaced by "quasi-projective" in the conclusions and $\underline{\mathrm{Quot}}_{\mathcal{F}/X/S}(S')$ now denotes the set of coherent quotients $\mathcal{G}$ of $\mathcal{F}'$, flat over S', whose support is proper over S'.

b. More generally, one may impose on the quotients $\mathcal{G}$ of $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$, flat over S', all kinds of additional natural conditions stable under base change, thereby obtaining subfunctors of $\underline{\mathrm{Quot}}_{\mathcal{F}/X/S}$ whose representability may be sought. In many cases, the usual criterion proves that these functors are again represented by open subsets of $\mathrm{Quot}_{\mathcal{F}/X/S}$. This is so, in particular, when one imposes one of the following additional properties:

1° The dimensions of the prime cycles associated with the Modules $\mathcal{G}_{s'}$ ($s' \in S'$)[17] induced on the fibers $X'_{s'}$ belong to a given set of integers.

2° (When $\mathcal{F} = \mathcal{O}_X$, so that $\mathcal{G}$ corresponds to a closed subprescheme Y of X'): Y is a smooth prescheme ([3], IV) over S'[18], respectively normal over S' (that is, the fibers $Y_{s'}$ are normal "over $k(s')$,"[19] meaning normal after every extension of the base field), respectively (when X is flat over S) an S'-local complete intersection[20] in X' over S' (that is, the fibers $Y_{s'}$ are local complete intersections in the $X'_{s'}$).

Other conditions would involve cohomological properties of the Modules $\mathcal{G}_{s'}$ induced on the $X'_{s'}$, etc. Of course, a conjunction of conditions each represented by an open subset U_i of $\mathrm{Quot}_{\mathcal{F}/X/S}$ is represented by the intersection of these open subsets. For example, assigning to every S' over S the set of closed subschemes Y of $X' = X \times_S S'$ that are étale coverings ([3], I) of S' of a given rank r gives a representable contravariant functor of S'.

[16] The 1962 erratum notes that Hartshorne proved the connectedness of the Hilb^P_r over an algebraically closed field and determined the pairs (r, P) for which $\mathrm{Hilb}^P_r \neq \emptyset$.

[17] FGA-CANON-ERR-0011. The printed text has $\mathcal{G}_s$ although the quantified point is s'.

[18] FGA-CANON-ERR-0016. In the six occurrences of this condition after base change, the printed text incorrectly retains S or X; the family lives on $X' = X \times_S S'$ relative to S'.

[19] FGA-CANON-ERR-0012. The printed text has $k(s)$; the residue field of the indexed point s' is $k(s')$.

[20] FGA-CANON-ERR-0013. The printed text has "k-local complete intersection"; the intermediate correction is relative to the base, and FGA-CANON-ERR-0016 requires the base-changed S' here.

c. The preschemes $\mathrm{Hom}_S(X,Y)$, $\Pi_{X/S}(Z/X)$, and $\mathrm{Isom}_S(X,Y)$ defined in [2], II, C, no. 2 exist under suitable projectivity hypotheses and occur as open subsets of suitable Hilbert preschemes. Since

$$\mathrm{Hom}_S(X,Y) = \Pi_{X/S}((X \times_S Y)/X),$$

the case of $\mathrm{Hom}_S(X,Y)$[21] reduces to that of $\Pi_{X/S}(Z/X)$. For every S' over S, the set of sections of $Z' = Z \times_S S'$ over $X' = X \times_S S'$ is in bijective correspondence with the set of subpreschemes Γ of Z'[22] (necessarily closed if Z is separated over X) such that the morphism $\Gamma \longrightarrow X'$ induced by $Z' \longrightarrow X'$ is an isomorphism. In this way, when X is flat and proper over S and Z 268
is quasi-projective over S, $\Pi_{X/S}(Z/X)$ exists and occurs as an open subprescheme of $\mathrm{Hilb}_{Z/S}$. Hence when X is projective and flat over S and Y is quasi-projective over S, $\mathrm{Hom}_S(X,Y)$ exists and occurs as an open subprescheme of $\mathrm{Hilb}_{(X\times_S Y)/S}$. When X and Y are both projective over S, it follows at once that $\mathrm{Isom}_S(X,Y)$ also exists and is represented by an open subset of $\mathrm{Hom}_S(X,Y)$. Similarly, when X is flat and projective over S and Y is quasi-projective over S, the S-prescheme $\mathrm{Imm}_S(X/Y)$ corresponding to the subfunctor of $\mathrm{Hom}_S(X,Y)$ whose elements are the S'-homomorphisms $X' \longrightarrow Y'$ that are immersions is also represented by an open subset of $\mathrm{Hom}_S(X,Y)$.

Let $\mathcal{L}$[23] (respectively, $\mathcal{M}$) be an invertible sheaf on X (respectively, Y), very ample relative to S. Then $\mathcal{L} \otimes_{\mathcal{O}_S} \mathcal{M}$ is a very ample sheaf on $X \times_S Y$ relative to S. Consequently, for every polynomial P with rational coefficients, $\mathrm{Hilb}^P_{(X\times_S Y)/S}$ is defined and is a quasi-projective prescheme over S. It therefore induces on $\mathrm{Hom}_S(X,Y)$ an open-and-closed subset quasi-projective over S, which we denote $\mathrm{Hom}_S(X,Y)^P$. Thus the sections of $\mathrm{Hom}_S(X,Y)^P$ over S are the S-morphisms $g : X \longrightarrow Y$ such that for every integer n,

$$\chi\Big(\big(\mathcal{L} \otimes_{\mathcal{O}_X} g^*(\mathcal{M})\big)^{\otimes n}\Big) = P(n).$$

In this way one obtains generalizations of Matsusaka's theorem asserting that the automorphisms of a "polarized" projective variety form an algebraic group. Here the assertion has a plainly more precise meaning, since this group is defined as the solution of a universal problem. Note also that over an algebraically closed field, the automorphism group considered formerly is obtained from the "true" one defined here by quotienting out the nilpotent elements. This explains why the older constructions have little chance of working over a nonperfect base field: the Ideal of nilpotent elements appearing after an extension of the base field need not be "defined over k." The same observation applies to most of the old-style constructions as well.

5. Infinitesimal study of Hilbert schemes

It follows from the next result: 269

PROPOSITION 5.1. — Let S be a prescheme, let S_0 be a subprescheme defined by a quasi-coherent Ideal $\mathcal{J}$ of square zero, let X be an S-prescheme, and let $\mathcal{F}$ be a quasi-coherent Module on X. Put $X_0 = X \times_S S_0$ and $\mathcal{F}_0 = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S_0}$, and let $\mathcal{G}_0 = \mathcal{F}_0/\mathcal{K}_0$ be a quasi-coherent quotient Module of $\mathcal{F}_0$, flat over S_0. For every open subset U of X, let $\mathcal{E}(U)$ be the set of quasi-coherent quotient Modules $\mathcal{G}$ of $\mathcal{F}|U$, flat over S, such that $\mathcal{G} \otimes_{\mathcal{O}_S} \mathcal{O}_{S_0} = \mathcal{G}_0$. As U varies, the $\mathcal{E}(U)$ are thus the sections of a sheaf $\mathcal{E}$ on U. The sheaf of groups

$$\mathcal{A} = \mathrm{Hom}_{\mathcal{O}_{X_0}}(\mathcal{K}_0, \mathcal{G}_0 \otimes_{\mathcal{O}_{S_0}} \mathcal{J})$$

[21] FGA-CANON-ERR-0014. The printed text has $\mathrm{Hom}_S(X, X)$ here, contrary to the preceding displayed identity.

[22] FGA-CANON-ERR-0015. The printed text has Z; these graphs are graphs of sections of $Z' = Z \times_S S'$.

[23] FGA-CANON-ERR-0017. The printed French uses plural « Soient » before a singular subject with a parenthetical counterpart; the English was already singular.

acts naturally on $\mathcal{E}$, which thereby becomes a sheaf "formally principal homogeneous under $\mathcal{A}$" (that is, for every open U in X, $\mathcal{E}(U)$ is empty or is a principal homogeneous set under $\mathcal{A}(U)$).

It follows that:

COROLLARY 5.2. — Suppose that locally on X there exists an extension $\mathcal{G}$ of $\mathcal{G}_0$ to a quotient of $\mathcal{F}$ flat over S (that is, the fibers of the sheaf $\mathcal{E}$ are nonempty). Then there is a canonical obstruction class

$$c(\mathcal{G}_0) \in H^1(X, \mathcal{A})$$

whose vanishing is necessary and sufficient for the existence of a global extension $\mathcal{G}$ of $\mathcal{G}_0$ to a quotient of $\mathcal{F}$ flat over S. If this class vanishes, the set $\mathcal{E}(X)$ of all possible extensions is a principal homogeneous set under

$$\mathcal{A}(X) = \mathrm{Hom}_{\mathcal{O}_{X_0}}(\mathcal{K}_0, \mathcal{G}_0 \otimes_{\mathcal{O}_{S_0}} \mathcal{J}).$$

[24] In particular, a global extension exists if $H^1(X, \mathcal{A}) = 0$.

COROLLARY 5.3. — [25] Suppose that $Q = \mathrm{Quot}_{\mathcal{F}/X/S}$ exists (cf. 4 (a)) – for example, that X is quasi-projective over a locally noetherian S and $\mathcal{F}$ is coherent. Let $x \in Q$ correspond to a
270 residue extension $K = k(x)$ of $k(s)$ ($s \in S$): thus x is defined by a coherent quotient Module $\mathcal{G}_0 = \mathcal{F}_0/\mathcal{K}_0$ of the Module $\mathcal{F}_0 = \mathcal{F}_K$ on the K-prescheme X_K. Let $\mathcal{A}$ be the coherent sheaf on X_K defined by

$$\mathcal{A} = \mathrm{Hom}_{\mathcal{O}_{X_0}}(\mathcal{K}_0, \mathcal{G}_0).$$

Then the Zariski tangent space to the fiber Q_s at x (the K-dual of $\mathfrak{m}/\mathfrak{m}^2$, where $\mathfrak{m}$ is the maximal ideal of $\mathcal{O}_{Q_K,x}$) is canonically isomorphic to $H^0(X_K, \mathcal{A})$.

The description of the Zariski tangent space admits the following generalization. Given an S-morphism $g : S' \longrightarrow Q$, that is, a section g' of $Q' = Q \times_S S'$ over S', the Module

$$\Omega = g^*(\Omega^1_{Q/S}) = g'^*(\mathcal{J}/\mathcal{J}^2)$$

(where $\mathcal{J}$ is the Ideal on Q' defined by the section g' of Q' over S') is characterized by the following formula, functorial in the coherent Module $\mathcal{M}$ on S':

$$\mathrm{Hom}_{\mathcal{O}_{S'}}(\Omega, \mathcal{M}) \simeq H^0(X', \mathcal{A}),$$

where $\mathcal{A}$ is the Module on $X' = X \times_S S'$ defined by

$$\mathcal{A} = \mathrm{Hom}_{\mathcal{O}_{X'}}(\mathcal{K}, \mathcal{G} \otimes_{\mathcal{O}_{S'}} \mathcal{M}).$$

[26] Here $\mathcal{G} = \mathcal{F}'/\mathcal{K}$ is the quotient Module of $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ corresponding to g. Indeed, apply (5.1) with S_0 replaced by S' and S by the prescheme $D(\mathcal{M}) = (S', \mathcal{O}_{S'} + \mathcal{M})$, where $\mathcal{M}$ is regarded as an Ideal of square zero.

When $\mathcal{F} = \mathcal{O}_X$ in (5.1), the datum of $\mathcal{G}_0$ corresponds to a closed subprescheme Y_0 of X_0, flat over S_0, defined by the Ideal $\mathcal{J}_0 = \mathcal{K}_0$. Formula (*) then gives

$$\mathcal{A} = \mathrm{Hom}_{\mathcal{O}_{X_0}}(\mathcal{J}_0/\mathcal{J}_0^2, \mathcal{O}_{Y_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{J}),$$

271 where $\mathcal{J}/\mathcal{J}^2$ is interpreted as the conormal sheaf of Y_0 in X_0, also denoted $\check{\mathcal{N}}_{Y_0/X_0}$. It is then useful to regard $\mathcal{A}$ as a Module on Y_0 and compute H^0 and H^1 on Y_0. Moreover, if Y_0 is locally

[24] In this last formula the printed text has $\mathcal{O}_X$ in place of $\mathcal{O}_{X_0}$, although the immediately preceding definition of $\mathcal{A}$ fixes $\mathcal{O}_{X_0}$.

[25] The printed text numbers this corollary "3.3"; it follows (5.1) and (5.2) in Section 5.

[26] The printed text puts $\mathcal{O}_S$ in this tensor product; here $\mathcal{G}$ and $\mathcal{M}$ are Modules over S', and the square-zero deformation construction requires the tensor product over $\mathcal{O}_{S'}$.

a complete intersection in X_0, with X flat over S, then local extensions in (5.1) are guaranteed to exist; furthermore, $\mathcal{J}/\mathcal{J}^2$ is locally free on Y_0, and one may write

$$\mathcal{A} = \mathcal{N}_{Y_0/X_0} \otimes_{\mathcal{O}_{S_0}} \mathcal{J},$$

where the first factor on the right is the normal sheaf of Y_0 in X_0. Using the fundamental criterion of smoothness ([3], III, 3.1), one obtains for example:

COROLLARY 5.4. — Under the hypotheses of (5.3), suppose that $\mathcal{F} = \mathcal{O}_X$, that X is flat over S, and that the closed subprescheme Y_0 of X_0 corresponding to $\mathcal{G}_0$ is locally a complete intersection. Then the Zariski tangent space to Q_s at x is canonically isomorphic to $H^0(Y_0, \mathcal{N}_{Y_0/X_0})$. If $H^1(Y_0, \mathcal{N}_{Y_0/X_0}) = 0$, then the Hilbert prescheme $\mathrm{Hilb}_{X/S}$ is smooth over S at x. ($\mathcal{N}_{Y_0/X_0}$ is the normal sheaf of Y_0 in X_0.)

REMARK 5.5. — This statement applies in particular when Y_0 is a complete intersection in X_0 defined by one equation, that is, a positive "Cartier divisor." Then $\mathcal{N}_{Y_0/X_0}$ is isomorphic to the sheaf on Y_0 induced by the invertible sheaf $\mathcal{J}^{-1}$ on X_0 defined by the divisor Y_0. This situation occurs especially in the study of families of positive divisors on a nonsingular projective variety X_0. The isomorphism between the Zariski tangent space at the point x of Q (or, equivalently, of the open subset D of Q corresponding to divisors) and $H^0(Y_0, \mathcal{N}_{Y_0/X_0})$ was known in classical algebraic geometry as the "characteristic homomorphism" from the former to the latter. It was defined only when x was a smooth point of the parameter variety T of a "complete continuous family" of divisors, that is, from our point of view, of an irreducible component of the scheme D endowed with its induced reduced structure. The tangent space to T at x is then a subspace of the tangent space to D at x, so the classical characteristic homomorphism is indeed injective, but it is surjective only under additional conditions, for example when D is integral at x. In fact, Zappa [8] constructed an example (with X a nonsingular projective surface over the complex numbers) 272
in which the characteristic homomorphism is not surjective even at the generic point of T. Thus D is not integral even at the generic point of the irreducible component under consideration. This shows especially strikingly how varieties with nilpotent elements are necessary for understanding phenomena of the most classical theory of surfaces.[27]

5.6

In (5.4) we gave a smoothness criterion applying in particular to schemes of divisors. Kodaira gave a different criterion in [6], namely the vanishing of $H^1(X_0, \mathcal{L})$, where $\mathcal{L} = \mathcal{J}_0^{-1}$ is the invertible sheaf on X_0 defined by the divisor Y_0. This criterion is valid when S is the spectrum of a field of characteristic zero and was proved in [6] by transcendental methods when the base field is **C**. More generally, with S arbitrary again, Kodaira's condition is sufficient for the canonical morphism

$$D \longrightarrow \mathrm{Pic}_{X/S}$$

from the prescheme of divisors to the Picard prescheme of X/S to be smooth at the point x in question (as follows readily from the usual smoothness criterion once the existence of $\mathrm{Pic}_{X/S}$ is known). If, in addition, $\mathrm{Pic}_{X/S}$ is smooth over S at the image of x (for example, if $\mathrm{Pic}_{X/S}$ is smooth over S), then D is smooth over S at x. On the other hand, Cartier proved that every group prescheme locally of finite type over a field k of characteristic zero is smooth over k. Combining these two results recovers Kodaira's result. These remarks also show that over a field K of characteristic $p > 0$,

[27] The 1962 erratum adds that Mumford had just constructed an irreducible component of the Hilbert scheme of $\mathbf{P}^3_{\mathbf{C}}$, whose general points represent nonsingular curves of degree 14 and genus 24, that is nonreduced at its generic point; blowing up one of these curves also gives a regular projective threefold whose formal moduli scheme is generically nonreduced.

when $\mathrm{Pic}_{X/S}$ is not smooth over k (as when X is the Igusa surface), the condition $H^1(X_0, \mathcal{L}) = 0$ implies, on the contrary, that D is not smooth at x, and is even nonreduced at x if K is algebraically closed.

Finally, let us record the following result, which plays an important role in the infinitesimal study of fiber spaces:

PROPOSITION 5.7. — Let X be a finite flat prescheme over a locally noetherian S, and let Z be a prescheme over S such that $\Pi_{X/S}(Z/X)$ exists (as it does if Z is quasi-projective over X). If Z is smooth over X, then $\Pi_{X/S}(Z/X)$ is smooth over S.

This follows immediately from the definition and the usual smoothness criterion ([3], III, 3.1). Note that when X is finite and flat over S, the existence of $\Pi_{X/S}(Z/X)$[28] can be treated very
273 elementarily, without using the theory of Hilbert schemes. For example, if X is radicial over S, then $\Pi_{X/S}(Z/X)$ exists without any restriction on Z. As another example, let T be an S-prescheme and let T_n be the "infinitesimal neighborhood of order n" of the diagonal of $T \times_S T$ in $T \times_S T$, equipped with the morphisms $p_1, p_2 : T_n \longrightarrow T$ induced by the two projections. Regard T_n as a prescheme finite over T via p_1, and suppose in addition that it is flat over T (as it is when T is smooth over S). For every prescheme X over T, put

$$(X/T/S)^{(n)} = \Pi_{T_n/T}(p_2^*(X/T)/T_n).$$

[29] This is a prescheme over T called the *bundle of germs of sections of order n of X over T* (relative to S). It depends functorially on X and is smooth over T if X is.

6. Relations with the norm and symmetric products

Let S be a prescheme, let X and Y be S-preschemes, and let

$$u : (X/S)^n \longrightarrow Y$$

be a symmetric S-morphism from the n-th Cartesian power of X/S to Y. For simplicity, suppose S locally noetherian and X and Y of finite type over S. To every coherent Module $\mathcal{F}$ on X whose support is finite over S, which is flat over S and has relative rank n over S (that is, $f_*(\mathcal{F})$ is a locally free Module of rank n on S), one can then naturally associate a section of Y over S:

$$\pi^u_{X/S}(\mathcal{F}) \in \Gamma(Y/S).$$

We shall not give the formal definition here, but merely note that the resulting formalism is a natural generalization of the usual formalism of norms and traces. When the n-th symmetric power of X over S exists (for example, when the orbits of the symmetric group $\mathfrak{S}_n$ acting on $(X/S)^n$ are contained in affine open subsets), one may take this symmetric power $\mathrm{Symm}^n_S(X)$ for Y, obtaining a canonical element

$$\pi_{X/S}(\mathcal{F}) \in \Gamma(\mathrm{Symm}^n_S(X)/S),$$

274 from which the $\pi^u_{X/S}(\mathcal{F})$ can be recovered. Another important case is that in which X is a commutative monoid over S, $X = Y$, and u comes from the composition law of X. We then write simply $\pi(\mathcal{F})$ for the section of X over S associated with the Module $\mathcal{F}$ on X.

Suppose now that $\mathcal{F}$ is a coherent Module on X such that $\mathrm{Quot}_{\mathcal{F}/X/S}$ exists, or at least such that the functor $\underline{\mathrm{Quot}}^n_{\mathcal{F}/X/S}$ assigning to every S' over S the set of coherent quotient sheaves $\mathcal{M}$ of

[28] The printed text has Z/K here; no K is defined, and the section functor under discussion since (5.7) is $\Pi_{X/S}(Z/X)$.

[29] The printed text places T_n/S under the symbol Π. Proposition 5.7 is applied here to the finite flat morphism $p_1 : T_n \to T$, and the result is explicitly a prescheme over T; hence the denominator is T_n/T.

$\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ that are flat over S' and have relative rank n is represented by an S-prescheme $\mathrm{Quot}^n_{\mathcal{F}/X/S}$. (When X is quasi-projective over S, $\mathrm{Quot}^n_{\mathcal{F}/X/S}$ does exist and, in the notation of Section 3, is $\mathrm{Quot}^P_{\mathcal{F}/X/S}$, where P is the constant polynomial n.) Since the formation of $\pi^u_{X/S}(\mathcal{M})$ is compatible with base change, one obtains a canonical morphism

$$\pi^u_{X/S} : \mathrm{Quot}^n_{\mathcal{F}/X/S} \longrightarrow Y,$$

and, in particular, when the n-th symmetric power of X over S exists,

$$\pi_{X/S} : \mathrm{Quot}^n_{\mathcal{F}/X/S} \longrightarrow \mathrm{Symm}^n_S(X).$$

The most important case is $\mathcal{F} = \mathcal{O}_X$, which gives a morphism

$$\pi_{X/S} : \mathrm{Hilb}^n_{X/S} \longrightarrow \mathrm{Symm}^n_S(X).$$

This is plainly an isomorphism for $n = 0$ or $n = 1$. But for $n > 1$[30], even when S is the spectrum of a field k and X is smooth over S, it is not in general an isomorphism, or even an injective morphism: a zero-dimensional subscheme of X (corresponding, for example, to an ideal $\mathfrak{i}$ primary for the maximal ideal in a local ring $\mathcal{O}_{X,x}$, for a closed point x of X) is not determined by the cycle it defines (in this case, by the codimension over k of $\mathfrak{i}$ in $\mathcal{O}_{X,x}$). One can say only the following, with S arbitrary again:

a. If X is smooth over S, the norm morphism induces an isomorphism from the open subset of $\mathrm{Hilb}^n_{X/S}$ classifying étale coverings of rank n contained in X (cf. no. 4, (b)) onto the open subset of $\mathrm{Symm}^n_S(X)$ corresponding to n-cycles without multiple components. 275

b. If, in addition, X has relative dimension 1 over S, then the norm morphism is an isomorphism from $\mathrm{Hilb}^n_{X/S}$ onto $\mathrm{Symm}^n_{X/S}$.

This is because a zero-dimensional subscheme of a nonsingular algebraic curve is determined by its corresponding cycle. The same observation applies more generally to positive Cartier divisors on a nonsingular algebraic scheme (and it is possible that in this very special case the Chow variety gives the same object as the Hilbert scheme).

7. Complements and questions

As J.-P. Serre observed, a well-known example of Nagata shows that one can find a scheme S, the spectrum of a field k; an S-scheme S', the spectrum of a quadratic extension k' of k; and a proper smooth (but nonprojective) S'-scheme X of dimension 3 such that $\Pi_{S'/S}(X/S')$ does not exist. It follows a fortiori that the Hilbert scheme $\mathrm{Hilb}^2_{X/S}$ does not exist (nor even the k-scheme that would represent the étale coverings of S of rank 2 contained in X, and a fortiori not the symmetric square of X; cf. the preceding section). This places serious limitations on the possibility of nonprojective constructions in algebraic geometry. (It is, however, plausible that no such limitations arise in analytic geometry, any more than they do in formal geometry (cf. [2], II).) On the other hand, if X is a proper scheme over $S = \mathrm{Spec}\, k$ and Z is quasi-projective over X, then $\Pi_{X/S}(Z/X)$ exists and is a scheme that is a disjoint union of a sequence of schemes quasi-projective over S (as in the projective case (3.1)). Indeed, one reduces to the case in which X itself is projective by dominating X by a projective S-scheme X'. We shall not give the details of the proof, which also uses the factorization result for a finite morphism recorded in [2], I, A, no. 2 (b). The method succeeds because, with S the spectrum of a field, the X' occurring in Chow's lemma is automatically flat over S.

[30] The printed text repeats $n \geq 1$, contradicting the isomorphism just asserted for $n = 1$.

I do not know whether the result remains true without a hypothesis on S, assuming only that X is proper and flat over S and Z quasi-projective over X. An important case in applications is when Z is a closed subscheme of X; if $\Pi_{X/S}(Z/X)$ exists, it is necessarily a closed subscheme of
276 S. It may be constructed directly and quite simply when X is projective over S, without using the theory of Hilbert schemes. More generally, the same method shows that if Z is affine over X, then $\Pi_{X/S}(Z/X)$ exists and is affine over S. It also shows that if X is proper and flat over S (not necessarily projective over S), then for every locally trivial vector bundle Z over X, $\Pi_{X/S}(Z/X)$ exists and is a vector bundle over S. It would be desirable to revisit and unify these results.

BIBLIOGRAPHY

[1] BOREL (A.) and SERRE (J.-P.). – Le théorème de Riemann–Roch, *Bull. Soc. math. France*, vol. 86, 1958, pp. 97–136.

[2] GROTHENDIECK (Alexander). – Technique de construction et théorèmes d'existence en géométrie algébrique, I, II, III, *Séminaire Bourbaki*, vol. 13, 1959/60, no. 190, 29 pp., no. 195, 22 pp.; and vol. 14, 1960/61, no. 212, 20 pp.

[3] GROTHENDIECK (Alexander). – Séminaire de Géométrie algébrique, I, II, III, IV. – Paris, Institut des hautes Études scientifiques, 1960/61 (mimeographed).

[4] GROTHENDIECK (Alexander). – Technique de construction en géométrie analytique, IV, V, *Séminaire Cartan*, vol. 13, 1960/61, nos. 11 and 12.

[5] GROTHENDIECK (A.) and DIEUDONNÉ (J.). – *Éléments de géométrie algébrique*, I: Le langage des schémas. – Paris, Presses universitaires de France, 1960 (Institut des hautes Études scientifiques, Publications mathématiques, 4); II, III, IV (forthcoming).

[6] KODAIRA (K.). – Characteristic linear systems of complete continuous systems, *Amer. J. of Math.*, vol. 78, 1956, pp. 716–744.

[7] SERRE (Jean-Pierre). – Exemples de variétés projectives en caractéristique p non relevables en caractéristique 0, *Proc. Nat. Acad. Sc. U.S.A.*, vol. 47, 1961, pp. 108–109.

[8] ZAPPA (G.). – Sull'esistenza, sopra le superficie algebriche, di sistemi continui completi infiniti, la cui curva è a serie caratteristica incompleta, *Pont. Acad. Sc. Acta*, vol. 9, 1945, pp. 91–93.

ADDENDUM

[added during proof correction]

It now appears that the conjectures of [2], III, no. 8 are false, even for nonsingular varieties over a field of characteristic zero, both as regards the existence and the quasi-projectivity of the quotient, and even when G acts with closed graph.

Erratum to Exposé no. 221

302

6. TDTE IV: Hilbert schemes.

[Bourbaki Seminar, vol. 13, 1960/61, no. 221, 28 pp.]

Page 6, Theorem 2.1, third line. — For "it is necessary," read "it is necessary and sufficient."
Page 15, line 13. — For "$P \neq Q$," read "$P < Q$."
Line 15. — For "If, on the contrary, $P = Q$," read "In the complementary case, $P(n) \geq Q(n)$ for large n."
Line 18, formula (*). — For "$<$" and "$>$," read "$\leq$" and "$\geq$."
Fifth line from the bottom. — For "r," read "Z."
Page 18, Remark 3.9. — Let us point out that Hartshorne has studied the connected components of Hilbert schemes over an algebraically closed field. He proves that the Hilb_r^P are connected and determines the pairs (r, P) for which $\mathrm{Hilb}_r^P \neq \emptyset$[4].
Page 23, fifth line from the bottom. — For "an irreducible component," read "of an irreducible component."
Pages 23 and 24, Remark 5.5. — In connection with Zappa's example, let us point out that Mumford has just constructed an irreducible component of the Hilbert scheme for $\mathbf{P}^3_{\mathbf{C}}$ (whose general points represent nonsingular curves of degree 14 and genus 24) that is nonreduced at its generic point. Blowing up one of the resulting curves, he also obtains a regular projective scheme of dimension 3 over **C** whose formal moduli scheme is nonreduced at its generic point or, equivalently, whose local moduli variety in the sense of analytic geometry over **C** (cf. Cartan Seminar, vol. 13, 1960/61) is nonreduced at all its points.

[4]Hartshorne (R.). — Connectedness of the Hilbert scheme (Thesis, Harvard, 1962).

Chapter 7

Exposé 232 – Picard schemes: existence theorems

143

Séminaire BOURBAKI
14th year, 1961/62, no. 232
February 1962

DESCENT TECHNIQUE AND EXISTENCE THEOREMS IN ALGEBRAIC GEOMETRY
V. PICARD SCHEMES: EXISTENCE THEOREMS

by Alexander GROTHENDIECK

1. Relative Picard groups and functors

For every prescheme (more generally, every ringed space) X, we call the group of isomorphism classes of invertible Modules (that is, Modules locally isomorphic to $\mathcal{O}_X$) on X the (absolute) *Picard group* of X, and denote it by $\mathrm{Pic}(X)$. Thus there is a canonical isomorphism

$$\mathrm{Pic}(X) \xrightarrow{\sim} H^1(X, \mathcal{O}_X^*), \tag{1.1}$$

where $\mathcal{O}_X^*$ denotes the sheaf of units of $\mathcal{O}_X$ (which is indeed identified with the sheaf of automorphisms of the standard invertible Module $\mathcal{O}_X$). Clearly $X \longmapsto \mathrm{Pic}(X)$ is a contravariant functor in X, and isomorphism (1.1) is functorial.

If X is a prescheme over another prescheme S, then as S' varies in the category $(\mathrm{Sch})/S$ of preschemes over S, the preceding construction gives a contravariant functor $S' \longmapsto \mathrm{Pic}(X \times_S S')$. This functor has no chance of being "representable" ([4], II, A), because the nontrivial automorphisms of the invertible Modules to be classified prevent it from being "local in nature" ([5], IV, 5.4). It must therefore be "made local." In general, for every relative prescheme X/S, introduce the relative group

$$\mathrm{Pic}'(X/S) = H^0(S, R^1 f_*(\mathcal{O}_X^*)), \tag{1.2}$$

where $f : X \longrightarrow S$ is the structural morphism (compare [4], II, C 3). In the cited passage this group is called the relative Picard group; for reasons that will appear, it is preferable here to call it

the *restricted relative Picard group* of X/S. As S' varies in $(\mathrm{Sch})/S$, the assignment $S' \longmapsto \mathrm{Pic}'(X \times_S S'/S')$ is a contravariant functor in S', also denoted $\mathrm{Pic}'_{X/S}$ and therefore defined essentially by

$$\mathrm{Pic}'_{X/S}(S') = \mathrm{Pic}'(X \times_S S'/S'). \tag{1.3}$$

144

This functor is now "local in nature," since the needed adjustment has been made. Intuitively, the right-hand side of (1.3) is the set of "algebraic families" of classes of invertible sheaves on (the fibers of) X/S, indexed by the parameter prescheme S'/S. When the functor Pic' is representable, the prescheme over S representing it is denoted $\mathrm{Pic}'_{X/S}$ and called the *Picard prescheme* of X over S; thus one then has

$$\mathrm{Hom}_S(S', \mathrm{Pic}'_{X/S}) \simeq \mathrm{Pic}'_{X/S}(S') = \mathrm{Pic}'(X \times_S S'/S'). \tag{1.4}$$

There are, however, important cases in which $\mathrm{Pic}'_{X/S}$ is not representable (for example, a "Brauer–Severi" variety over a field k with no k-rational point), although a relative Picard prescheme still has a natural definition. The reason is that the definition of Pic' from the absolute Picard groups $\mathrm{Pic}(X \times_S S'/S')$ has not yet been localized enough: more precisely, Pic' is not in general "compatible with faithfully flat descent." Let us make this explicit.

Let $\mathfrak{M}$ be the class of prescheme morphisms that are *faithfully flat and quasi-compact*; this class is stable under base change and composition. Let P be a contravariant functor from $(\mathrm{Sch})/S$ to the category of sets, and for every S-morphism $u : T' \longrightarrow T$ with $u \in \mathfrak{M}$, consider the diagram

$$P(T) \longrightarrow P(T') \rightrightarrows P(T' \times_T T') \tag{1.5}$$

obtained by applying P to

$$T \longleftarrow T' \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\leftleftarrows}} T' \times_T T'.$$

If P is representable, descent theory ([4], I, B, Th. 2) implies that (1.5) is exact for every $u \in \mathfrak{M}$. This is expressed by saying that P is compatible with $\mathfrak{M}$, in this case that P is "compatible with faithfully flat descent," or that the "presheaf" P on $(\mathrm{Sch})/S$ is a "sheaf" for the notion of localization supplied by $\mathfrak{M}$. For an arbitrary P, a standard procedure, familiar from ordinary topological localization, associates with it a "sheaf" $\mathcal{P}$ and a functor homomorphism $P \longrightarrow \mathcal{P}$ universal in the evident sense. The calculation of $\mathcal{P}$ can be made explicit as follows. To define $\mathcal{P}(T)$, for every T' over T such that $u : T' \longrightarrow T$ belongs to $\mathfrak{M}$, let

145

$\overline{H}^0(T'/T, P)$ be the subset of $P(T')$ consisting of elements ξ whose two images ξ_1, ξ_2 in $P(T' \times_T T')$ have the same image in $P(T'')$ for some morphism $v : T'' \longrightarrow T' \times_T T'$ with $v \in \mathfrak{M}$. [N.B. – The set $\overline{H}^0$ so defined is therefore larger than the set $H^0(T'/T, P)$ introduced in [4], I, A, 4(a).] As T' varies over a fixed T (still with $u \in \mathfrak{M}$), the $\overline{H}^0(T'/T, P)$ form an inductive system when the set of T' is given the preorder defined by domination, and one sets

$$\mathcal{P}(T) = \varinjlim_{T'} \overline{H}^0(T'/T, P). \tag{1.6}$$

The functorial law in T for this expression is immediate.

When

$$P(T) = \mathrm{Pic}(X \times_S T),$$

the contravariant functor on $(\mathrm{Sch})/S$ defined by (1.6) is called the *relative Picard functor* of X over S and denoted $\mathrm{Pic}_{X/S}$; the group $\mathrm{Pic}_{X/S}(S)$ is called the *relative Picard group* of X over S and denoted $\mathrm{Pic}(X/S)$. There is then an evident bijection

$$\mathrm{Pic}_{X/S}(T) \simeq \mathrm{Pic}(X \times_S T/T). \tag{1.7}$$

Thus an element of $\mathrm{Pic}(X/S)$ is defined by an element ξ' of a group $\mathrm{Pic}(X \times_S S')$, where $S' \longrightarrow S$ is faithfully flat and quasi-compact, such that there is a faithfully flat and quasi-compact morphism $S'' \longrightarrow S' \times_S S'$ for which the two inverse images of ξ' in $\mathrm{Pic}(X \times_S S'')$ coincide. An element ξ' of $\mathrm{Pic}(X \times_S S')$ and an element ξ_1 of $\mathrm{Pic}(X \times_S S_1)$ satisfying the conditions just stated define the same element of $\mathrm{Pic}(X/S)$ if and only if there is a faithfully flat and quasi-compact morphism $S_1' \longrightarrow S' \times_S S_1$ such that the images of the two elements in $\mathrm{Pic}(X \times_S S_1')$ are equal. It is often convenient still to work with the functor $P' = \mathrm{Pic}'_{X/S}$ introduced above; one checks at once that the canonical morphism $P \longrightarrow P'$ induces an isomorphism

$$\mathcal{P} \xrightarrow{\sim} \mathcal{P}', \tag{1.8}$$

which gives an often more convenient description of $\mathrm{Pic}_{X/S}$ in terms of $\mathrm{Pic}'_{X/S} = P'$. By 2.3
146 below, when P is replaced by P' in the description of $\mathrm{Pic}(X/S)$ just given, one may in fact take $S'' = S' \times_S S'$ and $S_1' = S' \times_S S_1$, at least under the conditions specified in the cited passage.

When the functor $\mathrm{Pic}_{X/S}$ is representable, one says that X/S admits a Picard prescheme, and the prescheme over S representing the functor is called the *Picard prescheme* of X over S, still denoted $\mathrm{Pic}_{X/S}$. It plainly suffices that $P' = \mathrm{Pic}'_{X/S}$ be representable, for then P' is already a "sheaf," and (1.8) shows that $P' \longrightarrow \mathcal{P}'$ is the canonical morphism

$$\mathrm{Pic}'_{X/S} \longrightarrow \mathrm{Pic}_{X/S}, \tag{1.9}$$

which is then an isomorphism. Thus our terminology agrees with that introduced above in (1.4). In general, when $\mathrm{Pic}_{X/S}$ exists, it is defined by the functorial isomorphism

$$\mathrm{Hom}_S(S', \mathrm{Pic}_{X/S}) \xrightarrow{\sim} \mathrm{Pic}(X \times_S S'/S'). \tag{1.10}$$

2. Relations among the various relative and absolute Picard groups

PROPOSITION 2.1. — Let $f : X \longrightarrow S$ be a morphism such that $\mathcal{O}_S \xrightarrow{\sim} f_*(\mathcal{O}_X)$. Then there is an exact sequence

$$0 \longrightarrow \mathrm{Pic}(S) \longrightarrow \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}'(X/S).$$

When X admits a section over S, the last morphism is surjective; that is, there is an isomorphism

$$\mathrm{Pic}'(X/S) \xrightarrow{\sim} \mathrm{Pic}(X)/\mathrm{Pic}(S).$$

The exact sequence may be regarded as the low-degree exact sequence associated with the Leray spectral sequence for f and $\mathcal{O}_X^*$. The second assertion is likewise formal.

PROPOSITION 2.2. — Let $f : X \longrightarrow S$ be a quasi-compact separated morphism such that $\mathcal{O}_S \xrightarrow{\sim} f_*(\mathcal{O}_X)$, and let $S' \longrightarrow S$ be faithfully flat and quasi-compact. Then:
(i) $\mathrm{Pic}'(X/S) \longrightarrow \mathrm{Pic}'(X \times_S S'/S')$ is injective;
(ii) if X admits a section locally over S (that is, every $s \in S$ has an open neighborhood U such that $X|U$ has a section over U), then the diagram

$$\mathrm{Pic}'(X/S) \longrightarrow \mathrm{Pic}'(X \times_S S'/S') \rightrightarrows \mathrm{Pic}'(X \times_S S''/S'')$$

147

where $S'' = S' \times_S S'$, is exact.

The first assertion follows from the elementary properties of faithfully flat descent and the following general remark. If $f : X \longrightarrow S$ satisfies $\mathcal{O}_S \xrightarrow{\sim} f_*(\mathcal{O}_X)$, then the functor $\mathcal{F} \longmapsto f^*(\mathcal{F})$ from locally free finite-type Modules on S to locally free finite-type Modules on X is fully faithful,

and its essential image consists of the Modules $\mathcal{G}$ on X such that $f_*(\mathcal{G})$ is locally free and the canonical homomorphism

$$f^*(f_*(\mathcal{G})) \longrightarrow \mathcal{G}$$

is an isomorphism. The second assertion was proved by descent theory in [4], I, B, 4.

The results of 2.2 may also be stated as follows.

COROLLARY 2.3. — Under the conditions of 2.2, the canonical homomorphism (1.9) $\mathrm{Pic}'_{X/S} \longrightarrow \mathrm{Pic}_{X/S}$ is injective, and is bijective if X admits a section locally over S. (Thus, in the latter case, the relative Picard group $\mathrm{Pic}(X/S)$ and the restricted relative Picard group $\mathrm{Pic}'(X/S)$ are identical.)

Combining this with 2.1 gives:

COROLLARY 2.4. — Under the conditions of 2.2, there is an exact sequence

$$0 \longrightarrow \mathrm{Pic}(S) \longrightarrow \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(X/S).$$

When X admits a section over S, the last homomorphism is surjective; that is, there is then an isomorphism

$$\mathrm{Pic}(X/S) \xrightarrow{\sim} \mathrm{Pic}(X)/\mathrm{Pic}(S).$$

REMARK 2.5. — Let $f : X \longrightarrow S$ be a morphism such that $\mathcal{O}_S \xrightarrow{\sim} f_*(\mathcal{O}_X)$, and let g be a section of X over S. For an invertible Module $\mathcal{L}$ on X, call an isomorphism $\mathcal{O}_S \xrightarrow{\sim} g^*(\mathcal{L})$ a g-rigidification of $\mathcal{L}$, and call an invertible Module $\mathcal{L}$ on X equipped with a g-rigidification a *g-rigidified invertible Module*. Every automorphism of such a structure is trivial, and $\mathrm{Pic}'(X/S)$ is identified with the set of isomorphism classes of g-rigidified invertible Modules on X[1]. (This is the fact that allowed descent theory to be used in proving 2.2(ii).) This gives a new interpretation of $\mathrm{Pic}(X/S)$, at least when f is also quasi-compact and separated, so that $\mathrm{Pic}(X/S) \simeq \mathrm{Pic}'(X/S)$ by 2.3. 148

REMARK 2.6. — Let $f : X \longrightarrow S$ be as in 2.2, and let $S' \longrightarrow S$ be faithfully flat and quasi-compact such that there is an S-morphism $S' \longrightarrow X$, or equivalently such that $X' = X \times_S S'$ has a section over S'. Put $S'' = S' \times_S S'$ and $X'' = X \times_S S''$, and consider the exact sequence

$$\mathrm{Pic}(X/S) \longrightarrow \mathrm{Pic}(X'/S') \rightrightarrows \mathrm{Pic}(X''/S'').$$

Applying 2.4 gives an exact sequence

$$\mathrm{Pic}(X/S) \longrightarrow \mathrm{Pic}(X')/\mathrm{Pic}(S') \rightrightarrows \mathrm{Pic}(X'')/\mathrm{Pic}(S''),$$

and in particular every element of the relative Picard group already "comes from" an element of $\mathrm{Pic}(X')$. This substantially simplifies the description of the relative Picard group given in the preceding section, and likewise that of the Picard functor of X over S, since for every T over S the preceding argument applies to $X \times_S T/T$ and the morphism $T' = S' \times_S T \longrightarrow T$. If, for example, f itself is faithfully flat, one may take $S' = X$. When f is moreover of finite type (respectively smooth, etc.), this permits one, in the description of the relative Picard functor $T \longmapsto \mathrm{Pic}(X \times_S T/T)$, to restrict to base changes $T' \longrightarrow T$ that are of finite type (respectively smooth, etc.). When f is projective and flat and S is locally noetherian, one proves that $S' \longrightarrow S$ above may be chosen so that S' is the disjoint union of flat coverings S'_i of open subsets S_i covering S; if f is also separable, S'_i may be taken étale over S_i.

[1] The printed text says "on S"; rigidification by a section $g : S \to X$ here concerns the Module $\mathcal{L}$ on X.

3. The main existence theorem: statement

There is not, even conjecturally, an existence statement for Picard preschemes that encompasses all known cases. A "practically necessary" condition, if one may say so, is that $f : X \longrightarrow S$ be proper, ensuring essential finiteness properties, and flat. These conditions are not sufficient, even if S is the spectrum of the algebra of dual numbers $k[t]/(t^2)$ over a field k (say the field **C** of complex numbers) and X has dimension 1. At the time of writing, the most important existence theorems for the Picard prescheme are deduced from the following theorem.

THEOREM 3.1. — Let $f : X \longrightarrow S$ be a morphism of locally noetherian preschemes. Suppose:
149 (i) f is projective;
(ii) f is flat;
(iii) the geometric fibers of f are integral.

Under these conditions, $\mathrm{Pic}_{X/S}$ exists.

The proof, which will be sketched in the next two sections, will show at the same time the following. Let ξ be the section of $\mathrm{Pic}_{X/S}$ corresponding to a very ample sheaf $\mathcal{O}_X(1)$ on X/S, that is, one induced by a projective immersion $X \longrightarrow \mathbf{P}(\mathcal{E})$. There is then an open subset U of $\mathrm{Pic}_{X/S}$, a disjoint union of open subsets quasi-projective over S, such that U is stable under translation by ξ, and $\mathrm{Pic}_{X/S}$ is the increasing union of the open subsets $U - n\xi$, all isomorphic to U. In particular, under the conditions of 3.1, $\mathrm{Pic}_{X/S}$ is separated over S.

REMARK 3.2. — Examples show (for instance with S the spectrum of a discrete valuation ring and X of relative dimension 1 over S) that if hypothesis (iii) in 3.1 is omitted and replaced by the weaker hypothesis that, for every $s \in S$, the homomorphism

$$k(s) \longrightarrow H^0(X_s, \mathcal{O}_{X_s})$$

is an isomorphism, then $\mathrm{Pic}_{X/S}$ need not be separated over S. This happens both when the geometric fibers of f are reduced but an integral geometric generic fiber "bursts" under specialization into two irreducible components, and when the geometric fibers of f are irreducible but an integral geometric generic fiber specializes to a "multiple fiber." The first case occurs, for example, with a conic degenerating into two intersecting lines; D. Mumford supplied an example of the second, with an elliptic curve degenerating into a double elliptic curve. These examples work in every characteristic.

REMARK 3.3. — Under the conditions of 3.1, I do not know whether $\mathrm{Pic}_{X/S}$ is a disjoint union of open subsets of finite type, and hence quasi-projective, over S.[2] The Hilbert polynomials $Q \in \mathbf{Q}[t]$, as in the case of Hilbert schemes ([4], IV), give a decomposition of $\mathrm{Pic}_{X/S}$ into a disjoint union of open subsets $\mathrm{Pic}^Q_{X/S}$, and it seems plausible that the latter are quasi-projective over S; this will at least be seen in the next exposé when f is smooth. Notice that if hypothesis (i) is replaced by the hypothesis that X is locally projective over S (which suffices for 3.1, since the existence of $\mathrm{Pic}_{X/S}$
150 is visibly local on S), it is, on the other hand, easy

to give examples in which $\mathrm{Pic}_{X/S}$ contains connected components that are not of finite type over S. For example, let X_0 be a nonsingular projective algebraic variety over an algebraically closed field k, equipped with an automorphism u and an element ξ of the Néron–Severi group of X_0 such that the $u^n(\xi)$ are pairwise distinct. One may take X_0 to be the product of an elliptic curve E with itself and u to be the automorphism

$$(x, y) \longmapsto (x, y + x)$$

[2]The 1962 erratum reports that D. Mumford answered this question affirmatively by means of the theory of Picard schemes developed here.

of $E \times E$. Let S be the union of two nonsingular irreducible curves meeting at two points a and b. There is a connected principal **Z**-covering P over S; using the actions of **Z** on X_0 defined by u, one obtains an associated fiber space over S with fiber X_0, trivial over $S - a$ and $S - b$ and, in the particular case considered, in fact an abelian scheme over S. It is readily seen that $\mathrm{Pic}_{X/S}$, which is also the fiber space associated with P and the actions of **Z** on $\mathrm{Pic}_{X_0/k}$ through u, contains a connected component isomorphic to

$$P \times \mathrm{Pic}^0_{X_0/k},$$

where Pic^0 denotes the connected component of the identity in Pic; this component is not of finite type over S. [Analogous phenomena also occur in various cases of Picard preschemes nonseparated over S, as considered in 3.3.]

4. Relative Cartier divisors and projective bundles

We shall need only positive divisors and will omit that additional qualifier throughout this section.

Let X be a prescheme. A *Cartier divisor* on X, or simply a divisor on X, is a closed subprescheme D of X defined by an Ideal $\mathcal{J}$ that is an invertible Module, that is, locally generated by a section that is not a zero divisor in $\mathcal{O}_X$. With D we associate the invertible Module

$$\mathcal{L}(D) = \mathcal{J}^{-1}.$$

The canonical injection $\mathcal{J} \longrightarrow \mathcal{O}_X$ gives a canonical homomorphism

$$s_D : \mathcal{O}_X \longrightarrow \mathcal{J}^{-1} = \mathcal{L}(D), \qquad \text{that is,} \quad s_D \in \Gamma(X, \mathcal{L}(D)).$$

Moreover, giving a divisor is essentially equivalent to giving an invertible Module $\mathcal{L}$ on X together with a section s that is nowhere a zero divisor, by associating conversely with such a pair $(\mathcal{L}, s)$ the "divisor" of s, denoted $\mathrm{div}(s)$. For a fixed invertible $\mathcal{L}$ on X, the set of 151

divisors D defining $\mathcal{L}$ is in bijection with the quotient set

$$\Gamma(X, \mathcal{L})^* / \Gamma(X, \mathcal{O}_X^*),$$

where $\Gamma(X, \mathcal{L})^*$ is the subset of $\Gamma(X, \mathcal{L})$ formed by sections that are nowhere zero divisors.

Now suppose given a locally finite-type morphism $f : X \longrightarrow S$, with S locally noetherian for simplicity. Let $\mathcal{J}$ be a coherent Ideal on X, let D be the subscheme of X that it defines, let $x \in X$, and put $s = f(x)$. The following conditions are equivalent:

(i) $\mathcal{J}$ is invertible at x (that is, $\mathcal{J}_x$ is generated by a regular element of $\mathcal{O}_{X,x}$), and D is flat over S at x;

(ii) X and D are flat over S at x, and D_s is a Cartier divisor on the fiber X_s at x;

(iii) X is flat over S at x, and $\mathcal{J}_x$ is generated by an element f_x inducing on X_s a germ that is not a zero divisor.

One then says that D is a *relative Cartier divisor* on X/S (or simply a relative divisor on X/S) at the point in question. Condition (i) shows that D is also a relative divisor at neighboring points of x. Thus, if X and D are flat over S and D is proper over S, the set of $s \in S$ such that D_s is a Cartier divisor in X_s (that is, such that D is a relative Cartier divisor at the points of X_s) is open in S. Furthermore, the preceding definition was made precisely so that relative Cartier divisors are stable under arbitrary base change $S' \longrightarrow S$. Let $\mathrm{Div}(X/S)$ be the set of relative divisors on X/S, and define a contravariant functor in S' over S by

$$\mathrm{Div}_{X/S}(S') = \mathrm{Div}(X \times_S S'/S').$$

Suppose X flat and proper over S. By characterization (ii) of relative Cartier divisors, $\mathrm{Div}_{X/S}$ may be regarded as a subfunctor of the functor $\mathrm{Hilb}_{X/S}$ defined in [4], IV, and the inclusion

$$\mathrm{Div}_{X/S} \longrightarrow \mathrm{Hilb}_{X/S}$$

is "representable by open immersions" (cf. [5], IV, 3.13), by the preceding remarks. The main
152 existence theorem of [4], IV, therefore gives:

PROPOSITION 4.1. — Suppose $f : X \longrightarrow S$ is projective and flat. Then the functor $\mathrm{Div}_{X/S}$ is representable; more precisely, it is represented by an open subset of $\mathrm{Hilb}_{X/S}$.

Using, for a fixed very ample sheaf $\mathcal{O}_X(1)$ on X/S, the canonical decomposition of $\mathrm{Hilb}_{X/S}$ into the open subsets $\mathrm{Hilb}^{Q}_{X/S}$ corresponding to the Hilbert polynomials $Q \in \mathbf{Q}[t]$, one also obtains the analogous decomposition

$$\mathrm{Div}_{X/S} = \coprod_{Q \in \mathbf{Q}[t]} \mathrm{Div}^{Q}_{X/S}$$

as a disjoint union of open subsets quasi-projective over S.

The assignment $D \longmapsto \mathcal{L}(D)$ also gives a functorial homomorphism

$$\mathrm{Div}_{X/S} \longrightarrow \mathrm{Pic}_{X/S} \qquad (+)$$

which we now study; under fairly general conditions it will prove relatively representable ([5], IV, 3). Begin with an element ξ of $\mathrm{Pic}_{X/S}(S')$, take $S' = S$ to simplify notation, and show that the corresponding subfunctor of $\mathrm{Div}_{X/S}$ is representable. First consider the case in which ξ is defined by an invertible Module $\mathcal{L}$ on X. Suppose that X is proper and flat over S and that its geometric fibers over S are integral. This also implies ([2], III, par. 7) that $\mathcal{O}_S \xrightarrow{\sim} f_*(\mathcal{O}_X)$ and that this relation remains valid after every base change $S' \longrightarrow S$. The relative Cartier divisors D on X/S for which $\mathcal{L}(D)$ and $\mathcal{L}$ define the same element of $\mathrm{Pic}(X/S) = \mathrm{Pic}_{X/S}(S)$, equivalently by 2.4 for which $\mathcal{L}(D)$ and $\mathcal{L}$ are isomorphic locally over S, are in bijection with the sections of the quotient sheaf $f_*(\mathcal{L})^*/\mathcal{O}_S^*$. This correspondence is compatible with base change. General considerations of the "Künneth" type from the cited passage (see also [6]) show that the properness of X/S and the flatness of $\mathcal{L}$ over S imply the existence of a coherent Module $\mathcal{Q}$ on S, defined up to unique isomorphism, and a sheaf isomorphism

$$f_*(\mathcal{L}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_S}(\mathcal{Q}, \mathcal{O}_S).$$

[3] The formation of $\mathcal{Q}$ is compatible with base change. Here $f_*(\mathcal{L})^*$ denotes the subsheaf of sets of $f_*(\mathcal{L})$ whose sections over U are the sections of $\mathcal{L}$ over $f^{-1}(U)$ that define relative Cartier divisors
153 on

$f^{-1}(U)/U$, that is, that induce sections which are not zero divisors on the X_s for $s \in U$. Since the fibers X_s are integral, this simply means that the induced sections on the fibers X_s are not identically zero or, equivalently, in terms of local homomorphisms $\mathcal{Q} \longrightarrow \mathcal{O}_S$, that these homomorphisms are surjective, by Nakayama's lemma. Thus the set of sections of $f_*(\mathcal{L})^*/\mathcal{O}_S^*$ is in bijection with the set of invertible quotient Modules of $\mathcal{Q}$, or, by the definition of the projective bundle $\mathbf{P}(\mathcal{Q})$ associated with the coherent Module $\mathcal{Q}$ (cf. [5], V, 2), with the set of sections of $\mathbf{P}(\mathcal{Q})$ over S. This description is compatible with inverse images, and hence yields the following theorem.

THEOREM 4.3. — Let $f : X \longrightarrow S$ be a proper flat morphism with integral geometric fibers, where S is locally noetherian, and let $\mathcal{L}$ be an invertible Module on X. For every S' over S, let $T(S')$ be the set of relative divisors D on $X \times_S S'/S'$ such that $\mathcal{L}(D)$ is isomorphic to $\mathcal{L} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ locally over S', that is, such that $\mathcal{L}(D)$ and $\mathcal{L} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ define the same element of $\mathrm{Pic}(X \times_S S'/S')$. Then

[3] The printed text places $\mathcal{O}_X$ under Hom; since $\mathcal{Q}$ is a Module on S, its dual is taken over $\mathcal{O}_S$.

there is a coherent Module $\mathcal{Q}$ on S, determined up to unique isomorphism, such that the functor T is represented by the projective bundle $\mathbf{P}(\mathcal{Q})$.

COROLLARY 4.4. — If f is moreover projective, then the functorial homomorphism $\mathrm{Div}_{X/S} \longrightarrow \mathrm{Pic}_{X/S}$ is representable by projective morphisms.

Indeed, when X admits a section (respectively, locally a section) over S, the preceding homomorphism is represented by projective bundles (respectively, local projective bundles), by 4.3 and 2.1. When f is quasi-projective, one readily reduces to the preceding case by a descent argument using finite flat quasi-sections of X locally over S.

REMARK 4.5. — Under the conditions of 4.3, the Module $\mathcal{Q}$ on S is not generally locally free, as expressed by the fact that the dimension of the reduced fibers of $\mathcal{Q}$, that is, the dimension of $H^0(X_s, \mathcal{L}_s)$ for varying $s \in S$, may jump. Given a coherent Module $\mathcal{Q}$ on a locally noetherian prescheme S, one readily checks that for a fixed $s \in S$, $\mathcal{Q}$ is free at s if and only if $\mathbf{P}(\mathcal{Q})$ is flat over S at the points above s; in that case it is even smooth over S at those points. Here, with $\mathcal{Q}$ defined in terms of $\mathcal{L}$ as above, this also means that formation of the direct image $f_*(\mathcal{L})$ "commutes with base change near s," or equivalently that $f_*(\mathcal{L})_s \longrightarrow H^0(X_s, \mathcal{L}_s)$ is surjective. This holds, for example, if $H^1(X_s, \mathcal{L}_s) = 0$. Subject to the existence of the preschemes involved, these criteria apply in particular to the universal situation $\mathrm{Div}_{X/S} \longrightarrow \mathrm{Pic}_{X/S}$ and give a necessary and sufficient condition, respectively a sufficient condition, for this morphism to be smooth at a given point of $\mathrm{Div}_{X/S}$. 154

5. Proof of the main existence theorem

Under the conditions of 3.1, choose a very ample Module $\mathcal{O}_X(1)$ on X/S, and let ξ be the corresponding element of $\mathrm{Pic}(X/S)$. For brevity put

$$P(S') = \mathrm{Pic}(X \times_S S'/S'),$$

and suppose first, for simplicity, that X/S admits a section. Let $P^+(S')$ be the subset of $P(S')$ consisting of the classes of invertible $\mathcal{L}$ on $X \times_S S'$ such that

$$R^i f'_*(\mathcal{L}(n)) = 0 \quad \text{for } i > 0 \text{ and every } n \geq 0, \qquad f'_*(\mathcal{L}(n)) \neq 0 \quad \text{for every } n \geq 0.$$

These conditions are stable under base change and hence define a subfunctor P^+ of P, plainly stable under translation by ξ. Using Serre's "Theorems A and B" ([2], III, 2) and the generalities in [5], IV, 5, one readily sees that P is representable if and only if P^+ is; in that case P^+ is represented by an open subset U of the prescheme $\mathrm{Pic}_{X/S}$ representing P, and the latter is the increasing union of the open subsets $U - n\xi$.

For brevity put $D = \mathrm{Div}_{X/S}$, and let D^+ be the inverse image of P^+ under the canonical morphism $D \longrightarrow P$. Thus there is a morphism

$$D^+ \longrightarrow P^+, \tag{+}$$

and D^+ is already known to be represented by an open subset D^+ of the prescheme $D = \mathrm{Div}_{X/S}$, whose existence is guaranteed by 4.1. It follows readily from 4.3 that (+) is represented by smooth projective surjective morphisms, more precisely by projective bundles associated with everywhere nonzero locally free Modules. Indeed, when $\mathcal{L}$ on $X \times_S S'$ is as at the beginning of this section, $f'_*(\mathcal{L})$ is a nonzero locally free Module whose formation commutes with base change; in the notation of 4.3, $\mathcal{Q}$ is then isomorphic to the dual of $f'_*(\mathcal{L})$. We shall deduce the representability of P^+ from the general criterion [5], IV, 4.7. In the cited passage, take $\mathfrak{C}$ to be the class of faithfully 155

flat quasi-compact prescheme morphisms, which are effective epimorphisms by [4], I B. Condition (a) of the cited passage, namely that (+) is representable by morphisms belonging to $\mathfrak{C}$, has

just been verified. So has condition (b), which says that the functor P^+ is compatible with faithfully flat quasi-compact descent; this is immediate. It remains to prove condition (c) of that passage: the equivalence relation R in the prescheme D^+ deduced from the $\mathfrak{C}$-representable morphism $(+)$ is $\mathfrak{C}$-effective, that is, it is effective and $D^+ \longrightarrow D^+/R$ belongs to $\mathfrak{C}$. First observe that the open subsets D^{+Q} of D^+ corresponding to the various virtual Hilbert polynomials $Q \in \mathbf{Q}[t]$ are stable under R, since the fibers of R are connected. It therefore suffices to show, for every Q, that the equivalence relation R^Q induced in D^{+Q} is $\mathfrak{C}$-effective. Now D^{+Q} is quasi-projective, while the equivalence relation R^Q is projective and flat. Thus the hypotheses of [4], III, 6.1 apply and give the required result.

In the general case in which X/S does not necessarily admit a section, one readily reduces to the preceding case by descent, or repeats the preceding proof with the required modification in the definition of P^+.

REMARKS 5.1. — The method used is essentially Matsusaka's method for the projective construction of Picard varieties. The result from [4], III invoked for passage to the effective quotient can also be deduced readily from the existence theorem for Hilbert schemes (see, for example, [6]); classically, these quotients were constructed with Chow coordinates. Notice that formation of the open subset $\mathrm{Pic}^+_{X/S}$ of $\mathrm{Pic}_{X/S}$ and its decomposition into open subsets $\mathrm{Pic}^{+Q}_{X/S}$ quasi-projective over S, according to the Hilbert polynomials of the divisors defining the invertible Modules in question, is compatible with base change, which permits the descent argument.

REMARKS 5.2. — It is possible that $\mathrm{Pic}_{X/S}$ exists whenever $f : X \longrightarrow S$ is proper and flat and the homomorphisms

$$k(s) \longrightarrow H^0(X_s, \mathcal{O}_{X_s}) \qquad (s \in S)$$

are isomorphisms; this last condition then also means that $\mathcal{O}_S \xrightarrow{\sim} f_*(\mathcal{O}_X)$ and that this relation remains valid after every base change $S' \longrightarrow S$. This is at least what is proved for analytic spaces when f is also projective, by a different method, Chow's method unless I am mistaken, explained
156 in [5], IX, 3.1.[4] In that method, passage to the

quotient by a proper flat equivalence relation in an open subset of the scheme of divisors is replaced by passage to the quotient by the projective group in the scheme of immersions of X in $\mathbf{P}^r_S$. This method can probably be adapted to schemes using Mumford's results on passage to the quotient by the projective group [8]. For the present, a written proof exists only when X has "many local sections" over S, for example when X is separable over a complete local ring with algebraically closed residue field.[5] In principle, this method would be more general, since it also gives the existence of Picard preschemes in cases in which they are nonseparated, where the first method must therefore fail. (Technically, the difficulty is that when the geometric fibers of f are no longer integral, the functor considered in 4.3 is represented not by the projective bundle $\mathbf{P}(\mathcal{Q})$ itself but by an open subset of it, leading to the delicate problem of taking the quotient by a flat but nonproper equivalence relation.)

REMARK 5.3. — Notice that the proof given here uses neither a prior construction of the Jacobians of curves or families of curves, nor the theory of abelian varieties or abelian schemes. It therefore differs essentially from traditional treatments such as Lang's book [7] or Chevalley's account [1], which follow the path sketched by A. Weil. Even for the Jacobians of nonsingular curves over an algebraically closed field, say the complex numbers, the construction given here is the only known one providing the very strong properties taken as the definition in Section 1: essentially Chevalley's, but allowing "parameter varieties" with nilpotent elements. That the construction of

[4] The 1962 erratum states that the existence conjecture advanced at the beginning of this remark is false, although Mumford's methods prove a slightly weaker theorem.

[5] The printed text omits "with algebraically closed residue field"; the erratum says that Mumford's counterexample makes this restriction indispensable.

Picard schemes should precede rather than follow the theory of abelian varieties is clear a priori, since in general Picard schemes are neither abelian schemes nor reducible to them. This is already seen for singular curves over an algebraically closed field, where Rosenlicht's "generalized Jacobians," which are not abelian varieties, occur. Moreover, the theory of abelian varieties, and more generally of abelian schemes, becomes much simpler once a general theory of Picard schemes is available. In particular, duality theory for abelian schemes, notably results of Cartier type, then becomes nearly formal (see, for example, [10]). 157

REMARKS 5.4. — Igusa's "compatibility principle" for the Jacobian of a curve degenerating to a singular curve can be properly understood only as an existence theorem for the Picard scheme of a relative curve scheme X/S, not necessarily smooth over S. It is therefore a special case of the main existence theorem 3.1 when the specialized curve is integral, that is, in classical language, irreducible of multiplicity 1. At present, the case of a reducible special curve, even when its components have multiplicity 1, that is, when the special curve is separable over the residue field, lies outside the known existence theorems, except over a complete discrete valuation ring with algebraically closed residue field; compare 5.2. This existence question will certainly arise in a scheme-theoretic geometric construction of the Baily–Satake "compactifications" of the moduli schemes of curves of genus g. (This compactification is currently known only for $g = 1$, through Igusa's work.)

6. Relative existence theorems

We sketch here some useful cases in which the existence of certain Picard schemes implies that of others, allowing various further existence theorems to be deduced from the main theorem 3.1.

PROPOSITION 6.1. — Let $f : X \longrightarrow S$ be a flat projective morphism such that in the Stein factorization $f = f''f'$, the morphism $f' : X \longrightarrow S'$ is flat with integral geometric fibers, and hence satisfies the hypotheses of 3.1, while the finite morphism $f'' : S' \longrightarrow S$ is flat. Then $\mathrm{Pic}_{X/S}$ exists and, with the notation introduced in [4], II, C 2, there is a canonical isomorphism

$$\mathrm{Pic}_{X/S} \xrightarrow{\sim} \Pi_{S'/S} \mathrm{Pic}_{X/S'} .$$

To prove this, one first establishes an isomorphism of functors

$$\mathrm{Pic}_{X/S} \xrightarrow{\sim} \Pi_{S'/S} \mathrm{Pic}_{X/S'},$$

uses 3.1, which implies that $\mathrm{Pic}_{X/S'}$ is representable, and then uses the structure of $\mathrm{Pic}_{X/S'}$ described in Section 3, which implies that every finite subset of a fiber of $\mathrm{Pic}_{X/S'}$ over S is contained in an affine open subset, to obtain the existence of $\Pi_{S'/S} \mathrm{Pic}_{X/S'}$. 158

For example, when X is a scheme that is the disjoint union of schemes X_i over S satisfying the conditions of 3.1, statement 6.1 reduces to the trivial statement

$$\mathrm{Pic}_{X/S} \xrightarrow{\sim} \prod_i \mathrm{Pic}_{X_i/S} .$$

COROLLARY 6.2. — Let $f : X \longrightarrow S$ be a projective flat morphism with locally integral geometric fibers (for example, a projective normal morphism). Then $\mathrm{Pic}_{X/S}$ exists.

In this case, moreover, S' is an étale covering of S; this holds whenever f is separable, that is, flat with geometrically reduced fibers. One checks that the structure theorem stated in Section 3 for $\mathrm{Pic}_{X/S}$ remains valid, by virtue of the analogous structure of $\mathrm{Pic}_{X/S'}$.

On the other hand, a descent procedure gives a relative existence theorem whose reach depends on the solution of the nonflat descent questions raised in [4], I, A (c). We spell out only the following particular case:

PROPOSITION 6.3. — Let $f : X \longrightarrow S$ be a proper morphism, and let X_1 and X_2 be two subschemes of X, flat over S, defined by two coherent Ideals $\mathcal{I}_1$ and $\mathcal{I}_2$ such that $\mathcal{I}_1 \cap \mathcal{I}_2 = (0)$ and $\mathcal{O}_X/(\mathcal{I}_1 + \mathcal{I}_2)$ is flat over S; that is, the subscheme of X that is the supremum of X_1 and X_2 is X, while their infimum Z is flat over S.[6] Suppose, in addition, that for every $s \in S$ the homomorphisms

$$k(s) \longrightarrow H^0(X_{i,s}, \mathcal{O}_{X_{i,s}}), \qquad i = 1, 2,$$

are bijective. Then the natural homomorphism of functors

$$\mathrm{Pic}_{X/S} \longrightarrow \mathrm{Pic}_{X_1/S} \times \mathrm{Pic}_{X_2/S}$$

is representable by affine morphisms. Consequently, if $\mathrm{Pic}_{X_1/S}$ and $\mathrm{Pic}_{X_2/S}$ exist, then so does $\mathrm{Pic}_{X/S}$, and the canonical morphism

$$\mathrm{Pic}_{X/S} \longrightarrow \mathrm{Pic}_{X_1/S} \times \mathrm{Pic}_{X_2/S}$$

is affine.

By faithfully flat descent one reduces to the case in which Z admits a section over S, thereby defining sections of X, X_1, and X_2 over S and allowing the automorphisms in the structures under consideration to be eliminated as explained in 2.5. The proof then consists in observing that the
159 data of an invertible

"rigidified" Module $\mathcal{L}$ on X are equivalent to the data of a triple $(\mathcal{L}_1, \mathcal{L}_2, u)$, where $\mathcal{L}_i$ is a "rigidified" invertible Module on X_i and u is an isomorphism from $\mathcal{L}_1|Z$ to $\mathcal{L}_2|Z$ compatible with the rigidifications. One need only verify that, for fixed $\mathcal{L}_1$ and $\mathcal{L}_2$, the datum of u is expressed by a section of a suitable scheme over S that is affine over S; this is easy. One readily deduces from 6.3:

COROLLARY 6.4. — Let X be a proper separable scheme over a field k, and let X_i be the irreducible components of X. If the $\mathrm{Pic}_{X_i/k}$ exist, then so does $\mathrm{Pic}_{X/k}$, and the canonical morphism

$$\mathrm{Pic}_{X/k} \longrightarrow \prod_i \mathrm{Pic}_{X_i/k}$$

is affine.

Combined with 6.2, this proves, for example, the existence of $\mathrm{Pic}_{X/k}$ whenever X is a projective separable scheme over a field k. When X is no longer separable over k, there is also a reduction result, using an argument due to Oort [11]. The method also applies over an arbitrary base scheme, a case useful, for example, in proving in the next exposé the finiteness result announced in 3.3. To avoid an overly technical statement, let us restrict ourselves to a base field:

PROPOSITION 6.5. — Let X be a proper scheme over a field k, and let X_0 be a subscheme with the same underlying set, hence defined by a nilpotent Ideal on X. Then the functorial morphism

$$\mathrm{Pic}_{X/k} \longrightarrow \mathrm{Pic}_{X_0/k}$$

is representable by affine morphisms. In particular, if $\mathrm{Pic}_{X_0/k}$ exists, then so does $\mathrm{Pic}_{X/k}$, and the morphism $\mathrm{Pic}_{X/k} \longrightarrow \mathrm{Pic}_{X_0/k}$ is affine.

Combining this with 6.4, one readily concludes:

COROLLARY 6.6. — Let X be a projective prescheme over a field k. Then $\mathrm{Pic}_{X/k}$ exists.

REMARK 6.6. — It is extremely plausible that $\mathrm{Pic}_{X/k}$ exists for every proper scheme X over a field k.[7] The preceding results reduce this question to the case in which k is algebraically closed and

X is integral. There is then an integral scheme X', projective over k, and a dominant morphism
160 $g : X' \longrightarrow X$ by Chow's lemma. It would therefore suffice to show that the
corresponding functorial morphism

$$\mathrm{Pic}_{X/k} \longrightarrow \mathrm{Pic}_{X'/k}$$

is representable, and no doubt representable by affine morphisms, since $\mathrm{Pic}_{X'/k}$ is already known to be representable. This raises nonflat descent questions that remain unresolved for the present. Notice that if one restricts the functor $\mathrm{Pic}_{X/k}$ to reduced preschemes, with X proper and integral over an algebraically closed field k, one does obtain a representable functor, as Chevalley [1] proved when X is normal and Seshadri [12] proved in general by a descent method. In our notation, however, the scheme constructed by these authors is not $\mathrm{Pic}_{X/k}$ but $(\mathrm{Pic}_{X/k})_{\mathrm{red}}$, the reduced scheme corresponding to $\mathrm{Pic}_{X/k}$.

REMARK 6.7. — More generally, let $f : X' \longrightarrow X$ be a surjective morphism of proper preschemes over k. Nonflat descent considerations lead one to conjecture that

$$\mathrm{Pic}_{X/k} \longrightarrow \mathrm{Pic}_{X'/k}$$

is an affine morphism.[8] This would imply in particular, after dividing by the connected components of the identity elements, that the corresponding homomorphism on Néron–Severi groups is injective modulo torsion. This can be checked by intersection theory when X and X' are nonsingular. The answer does not seem to be known in any other case.

REMARK 6.8. — Contrary to what one might think, Picard schemes of algebraic schemes with nilpotent elements are useful, and even indispensable, in various questions. Thus, when X is a projective scheme, say smooth over k,[9] and Y is a hyperplane section, one must consider the "infinitesimal neighborhoods" X_n of Y of all orders and the Picard schemes $\mathrm{Pic}_{X_n/k}$. When X is irreducible of dimension ≥ 4 (respectively ≥ 3), the canonical morphism

$$\mathrm{Pic}_{X/k} \longrightarrow \mathrm{Pic}_{X_n/k} \qquad (n \text{ large})$$

is an isomorphism (respectively induces an isomorphism on the inverse images of the torsion subgroups of Néron–Severi), a result that will be useful in the qualitative study of Picard schemes in the next exposé. Likewise, considering the Picard schemes of certain curves with nilpotent elements, together with the fundamental theorems of formal geometry [3], makes it possible to verify, in the equal-characteristic case, a conjecture of Mumford: for every complete normal noetherian local ring A of dimension 2, the divisor class group of A may be regarded as the set of k-rational points of an 161

algebraic group G over the residue field k; moreover, G is canonically determined once a coefficient field in A is chosen. When A has arbitrary dimension, it is plausible that there exists a pro-algebraic group over k playing the same role as G above. In the case in which $\mathrm{Spec}(A)$ can be "desingularized," it is constructed as an inverse limit of Picard schemes of suitable projective schemes over k with nilpotent elements.

[6] The final letter is printed "X," but the immediately preceding condition concerns the flatness of $\mathcal{O}_X/(\mathcal{J}_1 + \mathcal{J}_2)$ over S and therefore defines Z to be flat over S.

[7] The 1962 erratum reports that Murre had just answered this question affirmatively.

[8] The 1962 erratum reports that this question has an affirmative answer and refers to the last paragraph of the Comments on the next exposé.

[9] The printed text reads "regular"; the erratum prescribes "smooth over k."

BIBLIOGRAPHY

[1] CHEVALLEY (Claude). – Sur la théorie de Picard, *Amer. J. of Math.*, vol. 82, 1960, pp. 435–490.

[2] GROTHENDIECK (A.) and DIEUDONNÉ (J.). – Éléments de géométrie algébrique, Chapitre I et suivants. – Paris, Presses universitaires de France, 1960–1961 (Institut des Hautes Études Scientifiques, Publications mathématiques 4, 8, ...).

[3] GROTHENDIECK (Alexander). – Géométrie formelle et géométrie algébrique, *Séminaire Bourbaki*, vol. 11, 1958/59, no. 182, 28 pp.

[4] GROTHENDIECK (Alexander). – Techniques de descente et théorèmes d'existence en géométrie algébrique, I–IV, *Séminaire Bourbaki*, vol. 12, 1959/60, no. 190, 29 pp. and no. 195, 22 pp.; vol. 13, 1960/61, no. 212, 20 pp. and no. 221, 28 pp.

[5] GROTHENDIECK (Alexander). – Techniques de construction en géométrie analytique, I–X, *Séminaire Cartan*, vol. 13, 1960/61: Déformation des structures analytiques complexes, nos. 7–17.

[6] GROTHENDIECK (Alexander). – Séminaire de géométrie algébrique, edited by Lichtenbaum, Harvard University, 1961 (forthcoming).

[7] LANG (Serge). – *Abelian varieties*. – New York, Interscience Publishers, 1959 (Interscience Tracts in pure and applied Mathematics, 7).

[8] MUMFORD (D.). – An elementary theorem in geometric invariant theory, *Bull. Amer. math. Soc.*, vol. 67, 1961, pp. 483–486.

[9] MUMFORD (D.). – The topology of normal singularities of an algebraic surface and a criterion for simplicity. – Paris, Presses universitaires de France, 1961 (Institut des Hautes Études Scientifiques, Publications mathématiques, 9).

[10] MUMFORD (D.) and TATE (J.). – Séminaire de géométrie algébrique, Harvard University, Spring Term 1962 (forthcoming).

[11] OORT (F.). – Sur le schéma de Picard, *Bull. Soc. math. France*, vol. 90, 1962 (forthcoming).

[12] SESHADRI (C. S.). – Thesis, forthcoming in *Annali di Matematica*.

Erratum to Exposé no. 232

303

7. TDTE V: Picard schemes. Existence theorems.

[Bourbaki Seminar, vol. 14, 1961/62, no. 232, 19 pp.]

Page 7, Remark 3.3. — The question raised here was answered affirmatively by MUMFORD (see the Comments on the next exposé).
Page 13, Remark 5.2. — As I point out at the beginning of the next exposé, the existence conjecture advanced here is false; nevertheless, MUMFORD succeeds by his methods in proving a slightly weaker theorem.
Page 14, line 7. — After "complete," add "with algebraically closed residue field." The counterexample of MUMFORD just mentioned shows, moreover, that this restriction is indispensable.
Page 17, Remark 6.6. — As we point out at the beginning of the next exposé, the question raised here has just been answered affirmatively by MURRE.
Page 18, Remark 6.7. — The question raised here has an affirmative answer (cf. the last paragraph of the Comments on the next exposé).
Page 18, Remark 6.8, fourth line. — For "regular," read "smooth over k."

Chapter 8

Exposé 236 – Picard schemes: general properties

221

Séminaire BOURBAKI
14th year, 1961/62, no. 236
May 1962

DESCENT TECHNIQUE AND EXISTENCE THEOREMS IN ALGEBRAIC GEOMETRY
VI. PICARD SCHEMES: GENERAL PROPERTIES

by Alexander GROTHENDIECK

0. Complements to the preceding exposé ([3], V)[(1)]

There has been some progress on the existence questions for Picard preschemes raised in [3], V:

a. (Mumford). It is not true in general that when $f : X \longrightarrow S$ is a projective separable morphism (= flat with separable fibers), the Picard prescheme $\mathrm{Pic}_{X/S}$ exists, even if the fibers of f have dimension 1 and S is the spectrum of a complete discrete valuation ring. An example is obtained by taking $S = \mathrm{Spec}\,\mathbf{R}[[t]]$ and X to be the subscheme of $\mathbf{P}^2_S$ defined by the equation (where x, y, z are homogeneous variables)

$$x^2 + y^2 = tz^2.$$

This is a conic degenerating geometrically into two intersecting lines, while the special fiber over $\mathbf{R}$ is nevertheless irreducible (it is given by $x^2 + y^2 = 0$ over $\mathbf{R}$). One sees readily that after the étale extension $S' \longrightarrow S$, with $S' = \mathrm{Spec}\,\mathbf{C}[[t]]$, the Picard prescheme of X'/S' exists and has an explicit description as a disjoint union of copies of $\widetilde{S}$, where $\widetilde{S}$ is obtained from S by doubling the origin infinitely many times. The descent datum on $\mathrm{Pic}_{X'/S'}$ for $S' \longrightarrow S$ (given here by the action of the Galois group $\mathbf{Z}/2\mathbf{Z}$ of S' over S) is readily seen not to be effective: the group interchanges certain doubled points, so that some orbits are not contained in an affine open subset. Mumford can, however, show that if $f : X \longrightarrow S$ is a projective separable morphism such that, for every $s \in S$, the irreducible components of X_s are geometrically irreducible over $k(s)$, then $\mathrm{Pic}_{X/S}$ exists. His proof relies on a refinement of his quotient theorem; cf. [7]. Notice, on the other hand, that without any hypothesis on the irreducible components of the fibers X_s, it remains possible
222 that the scheme $\mathrm{Pic}^{\tau}_{X/S}$ (to be introduced below) nevertheless exists.

b. (Murre). Let X be a proper scheme over a field. Then $\mathrm{Pic}_{X/k}$ exists. The proof partly follows Chevalley's proof [1] and makes essential use of the group structure of the Picard functor.

For some complements concerning the theory of Picard schemes, especially in relation to abelian schemes, it will be useful to consult [7]. Finally, a notable gap in the present exposé is the absence of "equivalence criteria" permitting comparison of the Picard scheme of a projective scheme with those of its hyperplane sections. The key theorems for developing such criteria are in [5], to which the existence theorems for Picard schemes must be added.

1. Topological properties of commutative group preschemes

Let first k be a field and G a group prescheme over k. Since the identity element e, being rational over k, is necessarily closed, it follows at once that the diagonal of $G \times_k G$ is closed, and hence G is separated: every group prescheme over a field is separated. Denote by G° the connected component of the identity element e. Since e is rational over k, G° is in fact geometrically connected; that is, formation of G° is compatible with extension of the base field. It also follows that G° is stable under multiplication as a set, and when G is locally noetherian, so that G° is open, it may be regarded as an open subgroup of G. Henceforth suppose that G is locally of finite type over k. Then G° is geometrically irreducible and of finite type over k. Indeed, one may suppose k algebraically closed, and then also G reduced (for G_{red} is then a subgroup of G, since $G_{\mathrm{red}} \times_k G_{\mathrm{red}}$ is again reduced), hence smooth over k on a nonempty open subset and therefore everywhere, by translating that open subset. Thus G is locally irreducible, so its irreducible components coincide with its connected components, and G° is irreducible. Let U be an affine neighborhood of e in G°. Using the irreducibility of G°, one sees at once that $U \cdot U = G^\circ$, proving that G° is quasi-compact and hence of finite type over k.

For simplicity suppose that G is commutative. For every integer $n > 0$, let $G^{(n)}$ be the inverse image of G° under the nth-power homomorphism φ_n in G; thus $G^{(n)}$ is an open subgroup of G. Put

$$G^\tau = \bigcup_n G^{(n)}.$$

223

$$G^\sigma = \bigcup_{(n,p)=1} G^{(n)}, \qquad G^\rho = \bigcup_h G^{(p^h)},$$

where p is the characteristic exponent of the field k. These are open subgroups of G, satisfying

$$G^\sigma \cap G^\rho = G^\circ, \qquad G^\sigma G^\rho = G^\tau.$$

REMARK. — One may construct the quotient group scheme $G/G^\circ = N$ (cf. [3], IV), and then define G^τ, G^σ, and G^ρ as the inverse images in G of the torsion subgroup of N (respectively its p-primary component, respectively the natural complement of the latter, the sum of the q-primary components for primes $q \neq p$). Notice in this connection that N is a discrete separable group scheme over k. Thus, after choosing an algebraic closure $\overline{k}$ of k, with Galois group π, it is identified with an ordinary discrete group on which π acts by automorphisms. This gives the evident interpretation of the construction of the torsion subgroup and of its decomposition into q-primary components. When G is the Picard scheme of a proper scheme X over k, N may be called the (reduced) *Néron–Severi scheme* of X over k. When G°_{red} is a group subscheme of G°, which occurs in particular whenever k is perfect or G° is proper over k (for example, when X is geometrically normal), one should also introduce the quotient $N' = G/G^\circ_{\mathrm{red}}$, which tends to behave better than N under specialization, that is, when X varies in a family of algebraic schemes.

Now let S be a locally noetherian prescheme, and let G be a group prescheme over S, locally of finite type over S. We do not assume G to be of finite type over S, or separated over S. Put

$$G^\circ = \bigcup_{s\in S}(G_s)^\circ,$$

and, when G is commutative,

$$G^\tau = \bigcup_{s\in S}(G_s)^\tau, \qquad G^\sigma = \bigcup_{s\in S}(G_s)^\sigma, \qquad G^\rho = \bigcup_{s\in S}(G_s)^\rho.$$

224

These are subsets of G stable under multiplication in G, which plainly does not imply that they can be defined by group subpreschemes of G. In particular, it seems that in general there is no group subprescheme of G whose underlying set is G°. Of course, if one of these sets is open, then with its induced structure it is an open group subprescheme of G. We shall see that this is always so for G^τ. Thus, from the standpoint of representable functors, and especially from the standpoint of specialization, numerical equivalence behaves more satisfactorily than algebraic equivalence. Here are the main general properties of the sets just defined:

THEOREM 1.1. — The sets G°, G^τ, G^σ, and G^ρ are locally constructible. Moreover:
(i) G° is quasi-compact over S. When the G_s° are proper and G is separated over S, G° is proper over S and hence closed in G.
(ii) G^τ is open. If G° is closed, then so is G^τ.
(iii) If G° is closed, then so is G^σ, provided one is in equal characteristic, that is, all residue fields of S have the same characteristic. If G° is closed and $G \longrightarrow S$ is universally open at the points of G^σ (cf. Corollary 1.5 below), then $G^\sigma \longrightarrow S$ is universally open.
(iv) If G° is closed, then so is G^ρ. Suppose one is in equal characteristic and, for every integer $n > 0$ with $(n, p) = 1$, the nth-power homomorphism in G is open. Then G^ρ is open.

We give indications for the proof. The local constructibility of G° and its quasi-compactness over S are contained in the following lemma:

LEMMA 1.2. — Suppose $S \neq \emptyset$. Then there are a nonempty open subset U of S, a finite-type group scheme H over U with connected fibers, and a homomorphism of group schemes $H \longrightarrow G|U$ that is an open immersion with image $G^\circ|U$.

To prove the lemma, one may suppose S irreducible and let η be its generic point. After the base change $S' = \mathrm{Spec}(\mathcal{O}_{S,\eta}) \longrightarrow S$, one obtains a group scheme G' over the local artinian ring $\mathcal{O}_{S,\eta} = A$ with an open finite-type group subscheme G'° over A, as stated above. It therefore comes from a finite-type group scheme H over an open neighborhood U of η, and the canonical
225 immersion $G'^\circ \longrightarrow G'$ comes from an

open immersion $H \longrightarrow G|U$, which is a group-scheme homomorphism after U is made sufficiently small. If U is sufficiently small, the fibers of H are connected and all have the same dimension, equal to that of the fibers of G. It follows that for every $s \in U$ the image of H_s in G_s is exactly G_s° (after again shrinking U), proving 1.2.

The second assertion of (i) is contained in the following lemma, applied to a quasi-compact open neighborhood of G° in G:

LEMMA 1.3. — Let X be a finite-type separated prescheme over a locally noetherian S, let g be a section of X over S, and let X° be the union, in the X_s, of the connected components containing $g(s)$. If $s \in S$ and X_s° is proper over $k(s)$, then there is an open neighborhood U of s such that $X^\circ|U$ is proper over U, and hence closed in $X|U$.

By faithfully flat descent on the base, one reduces to the case in which S is the spectrum of a complete local ring and s is its closed point. Applying [2], III, 5.5.1, one sees that X is the disjoint union of two open subsets X' and X'', the first proper over S and satisfying $X'_s = X_s^\circ$. This

reduces to the case $X = X'$, that is, X proper over S. One then uses a standard argument based on the valuative criterion for properness of a subset (omitted from [2], Chapter II).

We now prove that G^τ is open. Equivalently, since formation of G^τ (as well as of G°, G^σ, and G^ρ) commutes with extension of the base, for every section g of G over S, the set $g^{-1}(G^\tau)$ is open. This means two things:

a. If $y \in S$ and $g(y) \in G^\tau_y$, then $g(y') \in G^\tau_{y'}$ for every $y' \in \overline{\{y\}}$ sufficiently near y.

b. If $y' \in S$ and $g(y') \in G^\tau_{y'}$, then $g(y) \in G^\tau_y$ for every generalization y of y'.

For (a), some integer $n > 0$ satisfies $g^n(y) \in G^\circ_y$. Since G° is constructible, $(g^n)^{-1}(G^\circ)$ is constructible, and hence $g^n(y') \in G^\circ_{y'}$ for $y' \in \overline{\{y\}}$ sufficiently near y. For (b), the powers $g(y')^n = g^n(y')$ remain in a quasi-compact open subset of $G_{y'}$, since they lie in finitely many classes modulo $G^\circ_{y'}$. Consequently there is a quasi-compact open subset U of G containing the $g^n(y')$ and therefore also their generalizations $g^n(y)$. Thus the powers of $g(y)$ remain in a quasi-compact open subset of $G_{y'}$, which readily implies $g(y) \in G^\tau_y$. 226

Suppose G° is closed; we prove the same for G^τ. Since G^τ is already known to be open and hence locally constructible, it remains to prove that it is stable under specialization. This follows because it is a union of closed sets, namely the inverse images of the closed set G° under the nth-power homomorphisms φ_n.

The same argument proves that G^σ and G^ρ are closed if G° is closed (in the first case subject to equal characteristic), once it is shown that G^σ and G^ρ are locally constructible. For $x \in G^\tau$, let $\nu(x)$ be the smallest integer $n > 0$ such that the nth-power homomorphism φ_n sends x into G°. Then G^σ (respectively G^ρ) consists of those $x \in G^1$ for which $\nu(x)$ is prime to p (respectively a power of p). The constructibility assertion now follows from the more precise statement:

LEMMA 1.4. — The function ν on G^τ is locally constructible.

Indeed, for every integer $n > 0$, the set of $x \in G^\tau$ such that $\nu(x) = n$ is locally constructible: it is the difference between $\varphi_n^{-1}(G^\circ)$ and the union of the $\varphi_d^{-1}(G^\circ)$, where d ranges over the proper divisors of n. Since G° is locally constructible, so are the $\varphi_d^{-1}(G^\circ)$, and therefore so is the preceding difference.

Suppose now that G° is closed and $G \longrightarrow S$ is universally open at the points of G^σ. We prove that $G^\sigma \longrightarrow S$ is open, that is, it maps a neighborhood in G^σ of $x \in G^\sigma$ onto a neighborhood of $y = f(x)$. Since G^σ is locally constructible, it suffices to prove that for every generalization y' of y there is a generalization x' of x in G^σ lying over y'. Base change reduces this to the case in which S is the spectrum of a discrete valuation ring and y, y' are respectively its closed and generic points. Since $G \longrightarrow S$ is open at x, there is a generalization x_1 of x in G above y'. After a further base change, one may also suppose that there is a section g of G over S with $x = g(y)$. If $k(y')$ has characteristic zero, take any generalization x' of x in G above y'. It belongs to G^τ because G^τ is open, hence to G^σ because $G^\sigma_{y'} = G^\tau_{y'}$. If $k(y')$ has characteristic $p > 0$, write

$$\nu(g(y')) = p^h m \qquad \text{with } (m, p) = 1;$$

227

choose integers a and b such that $ap^h + bm = 1$, and set

$$g_1 = g^{ap^h}, \qquad g_2 = g^{bm}, \qquad \text{so that} \qquad g = g_1 g_2.$$

By construction, $g_1(y') \in G^\sigma$ and $g_2(y') \in G^\rho$. Since G° is closed, it follows that $g_1(y) \in G^\sigma$ and $g_2(y) \in G^\rho$. But

$$g(y) = g_1(y) g_2(y) \in G^\sigma,$$

so $g_2(y) \in G^\sigma$, and therefore

$$g_2(y) \in G^\sigma_y \cap G^\rho_y = G^\circ_y.$$

[1]The printed text reads "X," which is not defined in this passage; all four loci under consideration are subsets of G.

The hypothesis, together with the fact that S is the spectrum of a discrete valuation ring, implies that $G - (G_y - G_y^\circ)$ is an open subset of G on which $G \longrightarrow S$ induces an open morphism. Thus a "quasi-section" passes through every point of G_y°. After making a further extension of the base S, one may consequently suppose that there is a section g_2' of G° over S such that $g_2'(y) = g_2(y)$. Set $g' = g_1 g_2'$. Then, by construction, $g'(y) = g(y) = x$, and

$$g'(y') = g_1(y')g_2'(y') \in G^\sigma.$$

Hence $g'(y') = x'$[2] is a generalization of x in G^σ lying above y'. This proves (iii).

It remains to prove the last assertion, (iv). We must show that if $x \in G^\rho$, then every generalization x' of x belongs to G^ρ. By taking images under φ_{p^h} for a suitable h[3], we may suppose that in fact $x \in G^\circ$. For every integer n prime to the characteristic, x then lies in the image of φ_n (because the nth-power map in a connected group of finite type over a field of characteristic prime to n is surjective). Since φ_n is open, x' also lies in the image of φ_n. More precisely, let U be a quasi-compact open subset of G^τ containing G_y°. Then $x' \in \varphi_n(U)$ for every n prime to p. Taking n to be a common multiple of the factors prime to p of the $\nu(z)$, for $z \in U$, gives $x' \in G^\rho$.

Assertion (1.1), (iii), is completed by the following:

COROLLARY 1.5. — Let $n > 1$ be an integer such that the nth-power homomorphism $\varphi_n : G \longrightarrow G$ is universally open (for example, étale), put

$$G^{(n)} = \varphi_n^{-1}(G^\circ),$$

228 and suppose that the connected fibers G_s° contain no "additive

component" (that is, after extending $k(s)$ to an algebraic closure, the resulting group contains no subgroup isomorphic to $\mathbf{G}_a$). Then $G \longrightarrow S$ is universally open at the points of the $G^{(n^h)}$. In particular, if the G_s° contain no additive component and if, for every integer $n > 1$, the homomorphism φ_n is universally open at the points of residue characteristic prime to n[4], then $G \longrightarrow S$ is universally open at the points of G^σ. Hence, by (1.1), (iii), if G° is closed, $G^\sigma \longrightarrow S$ is universally open. In these cases, $s \longmapsto \dim G_s$ is locally constant on S.

Indeed, the hypothesis implies that the kernel ${}_nG$ of φ_n is universally open over S. Hence $G \longrightarrow S$ is universally open at the points of ${}_nG$, and therefore, on replacing n by n^h, at the points of ${}_{n^h}G$. The hypothesis on the fibers G_s° says exactly that the points of order a power of n in G_s are dense in the union of the $G^{(n^h)}$. It follows readily that $G \longrightarrow S$ is universally open at every point of this union, in particular along the identity section, and hence that $s \longmapsto \dim G_s$ is locally constant.

REMARK 1.6. — Recall that a morphism $X \longrightarrow S$ is called universally open if it maps every open subset to an open subset and retains this admirable property after every base change $S' \longrightarrow S$. This also means (when S is locally noetherian and $X \longrightarrow S$ is locally of finite type) that every irreducible component of X dominates S, and that this property is preserved after every base change $S' \longrightarrow S$. In both assertions it is enough (under the finiteness hypotheses above) to test base changes $S' \longrightarrow S$ in which S' is the spectrum of a discrete valuation ring (complete with algebraically closed residue field, if desired ...). The definition extends in the evident way to a subset Z of X (such as the subset G^σ of G). It is a perfectly normal phenomenon, even for a projective smooth morphism $X \longrightarrow S$ with geometrically connected fibers (for example, the fiber square

[2]The printed text reads "$g(y') = x'$." The section constructed in the preceding sentence, whose generic fiber is being computed, is g'.

[3]The printed text reads φ_{n^h}. For $x \in G^\rho$, the definition on p. 223 gives a power of the characteristic exponent p that sends x into G°.

[4]The printed text reads "prime to p." In this statement quantified by n, the étaleness criterion and the argument using the kernels ${}_nG$ require residue characteristic prime to n.

of the modular family of elliptic curves over $S = \operatorname{Spec} \mathbf{C}[j]$), that $\operatorname{Pic}_{X/S}$ is not universally open over S: an irreducible component of $\operatorname{Pic}_{X/S}$ may lie entirely above a single point of S. This is connected with the fact that the rank of the Néron–Severi group of the fibers of X/S can jump upward ("complex multiplication" phenomena). On the other hand, (1.5) assures us that in favorable cases $\operatorname{Pic}^{\sigma}_{X/S}$ (and most often even $\operatorname{Pic}^{\tau}_{X/S}$, it seems) is universally open over S. 229

Finally, here is a useful case in which, exceptionally, G° consents to be open:

COROLLARY 1.7. — Suppose that $G \longrightarrow S$ is universally open at the points of G° (cf. (1.5)), and that the fibers G_s are separable and hence smooth over $k(s)$ (the latter condition is automatically satisfied in residue characteristic zero, by a theorem of Cartier). Then G° is open in G. If, in addition, S is reduced, then G° is smooth (in particular, flat) over S.

The first assertion can be sharpened as follows. If, for a given $s \in S$, $G \longrightarrow S$ is universally open at the points of G°_s and G°_s is separable over $k(s)$, then G° is a neighborhood of G°_s (and the hypothesis made at s will moreover remain valid at neighboring points). This statement, which is independent of any group structure, will be proved in [2], IV, §7. The last assertion in (1.7), also independent of any group structure or any question about a connected component, is a special case of a flatness criterion given in [IV], §5, which more generally implies:

COROLLARY 1.8. — Let U be an open subset of a fiber G_s such that $G \longrightarrow S$ is universally open at the points of U (cf. (1.5)). If G_s is separable over $k(s)$ and S is reduced at s, then G is flat over S at the points of U (and thus, in this case, smooth over S at the points of U).

REMARKS 1.9. — I do not know whether, when G is separated over S, G° or G^{τ} is always closed in G; this seems unlikely. There are, at any rate, obvious counterexamples if the separatedness hypothesis is dropped. For example, take the affine line with its origin doubled countably infinitely many times. This gives a group scheme G over the affine line all of whose fibers are reduced to the identity group except for one, whose fiber is $\mathbf{Z}$. This group scheme is an open subgroup scheme, namely the closure of the identity section, in the Picard prescheme of the S-scheme X corresponding to a family of conics degenerating into two intersecting lines.

Moreover, even starting with a finite flat group scheme G over the spectrum S of a discrete valuation ring V, so that G° is reduced to the identity section and hence closed, various conclusions become false if some hypotheses are dropped. Suppose that V has equal characteristic $p > 0$, and let G be the kernel of the homomorphism $\mathbf{G}_a \longrightarrow \mathbf{G}_a$ defined by the homomorphism of functors

$$f \longmapsto f^p - tf,$$

where t is a uniformizer of V. 230

(Recall that, by definition, the "additive group" $\mathbf{G}_a$ over S represents the functor $S' \longmapsto \Gamma(S', \mathcal{O}_{S'})$.) Thus, as an S-scheme,

$$G = \operatorname{Spec} V[x]/(x^p - tx)$$

is the union of p intersecting "lines," representing a group $\mathbf{Z}/p\mathbf{Z}$ that degenerates into an infinitesimal group of additive type. In this example, $G^{\circ} = G^{\sigma}$ (and it is one of the p lines), so it is not open in G. In mixed characteristic with residue characteristic $p > 0$, start with the group scheme $H = \mu_p$, the kernel of the pth-power map in $\mathbf{G}_m$. Thus

$$H = \operatorname{Spec} V[x]/(x^p - 1);$$

this is again the union of p intersecting lines, representing a group $\mathbf{Z}/p\mathbf{Z}$ (in characteristic zero) that degenerates into an infinitesimal group of multiplicative type in characteristic p. Let H' be the "constant group scheme" defined by the ordinary finite group $\mathbf{Z}/p\mathbf{Z}$, that is, the disjoint union of $\mathbf{Z}/p\mathbf{Z}$ copies of S, or equivalently $\operatorname{Spec} V^{\mathbf{Z}/p\mathbf{Z}}$. Then $G = H \times_S H'$ describes a group $(\mathbf{Z}/p\mathbf{Z})^2$ in

characteristic zero, degenerating in characteristic p into an infinitesimal group times $\mathbf{Z}/p\mathbf{Z}$. Here G^ρ is the union of the identity section and the special fiber, so G^ρ is not open, whereas G^σ is the union of the identity section and the generic fiber, so it is not closed, contrary to what is asserted in the equal-characteristic cases in (1.1), (iii) and (iv). These are, of course, phenomena connected with characteristic $p > 0$. The preceding results give:

COROLLARY 1.10. — Suppose that all residue characteristics of S are zero, so that $G^\tau = G^\sigma$ and $G^\circ = G^\rho$. Then $G^\tau = G^\sigma$ is open; it is even open and closed if G is separated over S and the G_s° are proper. Under the same hypothesis, $G^\circ = G^\rho$ is proper over S and hence closed in G. Finally, suppose that for every integer $n > 1$ the nth-power homomorphism in G is universally open. Then G° is open; and if, moreover, the G_s° have no additive component (for example, if the G_s° are proper, as above), then $G^\tau \longrightarrow S$ is universally open, and is even smooth if S is reduced.

We finally record the following easy result:

PROPOSITION 1.11. — There is an open subset U of S such that the set G' of points of G at which G is smooth (respectively flat) over S is the underlying set of an open subgroup scheme
231 induced from $G|U$. Furthermore, every section of G over U is a section of G' over U.

COROLLARY 1.12. — If G is smooth (respectively flat) over S at the points of the identity section, then it is so at the points of every section of G over S and at every point of G°. If, moreover, for every integer $n > 0$ the nth-power homomorphism $\varphi_n : G \longrightarrow G$ is étale at the points of characteristic prime to n, then G is smooth (respectively flat) over S at every point of G^σ.

2. Application to local properties of Picard schemes

THEOREM 2.1. —
(i) Let $f : X \longrightarrow S$ be a proper smooth morphism, and suppose that $\mathrm{Pic}_{X/S}$ exists (for example, suppose f is projective). Then $\mathrm{Pic}_{X/S}$ is separated over S, and every closed subset Z of $\mathrm{Pic}_{X/S}$ that is of finite type over S is proper over S.
(ii) Let X be a proper geometrically normal prescheme over a field k. Then $\mathrm{Pic}^\circ_{X/k}$[5] is proper over k.

Proof. (i) The valuative criteria ([2], II, §7) reduce us to proving the following. Let S be the spectrum of a complete discrete valuation ring and U the open subset consisting of the generic point of S. Every rational section of $\mathrm{Pic}_{X/S}$ over S, that is, every section over U, extends uniquely to a section over S. By the definition of $\mathrm{Pic}_{X/S}$, this is equivalent to the following statement: for every invertible Module $\mathcal{L}$ on $V = f^{-1}(U)$ there is an invertible Module on X extending $\mathcal{L}$, unique up to isomorphism. This follows readily from the description of invertible Modules on V, respectively on X, in terms of classes of Cartier divisors. Indeed, the local rings of X are regular (X being smooth over the regular scheme S), hence factorial by Auslander's theorem; thus every divisor on X[6] is a Cartier divisor. Every divisor on V extends to a divisor on X by taking its closure. □

REMARK 2.2. — The proof remains valid if one assumes only that f is flat and that its fibers X_s are locally complete intersections and smooth over $k(s)$ in codimension ≤ 2. This follows from the theorem proved in [5]: a noetherian local complete-intersection ring that is regular in codimension
232 ≤ 3 is factorial ("Samuel's conjecture"). Notice that the

result becomes false if "codimension ≤ 2" is replaced by "codimension ≤ 1," that is, by the hypothesis "normal," as is seen from a family of nonsingular quadrics degenerating into a quadratic cone.

[5] The printed text reads $\mathrm{Pic}^\circ_{X/S}$, although S does not occur in this statement over the field k.
[6] The printed text reads "on S." The factoriality invoked is that of the local rings of X.

(ii) Chow's lemma reduces us to the case in which X is projective and hence embedded in $\mathbf{P}_k^n$; we may suppose that X is connected. If $\dim X = 1$, then X is smooth over k, and (i) applies. If $\dim X \geq 2$, a variant of the known "equivalence criteria" shows that there are finitely many curves Y_i, smooth on X (obtained as intersections of X with suitable linear subspaces of $\mathbf{P}_k^n$), such that

$$\mathrm{Pic}^{\tau}_{X/k} \longrightarrow \prod_i \mathrm{Pic}^{\tau}_{Y_i/k}$$

is a monomorphism, and hence induces a monomorphism on connected components. The connected component of the right-hand side is proper over k by the preceding result, and the homomorphism of group schemes in question is necessarily a closed immersion. It follows that $\mathrm{Pic}^{\circ}_{X/k}$ is also proper over k. The delicate equivalence criteria can be avoided by using the structure of commutative algebraic groups over an algebraically closed field (due to Chevalley–Borel). One is reduced to proving that every morphism from the affine line with the origin removed[7] into $\mathrm{Pic}^{\tau}_{X/k}$ is constant. This is equivalent to saying that every invertible Module on $X[t, t^{-1}]$[8] comes from an invertible Module on X. This well-known elementary result does not even use the properness of X over k (the normality of X immediately reduces one to the case in which X is regular).

COROLLARY 2.3. — Let $f : X \longrightarrow S$ be a proper normal morphism (that is, flat with geometrically normal fibers), and suppose that $\mathrm{Pic}_{X/S}$ exists. Then $\mathrm{Pic}^{\circ}_{X/S}$ is proper over S and hence closed in $\mathrm{Pic}_{X/S}$. Moreover, $\mathrm{Pic}^{\tau}_{X/S}$ and $\mathrm{Pic}^{\rho}_{X/S}$ are also closed, as is $\mathrm{Pic}^{\sigma}_{X/S}$ in "equal characteristic."

Apply (1.1) and (2.1), (ii).

COROLLARY 2.4. — Let $f : X \longrightarrow S$ be a proper smooth morphism such that $\mathrm{Pic}_{X/S}$ exists and is a disjoint union of schemes $P^{(i)}$ of finite type over S (cf. [3], V, 4.1). Then each $P^{(i)}$ is proper over S.

This follows from (2.1), (i). As noted in (2.2), the result can be generalized by making less restrictive hypotheses on the fibers of f, but it becomes false if one merely assumes that f is normal. In that case, I do not know whether $\mathrm{Pic}^{\tau}_{X/S}$ is nevertheless proper over S, even assuming it to be of finite type over S. 233

THEOREM 2.5. — Let $f : X \longrightarrow S$ be a proper flat morphism such that $\mathrm{Pic}_{X/S}$ exists, and for every integer n let φ_n be the nth-power homomorphism in this group prescheme. Then φ_n is étale at every point $x \in \mathrm{Pic}_{X/S}$[9] whose residue characteristic is prime to n.

By the infinitesimal characterization of étale morphisms, this assertion is equivalent to the following:

LEMMA 2.6. — Suppose that S is the spectrum of a local Artinian ring A whose maximal ideal $\mathfrak{m}$ has zero $(\nu+1)$st power, and put

$$A_{\nu-1} = A/\mathfrak{m}^{\nu}, \qquad X_{\nu-1} = X \otimes_A A_{\nu-1}.$$

Let $\mathcal{L}$ be an invertible Module on X, and let $\mathcal{L}'_{\nu-1}$ be an invertible Module on $X_{\nu-1}$ whose nth tensor power is isomorphic to

$$\mathcal{L}_{\nu-1} = \mathcal{L} \otimes_A A_{\nu-1}.$$

Then there is an invertible Module $\mathcal{L}'$ on X whose nth tensor power is isomorphic to $\mathcal{L}$, provided that n is prime to the residue characteristic of $k = k(A)$.

We prove the lemma. Put

$$V = \mathfrak{m}^{\nu} = \mathfrak{m}^{\nu}/\mathfrak{m}^{\nu+1};$$

[7] The printed text reads "from the affine line." The linked erratum, p. 303, prescribes "affine line with the origin removed."

[8] The printed text reads $X[t]$; the linked erratum, p. 303, prescribes $X[t, t^{-1}]$.

[9] The printed text reads "$x \in X$." The map φ_n is the power endomorphism of the Picard prescheme.

this is a vector space over $k = k(A)$. First try to extend $\mathcal{L}'_{\nu-1}$ to an arbitrary invertible Module $\mathcal{L}'$ on X. The obstruction lies in

$$H^2(X_0, \mathcal{O}_{X_0}) \otimes_k V.$$

The hypothesis on $\mathcal{L}'_{\nu-1}$ and the extendability of $\mathcal{L}_{\nu-1}$ show that n times this obstruction is zero; the obstruction itself is therefore zero because n is prime to the characteristic. The ambiguity in the chosen extension lies in

$$H^1(X_0, \mathcal{O}_{X_0}) \otimes_k V,$$

and the discrepancy ξ between $\mathcal{L}'^{\otimes n}$ and $\mathcal{L}$ lies in the same Module. Correcting $\mathcal{L}'$ to make this discrepancy zero amounts to finding an η in that Module such that $n\eta = \xi$. This is again possible because n is prime to the characteristic.

COROLLARY. — Under the hypotheses of (2.5), suppose in addition that the Picard schemes of the fibers X_s contain no additive component (for example, that the X_s are geometrically normal; cf. (2.1), (ii)). Then $\mathrm{Pic}_{X/S} \longrightarrow S$ is universally open at the points of $\mathrm{Pic}^\circ_{X/S}$. If $\mathrm{Pic}^\circ_{X/S}$ is closed (for example, if the X_s are geometrically normal; cf. (2.3)), then $\mathrm{Pic}^\sigma_{X/S}$ is itself universally open over S. Finally, in equal characteristic, $\mathrm{Pic}^\rho_{X/S} \longrightarrow S$ is universally open.

234 Apply (1.5) and (1.1).

COROLLARY 2.7. — Let $f : X \longrightarrow S$[10] be a proper flat morphism such that $\mathrm{Pic}_{X/S}$ exists. Then the function

$$s \longmapsto \dim \mathrm{Pic}_{X_s/k(s)}$$

on S is upper semicontinuous (that is, it can jump upward but not downward), and is even continuous (that is, locally constant) if the $\mathrm{Pic}_{X_s/k(s)}$ contain no additive component.

The first assertion is trivial, or nearly so, for every group prescheme locally of finite type over a locally noetherian base, since it is enough to look along the identity section. The second assertion follows from (2.5).

REMARK 2.8. — Let $s, s' \in S$, with s a specialization of s'. Then (2.7) amounts to an inequality (respectively an equality) between the dimensions of the Picard schemes of $X_{s'}$ and of its "specialization" X_s. Even before Picard schemes were constructed, Serre had observed that, for a smooth morphism $f : X \longrightarrow S$, invariance of the dimension of the Picards of the X_s is a formal consequence of the specialization theory of the fundamental group ([4], X) and the classical Kummer relations between the points of finite order on the Picard variety and the abelianized fundamental group ([4], XI). In general, let α, μ, λ denote the dimensions of the abelian, multiplicative, and additive parts of $\mathrm{Pic}^\circ_{X_s/k(s)}$, and define α', μ', λ' similarly. The known relations are expressed by the following inequalities:

$$\alpha + \mu + \lambda \geq \alpha' + \mu' + \lambda' \qquad (*)$$

(valid provided that $\mathrm{Pic}_{X/S}$ exists, and therefore probably in all cases). When $\lambda = \lambda' = 0$, this inequality becomes an equality, valid under the same existence hypotheses:

$$\alpha + \mu = \alpha' + \mu';$$

on the other hand,

$$2\alpha + \mu \leq 2\alpha' + \mu' \qquad (**)$$

235

by Serre's argument if the X_s are separable (without even assuming the existence of $\mathrm{Pic}^\tau_{X/S}$), or if the ${}_n\mathrm{Pic}_{X/S}$ (the kernels of φ_n in the Picard functor) are separated over S (since they are

[10] The printed text reads "$f : X \longrightarrow Y$," although both the Picard prescheme and the following function have base S.

étale over S by (2.5)). One is inclined to conjecture that (*) is always an equality, or at least when the X_s are separable, and that

$$\alpha \leq \alpha', \qquad \lambda \geq \lambda' \tag{***}$$

holds whenever there is a group prescheme, locally of finite type over a locally noetherian S, whose fibers have constant dimension (see (1.3) for a positive result in this direction).

REMARK 2.9. — In every known case, $\mathrm{Pic}^{\tau}_{X/S}$ is universally open over S, but one should probably not let this encourage excessive illusions, even when $f : X \longrightarrow S$ is smooth. In any event, Mumford has constructed an example (admittedly with S nonreduced, in fact the spectrum of an Artinian ring) in which $\mathrm{Pic}^{\tau}_{X/S}$ is not flat over S, by varying the Igusa surface infinitesimally. The point considered by Mumford lies in $\mathrm{Pic}^{\rho}_{X/S}$, and it remains possible, for a smooth $f : X \longrightarrow S$, that $\mathrm{Pic}^{\tau}_{X/S}$ is flat over S at the points of $\mathrm{Pic}^{\sigma}_{X/S}$. The speaker nevertheless doubts that this is always so, even if one restricts to the points of $\mathrm{Pic}^{\circ}_{X/S}$ and assumes that S is the spectrum of a discrete valuation ring. The question is also connected with the study of fixed points of an abelian scheme under a finite group of automorphisms, a situation for which examples seem to be lacking.

It would seem that even when one restricts to a smooth projective morphism $f : X \longrightarrow S$, the local regularity results for $\mathrm{Pic}_{X/S}$ stated in this section, together with the conjectures in (2.8), nearly exhaust what can be said without more specific hypotheses on the nature of the fibers of f. Recall, however, that if the geometric fibers of $\mathrm{Pic}_{X/S}$ are reduced and have no additive component, then (1.8) and (2.5) imply that $\mathrm{Pic}_{X/S}$ is smooth over S at the points of $\mathrm{Pic}^{\sigma}_{X/S}$ when S is reduced. The same result holds, without any hypothesis on S, if $f : X \longrightarrow S$ is normal, as we shall see in (3.5). We record in this connection:

PROPOSITION 2.10. —
(i) If $\mathrm{Pic}_{X/S}$ is smooth (respectively flat) over S at the points of the identity section, then it is so at every point of $\mathrm{Pic}^{\circ}_{X/S}$ and at the points of every section of $\mathrm{Pic}_{X/S}$ over S. 236
(ii) Let $s \in S$ satisfy

$$H^2(X_s, \mathcal{O}_{X_s}) = 0.$$

Then there is an open neighborhood U of s such that $\mathrm{Pic}_{X/S}\,|U$ is smooth over U.
(iii) If X is a proper scheme over a field, then

$$\dim \mathrm{Pic}_{X/k} \leq \dim H^1(X, \mathcal{O}_X),$$

with equality if and only if $\mathrm{Pic}_{X/k}$ is smooth over k; this is always the case when k has characteristic zero.

Part (i) follows from (2.5) and (1.12), and (ii) follows from the infinitesimal criterion for smooth morphisms and a familiar obstruction calculation, since the "semicontinuity theorem" shows that the hypothesis at s remains valid at neighboring points. Finally, (iii) follows from the fact that $H^1(X, \mathcal{O}_X)$ is isomorphic to the Zariski tangent space of $\mathrm{Pic}_{X/k}$ at the identity. The last assertion is a special case of a theorem of Cartier saying that a "formal group" in characteristic zero is formally smooth over k.

3. The canonical abelian subscheme of $\mathrm{Pic}_{X/S}$, and the Albanese scheme

PROPOSITION 3.1. — Let k be a field and G a commutative group scheme of finite type over k with no additive component. Then G°_{red} is separable over k, and consequently is a smooth subgroup scheme over k.

This is trivial when k is perfect, in particular for $G_{\bar{k}}$, where $\bar{k}$ is an algebraic closure of k. It therefore suffices to see that $(G^{\circ}_{\bar{k}})_{\mathrm{red}}$ descends from a subscheme of G. The hypothesis on $G_{\bar{k}}$ (no

additive component)[11] readily implies that there is an integer m such that $(G^{\circ}_{\bar{k}})_{\mathrm{red}}$ is the scheme-theoretic image of the mth-power homomorphism in $G_{\bar{k}}$. Since this homomorphism comes from the analogous homomorphism for G, the scheme-theoretic image of the latter gives the required subscheme.

COROLLARY 3.2. — Let X be a normal proper scheme over k such that $\mathrm{Pic}_{X/k}$ exists. Then there is an abelian subscheme A of $\mathrm{Pic}_{X/k}$ whose underlying set is $\mathrm{Pic}^{\circ}_{X/k}$.

237 Indeed, by (2.1), (ii), $\mathrm{Pic}^{\circ}_{X/k}$ is proper over k and therefore satisfies the condition of (3.1).

Thus the preceding result shows that in certain cases the classical "Picard variety" (which in the present theory would be denoted $(\mathrm{Pic}_{X/k})^{\circ}_{\mathrm{red}}$) "is defined over k," without assuming that k is perfect.

Now let $f : X \longrightarrow S$ be a proper flat relative scheme, with

$$\mathcal{O}_S \xrightarrow{\sim} f_*(\mathcal{O}_X)$$

for simplicity, such that $\mathrm{Pic}_{X/S}$ exists and $\mathrm{Pic}^{\circ}_{X/S}$ is proper over S. For (3.3), (ii), we also assume that there is an open subset of $\mathrm{Pic}_{X/S}$ containing $\mathrm{Pic}^{\circ}_{X/S}$ that is quasi-projective over S. As seen earlier, this condition holds when f is projective with geometrically separable irreducible fibers. Recall that an abelian scheme over S is a group scheme over S that is proper and smooth over S and has geometrically connected fibers. We ask whether there is a subgroup scheme A of $\mathrm{Pic}_{X/S}$ which is an abelian scheme and whose underlying set is $\mathrm{Pic}^{\circ}_{X/S}$. We have just seen that one always exists when S is the spectrum of a field. Here is what can be said in the general case under consideration:

THEOREM 3.3. — Under the preceding hypotheses:

(i) If there is an abelian subscheme of $\mathrm{Pic}_{X/S}$ whose underlying set is $\mathrm{Pic}^{\circ}_{X/S}$, it is unique. Its formation is consequently compatible with base change.

(ii) For such an abelian subscheme to exist, it is enough that one exist after every base change $S' \longrightarrow S$ with S' local Artinian. If S is the spectrum of a local ring, it is enough to test the $S' = \mathrm{Spec}(A_n)$, where $A_n = A/\mathfrak{m}^{n+1}$. If S is reduced, it is likewise enough to test the base changes $S' \longrightarrow S$ in which S' is the spectrum of a discrete valuation ring (complete with algebraically closed residue field, if desired).

(iii) Suppose that A exists, and let $B = \mathrm{Alb}^0(X/S)$ be the dual abelian scheme (that is, $B = \mathrm{Pic}^{\circ}_{A/S}$ [7]). Then one can canonically construct a principal homogeneous space $P = \mathrm{Alb}^1(X/S)$ under B, together with an S-morphism $X \longrightarrow P$ universal for S-morphisms from X to para-abelian schemes (that is, principal homogeneous spaces under abelian schemes). The formation of $\mathrm{Alb}^0(X/S)$, $\mathrm{Alb}^1(X/S)$, and $X \longrightarrow \mathrm{Alb}^1(X/S)$ commutes with base change.

We sketch the proof.

Part (i) is a general rigidity property of abelian subschemes of commutative group schemes: if two such subschemes coincide set-theoretically at a point $s \in S$, they coincide over the entire

238 connected component

of s [7]. (This result generalizes a classical theorem of Chow.)

(ii) Hilbert schemes show that the functor associating to every S' over S the set (having either one or zero elements) of canonical abelian subschemes of $(\mathrm{Pic}_{X/S}) \times_S S'$ is represented by a scheme T of finite type over S. By (i), $T \longrightarrow S$ is a monomorphism, and by (3.2) it is surjective. The existence of a canonical abelian subscheme of $\mathrm{Pic}_{X/S}$ means that T has a section over S, equivalently that $T \longrightarrow S$ is an isomorphism. This is also equivalent to $T \longrightarrow S$ being étale, or, when S is reduced, to $T \longrightarrow S$ being proper. This proves (ii).

(iii) Directly from the definition of $\mathrm{Pic}_{X/S}$, for every abelian scheme C over S, giving an S-morphism from X to a principal homogeneous space under C is equivalent to giving a group

[11] The printed text has a malformed construction equivalent to "The hypothesis ... results." The linked erratum, p. 303, inserts the preposition needed for "It follows from the hypothesis ...".

homomorphism $C' \longrightarrow \mathrm{Pic}_{X/S}$, where C' is the dual abelian scheme of C. If the canonical abelian subscheme A of $\mathrm{Pic}_{X/S}$ exists, these homomorphisms necessarily factor through A (as can also be seen by using points of finite order). This proves (iii).

REMARKS 3.4. — We shall denote by $\mathrm{Pic}^{\circ\circ}_{X/S}$ the canonical abelian subscheme of $\mathrm{Pic}_{X/S}$, when it exists. Unfortunately it does not always exist, as is seen by varying the Igusa surface infinitesimally (through the first-order modular deformation). It may nevertheless be that $\mathrm{Pic}^{\circ\circ}_{X/S}$ exists at least when S is reduced, or equivalently, by (ii), when S is the spectrum of a discrete valuation ring. Let X_0, X_1 be the special and generic fibers of X, and put

$$A_1 = \mathrm{Pic}^{\circ}_{X_1/K},$$

where K is the fraction field of the valuation ring V. By Koizumi's theorem there is an essentially unique abelian scheme A over S whose generic fiber is A_1. As in (2.1), (i), one sees readily (now assuming that X is smooth over S) that the identity morphism of A_1 extends to a morphism

$$A \longrightarrow \mathrm{Pic}_{X/S}\,.$$

Hence there is a homomorphism

$$A_0 \longrightarrow \mathrm{Pic}^{\circ\circ}_{X_0/k}, \qquad (*)$$

which is readily shown to be surjective with kernel a finite p-group, where p is the characteristic exponent of the residue field k 239

(again by using points of finite order). With this understood, the following conditions on X/S are equivalent:

a. The preceding homomorphism (*) is an isomorphism (which Shimura would express by saying that formation of the "Picard variety" is "compatible with specialization").
b. $\mathrm{Pic}^{\circ\circ}_{X/S}$ exists (in which case it is none other than A).
c. (For reference) The $\mathrm{Pic}^{\circ\circ}_{X_n/S}$ exist.

By the observation about the kernel of (*), condition (a) holds when the residue characteristic is zero, but this result will be considerably generalized in (3.5).

Of course, if $\mathrm{Pic}_{X/S}$ is smooth over S at the points of $\mathrm{Pic}^{\circ}_{X/S}$, then the latter is open in $\mathrm{Pic}_{X/S}$ (cf. (1.7)). With the induced structure it is therefore an abelian subscheme of $\mathrm{Pic}_{X/S}$, hence equal to $\mathrm{Pic}^{\circ\circ}_{X/S}$, which exists in this case. But much more is true:

THEOREM 3.5. — Under the hypotheses of (3.3), let $s \in S$ be such that $\mathrm{Pic}_{X_s/k(s)}$ is smooth over $k(s)$ (or, equivalently,

$$\dim \mathrm{Pic}_{X_s/k(s)} = \dim H^1(X_s, \mathcal{O}_{X_s})).$$

Then there is an open neighborhood U of s such that $\mathrm{Pic}_{X/S}$ is smooth over S at the points of $\mathrm{Pic}^{\circ}_{X/S}\,|U$, which is consequently an open abelian subscheme of $\mathrm{Pic}_{X/S}\,|U$. A fortiori, $\mathrm{Pic}^{\circ\circ}_{X|U/U}$ exists.

We indicate the principle of the proof. The preceding discussion reduces us to the case in which S is the spectrum of a local Artinian ring A, and we argue by induction on the infinitesimal order of A. We may therefore suppose that $\mathrm{Pic}^{\circ}_{X_n/A_n}$ is smooth over A_n and need only prove that $\mathrm{Pic}^{\circ}_{X_{n+1}/A_{n+1}}$ is smooth over A_{n+1}. For this it is enough to construct an abelian scheme P_{n+1} over A_{n+1} extending

$$P_n = \mathrm{Pic}^{\circ}_{X_n/A_n},$$

together with an invertible Module $\mathcal{L}_{n+1}$ on $X_{n+1} \times_{A_{n+1}} P_{n+1}$ extending the invertible Module $\mathcal{L}_n$ on $X_n \times_{A_n} P_n$ that occurs in the definition of the Picard scheme Pic_{X_n/A_n} as the solution of a universal problem. (N.B. — We may suppose that X has a section over S ...) The construction uses the following key result: every abelian scheme defined over a quotient of a local Artinian ring can

be extended (in other words, the absolute "formal moduli scheme" ([3], II) of an abelian scheme over an algebraically closed field is smooth over the ring of 240

Witt vectors of k). This result itself follows from the general formal properties of the obstruction to lifting and from the group operations. Assuming it, first extend P_n arbitrarily to P_{n+1}. One then encounters an obstruction to lifting $\mathcal{L}_n$, lying in

$$H^2(X_0 \times P_0, \mathcal{O}_{X_0 \times P_0}) \otimes_k V \qquad (\text{where } V = \mathfrak{m}^{n+1}/\mathfrak{m}^{n+2}),$$

and in fact in the subspace

$$H^1(X_0, \mathcal{O}_{X_0}) \otimes H^1(P_0, \mathcal{O}_{P_0}) \otimes_k V$$

(because the restriction of $\mathcal{L}_n$ to each of the two factors X_n and P_n is trivial). This last space is precisely

$$H^1(P_0, \mathcal{T}_{P_0/k}) \otimes V,$$

where $\mathcal{T}_{P_0/k}$ is the tangent sheaf of P_0/k, and it is therefore also the space measuring the indeterminacy in lifting P_n to P_{n+1} ([3], II). Consequently this lift can be corrected (in one and only one way, as it should be) so as to kill the obstruction to lifting $\mathcal{L}_n$.

COROLLARY 3.6. — Under the hypotheses of (3.5), $R^1 f_*(\mathcal{O}_X)$ is a locally free Module on S near s, and its formation commutes with base change.

Indeed, this Module is the tangent Module of $\mathrm{Pic}_{X/S}$ along the identity section.

REMARK 3.7. — The same argument as in (3.5) shows that if S' is a subscheme of S defined by a coherent nilpotent Ideal, and if one assumes only that $\mathrm{Pic}^{\circ}_{X'/S'}$ exists and has smooth fibers, then $\mathrm{Pic}^{\circ}_{X/S}$ necessarily exists and is an abelian scheme over S. This permits Picard schemes to be constructed in some cases despite the absence of a projectivity hypothesis; for example, the dual abelian scheme of an abelian scheme over an Artinian ring always exists. On the other hand, applying (3.6) when X is an abelian scheme over S, and using the known exterior-algebra structure of $H^*(X_s, \mathcal{O}_{X_s})$ over $H^1(X_s, \mathcal{O}_{X_s})$ (Rosenlicht–Serre), one finds that for every i, $R^i f_*(\mathcal{O}_X)$ is locally free and, more precisely, isomorphic to the ith exterior power of $R^1 f_*(\mathcal{O}_X)$.

REMARK 3.8. — For a projective smooth morphism $f : X \longrightarrow S$, with S reduced and all residue characteristics zero, (3.6) was known by transcendental methods as a consequence of Hodge theory.
241 In fact, all the $R^p f_*(\Omega^q_{X/S})$ are then locally free. There are, on the other hand, counterexamples

in mixed characteristic to the local freeness of $R^1 f_*(\mathcal{O}_X)$, obtained from Serre varieties ([8], p. 50). There seems to be no counterexample in equal characteristic.

COROLLARY 3.8. — Under the hypotheses of (3.5), $\mathrm{Pic}_{X/S}\,|U$ is smooth over U at every point of $\mathrm{Pic}^{\tau}_{X/S}\,|U$.

Apply (2.10), (i). In particular, taking (2.10), (iii), into account:

COROLLARY 3.9. — Suppose that all residue characteristics of S are zero. Then $\mathrm{Pic}^{\tau}_{X/S}$ is smooth over S.

REMARK 3.10. — If $\mathrm{Pic}_{X/S}$ is also proper over S, it follows, for example, that the torsion subgroup of the Néron–Severi groups of the geometric fibers of f is constant on each connected component of S (as is also clear by transcendental methods when f is smooth and projective). A direct application of (2.5) shows more generally that if $\mathrm{Pic}_{X/S}$ is proper over S (for example, if $f : X \longrightarrow S$ is smooth and projective) and q is a prime distinct from all residue characteristics of S, then the q-primary component of the torsion subgroups of the Néron–Severi groups of the geometric fibers of X/S is constant on each connected component of S. In characteristic $p > 0$ this

is no longer true for the p-primary component of the torsion. It remains possible, however, that the rank over $k(s)$ of

$$\mathrm{Pic}^{\tau}_{X_s/k(s)} \,/\, \mathrm{Pic}^{\circ\circ}_{X_s/k(s)} = T_{X_s/k(s)}$$

is locally constant. When S is reduced, one can show that this is equivalent to the existence of $\mathrm{Pic}^{\circ\circ}_{X/S}$ together with the flatness of $\mathrm{Pic}^{\tau}_{X/S}$ over S, and it is enough to test the question when S is the spectrum of a discrete valuation ring. This is what I have verified in the few examples I have examined; but since the corresponding assertion with S Artinian is false (cf. Remarks (2.9) and (3.4)), one should probably not indulge in excessive optimism.

4. The finiteness theorem for the Picard scheme

Let $f : X \longrightarrow S$ be a projective flat morphism such that $\mathrm{Pic}_{X/S}$ exists. Consideration of the "Hilbert polynomials" $Q \in \mathbf{Q}[t]$ decomposes $\mathrm{Pic}_{X/S}$ into a disjoint union of open subsets $\mathrm{Pic}^{Q}_{X/S}$. Without further hypotheses, for example ones ensuring that $\mathrm{Pic}_{X/S}$ is separated over S, these open subsets need not be of finite type over S; a counterexample is obtained when X is "a conic degenerating into two lines." It is 242

possible, however, that they are of finite type when f is separable with geometrically irreducible fibers. The question is connected with whether $\mathrm{Pic}^{\tau}_{X/S}$ is of finite type over S, which may be true without any hypothesis on the fibers of X/S. When $f : X \longrightarrow S$ is smooth, note that $\mathrm{Pic}^{\tau}_{X/S}$ is contained in one of the $\mathrm{Pic}^{Q}_{X/S}$ (because on a nonsingular projective variety torsion equivalence is finer than—in fact, by Matsusaka, identical to—numerical equivalence for divisors), and is therefore of finite type over S if the $\mathrm{Pic}^{Q}_{X/S}$ are. The finiteness questions just raised retain an evident meaning even without assuming the existence of $\mathrm{Pic}_{X/S}$: they say that certain families of invertible Modules are "bounded" in the sense of [3], IV. For every algebraically closed field k, consider the integral subschemes of $\mathbf{P}^r_k$ of dimension n and degree d, and the invertible Modules on such preschemes with Hilbert polynomial Q, where r, n, d, Q are fixed. The problem is to show that the family of these Modules (viewed as coherent Modules on the $\mathbf{P}^r_k$) is bounded, that is, can be parametrized by a scheme of finite type over $\mathbf{Z}$.

Using Matsusaka's method [6], a rather technical proof (using the "equivalence criteria" in a suitable form) gives an affirmative answer when one restricts to nonsingular subvarieties of $\mathbf{P}^r_k$. More precisely:

THEOREM 4.1. — Let $f : X \longrightarrow S$[12] be a smooth projective morphism with geometrically connected fibers, with S noetherian. Let $\mathcal{O}_X(1)$ be a relatively very ample Module on X, and let E be a subset of $\mathrm{Pic}_{X/S}$ corresponding to a family $(\mathcal{L}_i)$ of invertible Modules on the geometric fibers $\overline{X}_{s_i}$ of X/S. Let D_i be a divisor (not necessarily effective) on $\overline{X}_{s_i}$ defining $\mathcal{L}_i$, let

$$a_n^{(i)} t^n + \cdots + a_0^{(i)}$$

be the Hilbert polynomial of $\mathcal{L}_i$, and let $\xi = \xi_i$ be a divisor defining $\mathcal{O}_X(1)$, that is, a hyperplane section. The following conditions are equivalent:

a. E[13] is quasi-compact, that is, contained in an open subset of finite type over S; equivalently, the family $(\mathcal{L}_i)$ is bounded.

b. E is contained in the union of finitely many of the sets $\mathrm{Pic}^{Q}_{X/S}$, $Q \in \mathbf{Q}[t]$; equivalently, only finitely many Hilbert polynomials occur among the $\mathcal{L}_i$.

b′. (If all fibers of X/S have the same dimension n.) The two coefficients $a^{(i)}_{n-1}$ and $a^{(i)}_{n-2}$ of the Hilbert polynomials of the $\mathcal{L}_i$ range over finite sets. 243

[12] The printed text reads "Let: $X \longrightarrow S$"; the morphism used below is f.

[13] The printed text reads "Q." The subset introduced in the statement, whose quasi-compactness expresses boundedness of the family, is E.

b″. (If all fibers of X/S have the same dimension n.) The integers

$$\xi^{n-1}D_i \qquad \text{and} \qquad \xi^{n-2}D_i^2$$

(the projective degrees of D_i and D_i^2) remain bounded above.

COROLLARY 4.2. — Let $f : X \longrightarrow S$ be a smooth projective morphism with geometrically connected fibers. Then the schemes $\mathrm{Pic}^Q_{X/S}$ and $\mathrm{Pic}^\tau_{X/S}$ ($Q \in \mathbf{Q}[t]$) are projective over S.

They are of finite type over S by (4.1), so (2.4) applies.

BIBLIOGRAPHY

[1] CHEVALLEY (Claude). – Sur la théorie de Picard, *Amer. J. of Math.*, vol. 82, 1960, pp. 435–490.

[2] GROTHENDIECK (A.) and DIEUDONNÉ (J.). – Éléments de géométrie algébrique, Chapter I and following. – Paris, Presses universitaires de France, 1960–1961 (Institut des Hautes Études Scientifiques, 4, 8, 11, ...).

[3] GROTHENDIECK (Alexander). – Technique de descente et théorèmes d'existence en géométrie algébrique, I–V, *Séminaire Bourbaki*, vol. 12, 1959/60, no. 190, 29 pp. and no. 195, 22 pp.; vol. 13, 1960/61, no. 212, 20 pp. and no. 221, 28 pp.; vol. 14, 1961/62, no. 232, 19 pp.

[4] GROTHENDIECK (Alexander). – Séminaire de géométrie algébrique, 1960/61. – Paris, Institut des Hautes Études Scientifiques, 1961.

[5] GROTHENDIECK (Alexander). – Séminaire de géométrie algébrique, 1961/62, edited by a group of auditors. – Paris, Institut des Hautes Études Scientifiques (forthcoming).

[6] MATSUSAKA (T.). – The criteria for algebraic equivalence and the torsion group, *Amer. J. Math.*, vol. 79, 1957, pp. 53–66.

[7] MUMFORD (D.) and TATE (J.). – Séminaire de géométrie algébrique, Harvard University, Spring Term 1962 (forthcoming).

[8] SERRE (Jean-Pierre). – Sur la topologie des variétés algébriques en caractéristique p, *Symposium internacional de topología algebraica* [1956, Mexico], pp. 24–53. – Mexico, Universidad nacional autónoma, 1958.

Erratum and supplement to Exposé no. 236

303

8. TDTE VI: Picard schemes. General properties.

[Bourbaki Seminar, vol. 14, 1961/62, no. 236, 23 pp.]

Page 12, line 16. — For "affine line," read "affine line with the origin removed."
Line 18. — For "$X[t]$," read "$X[t, t^{-1}]$."
Page 16, Proposition 3.1, sixth line. — For the malformed construction "The hypothesis ... results," read "It follows from the hypothesis ...."
Page 21, §4: The finiteness theorem for the Picard scheme. — Finiteness questions of the kind raised in this section have been almost completely resolved since the present exposé was written. We indicate the main facts now known in this direction. To simplify the statements, we implicitly assume that all the Picard preschemes occurring in them exist, although an evident modification of the statements removes every existence hypothesis. In what follows, S denotes a noetherian scheme and X, Y proper schemes over S.

(i) Let $f : X \longrightarrow Y$ be a surjective morphism. Then

$$f^* : \mathrm{Pic}_{Y/S} \longrightarrow \mathrm{Pic}_{X/S}$$

is a morphism of finite type. 304

(ii) Suppose that Y is projective over S and carries an invertible Module relatively ample over S. Let X be the zero scheme of an arbitrary section of this Module, and let $f : X \longrightarrow Y$ be the canonical immersion. Finally, suppose that the irreducible components of the fibers of Y/S have dimension ≥ 3. Then

$$f^* : \mathrm{Pic}_{Y/S} \longrightarrow \mathrm{Pic}_{X/S}$$

is of finite type.

(iii) Suppose that X is projective over S and all its geometric fibers are integral of dimension n. Let $\mathcal{O}_X(1)$ be an ample invertible Module on X, permitting Hilbert polynomials to be defined. If M is a subset of $\mathrm{Pic}_{X/S}$, then M is quasi-compact if and only if, in the Hilbert polynomials

$$a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots$$

of the elements of M, the coefficients a_1 and a_2 remain bounded.

(iv) For every integer $n \neq 0$, the nth-power homomorphism in the group prescheme $\mathrm{Pic}_{X/S}$ is a morphism of finite type.

Notice that (i) and (ii) also say that, under their respective hypotheses, a subset M of $\mathrm{Pic}_{Y/S}$ is quasi-compact if and only if its image in $\mathrm{Pic}_{X/S}$ is quasi-compact. It follows that an invertible Module $\mathcal{L}$ on Y is τ-equivalent to zero if and only if its inverse image on X is; in other words, $\mathrm{Pic}^{\tau}_{Y/S}$ is the inverse image of $\mathrm{Pic}^{\tau}_{X/S}$. In particular, to show that the former prescheme is of finite type

over S, it is enough to prove this for the latter, since $\mathrm{Pic}_{Y/S} \longrightarrow \mathrm{Pic}_{X/S}$ is of finite type. Applying (i), Chow's lemma, and (iii), one obtains:

(v) $\mathrm{Pic}^{\tau}_{X/S}$ is of finite type over S.

In general, combining (i) for a finite morphism with (ii) reduces most finiteness questions to the case in which X/S has geometrically integral normal fibers of dimension ≤ 2. Often, by applying (i) to a surjective morphism not necessarily finite and using resolution of singularities of algebraic surfaces (due in arbitrary characteristic to Abhyankar), one even reduces to the case in which X/S is smooth, and hence has nonsingular geometric fibers of dimension 2. Together with (v) and the Picard–Igusa inequality bounding the rank of the Néron–Severi group of a nonsingular projective surface, this proves the following generalization of Néron's finiteness theorem:

(vi) Let X/S be proper over S and otherwise arbitrary. Then the Néron–Severi groups

$$\mathrm{Pic}_{X_i/k_i} \,/\, \mathrm{Pic}^{\circ}_{X_i/k_i}$$

305 of the geometric fibers X_i/k_i of X/S

are of finite type, and their ranks and the orders of their torsion subgroups remain bounded.

The same reduction to nonsingular surfaces, together with the theorems known in that case (namely Néron's finiteness theorem and the nondegeneracy of the intersection form on the Néron–Severi group), implies:

(vii) Let X be a proper scheme over an algebraically closed field. There are finitely many closed integral curves C_i $(1 \leq i \leq r)$ in X with the following property: for every subset M of $\mathrm{Pic}_{X/k}$, M is quasi-compact if and only if the integers

$$\deg_{C_i'} \mathcal{L} \qquad (\mathcal{L} \in M)$$

remain bounded, where C_i' denotes the normalization of C_i.

Here r may be taken to be the rank of the Néron–Severi group. Once that group is known to be of finite type, (vii) reduces to the fact that the linear forms on the Néron–Severi group defined by curves C in X vanish simultaneously only on its torsion elements. When X is projective and nonsingular, this result, as well as (v), was due to Matsusaka. Using (vii), one readily obtains the following characterization of invertible Modules τ-equivalent to zero on X:

(viii) Let X/k be a proper scheme over a field, and let $\mathcal{L}$ be an invertible Module on X. The following conditions are equivalent:
a. $\mathcal{L}$ is τ-equivalent to zero;
b. For every coherent Module F,

$$\chi(F \otimes \mathcal{L}) = \chi(F),$$

where χ denotes the Euler–Poincaré characteristic;
b′. As in (b), but with $F = \mathcal{O}_Y$, where Y is a closed integral subscheme of dimension 1 in X;
c. For every such Y, writing Y' for the normalized curve,

$$\deg \mathcal{L}_{Y'} = 0.$$

If X/k is projective and carries an ample invertible Module $\mathcal{O}_X(1)$, the preceding conditions are also equivalent to:
d. For every integer m, $\mathcal{L}^{\otimes m}(1)$ is ample.
e. (If X is integral.) For every pair of integers m, n,

$$\chi(\mathcal{L}^{\otimes m}(n)) = \chi(\mathcal{O}_X(n))$$

306

(that is, (b) holds with $F = \mathcal{L}^{\otimes m}(n)$).

For the sufficiency of this last condition, observe that it says that all the Hilbert polynomials of the $\mathcal{L}^{\otimes m}$ are equal. Hence, by Mumford's criterion (iii), the $\mathcal{L}^{\otimes m}$ remain in a quasi-compact subset of $\mathrm{Pic}_{X/k}$, which is (a). Conditions (b), (b'), and (c) should be regarded as variants, on an arbitrary proper scheme, of numerical equivalence as customarily defined on nonsingular projective varieties. For the latter, the equivalence of (a) and (c) was of course well known (Matsusaka).

Criterion (e) also implies:

(ix) Let $f : X \longrightarrow S$ be a projective flat morphism whose geometric fibers are integral. Then $\mathrm{Pic}^{\tau}_{X/S}$ is open and closed in $\mathrm{Pic}_{X/S}$.

We confine ourselves to a few comments on the proof of the key results (i), (ii), and (iii). Result (iv) is somewhat separate and is proved using only (i) for finite surjective radicial morphisms, specifically for a Frobenius morphism. For (i), one makes essential use of the ideas of nonflat descent (see especially TDTE I, page 9). Since only finiteness results are sought, the lack of effectivity criteria for descent data is harmless. Mumford has recently proved a slightly weaker form of (iii), namely the criterion involving all coefficients of the Hilbert polynomial. His argument is extremely simple and is inspired by the proof of an ampleness criterion of Nakai (stated by Nakai for nonsingular surfaces and generalized by Mumford to arbitrary projective morphisms). It seems to me, however, that this argument is valid only under a slight additional restriction on the fibers of X/S, Serre's condition (S_2), which is satisfied when the geometric fibers are normal. This restricted criterion is then used in proving (ii): criterion (i) reduces one to the case in which Y/S is flat with geometrically integral normal fibers, and Mumford's criterion readily reduces one to the case in which X/S satisfies the same conditions. The dimension hypothesis then implies that the geometric fibers of Y and X have depth ≥ 2 at their closed points, which permits application of the "equivalence criteria" in the form given in SGA 1962 (Exposé XII), completing the proof of (ii). Once (i) and (ii) are available, it is not difficult to remove every normality hypothesis on the fibers from Mumford's criterion and prove it in the stronger form stated in (iii). 307

Finally, the proof of (i) and (ii) also shows that when S is the spectrum of a field k, the morphism

$$\mathrm{Pic}_{Y/k} \longrightarrow \mathrm{Pic}_{X/k}$$

is affine (and not merely of finite type).

Chapter 9

1962 Comments

297

BOURBAKI SEMINAR
14th year, 1961/62, Supplement
May 1962

FOUNDATIONS OF ALGEBRAIC GEOMETRY
COMMENTS

by Alexander GROTHENDIECK

Notice

We presented eight exposés in the Bourbaki Seminar, from 1957 to 1962, concerning the foundations of algebraic geometry. Except for the first, these exposés are written in the framework of schemes. All the results stated will find their place in the "Éléments de géométrie algébrique" of Jean DIEUDONNÉ and Alexander GROTHENDIECK. However, the substance of none of these exposés is currently covered either by one of the chapters already published or in preparation, or by any other book or article, and doubtless will not be for some years yet. This is the main reason that led us to gather these exposés, which will make available to users a certain number of key notions and results of scheme theory while awaiting a definitive account. Moreover, reading these texts will permit one to become familiar quickly with these results and notions, without being impeded by the necessarily cumbersome detail of a systematic treatise, and will provide the indispensable motivations for studying such a treatise.

For the reader's convenience, we have gathered here a number of comments and errata, grouped by exposé, which will in particular note in passing the progress made since the text was written and indicate certain supplementary references.

Various results appearing in these exposés have been treated in detail in the IHES Seminar on Algebraic Geometry, as well as in two consecutive seminars at Harvard in 1961/62, the first given by me and the second by MUMFORD–TATE, for which mimeographed notes (written by Mr LICHTENBAUM) are currently in preparation.

Abbreviations used

298 **SGA :** Seminar on algebraic geometry, Institut des Hautes Études Scientifiques, 1960–1962.

SHG : GROTHENDIECK Seminar, Harvard University, 1961/62.

SHMT : MUMFORD–TATE Seminar, Harvard University, 1961/62.

EGA : GROTHENDIECK (A.) and DIEUDONNÉ (J.). — Éléments de géométrie algébrique. — Paris, Presses universitaires de France, 1960, 1961, ... (Institut des Hautes Études Scientifiques, Publications mathématiques, 4, 8, 11, ...).

TDTE : GROTHENDIECK (Alexander). — Technique de descente et théorèmes d'existence en géométrie algébrique, I–VI, *Séminaire Bourbaki*, vol. 12, 1959/60, nos. 190 and 195; vol. 13, 1960/61, nos. 212 and 221; vol. 14, 1961/62, nos. 232 and 236.